

AGILE BUSINESS LEADERSHIP METHODS FOR INDUSTRY 4.0

Edited by
DR BÜLENT AKKAYA



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Chapter 1

A Review of Modern Leadership Styles in Perspective of Industry 4.0¹

Hasan Cinnioğlu

Abstract

The current Industry 4.0 era is considered not only as a process that dominates technological developments but also as a process that influences the leadership styles. Management 4.0 is essential for businesses to find and apply the appropriate technologies in the age of Industry 4.0. The leadership styles that business managers will adopt in order to be successful in this process and to survive in an intensely competitive environment can play an important role. At this point, a significant problem arises: identifying leadership styles that will bring success. In this context, the primary purpose of this chapter is to explain the modern leadership styles that business managers can adopt or follow in the age of Industry 4.0. In line with this purpose, the chapter first describes the historical development of leadership, leadership theories and modern leadership styles, such as transactional, transformational, technological, strategic, visionary and agile leadership, and all

these concepts are discussed based on the Industry 4.0 perspective.

Keywords: Industry 4.0; transactional/transformational leadership; technological leadership; strategic leadership; visionary leadership; agile leadership

Introduction

Globalization, changes and developments in information and communication technologies, increasing competition and the fact that the human element stands out in many areas have resulted in a profound change in the environment, in which businesses operate. Such an environment has forced businesses wishing to achieve their goals to give up their classical management approach and to use modern management techniques. As a result of these developments, the effectiveness of the classical management approach has decreased, which considers businesses as mechanical organizational structures, instead gives importance to effectiveness and efficiency and ignores the working human element or evaluates them as just a cost factor. For this reason, business executives must first create a common mission and vision. Then they must be able to guide or motivate their employees that can be qualified as viewers, for these purposes. In other words, business executives should be leaders rather than managers. Today, the concept of leadership emerges as an issue that has started to become much more critical than before. Because, human resource, which constitutes one of the essential elements of the businesses that desire to achieve a competitive advantage, plays a vital role in the Industry 4.0 era. The creation and sustainability of such a human resource are directly related to leadership ([Yüksel & Genç, 2018](#)). Industry 4.0 has brought along several opportunities and threats for businesses at the same time. Leaders should take the best advantage of these opportunities and take measures against threats. In other words, in order to benefit from the advantages of the fourth industrial

revolution, business leaders should put this concept at the top of their agenda ([Davutoğlu, 2018](#), p. 4041). The new era called Industry 4.0 is not only a result of the development of new technologies but also a result of a new entrepreneurial mindset ([Stere, 2017](#), p. 99). Creating a digital culture and acquiring the appropriate skills are as crucial as finding or implementing appropriate technologies in the age of Industry 4.0. For this reason, a leadership culture based on Industry 4.0 needs to be developed in enterprises ([Oberer & Erkollar, 2018](#), p. 5). This new mindset and technological developments have inevitably affected leadership theories. The main aim of this study is to examine modern leadership types within the scope of Industry 4.0. In this context, firstly, the concept of leadership, its characteristics and behavioral and contingency leadership theories are explained, and finally, modern leadership styles are discussed.

1. Concept of Leadership

Man is a social being, and people live together as a society. As a result, leadership is a concept that emerges from the need to guide the groups that make up the society ([Ercan & Sığı, 2015](#), p. 96; [İbicioğlu, Özmen, & Taş, 2009](#), p. 2). The concept of leadership has become even more important with the changes occurring in information and communication technologies, the increase in competition and the human element gaining importance for businesses. Leadership is one of the most important concepts within the scope of management and behavioral sciences, and it is one of the topics that have been researched a lot in parallel with the changes and developments started in the twenty-first century ([Telli, Ünsar, & Oğuzhan, 2012](#), p. 135). Leadership is a process related to the activities of the leader. When the literature is examined, it is possible to see that leadership is the subject of many pieces of research ([Shin, Heath, & Lee, 2011](#), p. 169). It is also worth noting that researchers define leadership in different ways ([Yukl, 2010](#), p. 20)

Leadership is the power to influence the actions and thoughts of others ([Abraham, 1992](#), p. 2), the ability to motivate and influence group members to contribute to the objectives and effectiveness of

the enterprise ([House, 1971](#), pp. 323–324) and the ability to build and develop a team that can outcompete in a competitive environment ([Hogan & Robert, 2005](#)). Leadership is defined as enabling one's followers to make emotional and physical efforts or influence and motivate followers to achieve the goals and objectives of the organization ([Koçel, 2014](#), p. 668; [Winston & Patterson, 2006](#), p. 7).

In general, the leader is the person who has the capacity to solve different problems in an organization or between people ([Mumford, Zaccaro, Harding, Jacobs, & Fleishman, 2000](#), p. 14; [Zaccaro, Mumford, Connelly, Marks, & Gilbert, 2000](#), p. 38), who creates, manages and sometimes changes the organizational culture ([Schein, 2004](#), p. 223), brings people together for a purpose, and has an influence on people thanks to his/her motivation and personal characteristics ([Dinçer & Bitirim, 2007](#), p. 61).

2. Theories of Leadership

It is seen in the literature that many leadership theories have been developed to date. When the historical development of leadership theories is examined, we encounter four main periods: trait theory, behavioral theories, contingency theories and modern theories ([Acar, 2013](#), p. 8). In this chapter, trait, behavioral, situational and modern theories developed based on the above characteristics will be explained.

2.1. Trait Theory

According to the trait theory, the most critical factor that determines the effectiveness of the leadership process is the traits of the leader ([Koçel, 2014](#), p. 676). Researches on the subject between 1930 and 1950 examined different leaders in terms of success and tried to determine which traits of leaders are essential on the way to success. Not only the physical properties of the leaders but also their psychological characteristics were examined ([Şahin, 2012](#), p. 145). According to the trait theory, leadership is not an acquired feature, but an innate feature of the person. These inherent traits of leaders

distinguish them from other people. Therefore, the only way to define the phenomenon of leadership is to determine the qualities of this type of person in detail ([Pazarbaşı, 2012](#), p. 23). In the trait theory, the physical characteristics of the leader, such as gender, age and height are discussed as well as his characteristics such as his sense of independence and self-confidence, and his inherent creativity and intelligence ([Demir, Yılmaz, & Çevirgen, 2010](#), p. 132).

The most significant criticism of this theory is that sometimes individuals among group members have better characteristics than the leader, but these individuals do not appear as leaders. This is a situation that does not comply with the trait theory. This has made it necessary to focus on different variables in addition to the features mentioned above to understand the leadership phenomenon fully. Another problem with this theory was the difficulty of measuring the characteristics of the leader and the understanding of some particular features different than they are ([Koçel, 2014](#), p. 678).

2.2. Behavioral Theories

Behavioral leadership theory is based on the approach that the main factor that makes the leader effective is the behaviors he/she exhibits in the leadership process, rather than his/her characteristics. The behaviors of the leader (such as his communication with his audience, the delegation of authority, planning and control and determining the goals) are important factors that determine the effectiveness of the leader ([Bakan & Büyükbeşe, 2010](#), p. 74).

The behavioral approach refers to the behaviors that the leader shows while leading in certain situations. Among the studies that have contributed significantly to the development of behavioral leadership theory are the leadership studies conducted at Ohio State University. Within the scope of these studies, leadership behaviors were tried to be determined by conducting many pieces of research about leadership ([Koçel, 2014](#), p. 679). Ohio State University leadership studies have demonstrated two basic leadership behaviors: behaviors that prioritize the individual and behaviors that focus on business. Leaders who prioritize individuals' exhibit behaviors such as taking care of people, caring about their feelings,

and taking into account their needs and wishes. On the other hand, leaders who focus on the business display behaviors toward achieving the given task or the specified goal. Performance, procedures or standards are more critical for such leaders than human relations (Yukl, 2010, p. 104).

Other studies that have contributed significantly to the development of behavioral leadership theory are leadership research at the University of Michigan. As a result of leadership studies in many industries, Michigan leadership studies have concluded that similar to studies at Ohio State University, leaders often display behavior that prioritizes individuals or businesses (Bojadjev, Kostovski, & Buldioska, 2015, p. 213). According to the studies of the University of Michigan, the business-oriented leader regularly supervises the employees in the working environment, uses the punishment as a motivating element and uses its status-based authority. On the other hand, leaders who prioritize individuals do not hesitate to delegate authority and try to improve the working conditions and personal qualities of employees (Koçel, 2014, p. 681).

Based on these two leadership behaviors, Robert Blake and Jane Mouton prepared a matrix called the Managerial Grid. In a Managerial Grid, the behaviors of leaders are grouped in two different dimensions: Concern for Production and Concern for People (Koçel, 2014, p. 682).

Another behavioral leadership theory is the X and Y theory developed by Douglas McGregor. According to McGregor's theory, employees are composed of two different classes with opposite behaviors. Employees in the X group are people who do not like to work, need to be checked continuously and escape responsibility. The employees in-group Y are people who enjoy working, do not need to be controlled and do not escape responsibility. According to this theory, if the leader thinks that the employees have the characteristics of the group X, he/she will be autocratic, and if he/she believes that they have the characteristics of the group Y, he/she will be more democratic (Kozak, 2008, p. 494).

In the model, he developed as a continuation of the studies of the University of Michigan, Rensis Likert has brought the behaviors of

leaders under four groups. The leadership variable was handled as trust in the subordinates, the degree of participation of the subordinates, and taking the opinions of the subordinates. According to this model, exploitative/authoritative and benevolent/authoritative leaders generally exhibit business-oriented behavior, while participative and consultative leaders exhibit people-oriented behavior (Sinha, 2008, p. 271).

2.3. Contingency Theories-Situational

The general assumption of the contingency theories is that different situations require different leadership styles. According to this approach, there is no single leadership behavior that can be effective anywhere and anytime (Kılınç, 1995, p. 60). According to this theory, the most appropriate leadership behavior will vary depending on the situation. According to this theory, the most significant deficiency of behavioral theories is that they define business-oriented or people-oriented leadership, but they do not specify in which cases people- or business-oriented leadership will be productive. However, according to the contingency theory, both types of behaviors can be equally effective under certain circumstances (Koçel, 2014, p. 687). There are many different theories within the context of contingency approaches. Some of these can be listed as Fiedler's contingency approach, path-goal approach, Hersey and Blanchard's contingency model, Reddin's three-dimensional leadership model and Vroom-Yetton leadership model (Peretomode, 2012, p. 14).

According to **Fiedler's Contingency Approach**, three different variables determine the effectiveness of leaders: leader-member relations, task structure and the positional power of the leader (Fiedler, 1972, p. 7; Justis, 1975, p. 160). These variables can reveal three different situations for the leader. Among these situations, *very favorable situation* refers to a situation where all three variables are very high; *moderately favorable situation* refers to a situation where one variable is low and other variables are high; and *unfavorable situation* refers to a situation where all three variables are very low (Peretomode, 2012, p. 16). According to Fiedler, who states that there are two different types of leadership for these environments

such as task- and relationship-oriented, and there is no single effective leadership style for all conditions ([Utech & Heider, 1976](#), p. 607).

Path-Goal theory is based on the expectancy theory of Vroom, one of the motivation theories. According to this theory, people's behavior is generally influenced by two factors: the *expectancy* for a particular output to be achieved as a result of a particular behavior, and the *valence* that expresses the individual's benefit or satisfaction ([House, 1971](#), p. 322). Leaders can motivate their employees by influencing their expectancies and valences ([Greene, 1979](#), p. 23; [House, 1996](#), p. 325). In the path-goal theory, leaders generally show four different leadership behaviors ([Malik, Aziz, & Hassan, 2014](#), p. 171): Directive leader, supportive leader, participative leader and achievement-oriented leader.

Hersey and Blanchard's contingency model focuses on two issues related to leadership behavior: task behavior and relationship behavior. Within the scope of task behavior, the leader determines the duties of individuals and how and by which method these tasks should be performed. In relational behavior, the leader continually communicates with individuals, listens to them, encourages them, clarifies their opinions and supports them ([Paksoy, 1993](#), p. 19).

The main point in *Reddin's 3-D leadership model* is managerial effectiveness. Reddin based his theory of 3-D on this basis. Reddin added effectiveness to Blake and Mouton and Ohio State studies and stated that leadership behavior varies on these three pillars. In other words, he added the effectiveness dimension of leadership to the relationship and task dimension. The effectiveness mentioned here is the degree to fulfill the tasks required by the position of a leader. It suggests that the effectiveness of the leader should be measured not by what he does but by what he has achieved or whether he is successful ([Ömürgönülşen & Sevim, 2005](#), pp. 92–93).

The last model we will consider within the context of situational leadership theories is the *Vroom Yetton leadership model*. According to this model, there is no single leadership style that can apply in all situations. A leadership style can only be determined as a result of the information obtained as a result of the analysis ([Reber, Auer, & Maly, 2004](#), p. 416). Vroom and Yetton have suggested decision

making and leadership processes that can be applied in different situations. This is a model that suggests the leader's relationship with his subordinates in the decision-making process or the degree of participation of the subordinates (Vroom & Jago, 2007, p. 20).

3. Modern Leadership Theories

Rapid changes in the world order due to globalization have brought many innovations into our lives. In this context, leaders in businesses have started to gain more importance, and a change in the understanding of leadership has emerged and became notable for businesses (Değirmenci, 2012, p. 26). Traditional leadership theories, which are insufficient to explain the concepts of leadership and their behaviors, have been replaced by new leadership theories (Ceylan, Keskin, & Eren, 2005, p. 33). It is seen in the literature that there are many modern leadership styles. However, Industry 4.0, which is based on a superior creative ability, requires participative management, organizational flexibility, a form of leadership that is based on the wishes and needs of individuals, supporting individuals, emphasizing merit and ethical behavior, continually increasing knowledge and putting science into practice. For this reason, modern leadership styles will be discussed that include such features, while modern leadership types are explained (Yüksel & Genç, 2018).

3.1. Transactional Leader

This leadership theory was introduced by J. M. Burns in the late 1970s and was later developed by different researchers led by B. M. Bass. The basis of this theory is to establish a business-related relationship between the leader and group members and the behaviors that the leader will exhibit while performing the task (Koçel, 2014, p. 695). Bass (1985, pp. 27–28) defines the transactional leadership as:

a leadership style where the leader informs his followers quite clearly about his expectations and also

explains what reward they can receive in return for their expected performance and effort.

[Antonakis, Avolio, & Sivasubramaniam \(2003, p. 265\)](#) describe transactional leadership as a process of change based on fulfilling contractual obligations, and generally setting goals, controlling and monitoring the results. In transactional leadership, the duties and powers of the employees, the performance expected from them, the obedience to their leaders, and the awards they will receive as a result of achieving the tasks are clearly stated. In other words, transactional leadership is based on the exchange between the leader and his followers ([Eren & Titizoğlu, 2014, p. 279](#)). Generally, different behaviors can arise between the leader and group members who come together to do a specific job in this type of leadership. According to [Bass \(1999\)](#), these behaviors can take place in four different dimensions: contingent reward, active management by exception, passive management by exception, and laissez-faire leadership ([Antonakis et al., 2003, p. 265](#); [Bass, 1999, p. 11](#)).

In the rewarding dimension, the leader rewards the employees as a result of their performance. These awards can have material or moral dimensions. Therefore, employees believe that they will be rewarded if they fulfill the expected goals ([Avolio & Bass, 2002, p. 2](#); [Bass, 1990, p. 22](#); [Kareem, 2016, p. 10](#); [Ravazadeh & Ravazadeh, 2013, p. 166](#); [Yavuz & Tokmak, 2009, p. 19](#)). According to [Bass \(1985, p. 34\)](#), if transactional leaders want to gain efficiency from their followers, they can reward in two different ways: praising a job well done and increasing wages or promotions. Management by exception, which is generally less useful than other practices, may be necessary in some cases. Management by exception is usually applied when the expected performance is not realized. Management by exception is divided into two as active and passive. In the active management by exception, the leader monitors the performance of the employees and makes corrective decisions by making interventions when necessary ([Avolio & Bass, 2002, p. 3](#); [Kareem, 2016, p. 10](#); [Koçel, 2014, p. 695](#)). It is important that the information should be distributed to employees by leaders with an

accurate perception of knowledge for corrective decisions (Akkaya & Üstgörül, 2020, p. 53). The leader focuses his attention on irregularities, errors and deviations from standards, complaints and failures (Bass, 1985, pp. 33–36; Karip, 1998, p. 449).

In the case of passive management by exception, the leader waits for the problems to arise in the business and does not take any measures to prevent the problems. He/she only receives solution suggestions from employees (Bass, 1990, p. 22; Bass, 1999, p. 11; Yavuz & Tokmak, 2009, p. 19). In passive management by exception, the leader does not interfere with problems until they become severe and believes that the problem must become chronic before attempting to solve the problem and advocates the belief of “if it ain’t broke, don’t fix it” (Bass, 1985, pp. 33–36; Karip, 1998, p. 449).

In Laisses-Faire leadership, which is the most passive form of leadership styles, the leader generally frees the employees about goals and performance standards and does not interfere in any way. The leader avoids interfering or making decisions when there are essential issues in the business. He/she usually does not appear when he/she is needed (Avolio & Bass, 2002, p. 4; Bass, 1990, p. 22; Koçel, 2014, p. 695; Ravazadeh & Ravazadeh, 2013, p. 166). Transactional leaders can successfully carry out many activities such as improving the effectiveness of the organization, avoiding unnecessary risks, achieving the goals and objectives and increasing the motivation of the employees (McCleskey, 2014, p. 122).

3.2. Transformational Leader

The transformational leadership concept, first introduced in Dawston’s (1973) Rebel Leadership, was later developed by James McGregor Burns in 1978 (Eraslan, 2004, p. 4). According to Burns, transformational leadership is a leadership style for motivating the followers using economic, political or similar forces and values to achieve the goals they form independently or mutually (Morçin & Çarıkçı, 2016, p. 99). Unlike other leadership styles that focus on

logical processes, transformational leadership focuses on emotions and values ([Akbar, Sadegh, & Chehrazi, 2015](#), p. 3490).

The transformational leader is the person who brings the organization to superior performance by performing change and innovation in an organization. To achieve this, he must have a leading vision and ensure that his followers adopt this vision ([Koçel, 2014](#), p. 696). The transformational leader provides innovation, improves performance, creates mission and vision and ensures the personal development of his followers ([Bass, 1985](#), p. 33). [Yukl \(1999\)](#) defines transformational leadership as:

a process of creating a commitment to realize the mission and goals of the organization by making some changes in the attitudes and behaviors of the members of the organization. ([Kent & Packianathan, 2001](#), p. 136)

Defining transformational leadership as a type of leadership that delegates authority and determines organizational goals with its followers, [Bass \(1990, p. 21\)](#) states that transformational leadership emerges when a leader expands and raises the interests of the employees, raises the group's awareness of goals and mission and creates a sense that he/she keeps the organization's interests in front of his/her interests. Bass stated that transformational leaders could achieve these results through charisma, the inspiration for subordinates and meeting the emotional needs and desires of subordinates ([Bass, 1990, p. 21](#)). Transactional leadership theory and transformational leadership theory are generally discussed together. This is because Burns suggests that the leader, who demonstrates transactional leadership behavior, can also display transformational leadership behavior ([Yavuz & Tokmak, 2009, p. 18](#)). In other words, transformational and transactional leadership styles reflect two different leadership styles in a complementary way ([Demir & Okan, 2008, p. 73](#)). Burns demonstrated the difference between transactional and transformational leadership as follows: While transactional enables both parties to fulfill their objectives in economic, political or psychological ways, transformational

leadership ensures that the objectives are fulfilled by transforming the ethical standards, behavior and motivations of the group members ([Simola, Barling, & Turner, 2010](#), p. 180).

Bass has presented seven different dimensions for transactional and transformational leadership models: Contingent reward, management by exception, Laissez-faire leadership, idealized influence, individualized consideration, inspirational motivation and intellectual stimulant ([Avolio, Bruce, Bass, & Jung, 1999](#), p. 441; [Bass, 1990](#), p. 22). The first three dimensions are transactional leadership; and the last four dimensions are transformational leadership dimensions.

In the transformational leadership, charisma is considered a divine gift ([Nikezic, Puric, & Puric, 2012](#), p. 286), and followers have strongly connected to the leader thanks to the charismatic characteristics of the leader ([Mittal, 2015](#), p. 27; [Yukl, 1999](#), p. 293). Charisma is the ability of a leader to spread his thoughts, behaviors, values, beliefs and performance to other people, and also to influence their thoughts, behaviors, values, beliefs and performance ([House, Spangler, & Woycke, 1991](#), p. 366). Idealized Influence leaders ensure pride in collaborating with them, give up their interests for the good of the group, act in a way that others respect them and show a sense of trust and strength ([Karip, 1998](#), p. 447). In the idealized effect behavior, leaders talk about the values and beliefs that they attach most importance, indicate the importance of having a strong sense of purpose and take into account the moral and ethical consequences of their decisions. This leadership dimension includes behaviors that emphasize the importance of having a common mission (sense of duty) ([Karip, 1998](#), p. 447). Individualized Consideration refers to that the leader deals individually with each member in the group and takes into account their needs, wishes and abilities. He/she also coaches them to ensure the personal development of his/her followers ([Avolio & Bass, 2002](#), p. 2; [Bass, 1990](#), p. 22; [Demir & Okan, 2008](#), p. 76). A transformational leader tends to the personal development of their followers. He/she evaluates the potential of his/her followers to fulfill their current duties and take more responsibility in the future. He/she gives individual examples or tasks to both meet the needs of the

organization and to significantly improve the abilities and motivations of the followers (Bass, 1985, p. 35). In Inspirational Motivation, the leader determines a vision desired by the group members and places high moral, ethical standards, a sense of purpose and purpose in the activities required for this vision to be realized. Thus, he/she inspires by motivating group members within the framework of this vision (Avolio & Bass, 2002, p. 2; Bass, 1990, p. 22; Demir & Okan, 2008, p. 76; Koçel, 2014, p. 696). In intellectual stimulant, the leader criticizes old traditions and beliefs and appeals to his followers' intellectual sides through introducing new perspectives, new standards of doing business and new business ideas. He/she also enables these new methods or ideas to be implemented by group members (Bass, 1990, p. 22; Demir & Okan, 2008, p. 76; Minister et al., 2015, p. 204; Yukl, 2010, p. 278). Transformational leaders create appropriate conditions for their followers to reveal their knowledge and skills and use them effectively and encourage their creativity to develop new methods and techniques (Acar, 2013, p. 9; Akkaya, Koçyiğit, & Tabak, 2018, p. 187).

3.3. Technological Leadership

The digital revolution has transformed the lives of people in the way that people communicate with technology rather than with each other. This change powers managers to comprehend these improvements in an environment and try to adapt them (Akkaya & Tabak, 2020, p. 1). Technological leaders are expected to produce strategies on how to use modern technologies to meet the needs of the organization. To meet these expectations, training managers create a vision, train staff, set priorities, allocate resources among staff and ensure organizational order. Setting up a technology-based assessment and evaluation system, following a comprehensive selection process to meet the specific needs of the training process and getting help from experts in and outside the training area are examples of technology leadership activities (Samancioğlu, Bağlıbel, Kalman, & Sincar, 2015, p. 79).

Bowen et al. (2013) stated that Daugherty et al. (2013) suggested a philosophical definition that the technological leader allows others

to use, manage, evaluate and understand the technologies of the designed world effectively and successfully. The technological leader is critical as our complex global society is increasingly dependent on technology. The technological leader is equipped in his own contexts to make informed and valued decisions and participate in leading technological development. [Tanzer \(2004\)](#) defines the technological leader as the person who manages and directs the employees of the institution for the efficient and effective use of technology in the organization and establishes the necessary coordination. On the other hand, [Valdez \(2004\)](#) defines technological leadership as coordination of strategies that include corporate leaders' understanding of developing technology strategies and practices to support the use of technology in organizations.

Visionary personality (developing a shared vision by creating a strategic plan that expresses how to use technology to improve learning), digital age learning culture (inspiring a culture of innovation and collaboration that enables time and space to be discovered and tested with digital tools), systematic development (ensuring that resources that support the effective use of technology for learning are sufficient and scalable to meet future demand), excellence in professional practice (developing the skills needed to lead change and drive systems, advance systems and develop a continuous improvement mindset on how technology can improve learning) and digital citizenship (ensuring that all students have talented teachers who actively use technology to meet their learning needs) are the concepts that technological leader should possess ([Hacıfazlıoğlu, Karadeniz, & Dalgıç, 2010](#), p. 537; [ISTE, 2009](#)). In other words, it is essential for school administrators, especially school principals, to have a high level of technological leadership characteristics in terms of reaching the information using technology, to maintain, synthesize, develop and spread to achieve the educational and training objectives ([Durnalı, 2018](#), p. 38). It would be beneficial for managers who want to lead employees successfully with technology to develop a technology leadership team, evaluate professional development practices, technical needs, employee learning and performance and address cybersecurity effectively. In other words, managers who have adopted the technological leadership

understanding should determine the technology goals and vision in the organization, lead the technology, be a model for their employees, provide the necessary resources to promote and support the use of technology in the organization, implement professional development activities and ensure the integration of technology with the organization ([Grady, 2011](#)).

People have to follow and learn about these developments in a society shaped as a result of continuous developments in the field of technology. Because information is continuously renewing and increasing, for this purpose, leaders who want to integrate their technological leadership understanding with the organization should realize in-service training to develop the competencies of the managers in the field of technological leadership ([Görgülü, Küçükali, & Ada, 2013](#), p. 69).

According to [Cantürk and Aksu \(2017\)](#), the understanding of technological leadership first begins with understanding and comprehending the importance of technology. If a leader cannot grasp the importance of technology, he cannot integrate the technology with his organization and cannot ensure that the employees benefit from the technology. Therefore, the leader must first grasp the importance and benefit of the concept of technology for the organization and determine what needs to be done to increase the performance of the organization and what technological tools are needed for it. Leaders should also encourage employees to use technological tools by providing in-service training.

3.4. Strategic Leadership

Strategic leadership is becoming the primary focus for business and academics and the most important subject of contemporary organizations, depending on the recent changes. Without effective strategic leadership, a company's reaching or maintaining its competitiveness will be significantly restricted ([Lear, 2012](#), p. 3). In other words, organizations need leaders who can think strategically, create a vision, predict the future and guide their employees in order to keep up with the dizzying change process in the world. Such leaders are called strategic leaders ([Akyüz, 2018](#), p. 45).

Although the strategy, which is defined as the art of providing and organizing action to reach a determined goal ([Aktan, 2008](#), p. 5), is mostly taken up in the military sense, it has been used frequently in the field of management recently. In the field of management, strategy refers to the ways an organization will follow to achieve its goal. There are various definitions regarding this concept in the literature.

[Ireland and Hitt \(1999](#), p. 63) define strategic leadership as the ability to predict, provide flexibility, think strategically and work with others. Effective strategic leadership practices can help companies improve performance while competing in turbulent and unpredictable environments. [Guillot \(2003](#), p. 10) defines strategic leaders as leaders with the highest conceptual ability in long-term planning, vision making and decision-making. [Uğurluoğlu and Çelik \(2009\)](#) define the strategic leader as a person who can create a realizable vision for the organization, think strategically and can predict the future. [Yang, Sun, and Eppler \(2009](#), p. 6) define strategic practices as a dynamic, repetitive and complex process consisting of a series of decisions and activities of managers and employees affected by interrelated internal and external factors, and the strategic leader is the person who turns strategic goals and plans into reality. Because today companies operating in emerging markets face with certain problems especially in turbulence, complex and changeable environment ([Akkaya & Tabak, 2017](#), p. 119). According to [B. J. Davies and Davies \(2004](#), p. 13), the strategic leader should have the ability to develop strategic and organizational processes, guide and develop people, develop culture and value systems, develop different institutional competencies and create active networks. [Boal and Hooijberg \(2000](#), p. 515) argue that the essence of strategic leadership includes learning capacity, capacity for change and managerial wisdom.

It can be said that the adoption of a strategic leadership approach in an organization improves the job performance of the employees ([Türk & Akbaba, 2017](#), p. 193). [Finkelstein and Hambrick \(1997](#), p. 221) argue that strategic leaders can be selected from managers such as general managers, front line managers, board members and department managers. Managers who adopt a strategic leadership

approach should have four leadership characteristics: managerial (ensuring the organizational order), political (ability to understand employees, manage and direct the employees to reach specified goals and objectives), transformational (paying attention to employees' opinions and motivating them) and ethical (expressing commitment to corporate values). The existence of the stated strategic leadership characteristics in a manager will provide the opportunity to be successful in cases of uncertainty (Uğurluoğlu & Çelik, 2009). It will be beneficial for subordinates to reach the goals and objectives of the institution, if the strategic leader follows the activities of the employees, wants them to act in accordance with hierarchical relations and takes responsibility for their work. Moreover, the leader, who can match the needs of the employees and the goals of the organization, implements a system that rewards successful activities and gives importance to the ideas of its employees, will be able to increase the service quality of the employees. One of the critical factors that increase the performance of the employees are the perception of strategic leadership by the employees (Türk & Akbaba, 2017, pp. 193–194).

3.5. Visionary Leadership

The concept of vision refers to thinking and designing the future. Visionary leaders are those who dream and design the future of the organization and society (Erçetin, 2000, p. 93). Vision enables people to act. Organizations that have a vision strive for a purpose. Organizations that do not have a vision cannot progress, because they do not know which way to go (Taylor, Cornelius, & Colvin, 2013, p. 567).

Visionary leadership is defined as the ability to create and communicate visions that can influence and mobilize society and organization (Çelik, 2000, p. 170). Leadership is to be far-sighted, to set vision and goals for the future of the organization to achieve those goals (Aksu, 2009, p. 2439). Visionary leadership is the ability to create an attractive future vision as well as being realistic and safe for the organization (Tekin & Ehtiyar, 2011, p. 4010).

Visionary leaders are people who can set new goals. Foresighted visionary leaders evaluate employees' abilities and make them believe they can do better (Özden, 1999, p. 48). Visionary leaders carry employees to the goals desired by the organization (Güçlü, 2016). In other words, it can be said that visionary leaders are far-sighted.

Studies on leadership show that the importance of the visionary leadership concept has increased after 1990. Visionary leaders are essential in terms of eliminating the uncertainties of the organizations regarding the future (Aksu, 2009, p. 477) and taking the necessary steps for possible crisis periods and ensuring the survival and sustainability of the organization.

In a rapidly developing and changing world, organizations must keep up with this change in order to survive. Visionary leaders are the leaders of the future as the source of this change and development (Çelik, 2000, p. 165). Visionary leadership can also be defined as the ability to create innovations. Visionary leaders, who inspire and motivate employees, and form the vision and mission of the organization, provide a high level of organizational harmony. Furthermore, they increase job loyalty, job performance and trust (Taylor et al., 2013, p. 568). Visionary leaders, who determine the plans that will bring the organization to success with the employees, are people who work hard to reach the plan, create high motivation in the employees, maximize employee performance and have the necessary awareness for the development of the organization (Tekin & Ehtiyar, 2011, p. 4010). While visionary leaders create a vision on the one hand, they also explain the vision to the employees and ensure that they adopt the vision. The route to be followed should be determined in order to reach the targets, and this route should be conveyed to the employees orally and in writing. Verbal and written ways are not sufficient in transferring the vision to the employees. Leaders should also be an example of employees with their behaviors (Robbins, 2000, pp. 144–145). The verbal and written approach is not enough to transfer the vision to the employees. The leader should be an example for the employees with their behavior (Robbins, 2000, pp. 144–145).

Visionary leadership has three main roles: showing the way, walking on the way and being the way (Çelik, 1997, p. 472). *Showing the way*, which is one of the roles of the visionary leader, is of great importance. The concept of way can be defined as the future image of vision. Visionary leaders determine the most appropriate way for the organization to achieve its goals. Thus, they direct the employees in that way. Organizations that do not make plans for the future cannot evolve. Visionary leaders are managers who set the vision and shape the future with their farsightedness and intuition (Yeyrek, 2018, p. 53). Visionary leaders create new ways (*being the way*) thanks to the vision they set. One of the essential features of visionary leaders is that they guide the employees in order to progress to the vision they have determined. Visionary leaders sacrifice and take risks to achieve their intended goals. Employees walk safely on the way, opened by visionary leaders and reach their goals at the end of the way (Çelik, 1997, p. 473). Another feature of visionary leaders is to lead employees to achieve goals (*walking on the way*). Walking on the way is as important as showing the way. If the predetermined way and the walked way are different, it is not possible to reach the targets. Moreover, despite the difficulties that may be encountered, it is crucial not to deviate from the specified way in order for the organization to achieve its goals (Acar, 2006, p. 19).

3.6. Agile Leadership

The demand for agile skills is increasing in organizations. It is quite clear that leaders around the world are increasingly using agile principles and practices to achieve their goals. Successful organizations have become aware of the necessity of not only agile principles but also agile leadership (Macintyre, 2017, p. 3).

Agile leadership can be defined as the leader who can find solutions for the problems encountered in the organization, who is creative, can adapt quickly to changes and developments, cooperate with employees, make mistakes, is reliable, trusting employees, and turns crisis into performance (Collyer & Warren, 2009, p. 359; Ferreira, Lima, & Costa, 2012, p. 712). Organizations should keep

their agility levels at the maximum level to ensure sustainable competitive advantage and customer satisfaction ([Akkaya & Tabak, 2018](#), p. 186). Agile leaders are leaders who can make fast, accurate decisions in uncertainties, prioritize innovation and science. They are role models for employees. In other words, they exhibit the behavior they expect from their employees first and lead by example. Agile leaders reassure the organization psychologically and delegate authority ([Acmagile, 2020](#)). In agile leadership, managers/leaders should empower their teams, inform them from technology and environmental changes, adapt them to those changes, provide self-training to employees and allow them to work on a specific project in order to get maximum profit and value in this competitive era ([Akkaya, 2020](#), p. 391). Agile leaders who can keep pace with change are capable of problem solving. They are also leaders who take precautions and adapt to changing conditions in any crisis in the organization. Agile leaders are leaders who can guide employees and enable them to define, spread and maintain the vision of the organization ([Parker, Holesgrove, & Pathak, 2015](#), pp. 118–119).

Agile leadership has nine main features listed below: ([Kaynakdergisi, 2020](#))

1. to guide through behavior, not words;
2. to improve the institution through a productive feedback culture;
3. to improve the results by improving the quality of thought;
4. to reveal important ideas at all levels of the organization;
5. to transfer authority and power appropriately;
6. to create positive emotions as the basis of creativity and discovery;
7. to cooperate;
8. to ensure that people find meaning in their work; and
9. to provide leadership behavior at all levels of the organization.

Organizations need to use dynamic, more flexible and agile abilities effectively. It has become inevitable for enterprises to change the organizational structure and become more flexible and agile organizations to keep up with technological innovations and to overcome competition ([Koçyiğit & Akkaya, 2020](#), p. 111). The

organizational structure of the organization, the goals of the organization and the organizational culture are at the core of agile leadership ([Scrum, 2020](#)). In short, agile leaders must first ensure that the organization is flexible, strive for the best of the organization with continuous improvement and ensure customer satisfaction. Employees must also support this mission. The most critical factors in agile leadership are to learn and support by experimenting, to provide more customer satisfaction and to seek better and faster ways to achieve these results ([Cinnioğlu & Salha, 2019](#), p. 94). Agile leaders develop a sense of vision in employees and hold group discussions all the time. They also help employees to grasp the vision firmly, determine business value and business priorities, make difficult decisions and focus on goals ([Parker et al., 2015](#), p. 119). Agile leaders who empower and motivate employees toward goals give confidence to employees to do their jobs, collaborate with them and enjoy the job ([Macintyre, 2017](#), p. 4).

There are four different types of agile leadership: context-setting agility, stakeholder agility, creative agility and self-leadership agility ([Joiner, 2017](#), p. 7). *Context-setting agility* determines the need for change and desired results. *Stakeholder agility* understands and adapts to stakeholders. *Creative agility* plans and provides analytical and creative thinking to solve problems. *Self-leadership agility* reflects itself and tries new behaviors.

In summary, leaders who demonstrate agile leadership behavior can be said to have the following characteristics: they can keep up with and adapt to the continually changing and developing world, turn crisis moments into opportunities, make mistakes but accept the mistake they make, cooperate with employees, set an example for the employee with their behaviors, they can make quick and accurate decisions during uncertainties, act as a role model for the employee and provide confidence in the organization.

Key Terms:

Core Competence: Capabilities that can create value cannot be imitated, have no substitution, are rare and outperform competitors.

Dynamic Capability: Dynamic capabilities are something that can make the firm dynamic and are regarded as the ability to change and reconfigure ordinary capabilities.

Transactional Leadership: It is a type of leadership that informs the followers about his/her expectations very clearly and explains what kind of reward they should expect in return for the high performance and effort.

Transformational Leadership: The person motivates followers to achieve the goals that people create independently or mutually by using economic, political or similar powers and values.

Technological Leader: The person supports the use of technology in the organization, establishes the coordination necessary for the efficient and effective use of technology in the organization, develops technology strategies and practices, manages and directs the employees.

Strategic Leader: The person creates a realizable vision for the organization, can think strategically and foresee the future and has the highest conceptual ability in decision-making.

Visionary Leader: The person can create a vision that can influence and mobilize the society and the organization and can set new goals and farsightedness.

Agile Leader: A person that can keep up with and adapt to the continually changing and developing world, turn crisis moments into opportunities, make mistakes but accept the mistake he/she makes, cooperate with employees, set an example for the employee with his/her behaviors, make quick and accurate decisions during uncertainties, act as a role model for the employee and provide confidence in the organization.

Conclusion

The concept of Industry 4.0, which emerged with the change of production processes and the reshaping of consumer demands in

parallel with this change, was used for the first time at the Hannover Fair held in Germany in 2011 (Ünlü & Atik, 2018, p. 432). The developments and changes that Industry 4.0 creates in the business environment may cause the classic leadership behaviors to be less effective. Based on these changes and developments, new organizational values are created in the most effective way of producing, using, sharing and disseminating information. This makes it necessary to understand the Industry 4.0 revolution and the leaders who comprehend this revolution (Yüksel & Genç, 2018). Organizations need leaders who can make quick decisions, give importance to innovation and creativity and analyze dynamic environmental conditions thoroughly, namely agile leadership characteristics in the environmental conditions created by Industry 4.0 (Cinnioğlu & Salha, 2019, p. 94).

Moreover, leaders need to have strategic and visionary leadership behaviors in such an environment, such as seeing the future, creating a vision, providing flexibility and thinking strategically, to initiate changes that will create a realizable future for their businesses (Uğurluoğlu & Çelik 2009, p. 121). It is an existential requirement for leaders to be visionaries in the period of Industry 4.0 and beyond. Besides, the development of businesses or societies and humanity reaching a higher level of prosperity may depend on the leading leaders (Yüksel & Genç, 2018). The leadership approach per Industry 4.0, which focuses on knowledge, learning and innovation, is created by combining knowledge-oriented leadership structure and transformational and operational style (Gray & Meister, 2006, p. 148). As learning and innovation are more specific, transformational leadership may be needed in the Industry 4.0 process (Davutoğlu, 2018, p. 4045).

In general, the success of business leaders in the Industry 4.0 period depends on their having some features. Some of these features are as follows: contextual intelligence, pragmatic ethics, ecosystem intelligence, narratives out of the paradox, emotional intelligence, constructive imagination, global intelligence, knowledge management capability, supportive behavior, systems thinking, powerful and agile emotions (Zakaria, Nasir, & Akhtar, 2019, p. 546). Production processes need to be redesigned in order to adapt to the

globalization and the increasingly competitive environment and to respond rapidly to continually changing and developing consumer demands. The continuous changes in customers' demands and needs make it inevitable for leaders to constantly adapt to changes to keep their firms in a competitive environment ([Akkaya, 2019](#), p. 137). For this reason, the process of transformation to Industry 4.0 is essential in both developed and developing countries ([Ünlü & Atik, 2018](#), p. 432). Leaders have a significant role in the successful realization of this transformation process. Leaders should analyze the Industry 4.0 process, the developments caused by this process and the effects that these developments may have on businesses. However, it is also essential which leadership behaviors should be demonstrated in this process. Therefore, leaders are now required to abandon their classic leadership behavior and adopt the modern leadership styles required by this new age.

Recommendations for Future Studies

Factors such as globalization, changes in information and communication technologies and increased competition have led to the replacement of traditional leadership styles with modern leadership styles in the field of business. However, even these elements are continually changing and developing. For this reason, many modern leadership theories that are currently applied may either be ineffective or lose their function soon. Postmodern leadership theories will likely emerge. However, at this point, a problem arises, such as determining the types of leadership required by the modern age.

There are dozens of modern leadership styles in the literature. These styles of leadership may be essential in organizational behavior issues or the internal environment of the business. In other words, these leadership styles can have an impact on organizational commitment, job satisfaction, organizational citizenship, costs and production. The important thing here is not only to adopt a style of leadership that is effective in the internal environment of the business but also to adopt a leadership style that can take into account the external environment of the business. Because the

external environment of the business is continuously changing and improving, these changes and developments may bring along some threats as well as opportunities for businesses. Business leaders should be able to take advantage of these opportunities and take measures against threats. Achieving this depends on the ability of leaders to follow technology carefully, to keep up with changes quickly, to be creative, innovative and innovation-oriented and to develop a vision that is appropriate for the age. Therefore, future studies need to focus on new types of leadership that can accommodate these elements.

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Chapter 2

Industry 4.0 and Marketing 4.0: In Perspective of Digitalization and E-Commerce

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Abstract

Today, with the introduction of concepts such as big data, social media, corporate social responsibility and e-commerce have become part of our lives, thus the transition to Marketing 4.0 is accelerating. One of the most vital activities that affect consumers' purchasing decisions is e-commerce. E-commerce plays a decisive role in purchasing stages and processes. For this reason, e-commerce has become an issue that increases the importance of individuals in the purchasing process and decision every day and needs to be emphasized. In line with all this, the concepts of digital marketing and e-commerce will be handled in a broad framework in this study. Moreover, the digitalization of e-commerce sites will be held in the review of literature beside the environment and techniques that can be used within the scope of digital marketing. New approaches and trends that guide modern marketing are changing day by day. The main reason for this change is based on the rapid transformation in information and communication technologies.

For this reason, it is important for marketing managers to adapt to these transformations and to use media and techniques in digital marketing.

Keywords: Industry 4.0; Marketing 4.0; digital marketing; e-commerce; customer; digitalization

Introduction

Human history is constantly changing, developing and advancing. Although these advances have necessitated technological advances, they have caused many interrelated changes in all areas, where people are actively involved. Especially globalization and developments in technology have caused the world to turn into a small village. All the developments that have emerged with the technology have shown their existence in the fields of industry and marketing as a result of which new waves have emerged. As information technologies evolve and the world changes, however, both industry and marketing differ. The increasing integration of the Internet of everything into the industrial value chain led to the birth of Industry 4.0. The marketing process has developed in parallel with the technology. Accordingly, the number of studies based on Marketing 4.0 has started to increase day by day.

In order to provide a sustainable competitive advantage and to adapt to the digital changes occurring in our age, businesses should follow the differences in marketing and update themselves according to these transformations. In order to survive through and overcome the competition, companies should increase their marketing performance ([Akkaya & Tabak, 2017](#)). Today, there is a transition from a traditional marketing approach to a marketing style that cares about digitalization. In the digital world, businesses want to promote their products and services, to survive in tough competition conditions and to differ from their competitors. For this purpose, it implements applications related to digital marketing in line with its own interests and purposes. In particular, the developments in

technology have increased recently. With the widespread use of communication tools such as the Internet, smartphones and tablet computers, communication methods between businesses and consumers have also started to change. As a result of this change, very important changes have been experienced in the understanding of digital marketing and continue to be experienced.

Thanks to the internet-based software and applications of digital technology, there is unlimited free and effortless connection between networks. In this way, the effect of every situation, strategy and activity in traditional marketing has been minimized. With the development and spread of internet technology, consumers have started to meet most of their needs via virtual means. In this context, the proliferation of internet technology and the presence of users in virtual environments have paved the way for the emergence of electronic commerce. Today, most of the shopping is done via the internet. In the future, consumers will meet almost all of the products they need over the internet, while manufacturers will sell more on the internet than before.

The impact of electronic commerce, which occurs with the rapid development on information and communication technologies along with globalization, on the economic life is quite high. E-commerce has found application and progress areas in all segments of economic life with its widespread accessibility and convenience. The main reason for these developments is that e-commerce is available on the internet and instantly. E-commerce is widely used in economic life thanks to its features. The number of uses of e-commerce has become more diverse as the number of people using desktop computers and especially mobile phones increases. It is possible to make transactions at any place and time with mobile applications. Due to the rapid developments in portable devices and technologies, the internet becomes available to a wide audience, and the spread of e-commerce increases significantly.

Background

In this study, firstly, the concepts of Industry 4.0 and Marketing 4.0 are mentioned and a broad explanation about e-commerce is made.

In the continuation of the study, the concept of digital marketing was discussed and information about the environment and techniques used in digital marketing is given today.

Industry 4.0

From past to present, the world has experienced three major industrial revolutions. The first of these is Industry 1.0, which started with the discovery of the steam machine for the first time in the industrial sense and aims to increase production. This revolution was followed by Industry 2.0, which existed as a transition to mass production in the early twentieth century and paved the way for benefiting from electrical energy. Subsequently, Industry 3.0, where production systems ceased to be analog and digital systems took place in the industry, emerged. These three industrial revolutions that took place were based on increasing productivity in production ([Can & Kıymaz, 2016](#)). As the developments in information technologies increase and the world changes, marketing and industry also change. The differentiation of the expectations and desires of the customers who purchased only the existing product in the 1960s complicated the production process of the enterprises. Thus, although businesses now need interdisciplinary work, they have introduced Industry 4.0, where all objects communicate and interact with each other over the internet.

Industry 4.0: It is planned that all units that have a direct or indirect relationship with production will work together and envisage that the software and information technologies of digital data will work in an integrated way ([Schuh, Potente, Wesch-Potente, Weber, & Prote, 2014](#)). The main goal of Industry 4.0 is the creation of smart products, procedures and processes. The main priority requirements within Industry 4.0 include real-time data monitoring, having instructions to control production processes, and monitoring the status of the product involved ([Almada-Lobo, 2015](#)). Industry 4.0 does not only consist of machines' communication, but also it has wider coverage. It is a comprehensive process that affects different scientific fields from genetics to information technology. Features that differentiate this revolution from other revolutions: It is the

interconnection of technological developments by triggering each other, acting in coordination and the development of all areas under the influence of each other ([Bulut & Akçacı, 2017](#)).

Industry 4.0 is a production understanding that must be integrated for businesses aiming to reach or hold competitive advantage in our world dominated by globalization. Enterprises that integrate with Industry 4.0 will be able to produce fast and high-quality production by taking the back of high-quality technology and employment ([S. Soyak & Soyak, 2018](#)).

Marketing from 1.0 to Marketing 4.0

The first phase of marketing is Marketing 1.0, which emerged in the industrial age. Marketing 1.0 mainly focuses on selling products, regardless of the needs and demands of the target market ([Jara, Parra, & Skarmeta, 2012](#)). For this reason, Marketing 1.0 considers it as product-based marketing. In this context, it aims to produce high quality products that provide functional benefits to customers ([Kotler, 2011](#)). Marketing is the heart of marketing products in 1.0. When it is evaluated in terms of communication, communication is provided in one way in traditional ways (such as TV, radio and e-mail) in Marketing 1.0 ([Erragcha & Romdhane, 2014](#)).

The second phase of the concept of marketing is the Marketing 2.0 era that comes with the first steps of the contemporary information age. Marketing 2.0 is a period that puts the customer in focus. In this period, detailed search and research about customers' needs and requests are at the forefront to reveal new target markets that turn into positive conditions ([Jara et al., 2012](#)). In the Marketing 2.0 concept, it is important to address and fulfill the needs and wishes of consumers ([Fucui & Dumitrescu, 2018](#)). In Marketing 2.0, besides receiving messages, customers are actively involved in all processes of marketing ([Erragcha & Romdhane, 2014](#)). In Marketing 2.0, businesses use the Internet as well as traditional media for communication and promotional activities ([Tarabasz, 2013](#)).

Marketing 3.0 is the value/people-oriented period in which the marketers are treated as human beings in integration with their mind, emotions and souls as well as the consumer attribute of customers

(Kotler, 2011). As it can be understood from here, in Marketing 3.0, his area of interest has completely shifted toward human (Varey & McKie, 2010). In this period, the demands and needs of consumers have changed constantly. For this reason, businesses have started to be more sensitive to this situation. Therefore, businesses have constantly conducted market research, followed technological changes and made them compatible with the values sought by consumers (Rahayu, Herawaty, Rahmawati, Prafitriyani, & Afini, 2018).

This process focused on more personalization of production and services based on big data analysis. As customers become social, they desire things that are “just done for them.” With this transition period, a new marketing approach was needed. This is “Marketing 4.0” (Kotler, Kartajaya, & Setiawan, 2017). Marketing 4.0 is about “a marketing approach that combines online and offline interaction between companies and consumers.” While blending machine or artificial intelligence to increase productivity of other technologies, it also strengthens the human-to-human connection to improve the customer interaction process (Fucui & Dumitrescu, 2018).

Fast process of technology and digital world are taking place in all organizational authoritative in all area and in all kind of sectors, because the business world is transformed by the postmodern revolution-Fourth Industrial Revolution (Akkaya, 2019). Today, with the introduction of concepts such as big data, social media, corporate social responsibility and e-commerce, our transition to Marketing 4.0 has accelerated (Taş & Şeker, 2017). Marketing 4.0, where digitalization, big data usage, content production on social media and successful social media management are important, will effectively create a strong relationship between the customer and the business (S. Soyak & Soyak, 2018). Since technology is constantly improving, business and social life are also affected by this situation. Marketing 4.0 basically aims to predict marketing activities based on future developments in the market. Therefore, predicting future trends with consumer trends, data management and advanced analytics are among the main objectives of Marketing 4.0 (Vassileva, 2017). In this period of increased competition in the market, businesses can create competitive advantage by using the

capabilities of Marketing 4.0. By placing customers in the new digital age, they can interact with them.

E-Commerce

With the development and spread of internet technology, consumers have started to meet most of their needs through virtual means. This situation paved the way for the emergence of electronic commerce. In this section, the scope, advantages, disadvantages and future of e-commerce will be discussed.

Definition and Scope of E-Commerce

Change is a fact of human lives and happens in all every part of the world where people live. Specifically, the nonstop changes in customers' requests and needs cause it inescapable for organizations to continually adjusting with changes to keep their organizations in a high competitive environment ([Akkaya & Tabak, 2020](#)). Human needs and expectations in the changing world are also changing very quickly. With this change, technological products and services are developing rapidly. Businesses aim to offer their products to consumers easily and quickly. Firms need to adapt to changes in both their internal and external environments and create new systems that let them focus on new trends, identify improvements and try to guess their potential impact on their manufacturing system ([Akkaya, 2020](#)). Traditional trading method has started to be replaced by electronic commerce. In this context, manufacturers of goods or services have started to offer their products or services with consumers, generally in electronic form ([Büyükseymen, 2018](#)). Accordingly, e-commerce has started to take an important place in our life. E-commerce can be defined in the most general sense as “any commercial business activity that the parties communicate electronically without direct physical connection or the need for a physical exchange” ([Yilmaz, 2018](#)).

E-commerce, which has been talked about lately, is closely related to the whole society, businesses and countries. E-commerce has become a remarkable element of economic practices. Today,

most businesses try to gain a place in a virtual environment by making use of e-commerce. As the number of businesses taking place in virtual environments increases, e-commerce has become an impetus in the world in terms of economy. The usage area of e-commerce, which is one of the economic developments, has started to become widespread day by day. The decrease in costs in the virtual store where businesses operate and the disappearance of intermediaries in traditional trade caused a decrease in prices. For this reason, it has been influential in the spread of e-commerce, as it is preferred by consumers and meets many expectations such as increasing profitability, competitiveness, market shares and brand recognition ([Akyazi, 2018](#)). The cheaper and widespread use of today's expensive technologies such as computers and the internet and their use by people rather than businesses have made a great contribution to e-commerce. Today, it is possible to see a computer or a mobile device in almost everyone. In this way, e-commerce can be realized between the company and the consumer by getting out of a form that is completely between companies.

Advantages of E-Commerce

With the increasing use of e-commerce, it has started to take an important place in the field of commerce. This area contains some positive and negative elements in itself. E-commerce has many advantages for consumers and businesses. The advantages of e-commerce for businesses are described below ([Akyazi, 2018](#); [Büyükseymen, 2018](#); [Çam, 2014](#); [Germir, 2015](#); [Gökmen, 2019](#); [Öztürk, 2019](#)).

- Thanks to e-commerce activities carried out on the internet, sellers have easier access to customers.
- Thanks to the ability to record online transactions and developing statistical software, customer actions, requests, interests can be followed. In this way, an important database about existing customers and potential customers can be obtained and personalized services can be offered to customers with data.

- Customers can report their problems during shopping via e-commerce sites. In this way, it can easily manage customer experiences.
- Advertising can be done through the e-commerce site. In this way, more effective brand perception can be created.
- Preparation or provision of orders after customer requests allows companies to work with minimum stock.
- Online stores can monitor their competitors who carry out their sales activities. In this way, it can effectively conduct market analysis and determine in which markets they can achieve rapid development.
- Buyer customers can be contacted directly using artificial intelligence.
- They can promote their products and services more easily in the national and international arena without any time and space restrictions. In addition, there is a chance to get customers from all over the world and take orders.
- With e-commerce, there is no holiday day shortage in stores. Online stores can be open 24 hours a day, 365 days a year.
- Gives small businesses the chance to grow and compete with their goals. E-commerce stores become suitable for the whole country and international sales as soon as they are established.
- Since many transactions can be carried out automatically through computers, the need for personnel can be reduced, thus saving costs.

The advantages of e-commerce for consumers can be listed as follows ([Akyazı, 2018](#); [Büyükseymen, 2018](#); [Öztürk, 2019](#)).

- Consumers can shop instantly from anywhere.
- It does not require high transportation budget to shop, find the right product or price.
- Due to the large variety of products, it is easier to find the right and most suitable product.
- Product and price comparison can be easily done.
- With the help of products, user experiences, comments, videos and visuals can benefit from the experience of consumers.

- For returns and after-sales transactions, the courier can be called easily by going to the store and proceeding easily.
- Since purchases from the relevant store are stored in the database, customers can have information about their order histories.
- It can make special production for them by being at the production stage of certain products.
- Thanks to the use of artificial intelligence, it can benefit from personalized offers and shopping opportunities.

Disadvantages of E-Commerce

E-commerce has major advantages for consumers and businesses as well as some disadvantages. These can be listed as follows (Akyazi, 2018; Büyükseymen, 2018; Öztürk, 2019):

- The seller and the buyer do not have the opportunity to meet face to face and to communicate.
- The customer does not have a chance to see and try the product he/she intends to purchase.
- Delivery times of the product purchased through e-commerce may be long or damage caused by the transport company may occur.
- The rate of consumers returning products is higher in Internet shopping than traditional retail stores. Consumers can return products within a certain period of time without showing any reason and paying shipping fees. This increases the logistics costs of the vendors in reverse significantly.
- No matter how many precautions are taken regarding security, there is a risk of hacking internet sites and personal information by hackers.
- Consumers make price researches on e-commerce sites before shopping. If there is no advantage in price, the chance of competition decreases.
- The doubts about the originality of the products on the e-commerce website, the complexity of the return process and the desire to visit the store prevent the shopping from being done on the internet.

The Future of E-Commerce

E-Commerce has increased its importance and has become an effective source of income today. It is anticipated that this development and progress will prevent retail sales and that e-commerce will be used in the future. The main reason for this prediction is that individuals experience time problems as they are active in working life.

It is estimated that an average of 50 billion devices will interact by 2020. It would merge with social network and e-commerce networks of systems such as smart city, home and device. However, the network in the ecosystem, which will be established through data, services, objects and individuals' internet environment, is expected to affect about 46% of the global trade volume in the next quarter century ([Gökmen, 2019](#)).

In the future, retailers with many small and medium economic powers will create their own virtual stores. Each business will reach the position where it can instantly answer the expectations of its customer portfolio with its own e-commerce website. In addition, it is anticipated that advanced programmed robot technology will be offered to customers in order to meet the needs of consumers. Today, e-commerce consumers desire to reach the products they purchase not within days but within hours. For this, e-commerce sites are focusing on new solutions such as drone, autonomous vehicles and courier delivery in terms of product deliveries in order to reduce delivery times ([Yilmaz, 2018](#)).

Digitalization and Digital Marketing

Advances in technology have had significant effects on consumers' daily lives. As the usage of smart devices, mobile technology and wireless network has increased, the consumer concept has started to be replaced by the digital consumer concept in today's technology age. Unlike traditional marketing methods, digital marketing is a marketing activity carried out on interactive platforms. In this section, the concept of digital marketing will be analyzed, the importance of digital marketing in e-commerce will be discussed, the differences between traditional and digital marketing will be examined, the distinctive features and benefits of digital marketing will be discussed

and information will be given on the methods and techniques used in digital marketing.

The Concept of Digital Marketing and the Importance of Digital Marketing in E-Commerce

The concept of digital transformation has found a place in the literature with the continuous development of software and hardware technologies, the widespread use of it and the possibilities brought by the electronic world by the users, throughout the period from the late 1990s until today. Digitalization: It has become an important force that regulates and changes human relations, consumer behavior and marketing channels of businesses in social and business life. Today, markets have expanded with digitalization and people have started to consume more. In addition, the demands and needs of consumers have also changed. In order to respond to all these requests, businesses have started to prefer consumer-oriented approach instead of product-oriented approach. Because adapting to the digital age, gaining and retaining customers is through having a digital marketing approach. Accordingly, it can be said that digital marketing is one of the most important elements of the century.

Digital marketing is defined as:

using the internet, social media, mobile and other interactive programs, as opposed to conventional methods such as radio, television, newspaper to support the brands and businesses of services or product manufacturers, to promote or advertise.
([Hopkins & Turner, 2013](#))

In [Girchenko and Ovsianikova \(2016\)](#), he refers to digital marketing as “a complex approach to the promotion of goods, services and brands by using a large number of digital channels or by integrating traditional channels into virtual marketing.” It is an umbrella concept used for the marketing of products and services through digital marketing, internet and digital technologies. This

umbrella concept also covers mobile phones, display advertising and other digital media ([Subbaiah, Reddy, & Kumar, 2018](#)).

Thanks to digital environment, businesses have become available to all people who can use the internet. Today, it is almost vital to use digital media for businesses. The digitalization of the marketing of goods and services and the increase in the use of the internet caused the spread of e-commerce. The introduction of brands, products and services in electronic environments and their sales are compulsory in terms of competition in the market. This imperative has forced some businesses to create their own e-commerce platforms, while others have started to operate in e-marketplaces. With the continuous development of technology, the communication between individuals and the masses shifting to e-platforms and the expansion of the e-commerce volume all over the world, e-marketing activities are now an imperative for businesses. With the digital transformation wave in the world, e-business and e-commerce are no longer seen only as a new sales channel in the life of companies. Digitalization and e-commerce also provide important opportunities for customer experience management, supply chain management, total quality management, sales channels optimization and efficiency management ([Süerkan, 2019](#)).

The Concept of Digital Marketing and the Importance of Digital Marketing in E-Commerce

Today, when the most digital form of marketing is experienced, the increase in web technologies and the expansion of mail networks and marketing activities have increased the usage areas of digital marketing. The concept of digital marketing is a phenomenon that has emerged with the widespread use of the internet, the increase of social media applications and the rapid development of technology.

One of the biggest reasons that digital marketing is different from other marketing applications is the disappearance of the concepts of time and space. Businesses can reach their customers in any place, time and situation via digital marketing. Because the digital world is an active and vibrant medium where users interact constantly with each other, digital marketing is different compared to other marketing

applications today and its domain is a broader marketing method compared to other marketing applications ([Aksoy & Sever, 2012](#)).

Consumers prefer the digital world, where many options are available while shopping and where they can access a lot of information about the product or service they will receive. Digital marketing activities are becoming more and more attractive day by day for companies. With the help of digital marketing tools, consumers can access information about the product or service they want by using digital platforms whenever and wherever they want. If they want, they can buy at any time they want with much lower costs ([Yasmin, Tasneem, & Fatema, 2015](#))

Personalized presentations are one of the important advantages that digital marketing offers to brands. In this way, brands can make their customers feel special and increase their brand loyalty. Because customers can find everything they are looking for and wonder with messages tailored to them. Digital platforms allow the message to be sent with many visual and audio riches. This increases the impact of the products and services offered in the perception of customers and makes them more interesting. With data analysis and search tracking on digital platforms, brands can make campaigns in accordance with their marketing strategies and carry out customer relations ([Büyükcelikok, 2018](#)).

Saving the production of digital products and services is one of the advantages of digital marketing. With digital marketing, the customer can access more information online about their preferences. Digital products and services are more efficient, stronger and more flexible than their traditional analog equivalents.

Businesses offer products and services to the consumer in a cheaper way with digital marketing tools. Thanks to digital marketing, the consumer has the opportunity to learn more about the product or service. In this way, it can make price performance comparisons between similar products or services. Live chat rooms have been established in order to resolve all questions, opinions and complaints of consumers regarding digital marketing regarding products or services. The consumer can obtain information about the product or service in the live chat rooms, and access all kinds of information regarding his order. The consumer can communicate any problem

faced by the business, product or service at the same time. In this way, the enterprise can take steps to solve this problem as soon as possible. In this process, digital marketing brings producers and consumers closer together.

Comparison of Digital Marketing and Traditional Marketing

With the advancement of technology, traditional marketing methods have been replaced by digital marketing methods. As digital platforms developed and growing with technology are facilitated to use, they are rapidly adopted and used by people of all ages and cultures. With the realization of this situation, a transition period has started to occur from traditional marketing to digital marketing. Digital marketing has enormous advantages compared to traditional marketing practices. Examining the differences between digital marketing and traditional marketing will enable a more detailed evaluation of the features of digital marketing. These differences are summarized in the table below ([Table 1](#)) ([İnanoğlu, 2009](#); [Yasmin et al., 2015](#)).

Table 1. Differences Between Digital Marketing and Traditional Marketing.

Traditional Marketing	Digital Marketing
The most important element is the consumer	The most effective factor is the internet and the internet related tools
It takes a long time to gather information about consumer characteristics and analyze the results	By looking at the content sharing, comments and likes of the Internet users, it is possible to get information about the features of the users immediately
Changes in marketing strategies in traditional settings are minimal	Marketing strategies and methods are constantly being renewed and changed according to the preferences of the consumer
Businesses cannot always reach the consumer. The producer can communicate with the consumer by meeting the consumer at a common place and time	The manufacturer can reach the consumer at any time, without any restrictions on space and time
The manufacturer is in communication only with the consumer	The manufacturer can communicate with both the consumer and other consumers to

The products offered to the customer are limited to those in the store	exchange ideas It is able to offer the customer a wide variety of products and services at the same time and effectively than at any other time
The return is only within working hours	The return can always happen
A campaign is valid for a long time	Campaigns can be easily changed and innovations can be used in any campaign

Media and Techniques Used in Digital Marketing

With the technological advances and changing lifestyle, digital life is rapidly becoming widespread. In digital marketing, many environments and techniques are used to transfer the promotional activities to more effective and wider audiences. In this section, general information about the environment and techniques used in digital marketing is presented.

E-Mail Marketing. With the widespread use of the Internet, businesses inform their customers about their campaigns, prices and activities via e-mail. In this process, businesses respond to the feedback of their customers and carry out works to ensure their demands and requests. E-mail marketing is an area that provides one-to-one communication between brands and customers using digital possibilities. Content, texts, brochures and all other images sent to customers in e-mail marketing are created with a marketing intelligence ([Büyükcelikok, 2018](#)). E-mail marketing generally includes a fiction that ideally includes a link to a detail page, where consumers click on this link to measure the effectiveness of the campaign or advertisement ([Breuer, Brettel, &Engelen, 2011](#)). E-mail marketing is a marketing method that can reach them directly instead of waiting for the target customer and has many advantages in it ([G. E. Belch & Belch, 2003](#)).

E-mail marketing has been the most used digital marketing method until recently in the field of personalized campaign and advertisement management. The effectiveness of this method has also increased in recent years with CRM (Customer Relationship Management) technologies and tools. E-mails previously shared with the masses are now shared with individuals with the effective use of

CRM tools. The ability of CRM tools to produce separate campaigns for each customer or potential customer has made e-mail marketing a stronger marketing tool for personalized marketing activities and enabled high returns ([Süerkan, 2019](#)). As many companies use e-mail to communicate with their existing customers, searches have become increasingly important to understand this process based on which e-mail campaigns affect the customer's attitudes and behaviors ([Cases, Fournier, Dubois, & Tanner, 2010](#)). E-mail marketing is a marketing channel that has a very high impact at a low cost with a properly established strategy. The most important thing is to send the right message to the right person at the right time and at the right frequency.

Mobile Marketing. Rapidly developing wireless communication technology and the advantages brought by the internet have revealed the concept of mobile marketing. Mobile marketing is “an interactive marketing tool used in the promotion of goods, services or ideas through smart phones to benefit the business and all stakeholders” ([Scharl, Dickinger, & Murphy, 2005](#)). Mobile marketing is not only a new evolutionary technology, but also a revolutionary technology. In addition, it is a rare change in the way customers connect with brands. Mobile marketing has a greater potential impact than the sum of radio, television and personal computers ([Hopkins & Turner, 2013](#)).

With the advancement of technology, the use of mobile devices has also increased. In addition, the limitations of the internet have disappeared quickly with mobile communication tools. Today, mobile cell phones, other personal and portable digital devices have become a necessity used in daily life, especially among young people. The more widespread and innovative features of mobile communication tools create opportunities for marketing managers that cannot be realized through traditional communication channels. Named with the term mobile marketing, this new channel is based on the promotion element of the marketing mix. It refers to a marketing communication using mobile communication channels in promoting goods, products, services and ideas ([Pousttchi & Wiedemann, 2007](#)). The main communication channels used in mobile marketing

are SMS, mobile advertising, QR code, mobile websites and mobile applications.

The most important advantage of mobile marketing is that it offers wide opportunities for innovation and space-time sensitive services in business processes due to any time and any place relationship. Mobile marketing combines the feature of using mobile phones to transfer and receive data from the internet ([Köse, 2017](#)).

Search Engine Marketing (SEM) and Search Engine Optimization (SEO). Search engines are one of the most used web tools today. A person browsing the internet or wanting to use the internet for any reason needs to use search engines to reach the product they are looking for. 90% of internet users use search engines to find information online. Therefore, it is important in terms of availability that the web page owned is one of the top 20 results in the search engine ([Atay, Alanyalı, Uyan, & Baş, 2010](#)). The benefits of these search engines, which direct the desired information to the relevant sites and rank about the information they contain, are very important. Businesses wishing to take advantage of this situation desire to be in the top ranks in the search lists. People who search in search engines generally prefer the top results. For this reason, companies should be involved in one of the SEM or SEO systems in order to get out in the first place. Businesses should use SEO and SEM marketing applications together to achieve superiority in an increasingly competitive environment, reach more consumers, manage marketing strategies effectively and achieve more effective results ([Sirovich & Darie, 2007](#)).

SEM is the marketing strategy conducted through search engines, and this is realized by using SEO, paid search engine ads and paid placement programs ([B. Mordkovich & Mordkovich, 2007](#)). For an effective SEM, it is necessary to know and use these marketing tools that exist in other search engines such as Google Adwords in Google, which searches using which words or phrases the target audience searches ([Büyükçelîk, 2018](#)). The search engine is one of the most important efforts to display without ads. Search engines review websites using certain algorithms and sort by starting from the most equipped and updated site according to the criteria they set. SEM applications are not limited to search results only. In

addition, it allows the display of advertising variations within sites that are relevant to the relevant market according to the elections (Nabout & Skiera, 2012). Today, modern businesses understand the importance of the search engine, and SEM continues its development rapidly. One of the reasons why SEM gains importance can be stated as the ability of search engines to give successful results in geographic targeting and supporting local businesses in regional areas (Dysart, 2004). Digital marketing communication provides a more competitive environment thanks to SEM. Businesses want to put themselves forward by collaborating with companies that provide SEM service in order to be ahead of their competitors (Moran & Hunt, 2014).

One of the digital marketing communication tools, SEO is the most basic way of organizing the content of search engines that are used in the internet environment. SEO: When searching on the search engine, it is defined as the works carried out to improve the general ranking of the website and to provide users access to the website. With SEO studies, it is aimed to make the websites accessible within the search engine and to increase the probability that the website can be found in the search engine (Mamaghani, 2009). Thanks to SEO, any website can be accessed more easily for the users and made more attractive according to the features and qualities sought (Puchkov, 2016). The main purpose of SEO studies is to enable customers to easily access brands' websites or the content they publish and to access these links from the first places of searches.

Influencer Marketing. Influencer marketing is the digital counterpart of word of mouth marketing. It is defined as "a type of marketing based on using key leaders to direct a brand's message to a larger market" (Byrne, Kearney, & Macevilly, 2017). Today, there is an important consumer group that uses social media. Enterprises competing in intense competition conditions want to accelerate their marketing processes. For this purpose, it has become almost imperative for businesses to use social media, in particular, to carry out marketing activities for the target audience, who spend quite a long time in digital media. At this point, businesses need social

influencers to help them share their marketing content online ([Singh & Diamond, 2012](#)).

People who are considered as phenomenon in social media are people who use social networking sites such as Facebook, Twitter, Instagram, Vine and Snapchat and attract social media users who follow them with the content they create and increase their popularity by spreading the content on social media ([Marwick & Boyd, 2011](#)). The role of influencer marketing in the digital marketing strategy is huge due to its ability to influence all audiences. Because all online customers or prospects read and follow them, act according to their positive and negative criticism, value their opinions and most importantly, trust them. Consumers are more likely to perceive and react positively to a sponsored post from a trusted friend or person. This is what makes influencer marketing attractive to consumers and brands ([Woods, 2016](#)).

According to data from 2017 in the research conducted by Creatorden Influencer Marketing firm, spending in this area has increased from \$ 1.1 billion in the previous year to \$ 1.5 billion ([Influencer Marketing, 2017](#)). The importance of influencer marketing today is related to the intensive use of digital channels. Consumers control other users' experiences and thoughts through social media. As a result of this control, it makes a decision whether to buy ([Yağci, 2019](#)).

Gamification. One of the new trends in marketing today is gamification. Gamification refers to the application of game items to any non-game situation or process ([Bitirim Okmeydan, 2018](#)). According to another definition, gamification is “the deliberate use of game items to make non-game tasks and contexts an enjoyable experience” ([Seaborn & Fels, 2015](#)). Gamification is expressed as the process, in which marketing activities for brands are carried out with the understanding of game mechanics and game play. It is a technique in which game methods are carried out to strengthen customers' relationship with the brand, to ensure and increase loyalty. The main purpose of the gamification method is not to play games, but to perform a business and marketing process by entertaining the customer ([Şahin, Tuna, & Tütüncü, 2014](#)).

It is seen that businesses have started to use gamification applications intensively in parallel with the development of mobile devices. One of the most important reasons for this increase can be shown to decrease the impact of traditional communication designs on consumers ([Nart & Güleç, 2018](#)). Gamification is used to create brand loyalty, increase the frequency of purchases and create customer satisfaction especially in the marketing area. Nowadays, entertainment is becoming widespread especially with mobile games and computer games. Playing a game in this period has become easier than ever thanks to the Internet ([Ögel Aydın, 2020](#)). One of the best known examples of gamification worldwide is the Foursquare/Swarm application. This application is designed to turn life into a game. Launched in March 2009 for the first time, the mobile application Foursquare is one of the examples of location-based social networks that enable people to create social networks and share their locations with their friends ([Frith, 2013](#)). The Foursquare-Swarm application acts by notifying the places visited by using the basic elements of gamification, commenting on these places and sharing them.

Three types of approaches have been determined according to the use of the concept of gamification. The first is internal gamification, which aims to achieve the targeted behaviors through the employees of the companies. The second is external gamification, which aims to achieve higher profits by improving the relationship between the business and the customer. Finally, it can be expressed as a behavior change aiming to develop targeted behaviors by taking into consideration the public interest ([Verbach & Hunter, 2012](#)).

Social Media Marketing. Until recently, media such as newspapers, magazines, radio, television and billboards have been among the tools that marketers use to promote their products and services in the market. However, the emerging Web 2.0 technology and its advantages have enabled marketers to shift to this new medium. Traditional marketing communication activities are seen as insufficient in many issues and cannot create sufficient opportunity doors for businesses. With the increasing use of the Internet, the increasing number of users in social media environments has

created new opportunity gates for businesses to carry out advertising and marketing communication activities. Thanks to these social media environments, businesses can both maintain their communication with their existing customers and reach their potential target audiences ([Şengüler, 2019](#)). In this direction, businesses use the opportunity to reach potential customers by being in social media environments and interacting with their existing customers ([Wiedmann & Buxel, 2005](#)).

Social media is defined as a platform that internet users create their own content and share with other users ([Feng & Li, 2009](#)). Social media sites are the channels where the content is produced by users. Data produced by millions of users, as well as personal information, demographic information, relationship and status sharing voluntarily processed into the system, turn these media into large data stores. Social media sites, which have reached a sufficient size in terms of prevalence and diversity, have become marketing channels that facilitate reaching the target directly as they provide detailed reporting and measurement opportunities for digital marketing ([Öztürk, 2019](#)).

With the advancement of technology and the increase in internet usage, people spend most of their time in social networks. It has become inevitable for brands to take place in social media platform. Brands have also started to communicate with their customers on social media. Therefore, social media marketing and mobile marketing areas have an important position in the digital marketing world. In particular, the effective use of social media and mobile technologies together has become more remarkable for digital marketing activities.

Social media has not only changed the way businesses and brands communicate with consumers, but also changed the way things are done in many ways. Businesses are increasingly using social media to reach existing customers, gain new customers, build trust, increase awareness and protect their brand image ([Mills, 2012](#)). The significant increase in the use of social networks, especially in the global sense, has made it a marketing area that social networks cannot neglect for businesses. Social media marketing, just like e-mail marketing, is able to reach advertising and

campaigns cheaper and more accurately. Another strength of social media marketing is that the feedbacks to the advertisements and campaign contents shown to customers or potential customers are much more. A customer who opens and read the e-mail should write a reply e-mail to the business to give feedback. However, “likes” and “sharing” activities on social media can be done by clicking a single icon. Therefore, feedback rates to social media posts are much higher ([Sürkan, 2019](#)).

Viral Marketing. Users on the digital platform take into account the comments of other users who share their experiences on social networks before purchasing a product or service. Businesses should enable more users to think positively about their businesses in order to make this effectiveness advantageous. In this case, viral marketing takes an important place. Viral marketing is one of the most effective and cost effective ways to create a buzz or rumor about products and services offered by businesses. It is an incentive strategy to send a marketing message to others by allowing effective and encouraging messages to be spread on the internet. Basically, viral marketing can be expressed as the technological dimension of word of mouth marketing ([Bilici, 2015](#)). The emergence of the word “viral,” which is mentioned in the concept of viral marketing, was thanks to the studies of word of mouth marketing in the internet environment. The Internet has become an important factor for companies to reach consumers and to strengthen their bonds with these individuals. The messages opened to circulation over the internet to carry out such activities form the basis of viral marketing ([Gündebahar & Kuş Khalilov, 2013](#)).

Viral marketing uses the network effect of the internet. A computer virus can be effective in reaching a large number of people quickly, which can affect many machines around the world ([Chaffey & Smith, 2008](#)). In viral marketing, the main purpose of the companies is to spread the message they want to give to their targeted customers on the targeted platforms with the right timing. Businesses enable customers who are satisfied with their products or services with viral marketing practices to tell other people about their content. Thus, it creates a self-growing advertising area and attracts customers ([Gülmez, 2011](#)).

On the basis of viral marketing a virus has the logic of spreading at the same time. When an individual is infected with an infected disease, he can infect this virus to the people he is in contact with. In viral marketing, the process progresses in this way. In other words, there is the spread logic of a digital virus ([Walter, 2006](#)). Today, different and advanced viral marketing techniques are applied. These include blogs, vlogs, e-mail mailing, advergaming, wikis, social media sites and websites with consumer reviews and reviews ([Seitel, 2016](#)).

Internet of Things. The Internet has brought a different dimension to people's abilities such as working, living and learning in a short time. Today, the use of high-tech devices that make the lives of societies easier has become indispensable. The operation of most of these devices is independent of each other. Thanks to the Internet of Things (IoT) technology, which is rapidly spreading, such devices communicate with each other and a smart communication ecosystem is created ([Akkuş, 2016](#)). The IoT concept was first introduced to the literature in 1999 in a presentation by Kevin Ashton, one of the founders of the Massachusetts Institute of Technology (MIT) Auto-ID Center ([Ashton, 2009](#)). "The communication network where physical objects can connect with each other or with larger systems through the Internet is called the Internet of Things" ([Rifkin, 2015](#)). According to another definition, the Internet of Things is defined as "Structures emerging with the development of modern wireless communication technologies and enabling the communication of objects with each other" ([Atzori, Iera, & Morabito, 2010](#)). Internet of things means that objects and especially machines in factories are connected to each other over the internet and therefore they are in communication ([Gubbi, Buyya, Marusic, & Palaniswami, 2013](#)).

The three concepts that do not change in defining the Internet of Things are "Object," "Internet" and "Inter-object connection." It is a physical object that can be defined by the concept of the object. The Internet refers to the communication infrastructure that holds and stores the virtual copy of objects. Thanks to this infrastructure, related data are connected through objects. One of the main purposes of the Internet of Things is to provide information about

physical objects identified with a single identity uniquely, when required, at the desired level of detail after passing the necessary, correct and appropriate filtering (Uckelmann, Harrison, & Michahelles, 2011).

Different applications have been developed by making use of the Internet of Things technologies. Examples of these are applications such as smart tools, online access to maps, internet access, audio-video content, information about a place, an anti-theft message and alert system and an assistant in the event of an accident (Gündüz & Daş, 2017).

It can be stated that the main benefit of internet of things for companies is data acquisition. The data collected are continuously and simultaneously accessible, and the process of obtaining and sharing data takes place in a powerful and efficient manner (Yang, Yang, & Plotnick, 2013).

Conclusion

Globalization and advances in technology have led to unprecedented increases in innovation, competitiveness and economic growth worldwide. Industry 4.0, called the digital industrial revolution, is a collective whole of information technologies and industry. It represents the fourth industrial revolution in manufacturing and industry. With the integration of technology completely into social life and corporate life, large scale changes and differences started to emerge. Many issues that have become a habit depending on traditional methods have started to change very quickly thanks to the unlimited possibilities created by technology. In this globalizing world, almost a new trend is emerging every day. Businesses should follow all the innovations in technology and improve themselves in order to survive both in the real and digital world. Digitalization is the fundamental dynamic of transformation in all conceivable sectors. Business models are being rewritten with digital transformation in all sectors, including those that are not considered to have a relationship with the digital world until today.

Today, the vast majority of the human population uses the internet and has a social media account. Aware of this, e-retail brands aim to

carry their marketing activities in traditional ways to digital platforms and reach more consumers. E-retailing is known today as a form of branding and marketing with a very large market share. Conducting marketing activities in this digital market, where consumers are concentrated, is of great importance for a brand.

Today, the consumer profile that visits the stores and places its shopping is replaced by the consumer profile, which purchases via social networks via virtual communication systems through its new communication systems. Consumers have the opportunity to access information whenever and wherever they want through digital channels. The consumer is not only what the brand says, but also has the opportunity to follow what the media and friends say about the brand. In addition, it is an important advantage that these channels make it possible to achieve measurable results.

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Chapter 3

Agile Leadership Model in Health Care: Organizational and Individual Antecedents and Outcomes

Safiye Şahin and Furkan Alp

Abstract

Nowadays, organizations have to resist the rising competition more effectively than their competitors and take a step closer to excellence in offering the product to customer demands. To do this, organizations need agile leaders in order to implement agility principles and practices. Especially in the health sector, health managers must be agile because of the specific characteristics of health services. From this view, this chapter aims to develop a theoretical agile leadership model in healthcare organizations. First, the authors define agile leadership and its sub-dimensions based on previous literature. Then, the antecedents and outcomes of agile leadership have been analyzed. “Drivers of agile leadership,” “organizational factors affecting agile leadership” and “individual factors affecting agile leadership” are identified as the antecedents of agile leadership. “Organizational outcomes” and “individual outcomes” are determined as the outcomes of agile leadership in the health sector.

Keywords: Agile leadership; quickness; flexibility; competency; collaboration; technology

Introduction

Today, organizations have become very complex due to changes in living conditions and competitive processes ([Joiner & Josephs, 2007a, 2007b](#)). In addition, due to the challenges posed by the global economy, organizations have to adopt faster decision-making processes in response to rapidly increasing customer demands ([Denning, 2016](#); [Maskell, 2001](#)). Especially, the age of technology and innovation have brought forward some concepts, such as creating value and leading to change. For this reason, the changes in recent years have forced organizations to conduct new research or develop new methodologies. Organizations wishing to succeed under these challenging conditions have to move away from traditional methods and continue to implement them more dynamically ([Akkaya & Tabak, 2020](#); [Mergel, 2016](#)). Therefore, they need leaders who can manage this change and adapt to the circumstances ([Akkaya & Tabak, 2020](#); [De Meuse, 2019](#)). Effective leaders do not see their position as sufficient due to their structure, but they are constantly changing and developing and have to express their potential skills. In this context, the concept of “agility,” which has recently gained importance, comes to the fore. In the definitions of agility, quickness, flexibility, uncertainty and change are emphasized ([Akkaya, 2019](#); [Sharifi & Zhang, 2001](#); [Sherehiy, Karwowski, & Layer, 2007](#)). To explain briefly, agility is more than speed. It is a very dynamic process involving change and transformation ([Kidd, 1994](#)).

Organizational agility is the ability of organizations to use their resources as innovative, proactive, flexible and cost-effective to meet customer demands in fast changing market conditions ([Holsapple, Li, Chung, & Goldsby, 2008](#); [Koçyiğit & Akkaya, 2020](#); [Narasimhan, Swing, & Kim, 2006](#); [Sharifi & Zhang, 1999](#)). Nowadays, when agility is applied to organizations by effective leaders, organizations will be able to resist the rising competition more effectively than their

competitors and take a step closer to excellence in offering the product to customer demands. Especially in the health sector, health managers must be agile because of the specific characteristics of health services, for example high uncertainty and urgency, rapidly changing technology and no room for errors in health services ([Balsa, Seiler, McGuire, & Bloche, 2003](#); [Begun & Kaissi, 2004](#); [Caymaz, 2015](#); [Chatfield & Reddick, 2018](#); [Poteat, German, & Kerrigan, 2013](#)). Today, in Industry 4.0 term, organizations operating in health institutions face some problems, especially in turbulence, complex and changeable environment, and it is important that the information should be distributed to by leaders with an accurate perception of knowledge for corrective decisions ([Akkaya & Üstgörül, 2020](#)). Health managers are required to make fast and accurate decisions.

However, when the literature on agility is examined, studies on agility and agile leadership are limited. Empirical studies and measurement tools are insufficient in number (e.g. [Charbonnier-Voirin, 2011](#); [Kassim & Zain, 2004](#)), where the concept of agility is mostly addressed in the context of manufacturing (e.g. [Abdallah & Nabass, 2018](#); [DeGroote, 2012](#); [Rane, Narvel, & Bhandarkar, 2019](#)) and information technologies (e.g. [DeGroote & Marx, 2013](#); [Swafford, Ghosh, & Murthy, 2008](#); [Van Oosterhout, Waarts, & van Hillegersberg, 2006](#); [Zain, Rose, Abdullah, & Masrom, 2005](#)). Especially, studies on agility in developing countries such as Turkey are less. However, the level of specialization of health care workers increases with the effects of globalization. So, managers have made it necessary to adopt new approaches. With this research, an agile leadership model in health care services will be developed and contributes to the literature with this aspect.

Agile Leadership and its Sub-Dimensions

Due to the innovations brought by living in a digital world, the leadership understanding has also changed ([Akkaya & Tabak, 2020](#)). The obligation to create talented, change-oriented, innovative and dynamic organizations and to have effective leaders in these organizations has become dominant all over the world in order to be successful regardless of the sector and to be more effective in this

digital environment. Although many studies have been done on agility and agile organizations in the literature, agile leadership appears as a new concept (Özdemir & Çetin, 2019).

The concept of agility was first introduced in 1991 by the Lighthouse Institute of Lehigh University in a report describing how US companies should proceed to become a production leader again (Nagel & Dove, 1991). However, in this report, organizational structures or project planning systems related to the concept of agility are not mentioned in detail (Özdemir & Çetin, 2019). In this context, agility and agile production were first used actively in the field of computer technologies (Akkaya, 2019). Bill Joiner and Stephen Joseph's (2007a, 2007b) studies are one of the pioneer studies about agile leadership. In their study, they provided detailed information about the definition, criteria and positive outcomes of agile leadership (Joiner & Joseph, 2007a, 2007b).

Agile leadership theory has its empirical origins in studies undertaken during the late 1990s (Akkaya, 2020). When the sub-dimensions of agile leadership are examined in the literature, there is no consensus on the sub-dimensions of agile leadership, as shown in Table 1. In this study, by analyzing the sub-dimensions of organizational agility and agile leadership in the previous studies (e.g. Henson, 2015; Joiner & Joseph, 2007a, 2007b; Kisperska-Moron & Swierczek, 2009; Meade & Sarkis, 1999; Sharifi & Zhang, 1999; Zhang & Sharifi, 2000) and taking into account the structure of health services, we conceptualize the agile leadership as a five-factor construct, namely, quickness, flexibility, competence, collaboration and technology tracking. Based on these sub-dimensions, we define agile leadership as a leadership approach that includes effective leadership skills, facilitating the adaptation of the organization, making quick and flexible decisions to ensure organizational development, supporting teamwork and collaboration within the organization and adapting to technological developments.

Table 1. Definitions and Sub-dimensions of Agile Leadership.

Authors	Definitions of Agile Leadership	Sub-dimensions of Agile Leadership
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Joiner and Joseph, (2007a, p. 6)	“Leadership agility is directly analogous to organizational agility: It’s the ability to take wise and effective action amid complex, rapidly changing conditions”	5 Sub-dimensions: Expert, achiever, catalyst, co-creator and synergist
Joiner and Joseph, (2007b, p. 36)	“In essence, it’s the ability to lead effectively under conditions of rapid change and high complexity”	5 Sub-dimensions: Expert, achiever, catalyst, co-creator and synergist
Henson (2015, p. 3)	“Effective leadership in an agile environment is less dependent upon seniority or privilege but more reliant on developing the skills to facilitate adaptation, attract and leverage diversity, foster collaboration and inspire outstanding performance”	7 Key Concepts: 1. Higher order thinking – not knowledge per se – is power 2. Knowing your stakeholders is key 3. New paradigms for collaboration and innovation 4. Organizations with resilient employees prosper 5. Creating the right environment to support different types of work 6. Never underestimate the ‘social brain’ side of leadership 7. Giving people the tools and resources to manage themselves
Hayward (2018, pp. 3-4)	“Agile leaders are connected leaders. They know how to connect with their team, customers, colleagues and wider stakeholders. They also know how to connect with societal trends that are shaping a new reality around us – a digitally accelerating, politically unstable reality that creates novel opportunities and raises the threat of obsolescence across products and whole sectors”	2 Sub-dimensions: 1. Enabler: Learning agility, clarity of direction, empathy and trust, empowering, working together 2. Disruptor: Thoughtfully decisive, digitally literate, questioning the status quo, creating new ways of thinking, close to customer trends

Kiely, Collins, Richard (2018)

“In the modern business world, many leaders emphasize the importance of being ‘agile.’ Borrowed from the software world’s concept of agile development, this management style requires a leader to be flexible, adaptable and fast in their decision-making”

13 Sub-dimensions:

1. Being present to gain insight
2. Inspiring creativity and innovation in your employees
3. Moving quickly
4. Adapting to any situation as it happens
5. Having the flexibility to take quick but sure actions
6. Being resilient and creative
7. Navigating new waters and charting new successes
8. Continuously learning and embodying a growth mindset
9. Embracing change by driving value
10. Letting go of what doesn’t work
11. Listening, learning and adapting
12. Connecting the dots
13. Shifting your management style to match your employees

Özdemir & Çetin (2019, p. 314) *Agile leaders are also expected to create synergies with the ability to collaborate and coordinate development teams, question the status quo, emotionally flexible, act as a catalyst between internal and external stakeholders, establish a communication-based, secure and empathic working environment*

5 Sub-dimensions:

1. Emotional agility
2. Digital literacy and technological agility
3. Synergetic agility
4. Shared responsibility and proactivity agility
5. Openness to innovations and adaptative agility

Leaders and managers worldwide are increasingly turning to agility principles and practices to achieve their strategic goals. Agile leaders are leaders who adapt to rapidly changing environmental conditions and can respond to this change quickly (quickness). In addition, agile leaders’ objective is continuous development and improvement. They aim to go beyond management functions and to adapt their team to a continuous change at all levels (flexibility). Therefore, agile leaders empower their employees. They typically not only focus on the goal of teams achieving their goals. They also cooperate more with the team

while achieving the goal and ensure that team members are more satisfied with their work (collaboration). As a result, efficiency increases. Agile leaders build the vision, support teamwork, so other employees support them (competence). Agile leaders are also responsible for ensuring innovation. They encourage their teams' development continuously (technology tracking). They lead their teams to work on improvements and provide the training their team to succeed.

Sub-Dimensions of Agile Leadership

In this study, agile leadership is examined under five sub-dimensions. These are quickness, flexibility, competence, collaboration and technology tracking.

Quickness

Today's companies operating in emerging markets face with certain problems especially in turbulence, complex and changeable environment ([Akkaya & Tabak, 2017](#), p. 119). Quickness, which can be explained as the ability of an enterprise to offer the product or service effectively in a short time, is very important for businesses to keep up with change and improve innovation ([D'Aveni, 1994](#); [D'Aveni, Dagnino, & Smith, 2010](#); [Wiggins & Ruefli, 2005](#)). Due to Industry 4.0 and the technological changes, quickness is one of the prominent concepts. Businesses, which provide products or services faster than competitors or respond to customer needs immediately, gain important advantages ([Bessant & Tidd, 2007](#)). Health services is a sector where technological developments are experienced intensively ([Zaman, Hossain, Ahammed, & Ahmed, 2017](#)). Technology that is developing day by day affects health policies and health services directly. Today, businesses with agile leaders who can adapt to speed and make faster decisions will have a more advantageous position compared to their competitors ([Wu, Tseng, Chiu, & Lim, 2017](#)).

Flexibility

The concept of flexibility for a business means adapting to environmental changes ([Sanchez, 1993](#)), as well as constantly responding to unexpected changes ([Kundi & Sharma, 2015](#)). Employees in flexible organizations focus more on fulfilling their responsibilities rather than rules. In this sense, the success of the employees in their jobs can create a source of inner satisfaction and happiness ([Nassiri, Shunko, & Mamani, 2019](#)). Technology is not just changing today. Due to an increase in dynamism, the businesses that remain in the traditional structure fail in the market. Organizations aiming at change and development must be flexible ([Akkaya & Tabak, 2018](#)). Leaders, who are flexible, will provide a serious advantage in competition.

Flexibility has been an ongoing agenda in healthcare services in recent years. The demand for products and services provided in health services is constantly increasing ([Morgans & Burgess, 2012](#)). Managers need to be flexible in order to cope with the constant change in the social, economic and political environment ([Mashange et al., 2019](#)). Flexibility, as part of the operational delivery of hospital services, is important for both functional and organizational compliance. As health services are located in an atmosphere that can turn into a crisis, flexibility is thought to be the main factor that will ensure the adaptation of healthcare managers.

Competency

Briefly, competency means the characteristics that a person should have for a successful business performance ([Levenson, Van der Stede, & Cohen, 2006](#)). Competency is considered as a set of capabilities that provide productivity, efficiency and effectiveness to firms aiming to achieve its goal ([Sharifi & Zhang 1999](#); [Tahmasebifard, Zangoueinezhad, & Jafari, 2017](#)). Agile leaders are concerned about the change, uncertainty and unpredictability in their business environment to respond accordingly. Therefore, agile leaders need to have a set of distinctive abilities to deal with change, uncertainty and unpredictability in their business environment. These abilities consist of four main elements ([Dahmardeh & Banihashemi, 2010](#)):

1. to identify changes reactively or proactively and react quickly to them;
2. to have the ability to ensure that businesses achieve their goals and objectives effectively;
3. flexibility/adaptation with the ability to process different processes with the same processes and achieve different goals; and
4. to have the ability of quickness to carry out activities as quickly as possible.

Collaboration

The ability to use the opportunities that a business has today is directly proportional to bringing together talented people and managing them effectively ([Richard, 1988](#)). Collaboration forms the basis of the idea of working toward gaining competitive advantage and customer satisfaction ([Hode, Noble, & Marshall, 2012](#)). Businesses need to integrate knowledge, experience and different perspectives to achieve their goals. Leaders contribute to the improvement of team performance by sharing new knowledge and experiences they have gained.

Healthcare is one of the areas where teamwork is imperative ([Gafa, Fenech, Scerri, & Price, 2018](#)). It is not possible to perform in health care services by a single professional group. This situation necessitates teamwork in health care services. Therefore, it is of special importance that healthcare teams perform their work effectively. Physicians, nurses, dieticians and other health professionals work collaboratively to improve the patient's treatment and health. Staffs working in administrative and support services are also a member of this team. They are responsible for the preparation of the environment necessary for the patient to receive the best quality service. Hence, providing collaboration is one of the most important features of agile leaders aiming to make their institutions successful.

Technology Tracking

Technology is inevitable for organizations in order to reach their goals and provide development (Patel et al., 2009). In an organization, leaders play the most important role in keeping up with rapid change and capturing technological development. Leaders should make the necessary coordination in the use of technology. Moreover, they should inform their team in this regard.

Especially following the technology and adaptation to it, hospitals are very important for the well-being of the patient (Gilvary, Madhukar, Elkhader, & Elemento, 2019; Reddy, Fox, & Purohit, 2019; Topol, 2019; Yu, Beam, & Kohane, 2018). For this reason, technology monitoring is also important in health services. So, healthcare managers need to be agile leaders who follow the technology.

Agile Leadership Model in Health Care Services

With Industry 4.0, agile production and agile leadership have increased in organizations. In order to adapt and respond to rapid change, leaders must also be agile. Due to the nature of agile leadership, organizational performance, customer and employee satisfaction, customer and employee commitment to organization, employee job performance, etc. are expected to increase. In addition, managers with high agile leadership skills are expected to have a high job performance. Due to its substantial outcomes, it is necessary to understand the antecedents of agile leadership. Factors affecting agile leadership are discussed in three dimensions within the scope of this study. These are environmental factors, organizational factors and individual factors. The agile leadership model developed as a result of the literature review is shown in Fig. 1.

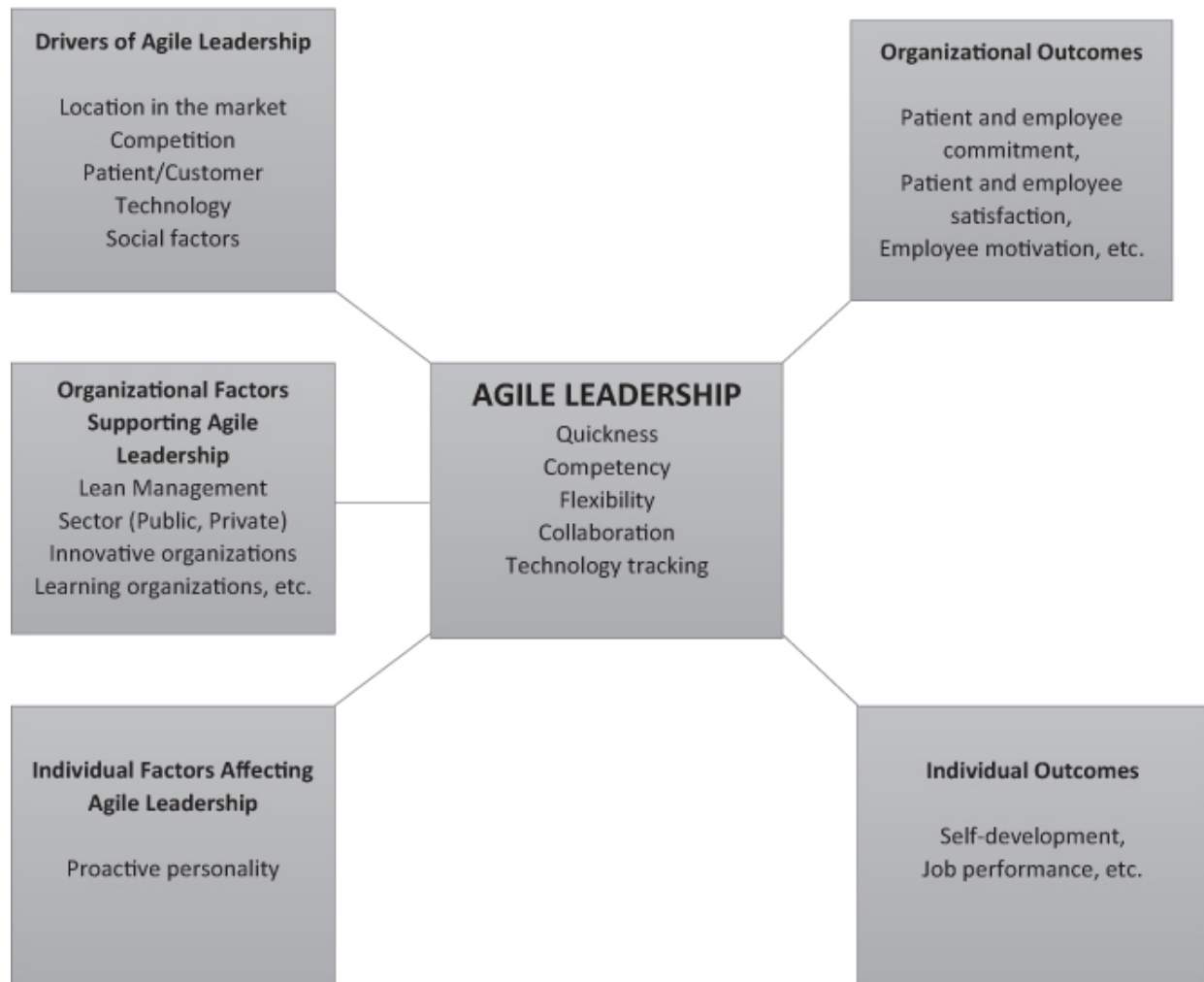


Fig. 1. Agile Leadership Model in Health Care Services.

As seen in [Fig. 1](#), environmental factors, organizational factors and individual factors play a role in the formation of agile leadership. According to our model, agile leadership is affected by changing situations over time (technological developments, increasing competition, changing customer expectations, changing market structure, differentiating community structure and desires, etc.). However, it has a positive impact on a number of individual and organizational outcomes.

Antecedents of Agile Leadership

According to a previous research, we examine the antecedents of agile leadership under three factors. These are:

- A. drivers of agile leadership;
- B. organizational factors affecting agile leadership; and
- C. individual factors affecting agile leadership.

A. Drivers of Agile Leadership

Environment can be defined as anything outside the organization's borders ([Hellreigel & Slocum, 1978](#)). In other words, environment is a general concept that expresses the sum of external and internal factors that can affect organizational activities. The most important feature of using the environment as a determining parameter is that it presents opportunities and threats. Therefore, understanding the environment in terms of organizations determines the degree of choice among the alternatives. [Zhang and Sharifi \(2000\)](#) developed a model for organizational agility. According to this model, they evaluated the environmental factors affecting the agility of organizations on seven items. These factors are:

1. position in the market/market structure;
2. competition;
3. customer requests;
4. technology;
5. social factors;
6. suppliers; and
7. internal complexities.

In their study, [Zhang and Sharifi \(2000\)](#) stated that environmental factors should be evaluated and scored in order to develop management and agility strategies. We draw on the study of [Zhang and Sharifi \(2000\)](#), since there is no such model in the literature on agile leadership. However, in this study, suppliers and internal complexities are not included in environmental factors. Only five factors are included (position in the market, competition, customer, technology and social factors).

Position in the Market. In order to stay out of the globalizing world and to take advantage of opportunities in ever-growing markets, one of the primary objectives of the enterprises is to strengthen its position in the sector and increase its market share (Woo, 1981). Both for the continuity of the business and for getting rid of the pressure of other companies are the most important reasons why businesses are permanent and grow today. In this context, enterprises need agile leaders to improve their market position and increase organizational performance (Kim & Thompson, 2012). The situations that increase the need for agile leadership in terms of strengthening the position of businesses in the market are as follows (Woo, 1981):

- In order to increase efficiency in the market and get rid of the pressure of competitors, the price balance should be analyzed well. This enables managers to adopt agile leadership. In particular, it is associated with the quickness, flexibility and competence sub-dimensions of agile leadership.
- Financial resources must be provided for the investments to excel in competition. This situation also increases the need of businesses for fast, flexible and highly competent leaders.
- Whether the competition in the domestic or foreign market, in both cases, delivering the goods ordered by the customer immediately is a very important factor to gain advantage over the competitors. This is closely related to the agile leadership's quickness, flexibility, collaboration, competence and technology tracking sub-dimensions.

Agile leaders analyze how to expand the production capacity of the enterprise, without causing a huge increase in costs. Agile leaders are more advantageous in dominating the market as they follow technological developments. Therefore, managers aiming to improve their position in the market are expected to adopt agile leadership.

Preposition Hypothesis 1a: As the importance of the organization's position in the market increases, agile leadership increases in organization.

Competition. Health care services are important in terms of employment and contribution to the economy. In the health care sector, there have been important strategic developments in recent years ([Anghel, Toroczka, Bassler, & Korniss, 2004](#)). With these developments in the sector, effective management of hospitals and determining goals for sustainable health are critical for all organizations. Due to the unique features of healthcare markets, competition occurs differently than most other markets. Competition in the health sector has increased especially with technological developments. Therefore, it is necessary to provide health services more effectively, efficiently and quality. While most researchers analyze health reforms and developments to improve competition in health markets, they argue that healthcare managers should understand institutional stakeholders' (employees, suppliers, customers, etc.) behavioral characteristics ([Westra, Angeli, Carree, & Ruwaard, 2017](#)). In such an environment, leaders need to make the right decision at the right time to protect their business from external threats. In addition, leaders should gain an advantage in the competition by managing the future crisis well. Healthcare organizations that want to gain superiority in today's tough competition conditions must meet their patients' needs and expectations in the best way and on time. Healthcare managers need agile leaders to ensure patient satisfaction, to use resources efficiently and to provide service in a sustainable, quality, economic and fair manner. Therefore, as competition increases, agile leadership is expected to increase.

Preposition Hypothesis 1b: As competition increases in the sector in which the organization operates, agile leadership increases in organizations.

Patients/Customers. Today, customers are the most important competitive tool of businesses ([Maas & Graf, 2004](#)). Especially regardless of the sector, providing services suitable for the customer demands in every sense is the primary duty of the enterprises ([Galbreath & Rogers, 1999](#)). Agility requires successful planning of resources to meet customer demands and expectations ([Ramasesh,](#)

Kulkarni, & Jayakumar, 2001). In order to meet customer demands in an agile environment, the five important features of the customers should be known (Van Oosterhout et al., 2006). These features are:

1. personalized product and service requests;
2. demand for shortening the delivery time;
3. increased quality expectations;
4. sudden changes in product/service quantity and features; and
5. basic changes in customer tastes.

In this context, changing competition conditions and rapidly differentiating customer demands reveal the necessity of agile leaders in businesses, because agile leaders work hard to gain and build a strong bond with customers beyond customer satisfaction.

Preposition Hypothesis 1c: Agile leadership in organizations increases as businesses increase their emphasis on customer needs and satisfaction.

Technology. Health services are a very rich environment in terms of information. This environment, where information is constantly revised, expanded and shared, is a very complex, diverse, interdependent and dynamic includes intense information discipline (Lintern & Motavalli, 2018) Today, healthcare managers and politicians encourage the development and use of information systems to improve the quality, efficiency and safety of healthcare services. Especially in recent years, technological developments in the health sector cannot be denied. Because technology has become a vital factor for hospitals which want to adapt to social and economic conditions. In this context, the use of technology, evaluation and analysis should be considered. Following the technology closely, knowing how to use the technology efficiently at the highest level and adopting it to the health institutions are among the behaviors expected from agile leaders. Hospitals are institutions that mistake or malpractice cannot be tolerated due to their characteristics. In this respect, agile leaders can prevent unnecessary and misuse of

technology. Therefore, as the dependence on technology increases in the sector, the need for agile leadership will increase.

Preposition Hypothesis 1d: As organizations' dependence for technology increases, agile leadership will increase in organizations.

Social Factors. No matter which geography they live in, the structure, shape and culture of each society are different from each other. Businesses that aim to meet the demands and needs of the society, where they live have always established themselves in the social field ([Ashraf, Razzaque, Liaw, Ray, & Hasan, 2019](#)). Along with globalization, it is very important for companies to know which society has what needs and to develop itself in this regard. Based on this:

Preposition Hypothesis 1e: Social factors affect the agile leadership in organizations in the business environment.

B. Organizational Factors Affecting Agile Leadership

Lean Management. Health systems are getting harder as the rapidly increasing aging in developed countries increases the demand for care ([Sloan et al., 2014](#)). Health systems need to increase the scope of social security and improve the quality of care, while meeting this increasing demand ([DelliFraine, Langabeer, & Nembhard, 2010](#); [Poksinska, 2010](#); [Sloan et al., 2014](#)). To achieve these goals, many organizations have turned toward industrial recovery approaches ([Kaplan, Patterson, Ching, & Blackmore, 2014](#)). In this sense, lean management system is an important improvement method. While it includes low costs and waste, promises improved quality, capacity and assurance ([Curatolo, Lamouri, Huet, & Rieutord, 2014](#); [Kaplan et al., 2014](#)).

Lean management is the simplification of the organizational structure in order to respond to the quality and standards that customers want more quickly and adequately. It is the elimination of unnecessary and non-added value transactions and positions. It is

also to bring the employees and the decision maker as close to each other as possible (Rotter et al., 2017). The main point is the establishment of a structure that is far from unnecessary details and can react immediately without getting caught in bureaucratic obstacles.

The need to improve service quality in the health sector has become increasingly important. Therefore, it is necessary to use the lean management approach to improve service quality and reduce waste in all clinical and support services of the hospital in general (Ben-Tovim et al., 2007). Success with lean depends on how the lean is applied daily. It also depends on ownership by senior management and employees. It is described as a lean journey. Because daily lean applications require time, effort and discipline throughout the enterprise. Each business process should be transformed into a customer-oriented process. Continuous improvement should be part of everyone's business. The whole company culture should change. And all this should also be done while trying to continue your business (Seres et al., 2014). It is known that leaders create appropriate conditions for their followers to reveal their knowledge and skills and use them effectively and encourage their creativity to develop new methods and techniques (Akkaya, Koçyiğit, & Tabak, 2018, p. 187).

Therefore, the lean management approach facilitates agile leadership practices. All health systems aim to control the ever-increasing costs, improve the quality of patient care, increase the number of patients served and reduce waiting times. The most important feature of the agile leader is to manage the changing conditions effectively and adopt the philosophy of continuous improvement and development by combining lean hospital practices with the institution.

Preposition Hypothesis 2a: Adoption of lean management increases agile leadership in organizations.

Sector (Public, Private). There are differences as well as similarities between the public and private sectors. On the one hand,

the public sector requires more analysis, auditing and hierarchy compared to the private sector. The private sector, on the other hand, is a sector with institutions that can make faster decisions due to their structures and aim to achieve a high level of profit ([Andrews, Boyne, Law, & Walker, 2009](#); [Boyne & Walker, 2004](#)). The public sector's hierarchy system and the government's decisive policies may restrict agile practices due to its bureaucratic structure. However, the private health organizations face competition and they have to be agile to survive. Therefore, the private sector may have a suitable environment in terms of agile leadership practices.

Preposition Hypothesis 2b: Level of agile leadership differs depending on the sector (public and private).

Innovative Organizations. Innovation is all kinds of scientific research and inventions that cover all areas and for commercial purposes ([Carr, Kendal, & Flynn, 2016](#)). First of all, innovation is an economic process. Health companies have to apply new techniques in their products in order to survive and gain an advantage in the market where there is intense competition. To do this, organizations must closely monitor the changes and developments in their sectors and keep up with the technology related to their fields. Because, achieving economic development and increasing the quality of life will be possible through innovation ([Oerlemans, Meeus, & Boekema, 2001](#)). The increasing competitive environment prompts healthcare organizations to provide quality service that will attract customers. Thanks to innovation, healthcare organizations meet customers' wishes and needs in the best way. Agile leaders support innovative ideas and practices.

Preposition Hypothesis 2c: Supporting innovative applications and having an innovative culture increases the agile leadership in organizations.

Learning Organizations. The rapid changes in the world naturally affect the organizations. Organizations that can keep up with change can survive. Today, it has become quite difficult to keep up with

speed. Institutions that cannot keep up with the changes may see themselves shrunk or deleted from the market after a while. So, learning is one of the most important advantages that exist in increasing competition and the future of the institution. Learning organizations are structures that not only respond to the change but also lead the change ([Tortorella, Fettermann, Cauchick Miguel, & Sawhney, 2019](#)). In learning organizations, agile leaders seek to create a proactive organization that constantly improves itself by maximizing learning.

Preposition Hypothesis 2d: Level of agile leadership increases in learning organizations.

C. Individual Factors Affecting Agile Leadership

According to [Joiner and Josephs \(2007a\)](#), there is no relationship between leadership agility and personality type. [Joiner and Josephs \(2007a\)](#) stated that regardless of the manager's personality type, every manager has a potential to be agile leader. However, when we consider the sub-dimensions of agile leadership – quickness, flexibility, competence, collaboration and technology pursuit – we expect it to be closely related to proactive personality. Because, [Hayward \(2018\)](#) reported that agile leaders are self-confident, proactive and technology-following individuals.

Proactive personality. [Grant and Ashford \(2008\)](#) defined proactive personality in a multidimensional way as it shows future-oriented actions and affects the individual and his/her environment. Proactive personality is related not only to intervening in the existing conditions, creating a better environment or introducing completely new conditions, but also to investigate learning opportunities and engage in learning activities ([Ashforth, Sluss, & Saks, 2007](#); [Morrison & Phelps, 1999](#)). [Huang, Tang, Lee, and Yang \(2017\)](#) found that proactive personality affects mobile learning performance. So, proactive people will be advantageous in following up technology and developing their competencies. Agile leaders are also expected to create synergies with the ability to collaborate and coordinate development teams, question the status quo, emotionally flexible, act

as a catalyst between internal and external stakeholders, establish a communication-based, secure and empathic working environment. Therefore, proactive personality has been identified as one of the individual antecedents of agile leadership.

Preposition Hypothesis 3a: There is a positive relationship between proactive personality and agile leadership.

Outcomes of Agile Leadership

Agile leadership is a recently studied construct in literature. Therefore, the number of studies on the positive outcomes of agile leadership is still limited. In this chapter, the outcomes of agile leadership will be examined under two subtitles. These are organizational outcomes and individual outcomes.

A. Organizational Outcomes of Agile Leadership

Every organization has a decision-making mechanism. Leaders' behavior is of great importance for the future of the organization. Agile leaders also contribute important developments in the organization with their behaviors. These are patient/customer and employee satisfaction, patient/customer and employee commitment, employee job performance, etc.

Customer/Patient Satisfaction and Commitment. The characteristics of the patient such as his/her past experiences, age, gender and education level may affect the patient's satisfaction ([McNeese, 1999](#)). Some of the situations positively affect the patient satisfaction in the health institutions. For example, the healthcare professional's taking care of the patient, the waiting time and a place where comfort and hygiene are important for patient satisfaction (Halil & Tarım, 2011). Agile leaders care about situations in which patients can be satisfied. So, patients are attracted to choose the health institution again which also means the patient's emotional commitment to the organization.

Preposition Hypothesis 4a: Agile leadership has a positive effect on customer/patient satisfaction and emotional commitment to organization.

Employee Satisfaction and Commitment. Employee satisfaction is briefly the level of meeting the expectations of the employee in a positive way ([Butcher & Davis, 1988](#)). Today, there are many factors that affect employee satisfaction. Leaders strive to maximize the level of satisfaction of their employees ([Özpehlivan, 2018](#)). Agile leadership has positive consequences for employee satisfaction. Agile leaders allow subordinates to improve themselves by providing them with mentoring support, for this reason, employees who are satisfied with their jobs also feel commitment to their organizations.

Preposition Hypothesis 4b: Agile leadership has a positive effect on employee satisfaction and emotional commitment to organization.

Motivation of Employee. Motivation is divided into two categories as instinct and extrinsic motivation.

Intrinsic motivation involves people doing an activity because they find it interesting and derive spontaneous satisfaction from the activity itself. Extrinsic motivation, in contrast, requires an instrumentality between the activity and some separable consequences such as tangible or verbal rewards, so satisfaction comes not from the activity itself but rather from the extrinsic consequences to which the activity leads. ([Gagne & Deci, 2005](#), p. 331)

In an organization with agile leaders, no matter how negative the conditions are, agile leaders work with great motivation and show high performance. It values them by striving for their employees to work in better working conditions. Agile leaders understand the expectations, needs and psychological conditions of their employees

and drive them in this direction. As a result of all this, agile leaders affect their organizations positively.

Preposition Hypothesis 4c: There is a positive relationship between manager's level of agile leadership and employee's motivation.

B. Individual Outcomes of Agile Leadership

Agile leadership requires not to be afraid of change, to renew itself regularly, to benefit from the opportunities brought by uncertainty, to protect against dangers, to provide competitive advantage and to be able to read customer expectations correctly. Due to these features, agile leadership may have a positive effect on a manager's performance and self-development.

Job performance of Agile Leader. Job performance is defined as the efforts made by individuals to achieve their organizational goals (Bağcı, 2014). In organizational terms, it refers to the mastery and professional aspects of the work. Agile leaders accomplish every task they do with great dedication and expertise. It is an indispensable policy of agile leaders to provide quality service for the organization and to provide an advantage in competition. In this context, agile leaders embrace the realization of their business performance with high quality and efficiency.

Preposition Hypothesis 5a: There is a positive relationship between manager's level of agile leadership and manager's job performance.

Self-development. People constantly change and strive to improve themselves. In this context, personal development is the change experienced by the individual as a result of his/her positive efforts to improve himself/herself (Reichard, Walker, Putter, Middleton, & Johnson, 2017). In this respect, the personal development levels of agile leaders will be very high. They reveal their hidden potential. They change the negative conditions in the organization and try to turn it into a positive one. Agile leaders do not

find the existing conditions sufficient and always think that there can be better. In short, they try to move themselves and their team further.

Preposition Hypothesis 5c: There is a positive relationship between manager's level of agile leadership and manager's self-development.

Conclusion

Today, competition is more based on the agility of existing products, services and business models (İleri & Soylu, 2010). In addition, managers and owners are also aware of the need to be agile (Akkaya, 2019). In order to apply agile management systems and be successful, companies need to follow technological developments and adapt them very quickly. Therefore, agile leaders are of great importance in this process.

With this chapter, a model has been developed in order to examine the agile leadership behaviors of health care managers in a comprehensive manner. As a result of the literature review, first of all, agile leadership's sub-dimensions are defined as quickness, competency, flexibility, collaboration and technology tracking. Then, the environmental structure that allows managers to exhibit agile leadership behaviors has been analyzed. "Drivers of agile leadership," "organizational factors supporting agile leadership" and "individual factors supporting agile leadership" are identified as the antecedents of agile leadership. According to the previous studies, agility contributes to healthcare institutions such as reducing errors and wastes, increasing safety and quality, increasing patient and employee satisfaction, etc. (Britt Pipe et al., 2012). From this point, "organizational outcomes" and "individual outcomes" are determined as the outcomes of agile leadership in the health sector.

Future Research Directions

Agile leadership emerges as a new concept in business management literature. Despite its importance, there is still limited research on agile leadership. Therefore, the antecedents and outcomes of agile

leadership are needed to be explored. In this chapter, the authors have developed prepositional hypotheses about agile leadership's antecedents and outcomes. The authors recommend that researchers should empirically test these prepositional hypotheses.

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Chapter 4

Industry 4.0 and International Trade: The Case of Turkey

Deniz Erer and Elif Erer

Abstract

The concept of Industry 4.0 emerged in Germany in 2011 for the first time causes production processes to be increasingly interconnected and to be more effective owing to its economic and social benefits. The structure of international trade is getting changed due to Industry 4.0. Manufacturing sector is one of the leading sectors in international trade. Manufacturing sector which has remarkable share of exports brings up against significant challenges aiming to competitively design and manage modern products, which are changing and accelerating the complicated multi-domain systems. That producing is more innovatively due to advantages provided by Industry 4.0 impact, positively international trade. The reason is that use of advanced robotic machines and automatization in production process reduces labour costs and thus increases competitiveness of a country in the international markets. The aim of this study is to exhibit a theoretical framework between the digitalization and international trade and to examine how Industry 4.0 impacts international trade in Turkey.

Keywords: Industry 4.0; automation; digital technologies; smart systems; international trade; competitiveness

1. Introduction

Industry is the important element of an economy which produces highly mechanized, automatized and value-added goods. Industrialization firstly began in the field of mechanization and it was named as first industrial revolution. Technological developments have proceeded over time, and second industrial revolution took place with the intensive use of electrical energy in production process and third industrial revolution occurred with the intensive use of the digitalization. Today, following third industrial revolution, the development toward Industry 4.0 (or fourth industrial revolution) has emerged in the manufacturing industry ([Lasi, Fettke, & Kemper, 2014](#)). The term “Industry 4.0” means “the organization of production processes based on technology and devices autonomously communicating with each other along the value chain” ([Rojko, 2017](#)). This concept was firstly introduced in the article by [Kagermann, Lukas, and Wahlster \(2011\)](#) at the Hanover Fair in Germany. Then, German National Academy of Science and Engineering published the Industry 4.0 manifesto in 2013.

Fast processing of technology and digital world is taking place in all organizational authorities in all areas and in all kinds of sector, because the business world is transformed by the postmodern revolution-Fourth Industrial Revolution ([Akkaya, 2019](#)). Industry 4.0 in production will arise from an integration of a variety of technologies which range from digital technologies such as the Internet of Things, 3D printing, advanced robotics to new processes such as artificial intelligence, data-driven production, to new materials such as bio-based or nano-based. This transformation in production processes will have a wide range of consequences for economic variables such as employment, income distribution, education, welfare, economic growth, and international trade ([OECD, 2017](#)).

The structure of international trade has to change due to Industry 4.0. With these advances in technology, international trade not only deals in commodities, but also trades in data driven goods. When production is more innovative due to the advantages provided by Industry 4.0, it affects international trade between two or more countries. Although production costs increase because of the investments to be used in the application of Industry 4.0, the use of advanced robotic machines and automatization in production processes reduce labour costs. It is expected for the importance of labour to decrease in production because of the fact that production exactly transforms to automatization with Industry 4.0. Therefore, competitiveness will be determined by technological advantages rather than cost advantages.

Smart factories supported by cyber-physical systems and the Internet of Things (IoT) will ramp up output. High-level investment costs relating to advanced technological machines initially reflect on the production costs; but then, costs are cut down owing to specialization and mass production. Moreover, Industry 4.0 applications are used adversely to influence export performances of the countries involved. The countries in question become more competitive through high-tech products in global markets and stimulate exports. The increasing of export figures creates surplus value in international trade of a country, and this situation leads to a rise in gross domestic product. Also, increasing exports appear as a factor raising not only total expenditures, but also employment.

Firms having the characteristics of a global brand in many sectors have shifted production lines to the countries with low labour costs over the last 20 years. Decreasing production costs by means of information and communication technology becomes more and more important in terms of avoiding the shifting of large parts of production lines to these countries. This transformation in industry provides important competitive advantages for countries. Industry 4.0 provides a number of important advantages for firms such as decreasing of quality assurance and carrying costs, shortening the time to market and increasing staff productivity. These acquirements in the most important elements of production will enhance total productivity. By

the productivity in question is growing, competitive power at the national level will increase (TUBITAK, 2016).

Technology is highly correlated with trade. Trade has become more modern owing to technology. Technological developments facilitate Global Value Chains by substantially decreasing communication, transport and co-ordination costs (OECD, 2017). Because of this relationship between technology and trade, policy makers should establish policies considering both (Fig. 1).

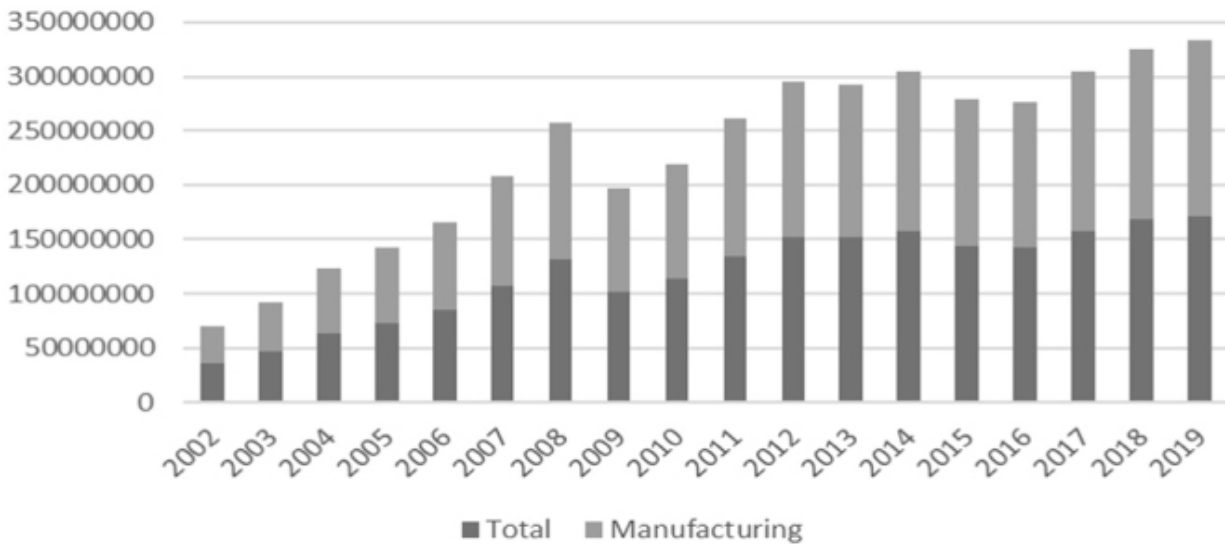


Fig. 1. Total Export and Export in Manufacturing Sector (Thousand US Dollar). *Source:* Turkish Statistical Institute.

Many countries around the world are accelerating preparations for Industry 4.0. Industry 4.0 technologies, which firstly started to be used in the automotive industry because of its position as fastest-growing industry in Turkey, seriously shorten the time products take to put on the market. In the present process, the most important mission of Turkey, the same as each country wanting to keep their competitive advantage and save their competitive power, is to carry out the requirements of Industry 4.0. Particularly Germany and the other industrial giant countries, which are leaders in the industrial area, have kept working within the framework of Industry 4.0 since 2013. However, interest relating to this concept in Turkey has only increased since 2015. Order of the meeting Supreme Council for

Science and Technology was determined as Industry 4.0 by Ministry of Science, Industry and Technology on February 2016. However, if Turkey cannot keep up with so-called transformation, Industry 4.0 will cause competitive pressure on Turkey's international trade to increase. In such a case, it is needed for new smart machines and systems which cannot be produced in the country to be exported. On the other hand, export volumes for Turkey are going to decrease unless it brings out the goods and services needed by the digital world. It is expected that half of the global trade volumes will use the communication of smart objects by 2030. It is clear that the shares of global trade volume from the countries not suiting up smart production systems will be lowered (Fig. 2).

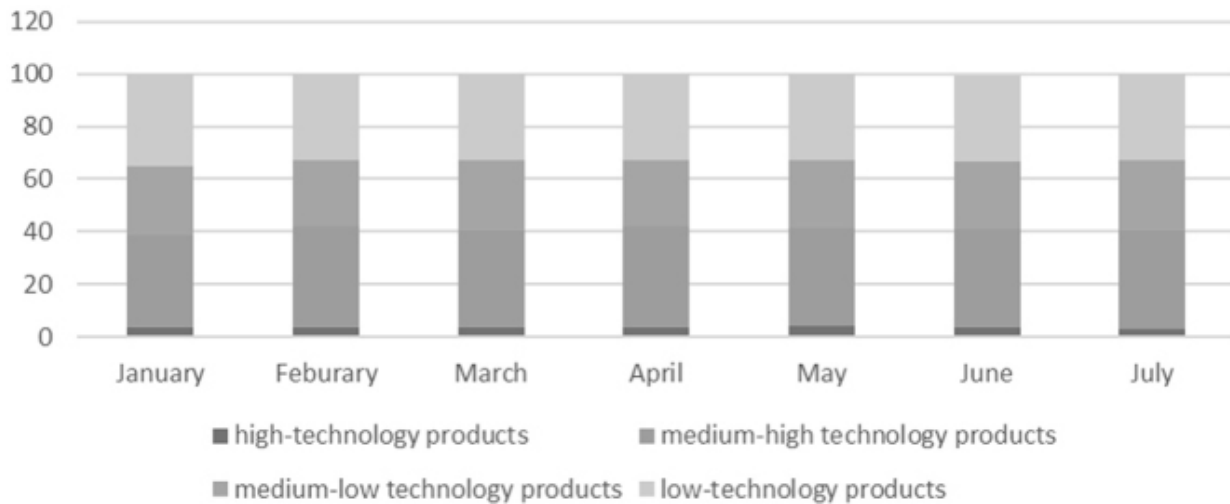


Fig. 2. Export of Manufacturing Products in Terms of Technology Density (Thousand US Dollar). *Source:* Turkish Statistical Institute.

In the light of this information, the aim of this study is to examine how Industry 4.0 affects international trade in Turkey. For this purpose, we will firstly explain Industry 4.0 and its main concepts in Section 2. Then, we will discuss how these technologies will contribute competitiveness of exporters in terms of theoretical framework between Industry 4.0 and international trade in Section 3. The development of Industry 4.0 in Turkey will be examined in Section 4. The effect of Industry 4.0 on international trade in Turkey

will be investigated in Section 5. Finally, we will discuss the obtained results and policy proposal.

2. The Main Concepts of Industry 4.0

The concept of revolution refers to radical and sudden change. Revolutions emerged during history as new technological advances cause an intensive change in economic systems and social structures. Since the beginning of the industrial revolution, technological advances have made it possible to complete three main stages, indicating a massive increase in industrial efficiency. The first industrial revolutions, which last over the period of 1760–1840, started with shift from muscle power to mechanical power. Steam-powered machines have started to be used in the factories. After the mechanization of production in the First Industrial Revolution, the technology more advanced and the transition to the Second Industrial Revolution. This period, which coincided between 1840 and 1870, is also known as the technology revolution. When the technology revolution arose, transportation developed positively with railways. Besides, in this period, the advance of electricity technology and the use of electricity in production process made mass production possible. Then, the Third Industrial Revolution started in the 1960s. This revolution also is known computer or digital revolution because of development of semiconductors, mainframe computing (1960s), personal computing (1970s and 1980s) and the internet (1990s). Automation became widespread in the industry with electronics and information technologies (IT) since the 1970s. Today, we witness the fourth stage of the industrial revolution, in which value chains are connected end-to-end through cyber-physical systems and dynamic data processing (Schwab, 2017, p. 10; [TUSIAD BCG, 2016](#)).

The concept of “Industry 4.0” is first emerged in Germany in 2011 and is now considered a rising trend all over the world due to the existing economic and social benefits ([TUBITAK, 2016](#)). Industry 4.0 is the rapid transformation that occurs in design, production, operation, and production systems service and products. As the German Chancellor Angela Merkel says, “Industry 4.0 is the comprehensive transformation of the whole sphere of industrial

production through the merging of digital technology and the internet with conventional industry.” In brief, everything in and around the production process is digitally related ([European Parliament, 2015](#))

We can summarize the basic concepts related to Industry 4.0 as follows ([Bilim, Sanayi ve Teknoloji Bakanlığı \(BSTB\), 2018b](#); [European Parliament, 2015](#); [TUSIAD BCG, 2016](#)).

Internet of Things: It refers to IT systems connected to all subsystems, processes, internal and external objects, producer, and consumer networks. The Internet of Things will allow more devices, even semi-products, to take advantage of integrated data processing by interconnecting with standard technologies. In this way, the equipment in the field will be able to communicate with each other and, if necessary, with central control systems. It will also make real-time decision-making possible, eliminating the need for central analysis and decision-making processes. The Internet of Things basically benefits from the physical systems communicating with each other. In this structure, data are collected from machines, computers, and automation systems used in production, and the collected data are transferred to local and global servers by wired and/or wireless communication methods. With the software solutions on the servers, current and historical data are evaluated, and actions regarding the physical system are implemented.

Artificial intelligence and smart software: Artificial intelligence is defined as technologies that imitate human intelligence by learning and making their deductions from what they learn. Thanks to artificial intelligence technologies, many systems, tools, and machines can become autonomously operating without human intervention. In the manufacturing industry, artificial intelligence or smart software solutions create benefits by ensuring the continuity of past learning and systems under changing and unexpected environmental conditions. Artificial intelligence systems have sub-headings such as machine learning, artificial neural networks, deep learning, pattern recognition, and application areas such as autonomous planning and natural language processing. Besides, artificial intelligence systems are seen as the most critical IT factor in digital transformation in the manufacturing industry, due to developing hardware infrastructures and cloud solution algorithms supported by data access. Therefore,

artificial intelligence is among the priority areas of all leading countries in digital transformation.

Cyber-physical systems: These systems consist of software embedded in hardware such as sensors, processors, and communication technologies, and they can autonomously change information, trigger actions, and control each other independently. Cyber-physical systems refer to all of the hardware and software systems that connect the physical world with the virtual world. These systems, which connect the virtual and the physical world, are composed of two crucial elements: a network of objects and systems that communicate with each other over the internet or with a designated internet address, and a virtual environment created by the simulation of real-world objects and behavior in the computer environment.

Cybersecurity technologies: Cybersecurity technologies are known as technologies that protect networks, computers, programs, and data from cyber-attacks. Cybersecurity infrastructure is of vital importance for the successful digital transformation in the industry and ensuring its sustainability.

Smart factories: Smart factories, also called “factories of the future,” represent some of the technical innovations under Industry 4.0, such as the integration of Information and Communication Technologies into the production process.

Big Data: The use of big data in production, the collection, and comprehensive evaluation of data from many different sources, such as production systems, as well as corporate and customer-based management systems, improves the quality of production, saves energy, and facilitates equipment maintenance.

Cloud Computing: Cloud computing is the collection of programs and data on a virtual server (in the cloud). Each device connected to the Internet can easily access information, data, and programs, and all of these services. Companies already use cloud-based software for some enterprise and analytics applications. With the increase in the performance of cloud technologies, the data and functions of the machines in the cloud platforms will increase, and more data-based services will be provided to the production systems.

Virtual reality and augmented reality: Simulating real-life or situations by computer-aided simulation and artificial reconstruction, presenting an artificial reality experience by abstracting the user from the physical world, is called virtual reality. Augmented reality is expressed as a situation where the users can interact with the environment around them, which is formed by combining the elements created by computers and enriched with sensory inputs in real-time. Using virtual reality and augmented reality in production processes will increase productivity and decrease costs.

Autonomous robots and mechatronic systems: Autonomous robots are machines with advanced and multidisciplinary elements that mimic the functions and lifestyles of living things, with programmable skills and intelligence. Robots have subgroups such as industrial robots and service robots. Mechatronic systems include smart machines, devices, systems, and processes. Mechatronic systems perceive their environment, can make decisions, and change their setting by commenting on the perceived environment.

Additive manufacturing systems: Additive manufacturing refers to the production by adding a certain material in layers on top of each other. Due to the developing technology, additive manufacturing has become possible with very thin layers. Additive manufacturing systems do not process any block pieces, because they produce the object by adding, and it works like computer numerical control machines. Another critical point is that there is no need for molding (as in traditional manufacturing techniques) to produce an object in the additive manufacturing industry. The 3D design of the object is transferred to the machine and can be produced one-to-one without any additional method. This method will be used more and more widely in the following periods, especially in areas such as complex and light designs, to produce small scales of special products. As a result, high performance and decentralized additive manufacturing systems will reduce logistics costs and stock levels. See Appendix (The Main Concepts) for the more detailed informations about Industry 4.0.

3. Theoretical Framework

The continuous changes in customers' demands and needs make it inevitable for organizations to constantly adapt to changes to keep their firms in a competitive environment ([Akkaya, 2019](#); [Akkaya & Tabak, 2020](#)) Because today companies operating in emerging markets face with certain problems especially in turbulence, complex and changeable environment ([Akkaya & Tabak, 2017](#)). Organizations need to use dynamic, more flexible and agile abilities effectively. It has become inevitable for enterprises to change the organizational structure and become more flexible and agile organizations to keep up with technological innovations and to overcome competition ([Koçyiğit & Akkaya, 2020](#)).

The production technologies change due to the dynamics of economic. In this point, the concept of innovation comes into prominence. In other words, the innovations play an important role in the development of economic and social lives ([Topleva, 2018](#)). Josef Schumpeter demonstrated that entrepreneurial creativeness presents innovations in the production process due to consumer needs and preferences in the 1940s ([Schumpeter, 2003](#)).

How fourth industrial revolution (or Industry 4.0) will affect the economy have discussed among economists. Some of economists who are known as techno-pessimists claim that the effect of digital revolution on productivity is almost over because of the contributions from digital revolution to have already been made. However, others who are techno-optimists argue that the technology and innovation in question will cause an increase in productivity and therefore it will provide higher economic growth (Klaus, 2016).

Another effect of the Fourth Industrial Revolution is on international trade. A country's ability to be more active in international trade depends highly on technological production, specialization, adaptation to change, and the potential to develop its innovation capacity. In today's world, where all sectors survive with the high-tech, countries will become more advantageous in international trade due to a transition from conventional industry to high technology ([Bilim, Sanayi ve Teknoloji Bakanlığı \(BSTB\), 2018a](#)).

The impact of Industry 4.0 on international trade stems from the comparative advantage of countries. The basis of modern foreign trade theory was put forward by David Ricardo. In his study,

“Principles of Political Economy and Taxation,” published in 1817, Ricardo stated his views on foreign trade. The theory of comparative advantages based on comparing a country’s foreign trade with international comparative costs. Ricardo firstly distinguished international trade from domestic trade via so-called theory ([Blaug, 1977](#)).

According to Ricardo, the rules relating to international trade differ from domestic trade. The mentioned distinction stems from the assumption of capital and labour not to move between countries. Thus, international trade is based on comparative production advantages, unlike domestic trade (Ricardo, [1817] [2004](#)). Ricardo stated that the countries have advantages. According to this theory, countries tend to purchase goods through importing that they cannot domestically produce or produce at a higher cost compared to the international market. Similarly, countries export goods and services that other countries cannot produce or produce at a higher cost than the international market. Every country tries to increase exports and decrease imports in order to achieve foreign trade balance. Competition in foreign trade arises here.

With the intensification of international trade and the increase in the diversity of commercial activities, capital movements among countries have accelerated. Accordingly, the rate of development in technology has increased immensely. In this context, the desire of the countries to have advanced technology in order to increase their competitiveness for foreign trade has become one of the main targets of both developed and developing countries.

In this sense, the digitization of the manufacturing industry is among the most important and priority policy areas for many countries. When country practices on the digital transformation of the manufacturing industry are analyzed, it is seen that four issues come into prominence, namely the development of educational infrastructure, supporting users, supporting national suppliers, and strengthening the data communication infrastructure ([BSTB, 2018a](#)).

Although initial manufacturing costs are high, Industry 4.0 enables manufacturers to produce quality goods at lower prices in the long run by providing low costs and labor cost savings. Therefore, countries with high labor costs will be able to compete with countries

with low labor costs. These countries will have a better competitive advantage, along with increased foreign trade potential. In other words, countries that go through the Industry 4.0 transformation will be much more competitive in international trade. As a result, one may infer that countries that produce Industry 4.0 appliances will have better foreign trade balance.

In order to effectively use the trade as an asset for growth and sustainable development, trade costs should be reduced in developing economies to participate in international manufacturing networks. Tackling non-tariff sources of trade costs, addressing cumbersome regulatory procedures, and documentation requirements are among the advisable solutions. Especially, trade facilitation, including paperless way of trade, has lately become a significant element.

Reducing trade costs and developing trade facilitation implementation can be accomplished by digitalization. Findings infer that digital technologies provide significant gains in the long run ([Digital Dividends, 2016](#)). However, one may encounter drawbacks estimating the impact on international trade due to the indirect growth and widely dispersed structure and complicated international supply chains. Although there exist certain drawbacks and challenges, research findings approve that transformation to digital technologies will eventually result in many benefits for countries.

One of the prominent concepts in increasing the competitiveness of countries in international trade is efficiency. Efficiency is defined as the output per input. Efficiency can be increased by increasing the amount of output per input or by reducing the amount of input per output. The efficiency of the country is directly related to its competitiveness. Studies examining the quality, efficiency, and operational effectiveness effects of Industry 4.0 infer that the optimization of automation, the development of product customization, along with the increase in efficiency, are among the benefits of Industry 4.0 ([Dalenogare, Benitez, Ayala, & Frank, 2018](#)).

The existing literature focused on Industry 4.0 transformation, which has become a trademark of Germany, started with the study by [Kagermann et al. \(2011\)](#), who are known as founding fathers of Industry 4.0. The authors in question represented how to transfer in

to Industry 4.0 and how to succeed in the changing paradigm shift via internet channels of objects in the study. In other words, it is explained the bridge between the cyber world and the physical world and synchronization between the digital world and the physical veracity in the study. They stated that Germany will have taken a leading role in digital transformation within ten years.

Some of the studies investigating the effects of digitalization on international trade in the literature have focused on the role of the Internet in so-called relationship. According to these studies, firms' potential to produce new goods and services and to serve new markets enhance substantially through the Internet. Besides, firms' transaction costs including communication, information and coordination diminish due to the one. The reason is that the use of websites, emails, dedicated platforms and online marketplaces smooth transaction of firms ([Tan, 2017](#)).

The other studies focus on the effects of digitalization on international trade research and how broadband users in a country impact a country's volume of international trade. These studies exhibit that an expected increase in broadband use causes trade openness to gross domestic product (GDP) to improve in the country. Also, the arguments related to digital trade facilitations focus on the application of modern information and communication technologies so as to simplify and automate international trade procedures ([Duval & Mengjing, 2017](#); [Mishra, 2017](#); [Papachashvili, 2018](#); [Riker, 2014](#)).

[Duval and Mengjing \(2017\)](#) investigated the paperless trade measures involving regional trade agreements. They indicated that since 2005 the amount of paperless trade measures in regional trade agreements put into action initially doubled. [Mishra \(2017\)](#) emphasized the sensitive and complicated relationship between the internet with digital transformation and international trade. In the study, it was shown that many issues relating to internet policy are central to trade in digital economy. Besides, cybersecurity and data protection act as barriers to electronic. [Papachashvili \(2018\)](#) stated that digital transformation encourages countries to improve paperless trading systems. Thus, countries can have broad platform so as to decrease trade costs. However, countries need to sort this with

issues concerning the cross-border data flows, privacy and competition conditions.

4. Development of Industry 4.0 in Turkey

One of the chronic problems of Turkey's manufacturing industry is that exports are highly dependent on the import ratio ([Kundak & Aydoğuş, 2018](#)). Also, the production and foreign trade structure of the manufacturing industry does not consist of high technology products ([Turkey Exporters Assembly, 2017](#)).

Although a few steps have been taken to solve the chronic foreign trade problems of Turkey, a radical transformation as the fourth industrial revolution, which is considered essential and highly innovative, has not been achieved yet. In this sense, digital transformation for Turkey's manufacturing industry should be considered as an excellent opportunity to implement necessary reforms.

Turkey's High Technology Transition Program was initiated in order to upgrade the Turkish manufacturing industry to high technology. It is foreseen that this transformation will take place with three elements. These three elements are identified as localization, the fourth Industrial Revolution and R&D, and the innovation ecosystem ([BSTB, 2018a](#)). Some focus sectors have been identified to increase the share of high-tech products in production and exports to get more shares from global trade and improve the industry's high technology production capacity. This situation is explained in the study conducted by [TUBITAK \(2016\)](#).

[TUBITAK \(2016\)](#) argues that the Turkish industry will benefit a lot in competitive advantage when the digital transformation takes place. However, the existing competitive advantages of Turkey, such as logistics and low labor cost, will not be valid if this digital opportunity is missed. In this sense, one may readily infer that the Turkish current account deficit will get worse if the digital transformation in the industry is not completed. According to the data of a [TUBITAK \(2016\)](#) study, the digital maturity of the Turkish industry is somewhat between Industry 2.0 and Industry 3.0. In other words, Turkish firms have not been transformed into automation technologies yet. The

three sectors with the highest digital maturity level are materials (rubber and plastic), computers, electronic and optical products, automotive and white goods sub-industry. Also, electronics, software, and materials are among the sectors with the highest awareness of digital transformation. In light of these findings, 8 critical technologies, 10 strategic targets, and 29 critical products were determined in three technology groups (Digitalization, Interaction, and Factories of the Future) on the technology roadmap developed by TUBITAK. Automotive, white goods, chemistry, and food sectors have been selected by taking into consideration the added value for priority sectoral applications. It has been noted that the selected sectors are already at a reasonable level of digitization. R&D studies are planned to be supported by TUBITAK in sectors that have the potential to make progress. TUBITAK continues its digitalization program in the industry with many supported projects and doctoral programs. On the other hand, it carries out the work of determining the digital maturity levels of public institutions. However, when considered the fact that Turkey's manufacturing industry consists mainly of SME's (Small and Medium Business)¹ that are not considerably supported with R&D by TUBITAK, granting mere focus sectors with digitalization is likely to postpone the overall transformation of Turkey. At this point, while TUBITAK highly and intensively supports focus sectors, the quality and quantity of incentive and support mechanisms aimed at moderate firms should be developed.

The research conducted by TÜSİAD (Turkish Industry & Business Association) in cooperation with Boston Consulting Group (BCG) in 2017 ([TÜSİAD & BCG, 2017](#)) supports the findings of [TUBITAK's study \(2016\)](#). According to research findings, 90% of Turkish companies have a high level of digital transformation awareness; however, only 61% of them are considered ready for this transformation. On the other hand, Turkish companies are yet in the process of implementing, pre-investment, and planning when compared to the companies in developed countries. Companies are in need of professional support, especially in developing strategies and roadmap, and managerial competencies. In this sense, developing mechanisms, which decrease Turkey's technology import and enable domestic technology production, will not only accelerate

the digital transformation but also ensure the domestic production and localization of transformation technologies within the country.

As Turkey's manufacturing industry consists mainly of SMEs, the way to capture the Industrial Revolution 4.0 is through the digitalization of SMEs. What is more important is that Turkey's determination to be the most prominent regional supplier of digital products and presenting digital services, products, and qualified labor to global markets rather than missing this drastic transformation. However, there exist severe concerns regarding the digital transformation among SMEs, the backbone of the Turkish industry. Chemistry and medicine, motor vehicles, machinery and equipment, semiconductors and electronics, and food and beverage are among the primary focuses of digital transformation support programs. SME's in such sectors are planned to be supported in their journey to digital transformation. The cluster model support program will also be included in the digitization process of SME's ([BSTB, 2018b](#)).

The biggest obstacle to digital transformation is determined as the lack of efficient and qualified workforce in the digital world, not only in Turkey but all over the world. The need for a workforce with advanced computing skills, producing, using, and managing smart machines and analyzing data from these technologies is rapidly increasing all over the world. Turkey is well aware of this situation and determined the first of six essential components of Digital Turkey roadmap as "human." Aiming to improve the educational infrastructure and skilled labor upbringing, Turkey will proceed in two ways from this point. The first of these is to train the workforce that the digital world will need and take the necessary measures, and the second is to improve the competencies of the existing workforce ([BSTB, 2018b](#)).

The objectives of the high technology transition program in the industry have been determined as reducing the foreign trade deficit, doubling the added value of the manufacturing industry, increasing the share of high technology products in exports, creating an industry with its own technology and increasing the competitiveness of the industry ([TUBITAK, 2016](#)).

Latin America and Middle East countries that provide incentives without any performance criteria do not seem to be successful in implementing their industrialization policies. On the other hand, it is

known that East Asian countries that provide industrial incentives based on the carrot-stick principle both withdraw support from companies that fail to meet their performance criteria and deprive them of any government support for a certain period of time ([Rodrik, 2009](#)). At this point, determining the focal sectors in the digitization process of Turkey's industry has been very accurate. The aim is to make a difference with these sectors and to ensure that companies that already produce medium-high and high technology are competitive on a global or regional basis. According to estimates, half of the world markets will trade smart products and services in 2030. In this sense, it is necessary to analyze failed and successful examples in implementing a series of incentives on digitization and create a proper model for Turkey that ensures the efficiency of resources aimed at supporting the focal sectors ([TUBITAK, 2016](#)).

5. The Effect of Industry 4.0 on International Trade in Turkey

Turkey has performed the transition from a low-middle technology group to a high technology group due to its investments in industry and technological transformation within the last 15 years. Since 2002, the share of products in the medium high technology category has increased by 50.4%, and the share of medium-high and high technology products has increased by 32.8% ([BSTB, 2018a](#)). According to 2019 data, manufacturing industry products stand for 94% of total exports of Turkey.

Within the scope of Transition to the High Technology Program in Industry, initiated in Turkey, it is aimed to increase the high technology production capacity of industry and be active in the global market with highly added value products. In this sense, focus sectors have been determined to double the added value of industry and increase the share of high technology products in production and exports. Chemistry and medicine, electronics and semiconductor, machine and equipment, and food and motor vehicle are among the aforementioned focus sectors. In addition, the information and software industry, which has a parallel relationship with other sectors, and the steel industry, which provides input to the sectors, are also

among the focus sectors. Due to technological transformation, it is expected that Turkey's industrial added value will reach 300 billion dollars, and the share of high-tech products will increase up to 15% in the following ten years ([BSTB, 2018a](#)).

The studies focused on the impact of Industry 4.0 on international trade for Turkey highlight that Turkey should improve immediately the global competitiveness index by complying with Industry 4.0 in manufacturing sector. [E. Nuroğlu and Nuroğlu \(2018\)](#) investigated the potential impacts of digital transformation to be happened in Turkey and around the world through Industry 4.0 on manufacturing sector and international trade in terms of a theoretical framework. In the study, it is indicated that the strategic documentaries focused on increasing share of medium-high and high technology products in export aim to solve the chronic problems relating to international trade, to increase technological density in Turkey, to decrease imported input ratio in producing process via naturalization projects and to double at least value added. [Özkan, Al, and Yavuz \(2018\)](#) evaluated how Industry 4.0 affects the economy in Turkey. In the study, it was stated that Industry 4.0 has positive effect export performance by means of using efficiently the applications of Industry 4.0 and augments competitive power in global markets. Besides, Industry 4.0 creates positive effect on consumption, investment, saving and employment due to the increasing of export and economic growth.

6. Conclusion

Since the beginning of the industrial revolution, technological advances have made it possible to complete three main stages, indicating a massive increase in industrial efficiency. At the end of the eighteenth century, steam-powered machines started to be used in the factories. Also, mass production was possible with electrical energy at the beginning of the twentieth century, and automation has become widespread in the industry with electronics and IT since the 1970s. Today, we witness the fourth stage of the industrial revolution, in which value chains are connected end-to-end through cyber-physical systems and dynamic data processing (TUSIAD).

The concept of “Industry 4.0” is emerged in Germany in 2011 and is now considered a rising trend all over the world due to the existing economic and social benefits. The New Industrial Revolution refers to the digital transformation of production systems with advances in information and communication technologies. The digitalization of every stage of the production chain, the development of the machine-human-infrastructure interaction, and the development of the “Intelligent Production Systems” have created a paradigm shift in the industry ([TUBITAK, 2016](#)).

- Digital transformation in industry enables many countries which assort with Industry 4.0 to have a competitive advantage in international trade. In other words, a country’s exports increase and its imports diminish owing to Industry 4.0. Moreover, the positive impact of Industry 4.0 on a country’s competitiveness extinguishes the advantages of countries with cheap labour cost in manufacturing sector.
- The quality of manufacturing improves via smart systems. Producing of faultless and higher quality products can increase due to creating information using big data analysis and smart software. Owing to the fact that losses in producing process are mostly put away by means of smart systems, time and costs in manufacturing diminishes. Therefore, the quality and quantity of products in manufacturing sector go up in a short time. Consequently, export increases and import decreases through Industry 4.0. These developments in manufacturing sector cause economic growth to push up.
- In this sense, the digitization of the manufacturing industry is among the most important and priority policy areas for many countries. In this study, it is evaluated the effects of digitalization and innovations having come out after Industry 4.0 on international trade in Turkey. Also, it is explained how the generated innovations especially in manufacturing sector impact competitiveness of the country.
- Turkey should improve immediately the global competitiveness index by complying with Industry 4.0 in manufacturing sector, which stands for 94% of total exports of Turkey, so as to decrease trade balance deficit. The training system should comply with Industry 4.0

in order to increase competitiveness in international market because of the fact that the existing training system is not convenient for 4.0 industrial revolution. The government is the leader to encourage the Industry 4.0 applications enable to increase export, to decrease import and therefore to close the current account deficit in Turkey.

Appendix (The Main Concepts)

Industry 4.0: This term also is called as the Fourth Industrial Revolution or the New Industrial Revolution. It is defined as the digital transformation in production process caused by advancements in technology such as information and communication, machineries which communicate with each other autonomously.

Internet of Things: Internet of Things is IT systems in which collected data from computers, machines, automation systems are transferred to local and global servers by wireless communication methods. Therefore, historical and current data can be evaluated and actions regarding the physical system are implemented.

Big Data: Big data are data which have increasing size and involve more varieties. Big data are larger and more complex data set than new data sources. Big data focus on what to do it rather than how much knowledge you have. It analyzes data obtained from any sources in order to provide saving of time and money, to develop new projects and to find optimal solution recommendations.

Cyber-physical systems: Cyber-physical systems are systems in which a physical mechanism is controlled and pursued by computer-based algorithms. They are composed of software embedded in hardware such as processors and sensors. Besides, they represent all of the software and hardware systems which link the physical world with the virtual one.

Artificial intelligence and smart software: Artificial intelligence refers machines and systems which imitate human intelligence and enhance themselves iteratively according to collected information. Artificial intelligence can be also thought as a sub-field of computer sciences in which either hardware or software smart systems are designed and improved. Also, the software developed by various methods and techniques to solve the real-world problems can be

called as artificial intelligence. The purpose of this technology is to create intelligence systems to solve real-world problems via quantitative systems.

Cloud Computing: Cloud computing is the collection of programs and data on a virtual server (in the cloud). Each device connected to the Internet can easily access information, data, and programs, and all of these services.

Virtual reality and augmented reality: Simulating real-life or situations by computer-aided simulation and artificial reconstruction, presenting an artificial reality experience by abstracting the user from the physical world, is called virtual reality. Augmented reality is expressed as a situation where the users can interact with the environment around them, which is formed by combining the elements created by computers and enriched with sensory inputs in real-time.

Autonomous robots and mechatronic systems: Autonomous robots are machines with advanced and multidisciplinary elements that mimic the functions and lifestyles of living things, with programmable skills and intelligence. Robots have subgroups such as industrial robots and service robots. Mechatronic systems include smart machines, devices, systems, and processes. Mechatronic systems perceive their environment, can make decisions, and change their setting by commenting on the perceived environment.

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¹SMEs are non-subsidiary, independent firms which employ fewer than a given number of employees. Although this number varies across countries, the most frequent upper limit designating an SME is 250 employees, as in the European Union. Financial assets are also used to define SMEs. In the European Union, a new definition came into force January 1, 2005. According to this definition, the turnover of medium-sized businesses (50–249 employee) should not exceed EUR 50 million (OECD, 2005).

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Chapter 5

Industry 4.0 and its Role on Labour Market: A Comparative Analysis of Turkey and European Countries

Elif Erer and Deniz Erer

Abstract

There is a close relationship between labour markets and technology. Technological development enhances labour productivity and creates new jobs in some industries requiring to be different skills while it destroys jobs in ones requiring to be low-skilled. Today, it is experiencing a deep digital transformation. It may be evaluated for Industry 4.0 to cause technological unemployment due to changes in the structure of employment and to bring about new structural problems in labour market. In addition, it is expected for technological progress such as automation and robotic in the production process to negatively influence employment of low-skilled workers. Industry 4.0 has generated a new production model, in which robotic technologies are effectively used in the production process. So-called new technology-based production process has started to change production, working relationship and daily life. Discussions about the effects of the developments in technology on the labour market and unemployment are separated two groups. While there exist optimist views

indicating that such development in technology will result in more productivity, pessimists believe that the use of robots, artificial intelligence, smart systems and algorithms in business life will eventually bring problems such as mass unemployment, mass poverty and social disruption. In this study, the authors aim to analyse the potential effects of Industry 4.0 on labour market in Turkey and European countries. From the findings of the study, the authors concluded that Turkey have a higher risk at automation than European countries.

Keywords: Industry 4.0; digital technologies; unemployment; labour market; Turkey; European countries

1. Introduction

The changes in production technologies, in other words technological innovations, have stimulated industrial revolutions since the 1990s ([Lasi, Fettke, & Kemper, 2014](#)). Industrial developments firstly started with the use of steam power and mechanisation in the eighteenth century. The first industrial revolution occurred between 1760 and 1840. This revolution caused a decrease of production costs and to be sold standardised and qualified products. The second industrial revolution started in the middle of the nineteenth century and into the early twentieth century. Therefore, it made mass production mass by means of using of electrical energy and invention of the telegraph and telephone. The third industrial revolution began in the 1960s. It is generally called as digital revolution since it was revealed by the development of semiconductors, mainframe computing (1960s), personal computing (1970s and 1980s) and the internet (1990s). Finally, fourth industrial revolution, or Industrial 4.0, is starting in today's world with the combination of physical, digital and biological worlds. The fourth industrial revolution can be characterised by artificial intelligence (AI) and machine learning, smaller and more powerful sensors, mobile internet ([Kurt, 2019](#); [Schwab, 2016](#), pp. 11–12).

The term of 'Industry 4.0' means use of interconnected and digital technologies facilitating new and more efficient processes in industrial production. This term was used firstly in a project digitalising manufacturing at the Hannover Messe in Germany in 2011 by the German government in order to provide the global competitiveness of German industry. This term defines the organisation of production processes, in which devices autonomously communicate with each other throughout the value chain. These technologies vary from digital technologies such as Internet of Things (IoT), 3D printing to new processes such as AI, data-driven production to new materials such as nano-based. These developments in industrial production affect productivity, trade, employment, income distribution as well as whole economy. Thus, governments should support to companies by considering so-called developments in production ([OECD, 2017](#)). (See Appendix (The Main Concepts) for the more detailed informations about Industry 4.0.)

The New Industrial Revolution refers to the digital transformation of production systems with advances in information and communication technologies. The digitalisation of every stage of the production chain, the development of the machine–human–infrastructure interaction and the development of the 'intelligent production systems' have created a paradigm shift in the industry ([TUBITAK, 2016](#)). These changes in technology will affect the next 10–15 years of production. These technological potentials in production are continually expanding due to mutually complementary technologies. For example, today, advances in software and data science enable the development of new intermediate goods ([OECD, 2017](#)). Reducing production costs through information and communication technologies has become essential to prevent international companies from shifting their production lines to countries that offer low labour costs. The realisation of this transformation in the industry will provide a significant competitive advantage to the countries. If this transformation is missed in our country, it is foreseen that the existing logistic advantage of our country in the production and the advantages of low labour cost will not be valid. Thanks to these technological developments, it is aimed to achieve a total productivity increase through gains such as reduced maintenance, quality and

stock holding costs of the companies, market release and machine operating efficiency (shortening when not mandatory) and increases in technical personnel efficiency. This increase of efficiency to be achieved in various sectors will benefit in the form of increased competitiveness at the national level. For these reasons, both developed countries and major information and communication technologies companies in the world have begun to rapidly develop the applications needed by the new industrial revolution ([TUBITAK, 2016](#)).

Industry 4.0 depends on many new and innovative technological developments: ([European Parliament, 2015](#))

- Application of *information and communication technologies* to digitise information and integrate systems at all stages of the production process and its use within and across factories.
- *Cyber-physical systems* that use information and communication technologies to control and manage physical processes and systems. These include embedded sensors, smart robots and additive manufacturing devices (3D printing).
- *Communication networks*, including wireless and internet technologies.
- *Simulation*: Modelling and virtualisation of production processes and designing products.
- *Big data and cloud computing*: Collection and analysis of large amounts of data.
- *Smart tools and augmented reality*.

Digitised production will result in extensive changes in production processes, outputs and business models. Smart factories provide increased flexibility in production. Automation of the production process, the transfer of data related to a product and the use of adjustable robots will enable different products to be manufactured in the same production facility. This customer-oriented manufacturing (mass customisation) enables small scale production due to the ability of adjustable machines to adapt quickly to customer-supplied features and additive manufacturing. This flexibility will encourage innovation because new products or prototypes can be produced

quickly without the need for new production lines. Digitalised designs and virtual modelling of the production process can reduce the time between the design and delivery of a product ([European Parliament, 2015](#)). The continuous changes in customers' demands and needs make it inevitable for leaders to constantly adapt to changes to keep their firms in a competitive environment ([Akkaya, 2020](#)). Smart automation of production processes and minimising human muscle strength will significantly reduce manufacturing errors. This also means reduced costs and increased customer satisfaction. At the same time, thanks to the coordinated use of digital technologies, an improvement in the manufacturing processes of enterprises, a decrease in costs, efficiency and an increase in competition in foreign markets are provided ([Gur, Unay, & Dilek, 2017](#)). This situation results in an increase in competitiveness in exports and thus a decrease in the foreign trade deficit. Digital transformation technologies in the industry have lately become more accessible and feasible because sensors, processors and cloud infrastructure of Industry 4.0 have become cheaper. Also, the data required for big data analysis have increased proportionally with the internet ([TUSIAD, 2016](#)).

While nine technologies are generally used in the elaboration of digital transformation technologies in industry, [TUSIAD \(2016\)](#) explains the digital transformation with 11 technologies: simulation, AI and smart systems, sensors, augmented reality, industrial internet, autonomous robots, additive manufacturing, horizontal and vertical integration, big data and analysis, cybersecurity and cloud. Due to the simulation, the data transferred instantly in the production processes in the physical world can be tested with virtual models ([Hermann, Pentek, & Otto, 2015](#)).

Industrial Internet allows machines, products, people and systems to connect and transfer data in real-time. Besides, the data are instantly analysed, and industrial processes are continuously optimised. The need for the IoT, which is the cornerstone of digital transformation in the industry, and the ability to process large amounts of big data produced for industrial purposes, have revealed the definition of the industrial internet. With the benefits it provides in production, logistics and the supply chain, it enables manufacturing

industry companies to be more competitive both domestically and internationally ([Boyes, Hallaq, Cunningham, & Watson, 2018](#); [IIoT, 2018](#)).

New technologies will interact with each other, and rapid improvements will occur by means of Industry 4.0 in industry and technology sectors. These new technologies in industrial production will have important effects in labour market. They lead to an increase of technology-intensive production. Improvements in industry lead to change production costs and productivity of labour, and therefore have several important consequences for labour market. Although so-called technological innovations frequently cause job losses for manual workers, they also create a global effect by affecting international economies.

Industry 4.0 carries with it some difficulties in contrast with its opportunities such as decreases in transport, production and communication costs, easier logistics and global supply chains. In respect to this, Industry 4.0 makes necessary for labour markets and production methods to transform. It is expected for transformations in production processes from Industry 4.0 to stimulate changes in both industrial relations and socio-economic and cultural structures.

There is a close relationship between labour markets and technology. Technological development enhances labour productivity and creates new jobs in some industries requiring to be different skills while it destroys jobs in ones requiring to be low-skilled. Today it is experiencing a deep digital transformation. Technological progress, such as robotisation, digitalisation, automation and computerisation, emerging in many industries has affected employment in terms of skills and occupations. Many of studies focusing on technological development-employment relationships in the literature have stated that technology completes skilled labour, and creates 'skill bifurcation' due to increases in the level of skill in complicated jobs; but, deskilling low-skilled jobs, and consequently generating a bias towards high-skilled labour force ([Acemoglu, 1998](#); [Autor, Katz, & Krueger, 1998](#); [Ben-Ner & Urtasun, 2013](#); [Bresnahan & Greenstein, 1999](#)). As such, technological developments enhance demand for educated employee ([Goldin & Katz, 2007](#)).

Both demand for high-skilled jobs and increases in low-skilled jobs have caused 'job polarisation', especially in developed countries ([Acemoglu & Autor, 2011](#); [Autor & Dorn, 2009, 2013](#)). Job polarisation arises from the difference between low-skilled, low-wage occupations and high-skilled, high-wage occupations ([Acemoglu, 1998](#)). Thereby, this situation causes wage inequality and the rise in the unemployment rate of low-skilled workers. Some of the studies in the literature have investigated the effect of technology on educational wage differentials in the developed countries and stated that skill upgrading is higher in technology-intensive industries ([Autor et al., 1998](#)). Also, so-called studies exhibited that higher job losses have occurred in industries where the share of R&D and new technology users are maximised ([Givord & Maurin, 2004](#)).

The skill-based technological change hypothesis, which explains the technology-employment, appears to ignore the presence of middle-skilled employee. However, Autor et al. (2003) suggested a different approach. Their approach is based not on tasks but rather unemployment or wages. They investigated the factors affecting job polarisation and set forth the substituting and complementing impacts of 'computer capital'. According to this approach, 'computer capital' replaces employees who carry out well-defined and limited tasks defined as routine. In other words, computers can perform the routine tasks in question. Moreover, 'computer capital' complements employees who perform operations which need creativity and problem solving, not well-defined communication skills. These operations are defined as non-routine tasks and thus, cannot be replaced by digital technologies.

Another hypothesis relating to technology-employment relationship is 'routinisation' hypothesis. The 'routinisation' hypothesis supposes that new technologies will substitute routine labour force, which are middle-skilled and have middle wage. That being said, the process of routinisation may happen both in low-skilled and high-skilled occupations. The new technologies may replace low-skilled or high-skilled employees ([Acemoglu & Autor, 2011](#); [Behaghel et al., 2011](#)).

As considered the above explanations, it may be evaluated for Industry 4.0 to cause technological unemployment due to changes in

the structure of employment and to bring about new structural problems in labour market. In addition, it is expected for technological progress such as automation and robotic in the production process to negatively influence employment of low-skilled workers. Industry 4.0 has generated a new production model, in which robotic technologies are effectively used in production process. So-called new technology-based production process has started to change production, working relationship and daily life.

From this point of view, the aim of the study is to investigate the potential effects of Industry 4.0 on labour markets in terms of the unemployment rates of low-skilled, middle-skilled and high-skilled employees. In the study, also, we tried to determine the advent of Turkey's Industry 4.0 by comparing with Turkey and European Union countries in terms of so-called technology-employment effects. Within this framework, we will firstly discuss theoretical framework explaining the relationship between Industry 4.0 and unemployment. We will explain unemployment rates in Turkey and European countries in Section 3. Then, we will examine the impacts of Industry 4.0 on unemployment in Turkey and European Countries, and we will compare Turkey with European countries in terms of the relationship in question in Section 4. Finally, Section 5 will show the conclusions.

2. Theoretical Framework

With the Fourth Industrial Revolution, discussions about the effects of the developments in technology on the labour market and employment are increasing among economists. Although technologic developments have positive impact on economic growth, it can negatively affect the labour market in the short term. For concerns about the effect of technology on jobs, in 1931, the economist John Maynard Keynes expressed the term of technological unemployment. He stated: 'due to our discovery of means of economising the use of labour outrunning the pace at which we can find new uses for labour' ([Schwab, 2016](#)).

Fast process of technology and digital world are taking place in all organisational authoritative in all area and in all kind of sectors, because the business world is transformed by the postmodern

revolution-Fourth Industrial Revolution ([Akkaya, 2019](#)). The Fourth Industrial Revolution will cause more upheaval than previous ones. New technology, such as IoT, 3D printing, AI, data-driven production to new materials, etc., will significantly change the nature of work and occupations in all industries. The main uncertainty is to what extent automation will substitute for labour. There are two destruction effects of technology on employment. The first one is technology-fuelled disruption and automation substitute capital for labour. This leads to unemployment or reallocate skills of workers elsewhere. The second one is capitalisation effect. By means of this effect, demand for new goods and services rises and it causes the creation of new occupations, business and industries ([Schwab, 2016](#)).

The use of machines based on information technologies and AI will have a substantial impact on the labour market in the twenty-first century. While there exist optimistic views indicating that such development in technology will result in more productivity, pessimists believe that the use of robots, AI, smart systems and algorithms in business life will eventually bring problems such as mass unemployment, mass poverty and social disruption. Economic structure, business relations, work profiles, working hours and wage models will considerably vary ([Wisskirchen et al., 2017](#)). Industry 4.0 will lead to changes in the structure of jobs. While technological changes lead to increased labour productivity, new technologies will replace the workforce in many industries and professions. Advances in machine learning, robotics and AI will cause changes in labour demand ([Brynjolfsson & McAfee, 2011](#)). However, automation will no longer be limited to physical or manual tasks and dangerous tasks, but will also jeopardise intellectual, cognitive or analytical white-collar jobs such as routine tasks, transport, office support, and consumer services ([Arntz, Gregory, & Zierahn, 2016](#)).

Until today, economists who have expressed their concerns about unemployment due to machines and computers have been labelled as enemies of technology ([Ford, 2018](#), p. 52). According to [Krugman \(2012\)](#), although some economists claim that there will not be such a problem, serious economists have been aware of the fact that technological development and innovation may negatively affect employers for almost two centuries. Two centuries ago, David

Ricardo developed an abstract model indicating the probability of technological unemployment. The main idea is that at some point, the wages for workers may fall below the level required for subsistence. Technology not only takes people's jobs away but also causes the remaining employers to work cheaper. A rational person will not see any advantage of getting a job for a low wage, so the worker will be unemployed, and the job will be done by a machine instead ([Brynjolfsson & McAfee, 2011](#)). Indeed, the wages of unskilled workers have tended to decrease in the US for more than 40 years.

Most of citizens used to work in primary sector offerings such as agriculture, mining, and manufacturing of raw materials. Right after the first Industrial Revolution, this tendency changed in a way that majority of citizens started to work in secondary sectors such as manufacturing. Nevertheless, there have existed new trends towards service sector since the ending of World Wars. For example, 7 out of 10 employees work in service sector. Recent studies highlight an emergence of fourth and fifth sectors. Accordingly, occupations related to recycling or requiring high intellectuality can be considered among such brand-new sectors. However, many experts argue the validity of such sectors. For them, it is useful to keep the conventional division as three sectors. They believe that regardless of the complexity of a job, it is still a service in which individual efforts are used for the sake of others.

The question arises as to which sector these jobs will be directed to, because automation and digitalisation jobs are eliminated, not only in the manufacturing industry, but also in the service industry, as presented above. Therefore, the only alternative will appear as unemployment or targeted education of affected individuals in the third sector. Education is likely to make sense only in the IT field. Not only employees, but companies that have nothing to do with IT and data processing in big data format will have to adapt to technical innovations to stay competitive. The new labour market is quickly approaching. Only the seller, who first discovered, developed, or even brought the new service model to the market, will make huge profits. Moreover, due to rapid technical development, new and profitable services will also become outdated very quickly. They will be replaced by other services that will be based on more advanced services, and

creative solutions will be found to meet the needs of customers that the old service does not serve.

However, the situation of the global labour market should not be overlooked. The famous economist Keynes, for example, predicted a similar development in the labour market before the 'second industrial revolution' in 1930. Despite crises and wars, and a 'third industrial revolution' that has gone through new areas of work and excess production profits, it has always been possible to counter the huge wave of unemployment. Employees will not be dismissed, as long as the productivity increase initiated by the introduction of AI and robotics leads to an increase in orders and profit. For many employers, a smart factory without human employees is not an alternative.

It is clear that a potential job loss will affect both the blue- and white-collar sectors, and the digitalisation (and automation) of services is a global phenomenon. However, this is a wide-ranging and diversified field of consulting services, particularly related to labour law. It is recommended that future laws to be guaranteed by uniform standards at the international level are geared towards technological advances and an increased need for flexibility.

The impact of technological change on employment and income inequality attracts scholars and policymakers. Despite the advantages of technological developments such as increasing productivity in production, there also exist some negative aspects. In particular, the fear of technological unemployment has spread worldwide. Much manual work done by people with digital technology can now be done autonomously. This will enable the same product to be made with fewer people. Therefore, although digital transformation contributes economically, it is thought that it may lead to an increase in unemployment. However, given that the number of people employed with additional employment opportunities has increased by providing economic growth for each industrial revolution that took place before the fourth industrial revolution, it is still too early to say that digital transformation will lead to unemployment.

In the period from the first industrial revolution to the present day, the economies of countries have undergone two structural transformation processes, namely, mechanisation and structural

transformation. Mechanisation has led to a revolution in the agricultural and manufacturing industries, causing rural workers to migrate to cities. The structural transformation that started in the 1960s decreased employment in the manufacturing industry, especially in developed economies, and enabled the service sectors to grow and create more jobs. Structural transformations experienced in the past show that economies adapt to technological shocks. However, it is not clear how long this adaptation process will take, and the increase in welfare provided by technological developments may not be shared equally by all segments of society.

As part of a research, scenarios on the technology adoption rate have been developed, and according to the technology adoption rate, it is predicted that between 400 million and 800 million people may lose their current jobs, and 75 million and 37 million of them will gain new skills and do other tasks, as of 2030. On the other hand, in the same report, it is stated that with the effect of six important catalysts that will increase labour demand, approximately 900 million new jobs can be created. Digitised estimates for the impact of digitalisation on the employment market differ.

A distinction is made between high, medium, and low-risk occupations that will be most affected by the fourth industrial revolution, depending on the possibility of computer-based automation. Those working in logistics and transportation, office and administrative support officers, and those working in production are considered under risk of losing their jobs. The comparative advantage of human labour gradually decreases in tasks related to mobility and skills. Digital transformation will require a workforce with new skills and competencies. Therefore, strategies and policies towards gaining new talent and skills are gaining increasing importance. Digital technologies are expected to reduce the demand for the workforce with low skill requirements and performing routine jobs ([BSTB, 2018](#)).

3. Technological Unemployment

The continuous changes in customers' demands and needs make it inevitable for organisations to constantly adapt to changes to keep their firms in a competitive environment ([Akkaya, 2019](#); [Akkaya &](#)

Tabak, 2020). Therefore, today companies operating in emerging markets face with certain problems especially in turbulence, complex, and changeable environment (Akkaya & Tabak, 2017) which cause unemployment.

Unemployment is defined by International Labour Organisation (ILO) (2013) that:

all persons of working age who were: a) without work during the reference period, i.e. not in paid employment or self-employment; b) currently available for work, i.e. were available for paid employment or self-employment during the reference period and c) seeking work, i.e. had taken specific steps in a specified recent period to seek paid employment or self-employment. Future starters, that is, persons who did not look for work but have a future labour market stake (made arrangements for a future job start) are also counted as unemployed, as well as participants in skills training or retraining schemes within employment promotion programmes, who on that basis, were 'not in employment', not 'currently available' and did not 'seek employment' because they had a job offer to start within a short subsequent period generally not greater than three months and persons 'not in employment' who carried out activities to migrate abroad in order to work for pay or profit but who were still waiting for the opportunity to leave.

Unemployment rate is obtained by dividing the number of unemployed people to total labour force. The labour force is expressed the sum of the number of employed and unemployed people. This ratio is considered as the underutilisation of the labour supply and it reflects that an economy cannot generate employment. Thus, it is indicator of the efficiency and effectiveness of an economy (International Labour Organization).

The development of technology with industrial revolution leads to appear the concept of technological unemployment. Technological

unemployment can be defined as the loss of people's jobs due to the technological change. The relationship between technological change and unemployment is an important issue for economists and policy makers, because it can lead to high unemployment rate since the beginning of industrialisation in the late eighteenth century. There were many examples in history such as the famous Luddite riot in England are gathered in Petit (1993) or Cohen (1995) ([Vinay, 2002](#)). In 1930, Keynes drew attention to the phenomenon of technological unemployment. He stated that:

We are being afflicted with a new disease of which some readers may not yet have heard the name, but of which they will hear a great deal in the years to come – namely, technological unemployment. This means unemployment due to our discovery of means of economising the use of labour outrunning the pace at which we can find new uses of labour. (Keynes, [1930](#), p. 364)

Although technological development can generate new jobs by providing process and production innovation ([Say, 2009](#); [Schumpeter, 2017](#)), it can rise unemployment ([Feldmann, 2013](#); [Wood, 2004](#)). [Liso and Leoncini \(2011\)](#) stated that technological development provides higher wage opportunity for skilled workers due to an increasing demand. However, [Piva, Santarelli, and Vivarelli \(2006\)](#) expressed that technological change negatively affects skilled and unskilled workers. Besides, [Alonso-Borrego and Colloda \(2002\)](#) indicated that technological development is one of the main sources for job creation and destruction. [Acemoglu and Autor \(2011\)](#) stated that technological change based on ability decreases the demand for computerised jobs.

4. The Impact of Industry: On Unemployment in Turkey and European Countries

Technological progress such as the use of AI and robots at work will cause that workers will be replaced by this technology. Thus, it is

expected that automation will lead to job destruction especially in low-skilled workers and that unemployment will increase in certain industries such as textile production and complex electrical equipment manufacturing (OECD, 2017). However, OECD Report (OECD, 2019) indicates that the digital revolution has made considerably contributions to job creation in digitally – intensive industries over the past decade.

Table 1 shows the stable, new, and redundant roles in occupations. According to Future and Jobs Survey (2018) published by *World Economic Forum*, new roles in future occupation are the roles such as data analysts and scientists, AI and machine learning specialists, e-commerce and social media specialists, big data specialists, software and operations managers which are based on technology, etc. However, it is expected to lose significance of the roles such as accounting, bookkeeping and payroll clerks, administrative and executive secretaries, accountants and auditors, business services and administrative managers, financial analysts, etc.

Fig. 1 exhibits share of jobs which are at a high risk of automation or a risk of change in Turkey and European countries (Finland, Sweden, United Kingdom, Denmark, Netherlands, Belgium, Ireland, Estonia, Austria, Czech Republic, France, Poland, Italy, Spain, Slovenia, Germany, Greece, Lithuania, and Slovak Republic). By estimating the number of jobs at risk of automation, we can examine the task content of individual jobs within each occupation. From Fig. 1, we may see that the estimate related to the share of jobs at high risk automation in Turkey is 16.4%. The maximum value of this ratio is 33.6% in Slovak Republic. The minimum value is 7.24% in Finland. These figures represent potential job destruction. Besides, 43% of jobs in Turkey have risk of significant change, in other words, it may see a large share of their tasks be automated while entirely new tasks may emerge. This ratio ranges from 25.9% in United Kingdom to 41.9% in Lithuania across European countries. These results indicate that low-skilled workers have a higher risk of automation. This situation leads to enhance disparities in the labour market.

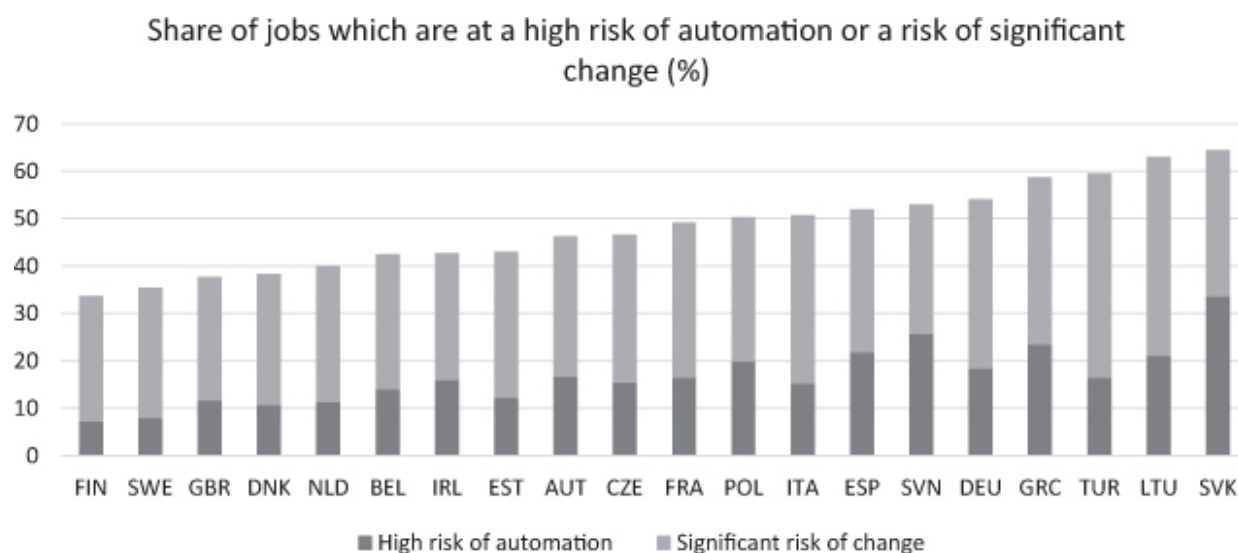


Fig. 1. Jobs at Risk Automation in Turkey and European Countries.
Note: Jobs are at high risk of automation if the likelihood of their job being automated is at least 70%. Jobs at risk of significant change are those with the likelihood of their job being automated estimated at between 50% and 70%. *Source:* [OECD Employment Outlook 2019](#) (OECD, 2019).

Table 1. Stable, New and Redundant Roles.

Stable Roles	New Roles	Redundant Roles
Managing Directors and Chief Executives	Data Analysys and Scientists*	Data Entry Clerks
General and Operations Managers*	AI and Machine Learning Specialists	Accounting, Bookkeeping and Payroll Clerks
Software and Applications Developers and Analysts	General and Operations Managers*	Administrative and Executive Secretaries
Data Analysts and Scientists*	Big Data Specialists	Assembly and Factory Workers
Sales and Marketing Professionals*	Digital Transformation Specialists	Client Information and Customer Service Workers*
Sales Representatives, Wholesale and Manufacturing, Technical and Scientific Products	Sales and Marketing Professionals*	Business Services and Administrative Managers
Human Resources Specialists	New Technology Specialists	Accountants and Auditors
Financial and Investment Advisers	Organisational Development	Material-Recording and Stock-Keeping Clerks

Database and Network Professionals	Specialists* Software and Applications Developers and Analysts*	General and Operations Managers*
Supply Chain and Logistics Specialists	Information Technology Services	Postal Service Clerks
Risk Management Specialists	Process Automation Specialists	Financial Analysts
Information Security Analysts*	Innovation Professionals	Cashiers and Ticket Clerks
Management and Organisation Analysts	Information Security Analysts*	Mechanics and Machinery Repairers
Electrotechnology Engineers	Ecommerce and Social Media Specialists	Telemarketers
Organisational Development Specialists*	User Experience and Human-Machine Interaction Designers	Electronics and Telecommunications Installers and Repairers
Chemical Processing Plant Operators	Training and Development Specialists	Bank Tellers and Related Clerks
University and Higher Education Teachers	Robotics Specialists and Engineers	Car, Van and Motorcycle Drivers
Compliance Officers	People and Culture Specialists	Sales and Purchasing Agents and Brokers
Energy and Petroleum Engineers	Client Information and Customer Service Workers*	Door-To-Door Sales Workers, News and Street Vendors, and Related Workers
Robotics Specialists and Engineers	Service and Solutions Designers	Statistical, Finance and Insurance Clerks
Petroleum and Natural Gas Refining Plant Operators	Digital Marketing and Strategy Specialisys	Lawyers

Note: Roles marked with * appear across multiple columns. This reflects the fact that they might be seeing stable or declining demand across one industry but be in demand in another.

Source: Future of Jobs Survey 2018, *World Economic Forum*.

Emerging economies have different economic conditions including higher costs of information and communication technologies capital and greater skills shortages ([Maloney & Molina, 2016](#)). Technological

progress leads to shift labour from low-productivity activities such as especially agriculture to high-productivity activities such as mostly manufacturing. In the report of [The World Bank \(2016\)](#), it is indicated that a large share of employment work agriculture and low value-added industries. For this reason, emerging economies have a higher risk of automation than advanced economies ([McKinsey Global Institute, 2017](#); [Nedelkoska & Quintini, 2018](#); [The Work Bank, 2016](#)). Therefore, Turkey faces a higher share of jobs which are potentially at risk of automation.

[Fig. 2](#) shows the unemployment rate, persons aged 15 and over, in Turkey and European countries during the periods of 2010 Q1–2019 Q3.

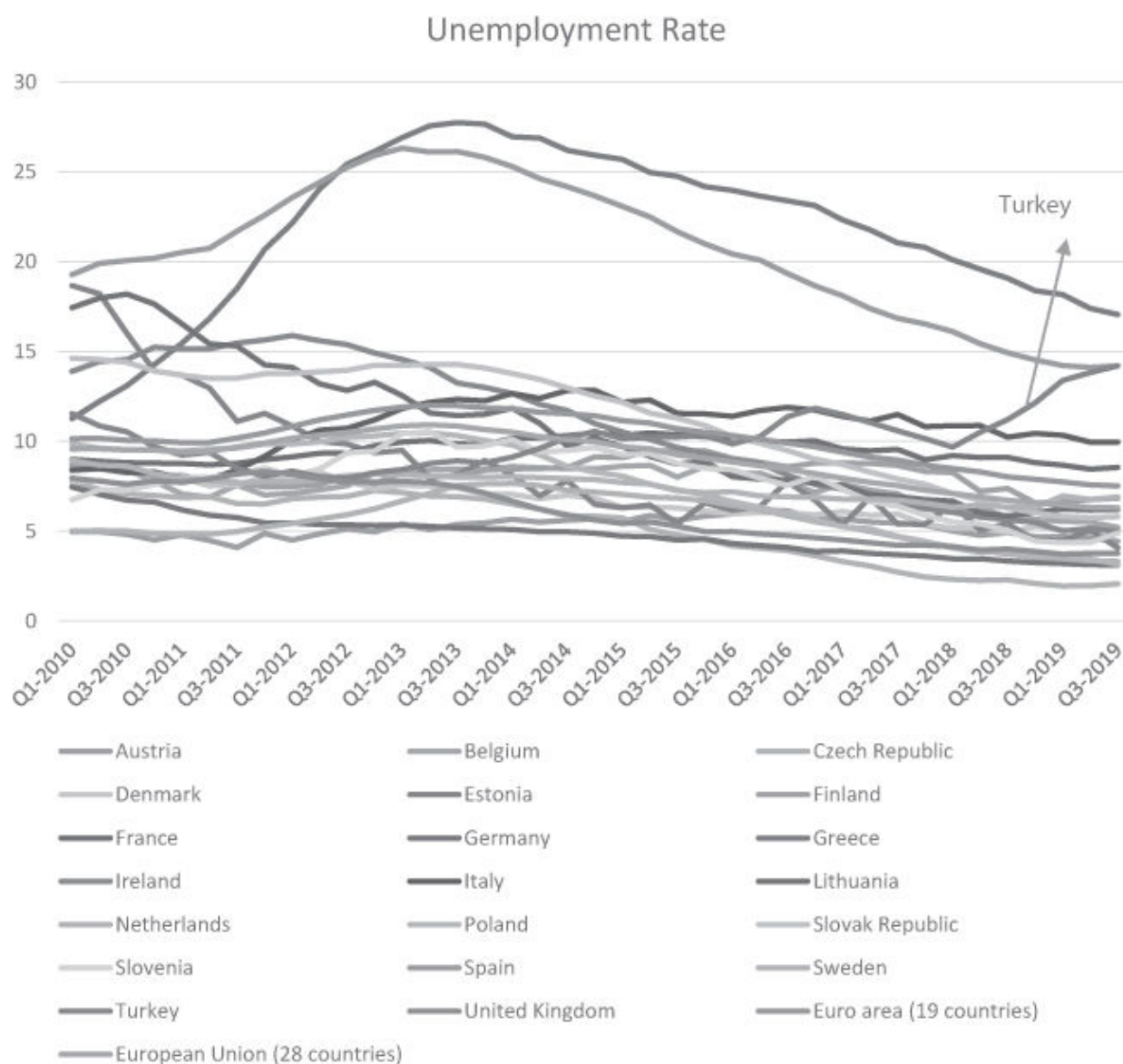


Fig. 2. Unemployment Rate, Aged 15 and Over, in Turkey and European Countries. *Source:* IMF International Financial Statistics.

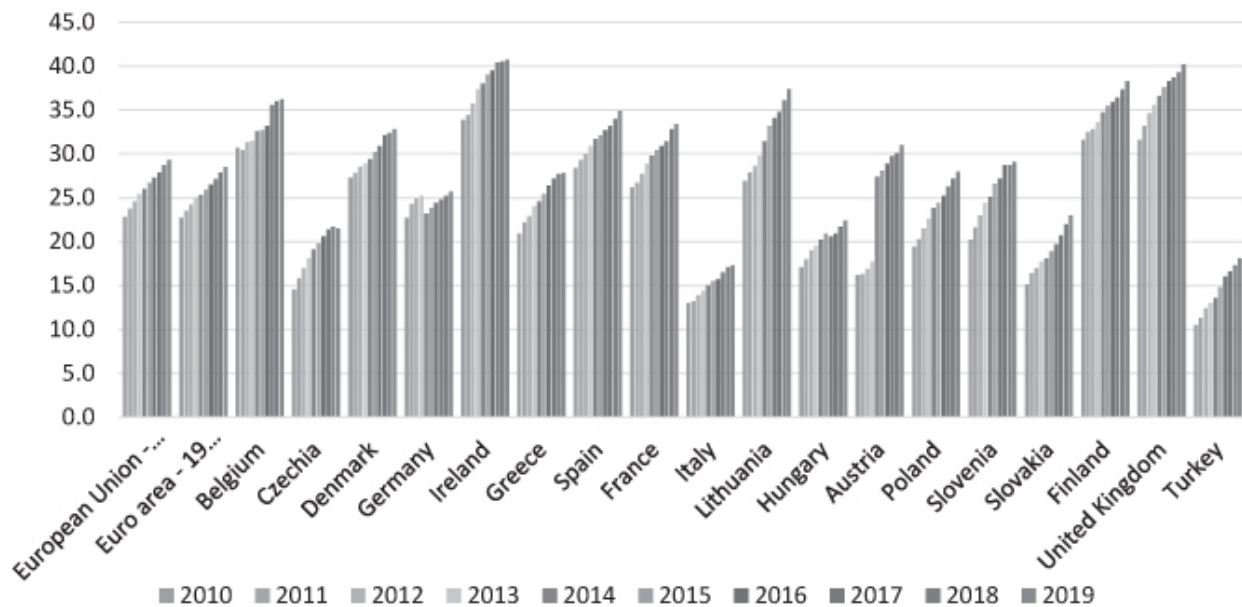


Fig. 3. Population by Educational Attainment Level, From 15 to 64 years, %, For Turkey and European Countries. Source: EuroStat Database.

As seen in Fig. 2, the countries which have highest unemployment rates are Spain and Greece. Unemployment rate in Turkey has shown a significant increase while overall rates in European countries have decreased since 2018.

The percentage of population by educational attainment level, from 15 to 64 years, for Turkey and European countries is shown in Fig. 3.

As examined in Fig. 3, it is seen that the percentage of population by educational attainment level, from 15 to 64 years, for Turkey is lower than European countries. (It changes from 10.5% in 2010 to 18.1% in 2019.) The average percentage of population by educational attainment level, from 15 to 64 years, for European Union (19 countries) is changed from 22.7% in 2010 to 28.5% in 2019. Similarly, so-called average percentage for European Countries (28 countries) is from 22.8% in 2010 to 29.3% in 2019.

5. Conclusion

Industry 4.0 refers to the digital transformation of production systems with advances in information and communication technologies. The digitalisation of every stage of the production chain, the development

of the machine–human–infrastructure interaction, and the development of the ‘intelligent production systems’ have created a paradigm shift in the industry. So-called digital transformation technologies in the industry are reshaping the labour market. Technological developments enhance labour productivity and create new jobs in some industries requiring to be different skills, while it destroys jobs in ones requiring to be low-skilled. Today, it is experiencing a deep digital transformation. Technological developments, such as robotisation, digitalisation, automation, and computerisation, emerging in many industries have affected employment in terms of skills and occupations. In other words, particular jobs and tasks which require to be low-skill and performing routines (i.e. agricultural labourers, workers in textile production, logistics and transportation, and office and administrative support officers) are disappearing while new jobs and tasks are created by new technologies. Thereby, strategies and policies towards gaining new talent and skills are gaining increasing importance.

In this study, we analysed effects of digital transformation in industry on labour markets of Turkey and European countries. The findings of the study indicated that the share of jobs at high risk automation in Turkey is 16.4%. The maximum value of this ratio is 33.6% in Slovak Republic while the minimum value is 7.24% in Finland. 43% of jobs in Turkey have a risk of significant change. This ratio ranges from 25.9% in United Kingdom to 41.9% in Lithuania across European countries. These results show that low-skilled workers have a higher risk of automation, and Turkey faces a higher share of jobs which are potentially at risk of automation. This situation enhances disparities in the labour market.

Consequently, the advance in technology caused by the Fourth Industrial Revolution such as AI, IoT, autonomous robots, machine learning, big data, etc., will cause robots to replace human labour, especially in the jobs which are at high automation risk. Therefore, it is expected that so-called changes in technology will lead to an increase in unemployment. So, the policies which will decrease unemployment should be proposed. For this purpose, primary importance should be given to education such as vocational and technical training, coding, etc.

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Appendix

Key Terms and definitions

Industry 4.0: The organisation of **production** processes in which devices autonomously communicate with each other throughout the value chain. The term of 'Industry 4.0' means use of interconnected and digital technologies facilitating new and more efficient processes in industrial production.

IoT: Information technology systems related to all subsystems, processes, internal and external objects, producer, and consumer networks. The IoT basically benefits from the physical systems communicating with each other. In this structure, data are collected from machines, computers, and automation systems used in production, and the collected data are transferred to local and global servers by wired and/or wireless communication methods.

AI and smart software: AI is defined as technologies that imitate human intelligence by learning and making their deductions from what they learn. AI systems are seen as the most critical information technology factor in digital transformation in the manufacturing industry, due to developing hardware infrastructures and cloud solution algorithms supported by data access. Therefore, AI is among the priority areas of all leading countries in digital transformation.

Autonomous robots and mechatronic systems: Autonomous robots are machines with advanced and multidisciplinary elements that mimic the functions and lifestyles of living things, with programmable skills and intelligence. Mechatronic systems include smart machines, devices, systems, and processes. Mechatronic systems perceive their environment, can make decisions, and change their setting by commenting on the perceived environment.

Cyber-physical systems: These systems consist of software embedded in hardware such as sensors, processors, and communication technologies, and they can autonomously change information, trigger actions, and control each other independently.

Cybersecurity technologies: Cybersecurity technologies are known as technologies that protect networks, computers, programs, and data from cyber-attacks.

Autonomous robots and mechatronic systems: Autonomous robots are machines with advanced and multidisciplinary elements that mimic the functions and lifestyles of living things, with programmable skills and intelligence.

Unemployment Rate: Unemployment is defined as labour force who have working power and desire, but they cannot find a job although they are willing to work at the current wage. Unemployment rate is the ratio of unemployed population to total labour force.

Technological Unemployment: The inclusion of technological advance to production process began automation time and decreased using of labour. Therefore, it reduced level of employment and appeared technological unemployment.

Chapter 6

Digital Taylorism as an Answer to the Requirements of the New Era

Ayşe Günsel and Mesut Yamen

Abstract

At the doorsteps of the fourth wave of the industrial revolution, it is compulsory to develop a new understanding regarding the future of human labor based on “Industry 4.0” for German manufacturers, and two American concepts: “The Industrial Internet” and “The Internet of Things.” How will the nature of human work be in the digital economy of the forthcoming future? The problem of unemployment and the composition of the labor market, in terms of professional skills, are yet to be waiting for answers. Scientific management is also transforming to answer the emerging requirements of this new era, as “Digital Taylorism” to re-organize work in a technocentric manner. Accordingly, the aim of this chapter is to examine the nature and the possible opportunities and threats of the digital age and try to develop a digital Taylorism understanding to minimize the negative impacts of digitalism on both individual workers and society in a way that all parts including the manufacturers can fully take the benefit of potential advantages of this new era.

Keywords: Digital taylorism; Industry 4.0; artificial intelligence; the internet of things; deep learning; machine learning

Introduction

At the doorsteps of the fourth wave of the industrial revolution, the term “Industry 4.0” is just launched and widely accepted in the industrial society and business environments as well as in academic literature. Whereas academic studies generally try to understand the concept and to develop relevant systems, business models and methodologies, the industry side particularly underlines a transformation from machine dominant manufacturing to digital manufacturing such as “The Industrial Internet” and “The Internet of Things” (IoT) ([Oztemel & Gursev, 2020](#), p. 128). Moreover, present and forthcoming organizational strategies are giving great attention to machines, robots, and Artificial Intelligence (AI). Automation to lessen basic or unskilled jobs, digitization of work to make remaining employees more efficient, and AI to make available more reliable and productive top-end professional work are all linked with what we call as Industry 4.0 ([Holford, 2019](#)). Machines are starting to act as decision-makers. 3D printing is progressing even more than expectations. Smart cities are broadening all over the world. The progress along this line raises the wish to create smart factories more and more every day. There is a huge possibility that more than 6 billion connected devices will require support in 2018. Global spending on big data is considered to exceed 200 billion dollars in 2020. By 2020, more than 3 billion workers are to be managed by a robo-boss, and 59% of US manufacturers will be exploiting robotics technology and AI ([WEF, 2015](#)). So what will the nature of human work be composed of in the digital economy of the forthcoming future? Will new technologies substitute jobs? If not, how will they be restructured? What will the broad implications be, both for individual workers, and legal regulation? Once codified and digitalized, automatic programs with computerized decision protocols can conduct many tasks, thereby replacing human decisions and

judgments. The problem of unemployment and the composition of the labor market, in terms of professional skills, are yet to be waiting for answers.

Scientific management is also transforming to answer the emerging requirements of this new era, as “Digital Taylorism” to re-organize work in a techno-centric manner. If the twentieth century gave birth to mechanical Taylorism described by the Fordist production line, where the knowledge of craft workers was caught, codified and re-used in the shape of the moving assembly line by management, no doubt the twenty-first century will be the era of Digital Taylorism ([Brown, Ashton, & Lauder, 2010](#)). Accordingly, the aim of this chapter is to examine the nature and the possible opportunities and threats of the digital age and try to develop a digital Taylorism understanding to minimize the negative impacts of digitalism on both individual workers and society in a way that all parts including the manufacturers can fully take the benefit of potential advantages of this new era. The rest of the chapter is structured as follows: the next section demarcates the theoretical framework for Industry 4.0 with its most highlighted components as IoT and Internet of Industrial Things. We will also discuss AI, machine learning, and deep learning. The second section will examine Digital Taylorism and discuss how Digital Taylorism can answer the requirements of the digital era. The concluding remarks are summarized in the conclusion section.

Industry 4.0

All technological advances, from water and steam-powered machines that brought the first industrial revolution of electrical and digital production, have made production processes automatic and sustainable, greatly increasing their capacity and efficiency. In these processes, the second industrial revolution revealed mass production with the use of electrical energy, and the third industrial revolution revealed the development of electronic and information technology and automation systems ([Vaidya, Ambad, & Bhosle, 2018](#), p. 233). With technological advances, the increasing use of machinery in the industry required the factories to be equipped with

more mechanized production systems ([Buchanan, 2005](#), p. 53). However, industrial revolutions that facilitated production at every stage brought the formation of large industrial factories that used the power of machines ([Makridakis, 2017](#), p. 54). Thus, the increasing competitiveness, market development, globalization, and advances in information technology have led to the emergence of the fourth wave of industrial revolution, referred to “Industry 4.0” ([Piccarozzi, Aquilani, & Gatti, 2018](#), p. 2).

The concept of Industry 4.0 was first introduced in the definition of networked smart systems at the Hannover Fair organized in 2011 in Germany ([Lezzi, Lazoi, & Corallo, 2018](#), p. 106). It was also used in 2012 to describe the high-tech strategy of the German government ([Strange & Zucchella, 2017](#), p. 174). Industry 4.0 is expressed as the transition to networked cyber-physical production system, in which all technological equipment within a production system is integrated ([Uriarte, Ng, & Moris, 2018](#), p. 587). Fast process of technology and digital world are taking place in all organizational authoritative in all area and in all kinds of sectors, because the business world is transformed by the postmodern revolution – Fourth Industrial Revolution ([Akkaya, 2019](#)).

The essence of Industry 4.0 is to create networked smart systems in an industrial process, where people, machines, equipment, and smart products can communicate with each other and achieve production by self-regulation ([Piccarozzi et al., 2018](#), p. 2). In this context, Industry 4.0 aims to create smart factories where production technologies are developed and transformed by Cyber-Physical systems (CPSs), the IoT, Industrial Internet, and cloud computing. Thus, production systems can monitor physical processes, create the digital twin or cyber twin of physical machines, and make smart decisions through real-time communication and collaboration with people, machines, sensors, and all other related parties ([Zhong, Xu, Klotz, & Newman, 2017](#), p. 616). Communication between all those parties within the context of Industry 4.0 enables data collection and analysis, development of real-time data tracking systems about the status and positions of products for the control of production processes, development of self-familiar and self-management to improve overall performance, and maintenance management of the

machines. The introduction of learning machines will enable the creation of a faster, more flexible, and more efficient production process to produce higher quality products at a lower cost (Vaidya et al., 2018, p. 234).

In general, there are nine technological advances that provide the basis for the formation of the fourth industrial revolution with the development of CPSs. These technologies, which are expressed as a base within the scope of Industry 4.0, are Big data and analytics, simulation modeling, augmented reality, layered production, horizontal and vertical system integration, autonomous robots, cybersecurity, the Industrial IoT, and cloud technologies (Rodic, 2017, p. 194; Strange & Zucchella, 2017, p. 181; Uriarte et al., 2018, p. 587). Even though all of the given components are essential for Industry 4.0, we will focus on IoT, industrial internet–industrial internet of things, and AIs, those have the most significant impacts on labor in the near digital future.

Internet of Things

The IoT refers to an internet-based technological approach in which every object within a network can provide mutual data transmission with all other objects. It is aimed to establish an uninterrupted and automatic connection between objects, systems, and services by providing communication and data sharing between the IoT and physical products (Zhong et al., 2017, p. 617). Objects that can capture and process data in the application processes are equipped with sensors to share large amounts of data production and transmission (Boyes, Hallaq, Cunningham, & Watson, 2018, p. 3). With the use of the real-time data processing capability of the sensors, as well as the ability to connect each equipment or product in production processes, to determine the need for preventive maintenance and repair, for example, to monitor the inventory levels of each object, to provide auto-control of the stock level, companies, suppliers, and customer intelligent systems have been developed that lead to a reduction in the need for intermediaries for data sharing between them (Strange & Zucchella, 2017, p. 175). The self-learning ability of intelligent machines connected to a network has

made it possible to operate with collective AI by increasing the level between them (IoT) ([Huang & Rust, 2018](#), p. 161).

Industrial Internet of Things

Industrial IoT is a concept used for the connection of machines used in production, based on the same principles as the IoT approach ([Boyes et al., 2018](#), p. 3). Industrial Internet has the ability to receive continuous information from various sensors and objects, securely transmit sensor readings to cloud-based data centers, and seamlessly update related parameters in the form of a closed-loop system. In this way, it can detect faults effectively and trigger maintenance operations. In an Industrial IoT system, much new Information and Communication Technologies such as Industrial Wireless Networks and IoT are included in a single system ([Gierej, 2017](#), p. 207). For example, control systems manufacturer Bosch Rexroth maintains a semi-automated, decentralized production process in a valve manufacturing facility. In this process in which the products are marked with radiofrequency codes, each product can know which production stages it should go through and act according to what needs to be done ([Tansan, Gökbulut, Targotay, & Eren, 2016](#), p. 28).

Artificial Intelligence

AI is the intelligence exhibited by machines ([Wirth, 2018](#), p. 436). It is aimed to display the biological intelligence and behavioral characteristics of living things by a machine or an inanimate entity through AI researches conducted in computer science ([Farrow, 2019](#), p. 61). For this purpose, AI can be described as the science of making smart machines ([McCarthy, 2007](#)).

AI consists of many technical applications to enable machines to imitate the form of biological intelligence ([Lexcellent, 2019](#), p. 5). It is closely related to the ability of a machine to perform human-like tasks such as image recognition and natural language processing ([Abellera & Bulusu, 2018](#), p. 2). It deals with the events that occur while performing tasks and requiring smart thinking ([Simon, 1995](#), p.

95). So, AI includes a heterogeneous set of tools, techniques, and algorithms ([Jarrahi, 2018](#), p. 577). Technically, it is a large collection of computer-aided systems for task performance, including but not limited to machine performance, automated reasoning, knowledge pools, image recognition, and natural language processing. Such systems include task input for voice, text, image, and numerical data, algorithms, and task output for decisions and solutions ([von Krogh, 2018](#), p. 405).

AI is a cognitive science related to a variety of research fields like image processing, natural language processing, robotics, machine learning and so on ([Lee, Davari, Singh, & Pandhare, 2018](#), p. 20). It can be defined as intelligent systems capable of understanding, thinking, and learning ([El Namaki, 2016](#), p. 77; [Jarrahi, 2018](#), p. 577). This aspect is also a part of psychology and cognitive science ([Simon, 1995](#), p. 95). Besides, engineering (Norbert Wiener's studies on cybernetics, including feedback and control), biology (e.g., W. Ross Ashby and Warren McCulloch and Walter Pitts's work on neural networks in simple organisms), experimental psychology, communication theory (e.g., Claude Shannon's theoretical work), game theory (especially by John Von Neumann and Oskar Morgenstern), mathematics and statistics (e.g., Irving J. Good), logic and philosophy (e.g., Alan Turing, Alonzo Church, and Carl Hempel), and linguistics (Noam Chomsky's grammar study) were influenced by the ideas from many disciplines ([Buchanan, 2005](#), p. 56; [El Namaki, 2016](#), p. 77).

The most comprehensive aim of AI is to create smart machines that think and act like a human with the ability to simulate intelligence and make decisions through processes similar to human mind ([Salvaris, Dean, & Tok, 2018](#), p. 3). In this sense, AI can also be defined as the simulation done by computer systems; as it imitates the human intelligence processes ([Lexcellent, 2019](#), p. 5).

Expressed by [Norvig \(2012, p. iii\)](#) as the most important general-purpose technology of our time, the importance of AI is the ability to do the right thing when a person does not know what to expect ([Norvig, 2012](#), p. iii). In other words, it is the ingenuity of continuing to improve the performance of a machine without having to explain

exactly how all the work is done without people ([Brynjolfsson & McAfee, 2017](#), p. 2).

AI is already being developed by large companies and is followed by governments all around the world ([Carriço, 2018](#), p. 29). Today, organizations are increasingly involved in many processes, such as choosing the appropriate applications for artificial organizational positions, consulting customers about financial products, performing financial transactions, insuring customers, planning complex logistics, diagnosing patients and recommending treatments, predicting technological development, and even following criminal activities ([von Krogh, 2018](#), p. 404).

Development of AI. The foundations of the idea of AI are based on philosophy, fiction, and imagination. Topics addressed by many science fiction writers, including Lyman Frank Baum and Isaac Asimov in the twentieth century, such as Jules Verne's Journey to the Moon, especially in the nineteenth century, and space technology, are now possible to consider the possibility of intelligent machines with human features in real life ([Buchanan, 2005](#), p. 53). For instance, in 1818, Mary Shelly's "Frankenstein" outlined how intelligence and life could be created through scientific experiments, while in 1872, Samuel Butler dreamed about machines that can develop consciousness by Darwinist selection in his novel "Erewhon" ([Farrow, 2019](#), p. 63). Lyman Frank Baum, in 1907 in his novel "Tik-Tok of Oz," described the mechanical man named "Tiktok" as a sensitive, thoughtful, wonderful-speaking entity that can live like a person ([Buchanan, 2005](#), p. 53).

In science fiction novels, the terms Android and Robot are defined as mechanisms that combine mechanics, biology, and intelligence ([Farrow, 2019](#), p. 63). The term Android was first depicted in 1886 by the name "Hadaly" in the novel "L'ève Future" by Mathias Villiers de l'Isle-Adam ([Gerould, 1984](#), p. 318; [Hashimoto et al., 1997](#), p. 106). On the other hand, the term robot first appeared in 1921 with Karel Capek's play Rossum's Universal Robots ([Gladden, 2016](#), p. 314). Those novels, combined with industrial developments, inspired scientists in the field of robotics, AI, and computer science for generations ([Haenlein & Kaplan, 2019](#), p. 2).

Birth of AI: The first scientific studies on AI were started by Alan Mathison Turing. Turing developed a code-cracking machine for the British government called Bombe to decipher the Enigma code used by the German army to hide radio communications in the Second World War. Having a weight of approximately one ton, Bombe is generally considered to be the first working electro-mechanical computer. The powerful way that Bombe could break the Enigma code that even the best human mathematicians couldn't decipher Turing to work on the intelligence of such machines ([Haenlein & Kaplan, 2019](#), p. 2). From this point, AI started as a study on the nature of intelligence in the 1950s ([Simon, 1995](#), p. 95).

Turing test: Turing's mind-opening article in the philosophical magazine *Mind* in 1950 is an important turning point in the history of AI ([Buchanan, 2005](#), p. 56). [Alan Turing \(1950\)](#) published an article under the title of "Computing Machines and Intelligence." The first two sentences in Turing's article are remarkable ([Carriço, 2018](#), p. 29; [Wirth, 2018](#), p. 436). "Can machines think? I propose to consider the question" ([Turing, 1950](#), p. 433).

Alan Turing suggested that if a computer can make a conversation with a human being, we should accept that it will "think" ([Norvig, 2012](#), p. ii). The Turing Test developed in this direction is designed to provide a satisfactory functional definition of intelligence. In this test, a computer passes a computer test if a person is unable to tell whether the written answers come from a person or a computer after asking some written questions ([Russell & Norvig, 2010](#), pp. 2–3). In other words, if a person interacts with a machine, and cannot distinguish the machine from the human, then it can be said that the machine is smart. Turing test has been accepted as a criterion for defining the intelligence of an artificial system ([Lexcellent, 2019](#), p. 5).

Dartmouth Conference: The work in the field of thinkable machines began with Turing, but the term AI was first formalized by a conference organized by Professor John McCarthy of Stanford University in 1956 and hosted the Dartmouth Artificial Intelligence Summer Research Project ([Haenlein & Kaplan, 2019](#), p. 3). With this conference, funded by the Rockefeller Foundation, McCarty brought together researchers from various fields, scientists from multiple

academic disciplines, such as cybernetics, psychology, and computer science, to create a new research area for building machines capable of imitating human intelligence. Thus, AI emerged as an academic field ([Haenlein & Kaplan, 2019](#), p. 3; [von Krogh, 2018](#), p. 404).

AI Learning Models

Today AI encompasses a wide range of fields, including machine learning and deep learning. However, the incomprehension of those concepts can cause them to be used interchangeably. In this context, AI covers machine learning and deep learning, while machine learning has a concept that includes deep learning ([Salvaris et al., 2018](#), p. 9).

While AI is concerned with making machines intelligent using multiple approaches, the basic approach of machine learning aims to produce machines that learn to perform the functions needed to achieve AI. Deep learning, on the other hand, can be expressed as a technique to apply machine learning ([Rebala, Ravi, & Churiwala, 2019](#), p. 3). In this sense, we can say that deep learning is the sub-branch of machine learning, and machine learning is the sub-branch of AI ([Salvaris et al., 2018](#), p. 25).

Machine Learning. Machine learning is a computer science field that examines algorithms and techniques to automate solutions to complex and difficult problems too ([Rebala et al., 2019](#), p. 2). The term machine learning was first used by Arthur Lee Samuel in 1959 ([Akerkar, 2019](#), p. 19).

The ability of AI systems to meet the requirement to get their own information by extracting patterns from raw data is expressed as machine learning ([Goodfellow, Bengio, & Courville, 2016](#), p. 2). In other words, machine learning can be expressed as an algorithm or a set of algorithms that enable a computer to recognize patterns in a data set and interpret them in actionable ways ([Abellera & Bulusu, 2018](#), p. 2). The term algorithm here refers to a learning algorithm. The learning algorithm is used to train, validate, and test the model using a specific dataset to find, validate, and evaluate its performance ([Suthaharan, 2016](#), p. 7).

Machine learning involves the development of computational approaches to automatically interpret data. It is used to create models by learning from existing datasets to predict results or behaviors ([Rebala et al., 2019](#), p. 4). This technology draws on the idea that learning is a dynamic process. As a result of the development of machine learning, it has become possible to create new data with examples and experiences, contrary to the pre-determined rules. Like a person, a machine can hold information and become smarter over time ([Akerkar, 2019](#), p. 5).

The most important thing to know about machine learning is that it represents a different approach to software creation. With machine learning, the machine can learn from examples instead of being clearly programmed for a particular result. This development creates a significant difference from previously used applications ([Brynjolfsson & McAfee, 2017](#), p. 8). In this process, the human developer defines the algorithm or algorithms to be used, and then the machine uses the data to create a unique solution ([Burgess, 2017](#), p. 20). For example, people are not born with the knowledge that a flower is a flower. They learn by seeing flowers repeatedly during infancy and after and by recognizing different characteristics such as stem, leaves, and circular symmetry. Machine learning works in a similar way. It develops itself further by learning to do a job from experience. The key point here is that the recognition of an algorithm is not specifically designed by the programmer. On the contrary, machine learns with repeated data and statistical methods. AI agents need to be trained for the development of machine learning. As a part of this training, large amounts of data should be provided ([Abellera & Bulusu, 2018](#), p. 4).

The machine learning field allows machines such as computers to discover models and algorithms that can help them to learn from data ([Suthaharan, 2016](#), p. 123). From this point of view, it enables computers to learn by trying to find a function that can predict the results ([Abellera & Bulusu, 2018](#), p. 4). In order to create such an intelligent system, two main components that can help machine learning approaches achieve this goal are learning models and learning algorithms ([Suthaharan, 2016](#), p. 123).

Deep Learning. Machine learning systems are used to identify objects in images, convert speech to text, match news items, publications, or products with users' interests, and select relevant search results ([LeCun, Bengio, & Hinton, 2015](#), p. 436). Algorithms guiding most of this success have been made possible by deep learning using neural networks ([Brynjolfsson & McAfee, 2017](#), p. 11). Deep learning is a branch of machine learning, inspired by how the human brain works, using many nonlinear information processing layers that are either controlled or uncontrolled ([Akerkar, 2019](#), p. 30; [Bridgwater, 2019](#)). In this sense, it can be stated that deep learning models are engineering systems inspired by the biological brain ([Goodfellow et al., 2016](#), p. 13).

The first signs of the development process of deep learning are based on ideas about how the human brain processes information and can be modeled and called artificial neural networks ([Salvaris et al., 2018](#), p. 15). Neural networks in biology are interconnected, inter-exchanged units, and each connection has a weight associated with it. This idea was called artificial neural networks when applied to machine learning models. With several artificial neural networks lined up one after the other, they can be designed in layers, and deep learning models have been created ([Akerkar, 2019](#), p. 30). Neural network researchers have used the term deep learning to emphasize that they can train deeper neural networks than previously possible and to draw attention to the theoretical significance of depth ([Goodfellow et al., 2016](#), p. 19).

The most important feature of deep learning is that these feature layers are not designed by humans ([LeCun et al., 2015](#), p. 436). Those neural networks learn using educational data and backpropagation algorithms ([Chui, Manyika, & Miremadi, 2018](#), p. 3). However, this process requires extremely wide training sets since it may involve millions or even billions of parameters ([Salvaris et al., 2018](#), p. 15).

Deep learning and traditional machine learning are data-driven AI techniques that model the complex relationship between the input and the output. Compared to deep learning, traditional machine learning is limited to its ability to process natural data in its raw forms ([LeCun et al., 2015](#), p. 436). Machine learning models usually have

3-layer shallow structures, namely input, output, and a secret. However, there may be many more layers in deep learning models ([Wang, Ma, Zhang, Gao, & Wu, 2018](#), p. 146).

Today, artificial neural networks are yet inadequate to handle large amounts of complex data of human speech, image, and information processing ability. On the other hand, with the development of deep learning, it has become much easier to recognize, classify, and categorize the patterns in a machine ([Akerkar, 2019](#), p. 33). In this sense, deep learning algorithms have the advantage of using larger data sets better than machine learning algorithms ([Brynjolfsson & McAfee, 2017](#), p. 11; [Gunning, 2017](#), p. 4). In a deep learning process, the image given to the model comes in the form of a series of pixel values, and simple patterns such as edges and simple shapes are formed in line with the learned features in the first representation layer. The second layer typically detects motifs by detecting certain edge arrangements, regardless of minor changes in edge positions. The third layer can combine motifs with larger combinations corresponding to parts of known objects, and subsequent layers perceive objects as a combination of these parts. As a result, the most likely predictive inference is achieved. In this process, deep learning models learn the most appropriate weights of functions that match input data with output classification by analyzing large amounts of data and gradually correcting the predicted results against real labeled data. In this sense, the model automatically learns the attributes of the input data. Thus, in the case of processing new data, raw inputs such as pixel values in this image are fed directly to the model before, so there is no need to create properties manually ([Salvaris et al., 2018](#), p. 15).

With deep learning, automatic feature learning, and high-volume modeling capabilities, it is an advanced logical analysis tool for intelligent production in the big data age ([Wang et al., 2018](#), p. 145). Deep learning has proven itself to be useful in many software disciplines such as computer vision, speech, and audio processing, natural language processing, robotics, bioinformatics and chemistry, video games, search engines, online advertising, and finance ([Goodfellow et al., 2016](#), p. 10). Neural networks, which form the basis of deep learning capabilities, are still more efficient thanks to

the access to big data, the last two crucial technological advances, and an increase in computing power ([Akerkar, 2019](#), p. 5).

Industrial AI

Industrial AI functions as a systematic discipline that focuses on the use of AI technologies in industrial applications ([Lee et al, 2018](#), p. 20). In this sense, the integration of AI technologies with industrial processes is handled within the scope of industrial AI in order to achieve smart processes in all stages of industrial value chains such as business management, R&D design, production, service, and other business activities ([Zhang et al., 2019](#), p. 2369).

Historically, the use of AI in the industry has been largely dependent on the performance of machine learning algorithms, the developer's experience, and preferences, and the success of AI in industrial applications has been limited. However, with the development of Industry 4.0, the integration of AI technologies with the new technological advances such as the Internet of industrial objects, big data analysis, cloud computing, and CPSs enable the industry to work more flexibly and efficiently. In this context, it can be stated that the use of AI technologies in the industry has gained meaning with Industry 4.0 ([Lee et al., 2018](#), p. 20).

Digital Taylorism

With the hyper-dynamic technological progress, Taylorism has reached far beyond the factory floor in the digital age. Scientific management is also transforming to answer the emerging requirements of this new era, as "Digital Taylorism" to re-organize work in a techno-centric manner. If the twentieth century gave birth to mechanical Taylorism described by the Fordist production line, where the knowledge of craft workers was caught, codified and re-used in the shape of the moving assembly line by management, no doubt the twenty-first century will be the era of Digital Taylorism ([Brown, Lauder, & Ashton, 2010](#)). Today, it is becoming more and more common for managers to collect, process, evaluate, and act on a large amount of information quickly and inexpensively ([Frischmann](#)

[& Selinger, 2017](#)). Organizations need to use dynamic, more flexible and agile abilities effectively to keep up with technological innovations and to overcome competition ([Koçyiğit & Akkaya, 2020](#)). Based on those developments, the management and control processes for the surveillance and measurement of labor, in line with Taylorism's principles, such as rationalization, standardization, and business segmentation, are defined as Neo-Taylorism or Digital Taylorism which involves various forms and combinations of software and hardware ([Altenried, 2020](#), p. 5).

Business processes that can be digitally coded and tasks that can be computerized by decision protocols –AI–; now are beginning to take the place of human decisions in management ([Holford, 2019](#), p. 5). In addition, the professional and technical knowledge of the employees can be easily caught, stored, and made accessible for re-use by digital systems, database, and memories. So that information is no longer just being possessed by human beings but also digital minds ([Brown, Ashton et al., 2010](#)). As mechanical Taylorism allowed companies to catch the knowledge of manual craft workers and re-configure it throughout the use of assembly lines to decrease the costs, Digital Taylorism is offering analogous opportunities for companies to decrease the cost of many kinds of knowledge work currently carried out by middle class managers and professionals. Here, progress in computing power and software design are providing companies to digitalize knowledge which can be re-used repeatedly worldwide ([Chapman, 2012](#)). That is exactly the reason for authors such as [Au \(2011\)](#), [Irani \(2015\)](#), and [Moore and Robinson \(2016\)](#), to claim that digital technologies seem to have either brought back Taylorism or basically made it more apparent in many contemporary management practices within the workplace.

Brown et al. (2011, pp. 7–9) define “Digital Taylorism,” as a system based on the comprehensive organization of routine and “knowledge work” together, whereby the latter involves creative and intellectual tasks being subject to the same process as chain work. Once codified and digitalized, those tasks can be implemented by automatic programs with computerized decision protocols, in a way that substituting and event taking the place of human decisions and judgements. Such processes, without a doubt, can be relocated

across computerized global networks, thus rendering many jobs effortlessly to export, change, or replace. Digital Taylorism fundamentally follows at least one or more of Taylor's original principles by ([Holford, 2019](#)):

1. breaking down complicated tasks into simple and standardized ones;
2. monitoring everything those workers do; and
3. linking pay to performance.

Most of the processes operate automatically with the standardization of labor on most platforms, thanks to precise technologies that track employees with various algorithmic monitoring and grading technologies -Digital Taylorism-. Algorithmic management and surveillance tools provide the measurements of the results and feedback that are the main features of Digital Taylorism ([Altenried, 2020](#), p. 6).

Under Digital Taylorism, surveillance systems are increasingly used to ensure that workers follow standard tasks and procedures for the measurement and follow-up of labor and to meet their performance goals ([Holford, 2019](#), p. 5). In this context, many companies today use wearable tracking technologies as part of their new time and movement efforts. These technologies have detection capabilities designed to collect data about real-time communication and interaction to create continuous data streams for employee evaluation ([Frischmann & Selinger, 2017](#)). For example, the Amazon Company has a wearable technology that can track the movement and productivity of warehouse workers with a type of wristband that has its patented tactile notification feature and provides the function of alerting and directing workers with ultrasonic sound pulses and radio broadcasts through wristbands if they make a mistake. This technology owned by Amazon is not only following the actions of workers but also facilitating time-consuming tasks, such as responding to orders and packing them for fast delivery ([Yeginsu, 2018](#)).

Technological developments not only led to business concentration but also made it possible to monitor the work of the

employees more closely ([Brown, Lauder et al., 2010](#)). In Taylor's analog period, where the principles of scientific management spread rapidly, workers were aware that they were observed by management with chronometers and devices, while surveillance tools were less noticeable by the employees in the digital period of Taylorism. For example, a cashier, who makes purchases with a virtual cash register application on his tablet, was allowed to be hidden under the supervision of digital Taylorism, even though he was not aware that his actions were tracked with the program running secretly in the background, or the fact that conversations and geolocation movements were tracked with the smart phones given to employees ([Frischmann & Selinger, 2017](#)). However, this also increases the information asymmetry between platforms and employees in one way ([Altenried, 2020](#), p. 6). In addition, the addition of peer supervision to continuous monitoring and auditor feedback can create dynamics that can create excessive competition and conflict among employees ([Frischmann & Selinger, 2017](#)). It effectively encourages their creativity to develop new methods and techniques ([Akkaya, Koçyiğit, & Tabak, 2018](#)). This raises concerns that employees may be treated more like robots than people. And robots work without making humanly mistakes. They don't want wage, they don't require health insurance, and other employee costs ([Yeginsu, 2018](#)). What will the nature of human work be composed of in the digital economy of the forthcoming future? Will new technologies substitute jobs? If not, how will they be restructured? What will the broad implications be, both for individual workers, and legal regulation? Yet there are so many unanswered questions. 2017 McKinsey and Co. report states that about half the today's jobs carried out by workers are very likely to be automated, however for most occupations, partial automation is expected than full automation in the medium term, and the technologies will create new job opportunities in terms of Creative Destruction ([Holford, 2019](#)).

From a similar point of view, [Schumpeter \(2015\)](#) claims that 'digital Taylorism' or scientific management's will provide a much wider range of jobs further than the traditional industrial context, such service workers, knowledge workers, and managers themselves. Technology now plays a cutting-edge role in dividing

work across a wider range of jobs, conducting time-motion studies at more digital levels, in addition to linking pay to performance as an unbounded assessment ([Holford, 2019](#)).

Despite having disadvantages for employees, contemporary digital Taylorism allows for algorithmic management of labor by fully standardizing a large number of workers participating in various forms of labor on a large variety of situations and organizational synchronicity in a wide variety of situations ([Altenried, 2020](#), p. 8). Accordingly, digital Taylorism provides companies with a powerful opportunity to compare the performance of facilities, offices, suppliers, managers, and workers in many parts of the world, at least for now ([Brown, Ashton et al., 2010](#), p. 74).

Conclusion

Today, world is at the thresholds of a second phase of globalization, which is hallmarked by fourth wave of industrial revolution in terms of Industry 4.0, IOT, and Industrial internet. Transition to networked smart systems, automation AI creates smart factories with almost no need to unskilled or even semi-skilled employees. We also call those smart factories and “dark factories” as they can operate without any human worker in the dark. AI also seems to have a potential to replace the low and even middle management in decision making processes. Considerably amount of the intellectual labor is now “outsourced” to places from East Asia such as China and India, where extremely qualified university graduates are doing jobs with lesser costs than Western Europe or North America. In past, it was just the unskilled jobs that were outsourced to the Western world. However, all of these functions can be found worldwide particularly based on the technological progress. Control is remote based on software, to facilitate the monitoring of activities from a distance. So we have many questions about what the production systems are evolving to? What will the nature of human work are composed of in the digital economy of the forthcoming future? Will new technologies substitute jobs? If not, how will they be restructured? What will the broad implications be, both for individual workers, and legal regulation? In order to find an answer to those questions, the

beginnings of the twenty-first century witnesses a renaissance in Taylorism from a digital perspective: Digital Taylorism.

Just as Classical, or in other words, mechanical Taylorism provided standardization, job sharing, and specialization, control mechanisms, and allowed organizations to catch the knowledge of manual craft workers and reuse it through the assembly lines in order to increase productivity and to reduce the costs; Digital Taylorism is providing parallel opportunities including standardization through automation, specialization -in terms of knowledge workers-, digital surveillance systems, and databases to decrease the cost of many kinds of knowledge work presently carried out via middle class managers and professionals. Here, the employment problem may seem to arise; but digital revolution, as any other radical innovation wave, creates its own Creative Destruction. So many jobs will probably perish but many others may emerge too, such as gig economy with freelance working. No one can build a wall against change. What individual employees and companies should do is to try to understand the dynamics of the digital age, build a new system based on digital Taylorism, and try to explore and exploit the digital opportunities. As Louis Pasteur suggests: “Chance favors only the prepared minds.”

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Key Terms and Definitions

Artificial Intelligence: AI is the intelligence of smart machines.

Machine Learning: Machine learning is a kind of learning exhibited by computers. In this learning type, computers develop models based on existing datasets to predict the future results.

Deep Learning: Deep learning is a specific type of machine learning in which computers try to imitate biological brain composed of artificial neural networks.

Digital Taylorism: The transmission and usage of Taylor's basic principles into digital age based on computer technologies.

Creative Destruction: The launch of radically new technologies creates new industries based on those new Technologies, while exterminating the old industries. This is called creative destruction.

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Chapter 7

Leadership Styles and Female Managers in Perspective of Agile Leadership

Bulent Akkaya and Sema Üstgörül

Abstract

Female leader is a major topic in the male of today that's hardly getting attention. World today needs women in more positions of leadership. Today's organizations need effective and agile leaders who understand the complexities of the rapidly changing global environment. This situation requires urgent attention as the world needs female as well as male characteristics to address global issues. Women represent half of the modern world's abilities and population. They are important for economic and social prosperity for the world. Women carry a viewpoint that respects not only competitiveness but also organizational and team cooperation. Their female ideals are a functioning structure of new, cooperative and open economy. Eventually, the leadership of women will not only boost business, family and culture, but also the environment that will become more prosperous and peaceful as a result. In today's modern world many researches are conducted on global, social and cultural forces, such as

globalization, e-commerce, changing markets, the spread of technology and the need for teamwork, alliances and partnerships, show a high need for women leaders. But what is the role and leadership style of female leaders here? In this context, the purpose of this study is to discuss the leadership styles and what kind of leadership style female managers exhibit in line with the literature. The research showed that female managers have more agile leadership qualities and the authors suggested that female leaders should be brought forward for the opportunity to lead others toward a better future.

Keywords: Female leader; leadership theories; women managers; agile leadership; strategic leader; leadership styles

Introduction

Today's organizations need effective leaders who understand the complexities of the rapidly changing global environment ([Nanjundeswaraswamy & Swamy, 2014](#)). No matter how advanced science and technology, people today do not reach all the goals and objectives alone (Arslan & Uslu, 2014). For this reason, people who are social entities should cooperate to meet their own needs and achieve the goals they cannot achieve alone. Organizations have formed as a result of people coming together and cooperating. Nevertheless, there is a need for managers who will lead them in achieving the goals and objectives of the organization. An effective manager should be able to adopt different leadership styles ([Ojokuku, Odetayo, & Sajuyigbe, 2012](#)).

Leadership is an ongoing process that involves achieving the goals of the organization by affecting the behavior of people in the organization ([Packard, 2009](#)). Leadership is the art of motivating a group of people to act toward achieving a common aim ([Depree, 2011](#)). The leader is the inspiration for and director of the action. They are the person in the group that possesses the combination of personality and leadership skills to make others want to follow their

direction. Looking at these definitions of leadership, it can be said that leadership is a political–cultural concept that does not depend on formal positions, and leadership is largely composed of learned knowledge, attitudes and skills ([Winston & Patterson, 2006](#)).

Various leadership approaches and theories have been developed in literature such as *Great Man Theory*, *Trait Theory* and *Transformational Leadership Theory*. What type of leadership do women leaders use? It is stated that male and female leaders use almost the same leadership ability and style, but individuals use different leadership styles depending on situational factors regardless of gender ([Merchant & Karima, 2012](#)).

The aim of this study is to discuss the leadership styles and what kind of leadership style female managers exhibit in line with the literature. In this context, leadership theories and styles will be discussed then women managers and leader will be explained in the light of literature. Finally, a new leadership style “agile leadership” will be introduced, and the chapter will be concluded with some suggestions for future researches.

Leadership Theories

There is no universal definition of leadership, which is one of the most important subjects of research in the field of management. Leadership is seen as the most important and effective response to the challenges and opportunities brought by globalization. Determining values, culture, change, tolerance and motivation of employees are very important in every organization.

The role of leadership is to determine the mission and vision of a company, develop appropriate strategies, motivate and positively influence employees, inspire employees and create a corporate culture. They analyze both internal and external factors for the company’s vision (Ahmed, 2012). For many years, many researchers have conducted various researches and studies to discover these factors. In this process, the following leadership theories developed:

Great Man Theory: According to this theory, some people are born to be leader and leadership is a heroic act ([Maloş, 2012](#)).

Trait Theory: According to this theory, the individual must have the characteristics of being self-confident, honest, extroverted, reliable, intelligent, determined, sincere, collaborative, creative, charismatic, harmonious, success-oriented and socially active to be an effective leader ([Khan, Nawaz, & Khan, 2016](#)).

Behavioral Theory: According to this theory, leaders have four behaviors. They are support, goal emphasis, work facilitation and interaction facilitation ([P. S. Aithal & Aithal, 2019](#)).

Contingency Theories: According to this theory, leadership consists of three factors: characteristics, behavior and situation. The behavior of a leader depends on the situation ([Khan et al., 2016](#)).

Charismatic Leadership Theory: According to this theory, leader should have some extraordinary and exceptional qualities such as forecasting, energizing and activating ([Anderson & Sun, 2017](#)).

Transactional Leadership Theory: According to this theory, leaders use reward and punishment methods to motivate his/her followers to achieve his/her goals ([Anderson & Sun, 2017](#)).

Transformational Leadership Theory: According to this theory, leaders should transform or change the perceptions, behavior and expectations of the followers and direct them toward a common goal which will lead to the accomplishment of the leader's vision ([Anderson & Sun, 2017](#)).

Leadership Styles

Leadership style in terms of manager's style appeared by Ohio and Michigan Universities in USA in 1940 ([Sellgren, Ekvall, & Tomson, 2006](#)). [Max De Pree \(2011\)](#) explained leadership in introduction of his book *Leadership is an Art*. The most important aspect of leadership is about employees of the organization. Max De Pree added that they should be a part of the decision-making process and all information should be shared with the workforce which is stated by him as "the best management process for today's environment is participatory management based on moral relations." He says that leader must have good communication and "without effective communication, actively practiced, without the art of scrutiny, those values will disappear in the sea of trivial memos and impertinent

reports.” This makes managers and leaders more successful. Successful leaders and managers are always focused on learning based on field experiments and argue that leadership skills emerge as a result of constant observations and experiments (Ahmed, 2012).

Leaders can carry a mix of the different leadership styles depending on their industry and the obstacles they face. In this context, there are eight leadership styles.

Democratic leadership: It is known as a common leadership or participatory leadership, because this kind leader encourages employees to be a part of the decision-making process (Khan et al., 2015). Democratic leaders offer guidance to group members. They usually inform their employees about functioning and share their responsibilities for decision making and problem solving. The democratic leader is responsible for deciding who is in the group and who will contribute to the decisions taken (Iqbal, Anwar, & Haider, 2015). In the literature, it is stated that democratic leadership style is one of the most effective leadership types that provides higher productivity, and group members work more with high motivation and remarked that good democratic leaders are with honesty, creativity, courage, competence and fairness specific traits. These leaders tend to seek different opinions and do not try to silence those who offer dissenting voices or a less popular perspective. Hence, strong democratic leaders create trust and respect among group members, where they are talented and eager to share the knowledge of the group members (Arabaci, 2019; Uluköy, Kılıç, & Bozkaya, 2014).

Autocratic leadership: It is also known as authoritarian leadership. Their own opinions and thoughts prevail over all decisions; the opinions of the group members are invalid (Cherry, 2019). The sole responsibility for the decision and the result is the leader. However, autocratic leadership also has some advantages. For example, quick decision making, control over a company’s processes and operations, etc. Autocratic leaders can react quickly to events due to their ability to make quick decisions. Autocratic leaders have some characteristics which are listed below (Rüzgar, 2018).

- Leaders take almost all decisions on their own.

- Leaders evaluate all working methods and processes.
- Leaders do not trust group members about decisions or important tasks.
- Leaders are highly structured and very rigid behaviour in work.
- Leaders tend to discourage the creativity and readiness of group members.
- Leaders have some certain rules which are important and clearly stated.

This type of leadership style is seen mostly in businesses which are relatively small with fewer employees. The autocratic style can be useful in some environments, but it also has traps and is not suitable for all environments and groups. In order to expect team members to follow the rules, these guidelines must first be clearly defined and all members of the team informed. If the leader criticizes the team members only when they make mistakes and never rewards the event from their success, the group will lose motivation. Because listening to people's thoughts makes them feel as if they are making an important contribution to the mission and vision of the institution.

Laissez-Faire or Free-Rein leadership: The Laissez-faire leadership style has a favorable approach to management. They allow group members to express their ideas to support decision making, and they do not give direction. Today, it gives autonomy to employees, which is a greater motivation factor for people in organization, and changes survival and rewarding as the primary motivation factor ([Zareen, Razzaq, & Mujtaba, 2015](#)). They allow group members to take responsibility. Consequently, they attach great importance to the training and experience of institutional members. Therefore, these leaders support members to get the training they need ([Jones & Rudd, 2008](#)).

Strategic leaders: It is defined as the ability to see the future, create vision, provide flexibility, think strategically and work with others to initiate changes that will create an achievable future for the organization ([Hitt, Ireland, & Hoskisson, 2012](#); [Rahman et al., 2018](#)). Strategic leadership theory has seen organizations as a reflection of senior executives, especially the CEO (Chief Executive Officer).

Numerous CEOs have accepted their strategic leadership responsibilities on their own. As the strategic leaders of organizations, the main task of CEOs is to choose a vision for their organizations and create the conditions to achieve this vision. Strategic leaders have been recognized as heroes of companies, especially when these elections have resulted in financial success (Uğurluoğlu & Çelik, 2009). However, with the globalization of the twenty-first century, the stagnant and predictable environmental conditions of the organizations have changed dramatically. The new competitive structure of the global economy has prevented the direction of the organization from being determined from the perspective of a single individual and that twenty-first century organizations differentiated its structure and strategic leadership practices. Strategic leadership focuses on people who take full responsibility of the organization, but includes not only the title-winning leaders of the organization, its top management teams, members such as board members and department general managers as well (Hitt, Hoskisson, & Ireland, 2007).

Basically, three roles are expected from a strategic leader in achieving a common goal, creating teams and developing manpower. Strategic leadership brings the basic capabilities of the organization to the forefront in the face of uncertainties in the environment (Akyüz, 2018). It is a leadership style desired by many institutions as it supports multiple types of employees at the same time. However, strategic leaders decide how many people they can support at the same time and what is the best direction for the company if everyone is always on track. They use the SWOT analysis method to systematically evaluate the information collected to understand and manage the environment in which the organization operates (Daniels & Ramey, 2005). With the results obtained from the SWOT analysis, when institutions work, their strengths and weaknesses, the opportunities and threats they face, their situation becomes clearer and provides leaders with a clearer perspective and a healthy perspective.

Strategic leaders do not satisfy the current situation. Strategic leaders are constantly working to catch the best and the newest. They always want to be one step ahead. Success and permanence

are indispensable for strategic leaders ([Akyüz, 2018](#)). Instead of saying “this is enough” when they achieve a certain degree of success and stability, they aim to go one step higher. In addition, strategic leaders must have their own value judgments, have the ability to mobilize employees, have social intelligence and be ambitious enough to fulfill all these functions ([Ülker, 2009](#)).

Transformational leadership: There is no question that more efficient use of human capital and innovation in the enterprise is closely linked to the existence of an effective leader within the organization. Transformational leadership is a new leadership philosophy needed for effective leadership in today's world. [Yukl \(1999\)](#) describes transformational leader as the leader who realizes the great change process in attitudes and behaviors among the members of the organization and contributes greatly to the mission and vision of the organization. Transformational leaders avoid individual-oriented decisions; they focus on the needs and interests of the organization by giving importance to their thoughts in their subordinates. This type of leadership is a leadership that ensures human development and mutual trust, strengthens the communication system and ensures mutual satisfaction for both parties and eliminates confusion. The transformational leader is the master of change, designs a better future, has foresight, creates a vision and arouses the desire to effectively adopt this vision for everyone ([Karakaya, 2005](#)). They pretend to be coaches or mentors to show the special attention individuals need for the development and success of their employees. He/she knows how to listen very well and hands out tasks to developmental characteristics among employees. It tracks and helps the leaders of the organization, if necessary, after the allocation of duties. Transformational leaders also have very good empathy skills. Thanks to these skills, transformational leaders make every employee feel valuable for the organization and help improve their performance ([Özarallı, 2003](#)). It is a kind of leadership that is highly preferred by growth-oriented institutions. They motivate group members to see what they can do. They organize the reports and guide the members to take on new responsibilities ([Stone, Russell, & Patterson, 2004](#)).

Transactional leadership: Transactional leadership is based on bureaucratic authority and organizational legitimacy, based on behavior such as rewarding performance and achievements, researching deviations from rules and principles and taking flawless actions. Sustainable leadership is a leadership approach with short-term orientation, concrete goals, power from the office, extreme compliance, rules and guidance and a strict reward – penalty system for completing tasks and obeying members of the organization (Bass, Avolio, Jung, & Berson, 2003).

Avolio, Bass, and Jung (1999) stated that transactional leaders exhibit three types of management style:

- Contingent reward; leaders use their powers to reward high-performance employees. Rewards are in the form of a currency or a status. Employees are well aware of what is expected of them.
- Management by-Exception (Active); Leaders have chosen to do their job to make past activities of employees more efficient and effective. At the beginning of the work, a certain standard is determined, and they do not intervene until a problem arises. Focusing on error and enforcement here can create tension for employees.
- Management by-Exception (Passive); leaders do not deal with employees in any way, but they intervene when the targeted standards are not reached. This is an effective management style only for those who are advanced in self-management.

Coach-style leadership: Coaching leadership style is characterized by partnership and collaboration. As leaders act as coaches, they contribute to teamwork and ingenuity. Short-term fire fighting reactions are replaced by long-term strategic thinking. Leaders who adopt the coaching style increase their responsibilities and beliefs toward their employees by choosing wherever possible and providing a good balance of support and difficulty. This means less stress for workers who feel in control and the leader who doesn't feel pressured to steer the ship. These leaders focus on identifying and nurturing each member's strengths in their team. They also focus on strategies that will make their teams work better

together. Besides, they put more emphasis on the growth and success of individual employees ([Harper, 2012](#)).

Bureaucratic leadership: The bureaucratic leadership style focuses on rules and procedures for managing teams and projects ([Lieberthal & Lampton, 2018](#)). This is a classic leadership style. These leaders are resistant to innovation and change, and great opportunities can be lost during the process. And considering how highly competitive the business environment is becoming, having a leadership style that is not flexible isn't a great idea for business growth. Bureaucratic leaders take decisions in accordance with the rules, taking into account the hierarchical system. This leadership style can be advantageous in well-organized business lines and can be an effective management style in companies that do not require more creativity or innovation than employees. They are well aware of policies and guidelines. This leadership approach is often used in public services (<https://www.managementstudyhq.com/> Date of access: 20,02,2020).

The general characteristics of bureaucratic leadership are as follows ([Marquis & Huston, 2009](#)):

- Bureaucratic leadership is based on a strict and formal hierarchy that ensures that members within the organization are aware of the structure. Authority is regulated to ensure a higher level of leadership that controls each lower level, which is controlled by another leader layer.
- Official hierarchy rules cannot be followed. Within the bureaucratic system, there are rules for maintaining and maintaining control. These rules ensure consistency.
- Bureaucratic leadership is not always associated with high-level skills, knowledge and expertise are important. The leader always aims to combine the best talents in the right position. In addition, the bureaucratic leadership framework clearly reveals roles and units. The job expertise and skill set required to accomplish these are defined to find the right person as smooth as possible.
- Bureaucratic leadership does not focus on people. Individuals and individual achievements are not at the center of the institution. Therefore, individuals play less role.

- Recruiting in the bureaucratic leadership model relies solely on technical competence, just as it is directed to roles that fit the individual skills of subordinates. This means that the person who has the skills necessary for a particular role will always play the role.
- Although the focus is on performance, it is successful if it distributes tasks to individuals in line with the abilities of the leading members.

Women Managers and Leaders

The literature searches for answers to the following questions about leadership ([Vasilescu, 2019](#)).

What exactly produces a remarkable leader?

Do we need specific personality traits to be more suited to leadership roles?

Does being a leader have gender, age, culture?

Because of the tendency of the sectors to prefer men and cultural norms for leadership positions, women do not work in managerial positions sufficiently ([Deloitte Development LLC, 2015](#)). Gender discrimination is in its most general form. It can be seen in entry to professions, wages and promotion (Alparslan, Bozkurt, & Özgöz, 2015). A number of jobs are determined from the outset as men's work or women's work. Many work-related traits are frequently only applied to one species, whereas the other breed is believed to be totally devoid of such qualities. In addition, the idea that women cannot pay enough attention to their jobs due to their family responsibilities or that woman will work temporarily. Even if women work hard to come to senior management, they try to hit an invisible glass ceiling that prevents them from going up somewhere (Alparslan, Bozkurt, & Özgöz, 2015).

Today, although the number of women on the working floor has increased, few have fulfilled their senior management positions which are described as "glass ceiling syndrome" in management literature. "Glass Ceiling Syndrome" refers to any invisible obstacles that cannot be explained between women and senior management positions ([Korkmaz, 2016](#)). This ceiling can be demonstrated by the

fact that, although women own 44% of senior management jobs, they account for only 5% of senior managers (Corporate Leadership Council, 2002). This situation negatively affects the career development of women and creates serious problems. These barriers prevent women from moving forward in their careers, that is forming a glass ceiling, lack of role models, significant lack of experience, stereotypes and prejudices about roles and talent, exclusion from informal networks and family responsibilities. [Ustabaş and Fındıklı \(2017\)](#) studied on this situation by interview with some female employees. Some of them said that “Generally, male candidates are brought to Manager positions. The management of women is delayed due to marriage and pregnancy issues.” “I have had this problem many times. I saw that a man was brought to the position I deserve every time.”

At the present time, as training opportunities for women increase, opportunities to work in a professional job also increase. Among the G7 countries, the worst performers are Japan with 7% female managers and Germany with 18% female managers (<https://www.goldmansachs.com/>. Accessed on February 18, 2020). African-American women and Asian-American women make up only 5.3% and 2.7% of all people employed by management, professional and related professions ([US Bureau of Labor Statistics, 2011](#)). In 2016, 65.1% of men in the work rate, 28% of women, however, women’s labor force participation rate realized as 71.3% of university graduates in Turkey (<https://www.ailevecalisma.gov.tr/uploads>). In addition to the fact that women’s employment is lower than that of men in executive and decision-making positions, it is also exposed to inequality in terms of wages. It is stated that there is a discrimination between male and female employees, especially in the industry and service sector, and the woman receives lower wages despite doing the same job as the male ([Bilgin, 2012](#)). However, low wages due to gender of women have been the subject of debate, as worker women play an active role in working life over time. In 1951, Equal work by the International Labor Organization has been adopted the principle of equal pay between male and female workers. But today, still, women employees face discrimination about wages even if they progress equally with men in their careers. There are 14.4% women

and 85.6% men in the executive positions of 500 companies in USA. Regarding manager earnings, the average income is \$1,266 for men and \$939 for women (<http://www.bls.gov>). While more than half of the jobs are done by women in the world, the fact that women have 10% of the total income in the world and 1% of their assets is one of the best indicators of the need to solve the problems of women in business life (Bilgin, 2012; Ustabas & Findikli, 2017).

Although women are gradually taking part in the management positions of companies, the number of women in management positions is disproportionate to the number of men in leadership positions. However, research shows that global, social and cultural forces, such as globalization, e-commerce, changing markets, the spread of technology and the need for teamwork, alliances and partnerships, show that the need for women leaders is high (Valerio, 2009).

Female Leaders in Perspective of Agile Leadership

Some leaders still believe in the concept of male gender leadership, since the idea of leadership is based on the “Great Man” theory. However, with the publication of articles on the leadership characteristics of women in the 1990s, research on female leaders began (Rowney & Cahoon, 1990).

In the studies, it has been examined what men and women managers do differently. Helgesen (1990) observed that women framed a circular and inclusive “network of inclusion” rather than hierarchical and exclusive and women work at a steady pace, view unscheduled interruptions as a part of work flow, maintain a complex network of relationships, make time for activities not focused on work and focus on the “ecology of leadership.”

According to a report from Peterson International Economic Institute, there is a positive relationship between women’s leadership and profitability. It is determined that 30% of the leaders are more successful than women (Eades, 2019). Nevertheless, the disturbing truth is that there are still few women in the top leadership roles in firms. There may be because of some factors such as culture and prejudices that women are few in management. Due to this gender

discrimination in management, research has focused on the different behaviors of male and female managers. According to Liu's statement (2018), in some studies, the effect of gender on leadership was investigated, and it was found that the leadership tactics used by men and women are different. Studies have also indicated that women use more inspiring, attractive and business than male workers.

Male managers and leaders are more ambitious when trying to influence others, while women try to impress by giving advice and inspiration ([Merchant, 2012](#)). There are some situations that women are more successful.

In 2019, according to the study of Zenger and Folkman, the issues that women are more capable than men are:

- ✓ taking initiative (55.6% women, 48.2% men);
- ✓ practicing self-development (54.8% women, 49.6% men);
- ✓ resilience (54.7% women, 49.3% men);
- ✓ developing others (54.1% women, 49.8% men);
- ✓ displaying high integrity and honesty (54.0% women, 49.1% men);
- ✓ inspiring and motivates others (53.9% women, 49.7% men);
- ✓ driving for results (53.9% women, 48.8% men);
- ✓ building relationships (53.2% women, 49.9% men);
- ✓ adapting changes (53.1% women, 49.8% men);
- ✓ collaboration and teamwork (52.6% women, 40.2% men);
- ✓ establishing stretch goals (52.6% women, 49.7% men);
- ✓ communicating powerfully and prolifically (51.8% women, 50.7% men);
- ✓ connecting to the outside world (51.6% women, 50.3% men);
- ✓ solving problems and analyzes issues (51.5% women, 50.4% men);
- ✓ leadership speed (51.5% women, 50.5% men); and
- ✓ innovating (51.4% women, 51% men).

As understood from this study, women have more meaningful, polite and collaborative communication features. When given opportunities, female leaders may be more successful than male leaders with these characteristics ([Basow & Rubenfield, 2003](#)).

Agile Leadership and Female Managers

These leadership characteristics of women managers or leaders are closely related with agile leadership. Agile leaders focus on collaboration and communication with followers in order to survive the competition in the global world. Agile leadership is different from traditional leadership styles mentioned above. The most important difference is having never-ending desire and motivation to develop its competence and potential, because only such an understanding of leadership can mobilize the institution by adapting the rapid changes in a competitive environment.

[Popper, Power, & Stanson \(2013\)](#) listed nine principles for agile leaders. They are:

1. Leading through behavior, not words: Agile leaders take a humble, empathetic, kind, compassionate approach that takes into account others. They strive to improve these features.
2. Improving results by improving the quality of thought: Agile leaders look at problems and issues from different angles.
3. Improving the institution through an effective feedback culture: Getting feedback is perceived as a more or less negative experience in every culture. Agile leaders' attitude opens the way for courageous, useful, meaningful feedback. They pay attention to be open, honest and respectful for this to be effective when giving feedback.
4. Ensuring that people find meaning in their work: Agile leader helps employees develop a common purpose and bridge the work they do for life reasons. They do this both by creating a meaningful vision for the institution and by connecting with the mission. Thus, they address their employees' brains and hearts and ensure that their corporate values are reflected in the way they do daily business.
5. Creating positive emotions as the basis of creativity and discovery: Agile leader inspires people to come to the business environment in their best form. They create an atmosphere of trust so that positive emotions reflect human potential. Because creativity and new inventions are carried out by going beyond the

- comfort area and taking risks. For this, the confidence that the leader will provide is the basic condition.
6. To provide leadership behaviors at all levels of the organization: Leadership is to influence with its simplest definition. Therefore, leadership behaviors can be expected from employees of all levels of the institution. To this end, it encourages learning and adapting to new conditions by activating the leadership potential that exists within all employees. This is only possible if employees take responsibility.
 7. Transferring authority and power appropriately: Agile leader knows that people show their best performances with high energy when they come to work and dedicate. Empowering employees to solve the needs and tensions that arise within the organization is an important and necessary skill.
 8. Collaborate: Agile leader knows that forgiveness, positive approach, generosity and an understanding of the value of possessions are necessary for a healthy working environment. Thus, in an institution that goes to high goals, it is aware that the feeling of psychological trust encourages taking responsibility, learning and development in order to get out of the comfort area.
 9. Bringing out important ideas at all levels of the organization: Agile leader is open to different ideas regardless of their status. For this he stops, listens and gives enough time for his subordinates to develop and express themselves.

Instead of hierarchical structure and authority, more cooperation, teamwork, positive feedback and high motivation are observed in the institutions with the agile leadership approach. In a meta-analysis study comparing the characteristics of male and female leaders, it was stated that the transformational and conditional reward behaviors, which are among the agile leadership characteristics, are more common in female leaders. [Eagly, Johannesen-Schmidt, and Van Engen \(2003\)](#) remarked that male leaders, on the other hand, have a high focus on employees' mistakes and failures. They concluded that research on transformational, transactional, laissez-faire leadership and agile styles indicates that female leaders will be more successful, albeit a small advantage. The results of the

research of Pew Research Centre conducted with 1,060 men and 1,190 women; half of the participants stated that women are more honest, 38% of women are smarter, 80% of women are more compassionate and 62% of women are more creative ([Pew Research Center, 2008](#)). In a qualitative study comparing male and female leaders, women found that women were more inclusive, persuasive, learned from adversity, and took risks ([Caliper, 2020](#)). Another contemporary example of why women should be leaders is stated by former Austrian ambassador Swanee Hunt as follows: Whatever the official authority is, women are innovative, they do not discriminate politically, ethnically and culturally, they have a more harmonious perspective with society ([Hunt, 2005](#)).

There are plenty of stories of women spearheading social change. For example, according to Chandler, Wendy Kopp, founder and CEO of Teach For America, has established an organization that aims to eradicate education disparities around the US by recruiting two-year college graduates in low-income communities ([Chandler, 2011](#)). [Chandler \(2011\)](#) underlined that a study of 2,250 adults (1,060 men and 1,190 women) ranked women better than or equal to men in seven out of eight primary leadership positions.

As former United Nations Secretary-General Kofi Annan endorsed:

The work done after the study taught us that it is not a more effective means of development than girls' education and women's empowerment. As a result of the increased participation of women in working life and management, families become healthier, they are better fed, their income, savings and investments increase. (Annan, 2004: cited in [Chandler, 2011](#))

These results and expressions show that female managers and leaders have the characteristic of agile leadership.

Conclusions

One of the biggest components that contribute to the leadership styles is communication and social interaction between the leader and corporate members. The most important thing that women and men differ in leadership style is communication. In the work environment, women value communication more than men, and female leaders can establish closer communication links with members of their organizations. Men, on the other hand, use more power-oriented and controlling communication style (Foels, Driskell, Mullen, & Salas, 2000).

Today, institutions need “relational, nurturing and attentive” leaders and managers to ensure their sustainability in the globalizing world. Such leadership traits are often associated with women. When the data of *The Economist* (2015) magazine are examined, it is seen that women in the world have a 32% ratio of working in high management positions even in countries where the gender inequality is at a minimum level. Despite these views and observations, the lack of women in leadership positions continues to be reported across the world. In modern leadership styles, women have more characteristics of agile leadership. In order to adapt changes, to solve internal and external problem quickly, to benefit from competitive women should be participated in top level management. The below list may guide to achieve it.

- Women leaders should be increased both in the private and public sectors,
- equal pay and work;
- flexible employment contracts;
- support women’s career;
- more flexible working environments;
- performance based evaluations;
- women encouragement through media;
- increasing women labor and employment; and
- visibility, regular and systematic statistics of women’s labor.

It would seem, therefore, that although the leadership literature has played a significant role in raising the profile of women in management, further advances are required in order to advance the

careers of women in management. Other innovative contributions of women worldwide will dramatically impact social change through applying use of technology and economic transformation for women's empowerment and gender equality.

Finally, there are some researches on female roles. [Latu, Mast, Bombari, Lammers, and Hoyt \(2019\)](#) investigated the behavioral process through which visible female leader role models empower women in leadership tasks. Female role models can inspire girls and women in male-sectors ([Cheryan, Siy, Vichayapai, Drury, & Kim, 2011](#)), sports ([Greendorfer, 1977](#)) and administrative and political leadership ([Simon & Hoyt, 2013](#)). There are many factors that may account for these positive effects, including the increased sense of belonging and self- of women ([Dennehy & Dasgupta, 2017](#)). In this current research, we propose a different behavioral explanation that describes how female leaders empower women in agile leadership roles and behaviors.

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KEY TERMS:

Transactional Leader: A leader that tells followers very clearly of their expectations and describes what kind of reward they can expect in return for good performance and effort.

Transformational Leader: A leader that motivates followers to achieve the objectives that people establish individually or collectively through the use of economic, political or similar forces.

Agile Leader: A leader that can keep up with and adapt to the ever-changing and evolving world, transforms moments of crisis into opportunities.

Strategic Leader: A leader that provides a clear vision for the company, can think critically and foresee the future, and has the highest analytical potential in decision-making.

Coach-Style Leadership: A leader that acts like a coach who contribute to teamwork and creativity.

Chapter 8

The Effect of Industry 4.0 on Accounting in Terms of Business Management

Ece Çevik Özcan and Bulent Akkaya

Abstract

While the studies on the use of smart robots in the production, which is known as the Industry 4.0 revolution, continue rapidly, the competition race is in the right proportion with this and businesses that do not want to fall behind the age have to follow the developments in Industry 4.0 and adapt. This chapter examines the Industry 4.0 revolution and its effect on accounting. Incorrect information from one of the business departments affects the entire system. Especially in the accounting department, which records the financial movements of the business and reports the results of them, and which is effective in making decisions that affect the whole business based on these results, the use of smart systems helps to reduce human-made mistakes and the system to act faster while also the systems used in traditional accounting are abandoned. However, brain power is still required, which must analyze the results. While the revolution is still developing, there are a lot of theories about the future. The expected

outcome of these is that with the revolution, the need for brain power to analyze system outputs will increase, even if the need for human labour in the industry decreases. In this context, especially those who perform the accounting profession must constantly renew themselves.

Keywords: Industry 4.0; the cyber-physical systems; robotic process automation; block chain; accountant; accounting engineer

Introduction

While the studies on the use of smart robots in the production, which is known as the Industry 4.0 revolution, continue rapidly, the competition race is in the right proportion with this and businesses that do not want to fall behind the age have to follow the developments in Industry 4.0 and adapt. This chapter examines the Industry 4.0 revolution and its effect on accounting. Incorrect information from one of the business departments affects the entire system. Especially in the accounting department, which records the financial movements of the business and reports the results of them, and which is effective in making decisions that affect the whole business based on these results, the use of smart systems helps to reduce human-made mistakes and the system to act faster while also the systems used in traditional accounting are abandoned. However, brain power is still required, which must analyze the results. While the revolution is still developing, there are a lot of theories about the future. The expected outcome of these is that with the revolution, the need for brain power to analyze system outputs will increase, even if the need for human labor in the industry decreases. In this context, especially those who perform the accounting profession must constantly renew themselves.

Progress of Industry

When looking at the industrial developments from past to present, the first of the three major industrial revolutions is the first industrial revolution, which is the symbol of steam machines developed in England, covering the period toward the end of the eighteenth century, where water and steam were used as production power (Tekin & Karakuş, 2018). With the use of water and steam technologies in production, tools and manpower-oriented workshop-type production began to be abandoned and started to move to machine-oriented factory-type production.

After the first industrial revolution, the use of electricity, oil and gas as a production power took place, and the transition from steam power to electrical energy has taken place. This period is known as the second industrial revolution which toward the end of the nineteenth century and identified with mass production (Esmer & Alan, 2019). The production line design used by Henry Ford, the pioneer of the serial production system, has become an example for many sectors and started to be applied. Therefore, the second industrial revolution is generally known as Fordism (Gabaçlı & Uzunöz, 2017). After this trend, the transition to digital systems took place and the third industrial revolution, the age of digitalization in production has begun.

With digitalization, the phenomenon of globalization has gained importance and the rapid transfer of information has become possible. The need for labour power has decreased further, and the number of machines programmed for various production processes is increasing rapidly. The rapid development and integration of computer hardware, software, networks and digital technologies, which are the determinants of the industrial revolution, transformed the society and the economy. A new era which cyber physical systems, dynamic data processing and value chains are connected to each other has come due to the proliferation of information technologies and automation (Ozsoylu, 2017).

While the increasing demands and changing needs of industry and consumers and new searches are continuing, finally the concept of Industry 4.0 has emerged. This concept was first mentioned in Germany in 2011 (Derya, 2018).

Industry 4.0 is a strategic attempt by the German Government to continue growth of the industry. Germany's Industry 4.0 is an attempt to ensure its position as one of the leading actors in the automotive and machinery industry ([Rojko, 2017](#)).

The wind of innovation created by Industry 4.0 has made a global impact by crossing the borders of Germany at the points of the using the Internet of Things and Services and the Cyber-Physical Systems (CPS) at every stage of production, new generation ICT (Information and Computer Technology), technological opportunities and scientific difficulties. As a result, it is the fourth industrial revolution itself, befitting the name of Industry 4.0 ([Posada et al., 2015](#)).

CPS can be defined as a cooperation between the machines used in the enterprises along the production line, storage solutions and production facility. In such a system, the virtual and physical world are intertwined. Processes such as planning, analysis, modeling, design, implementation and maintenance work collaboratively. CPS system which the production process works in cooperation maximizes the efficiency ([Badem & Kılınç, 2019](#)).

Industry 4.0, where economic power is carried to digital environment, takes its power from new technologies such as cloud computing, project analysis, machine learning, robotics, Artificial Intelligence (AI), Distributed Ledger Technology and mass data processing ([Antoney & Augusthy, 2019](#)).

With this technology breakthrough developed with Industry 4.0, while large firms calculate the size of the slice they will receive from the global economy cake, smaller national firms can focus on developing their competitiveness to become international players. Especially if these local firms are in high-cost trade zones, their survival skills are directly proportional to their competitiveness ([Bettiol, Capestro, De Marchi, & Di Maria, 2020](#)).

Industry 4.0, which means smart machines for production and service processes, also creates value thanks to the chain of smart robots created along the production line. The impact of Industry 4.0 will be unquestionably great considering the development of the global economy. The emergence of Industry 4.0 has a major effect on the global economy and affects inter-business which is shifting consumer preferences, improving asset quality by increasing data

output, new relationship establishments as learning about the value of new ways of collaboration, digital transformation of operating models into new business models, especially, open web-based platforms generate new opportunities and increase the competition (Di Vaio & Varriale, 2019).

The Industry 4.0 concept can be called a smart factory. This smart factory has a decentralized communication network thanks to smart robots that are effective throughout all production and service processes. With this network that requires advanced technology and big data management, robots can interact with each other and people in real time (Morrar, Arman, & Mousa, 2017).

A new paradigm that is decentralized, based on the communication of intelligent systems in production processes with the strengthening of communication between people, machines and resources is shaping the fourth industrial revolution with the loss of power of the central control mechanism. In such a factory that has evolved with the fourth industrial revolution, all production lines will be smart. Since the products are capable of reporting, they can directly participate in the production processes by providing information such as production history, order status and stock status to the machines on the production lines (Hermann, Pentek, & Otto, 2016). Smart product is aware of about their whole production stages. They can not only give information about their past orders, but also make inferences about the future and shape the production line.

The fact that smart products can determine their own physical properties gives the factory the ability to meet any specific demands of customers. Thanks to their ability to interact with their environment, they maximize efficiency throughout the production line (Schmidt et al., 2015).

Industry 4.0 in Enterprises

Industry 4.0 is a process which a factory “intelligently” interacts directly with machines in the production of finished and semi-finished products typically used during the production stages. This scenario enables customer-oriented production in a cheap and efficient way.

Industry 4.0 vision goes beyond customer-centric production and promising real-time control, integrated maintenance, better adaptability, increased collaboration across the supply chain, better monitoring possibilities and smarter products and new business models ([Koca, 2018](#)).

Industry 4.0 is an interactive, smart concept that is constantly changing and developing rather than a closed system and should be defined as the cornerstone of a larger structure. This structure will become a structure far beyond the factory and will be a system that works like a single body. Thanks to its advanced technologies, it connects the major sectors in the world. Especially with the technology of the internet of things and services, the systems that are interconnected within the factory can be connected to a higher structure and ultimately to the whole world. Thus, a wide range of integrity arises from energy to health, from logistics services to the construction industry. Many systems such as smart buildings and residential areas, continuous and sustainable energy, much faster and mobility logistics services, quality and smart health services are some of the instruments of this integrity ([Bartodziej, 2017](#)).

Key features of Industry 4.0 can be shown as follows ([Ozsoylu, 2017](#)):

- Interoperability: CPS provide ability to connect and communicate between people and smart factories. CPS enable people and smart factories to connect and communicate. A pilot project implemented at Bosch Rexroth Hamburg factory, where hydraulic valves are produced for tractors and similar agricultural vehicles can be given as an example. Within the scope of the project, people, machines and products work together and give an idea of how the production environment will be in the future.
- Virtualization: It is the creation of a virtual copy of the Smart Factory by combining sensor data with virtual models and simulation models.
- Decentralization: A replacement of the hierarchical system involving managers who have the power to make decisions in any problem with the bottom-to-top control approach is managed by CPS ([Unlu & Atik, 2018](#)).

- Modularity: It states the flexibility of smart factories. It is the ability to adapt to changing conditions over time. These factories are designed to be flexible enough to integrate changes in consumer demands into production.
- Real Time Production: Conducting the material flow according to needs, minimizing production and stock times along value chains.

Another concept mentioned with Industry 4.0 concept is “Digitalization” or “Digital Transformation.” Digitalization is defined as changing business models using digital technologies and creating opportunities to provide new sources of income or generate value ([Dengiz, 2017](#)).

Even if the skills that employees have are necessary for the particular job, Industry 4.0 requires employees to have ICT skills as well. Due to the Industry 4.0 uses high-tech systems, full compatibility is expected from the employees. Employees must have the ICT skills equal to the basic skills required for the job to minimize the impact of the human factor. In businesses where hybrid works are carried out, it is necessary for the works to proceed without disruption, and employees should be equipped with skills such as cooperation, autonomy and high communication ([Karacay, 2018](#)).

In today’s world, machines are operated only by human commands with under the human control and observation. This is exactly what is desired to be replaced by Industry 4.0. Machines are emerging that can operate independently of humans, with the help of prognostic-monitoring systems.

In the Industry 4.0 concept, a dynamic working environment is created with machines that can think like humans, have the ability to act flexibly and comment on production processes according to the demands of customers. Factories producing with smart machines will be able to respond instantly to the demands of customers and produce smart products with the help of communication provided along the production line. In smart factories where smart products are produced, the communication network needs to be analyzed effectively and the data obtained must be analyzed quickly. Problems in communication and timely analysis of data will affect management and direct production processes.

At the same time, the factory network and the processed data will require high levels of security. Some vital information for the company and important information about products and production planning will not be protected by traditional methods, which will lead to major developments in the field of cybersecurity (Benešová & Tupa, 2017).

Industry 4.0 indicates the transition from solid and central factory control systems to common rational systems, while the importance of the human factor has continued. In addition, the idea of transitioning to an organization where factories and machines will shape their automation is dominant instead of machines that produce today's products that have difficulties in meeting consumer preferences in order to meet consumer needs simultaneously in Industry 4.0 (Alcin, 2016).

What has been done in order to make Industry 4.0 dream come true from past to present, is to catch zero error goal in production processes, to achieve maximum quality in manufactured products, to ensure work safety of employees and to deliver the desired products to the customer (Burrit & Christ, 2016).

Industry 4.0 has been on the agenda of governments and companies for a while and is widely debated. The potential promised by Industry 4.0 brings with it a lot of needs. It is not enough for companies to be technologically sufficient to capture and implement it. Both today's companies' management models, visions and strategies should be able to evolve accordingly to implement Industry 4.0 (Villa, 2018).

Despite all the developments and discussions on it, Industry 4.0 is still at the end of the tunnel. Although the technologies and services it promises are exciting, it is not yet possible to apply them in today's possibilities. Industry 4.0, which is seen as the target to be reached and which is getting closer day by day, determines the character of companies, states and global economy. The impact of Industry 4.0 technologies and preparatory work is actually the main force that shapes the changing world (Retrieved from https://arro.anglia.ac.uk/701766/1/Zucchella_2017.PDF. Accessed on January 5, 2020).

Companies have to undergo major changes in order to have high technologies targeted by Industry 4.0, which is actually Industry 4.0 itself. IT, is undoubtedly one of the main instruments of major industrial transformation. If the structure of companies is not technologically good enough, they will have to make great sacrifices to capture what IT brings. These sacrifices mean huge costs for companies. Moreover, the investments required for the transition to this technology required for Industry 4.0 can be very high. Companies do not want to jeopardize their already fragile economic power to continue their activities unless they clearly see the benefits of these technologies ([Hamada, 2019](#)). Currently, Industry 4.0 is inevitably approaching. All technological developments make this reality even more radical. As in the past, survival in a competitive economy lies in being prepared for what is coming. Therefore, the fact that companies are prepared for Industry 4.0 emerges as a necessity in a logical framework ([Ghobakhloo, 2018](#)).

With Industry 4.0, companies can make customer oriented production instead of mass production with the same cost. Moreover, the product range can be customized according to the demand of the customers. Working environment is more flexible and efficient since production can be made according to demand. Advantages such as conservation and proper use of resources, energy efficiency and rapid production are the developments made possible by Industry 4.0 ([Badem & Kılinc, 2019](#)).

Industry 4.0 envisages the integration of software and information technologies with digital data by designing the collaboration of all units that are directly or indirectly related to the production department in the enterprises ([Sener & Elevli, 2017](#)). This is seen as an important strategy to take part in the competitive environment of the future. Industrial companies are currently researching the Industry 4.0 term to tackle challenges such as increased individualization of products, increasing resource efficiency and shortening time to market penetration ([Yıldız, 2018](#)). In this context, considering the new production system the main factor to adapt is the innovation that emerged with research and development. Research and development activities will be the determinant of the company's competitive behavior. It is thought that the

competitiveness of firms will depend on allocating significant resources to research and development activities ([Bagci, 2018](#)).

Industry 4.0 requires innovation and change in all areas of life. What individuals need to have in order to keep up with these changes can be listed as personal and social competencies. Cognitive abilities and ability to develop value mechanisms are two types of personal competencies that individuals must have in order to keep up with technological advances in Industry 4.0. Today, with increasing digitalization, the way we work is changing. Developments such as the internet of things, smart factories, digital systems and robotic in automation are gradually reducing the role of human beings in business life. As social beings, we tend to establish relationships with other individuals in society and business life. Personal skills shape the individual in the point of establishing relationships with other individuals. In a social environment, communication skills of individuals gain importance and therefore trigger individuals to develop social mechanisms such as teamwork, communication with other individuals and activity within the group. ([Wardini, 2018](#)).

Increasing competition with the globalization brings difficulties and some obligations for companies. The rule of survival in a global economy is to keep up with competition by innovation. At this point, innovation can be seen as an obligation. Companies can continue to stay competitive by globalizing with innovation. Companies should adapt to innovation in their production systems in order to reduce errors in production and costs and respond quickly to the demands of customers. Along with innovation, companies can develop their features such as flexible production, adaptability and agility ([Mohamed, 2018](#)).

Industrial policy decisions that serving the business needs correctly and compatible with research and development activities, should be made for Industry 4.0 applications. It is suggested that the policies established with this target should be within the scope of following articles ([Arslan & Demirkan, 2019](#)):

- standardization and reference architecture;
- managing complex systems;

- a comprehensive broadband infrastructure for the industry;
- safety and security;
- work organization and design;
- education and continuous professional development;
- regulatory framework; and
- resource efficiency.

Business management plays an important role in these decisions.

Real-time evaluations and decision making will become standard thanks to high-tech analysis systems in Industry 4.0. In the light of information coming from three different sources as production line, customer and management, inferences can be made instantly and production can be made accordingly ([Rüßmann et al., 2015](#)).

Expected Effects of Industry 4.0 on the Accounting

With Industry 4.0, independent systems in the industry are planned to act in a holistic way. Integration of systems in this way will ensure vertical integration (R&D, production, data processing, financial affairs, sales and marketing, etc.) and horizontal integration (seller and logistics companies) of non-production departments within the enterprise. Thus, all data created by the system related to production processes will be integrated into planning and control processes thanks to the methods such as cloud computing and data analysis ([Yurekli & Sahiner, 2017](#)). As developments continue with Industry 4.0, the demand for brain power that will be able to understand and use this technology will increase.

Some advantages brought by Industry 4.0 will also bring different problems. The use of robot technology also raised the issue of taxing on these robots. Thanks to the monitoring of the production system, it is easier to diagnose faults, to gain self-awareness of the systems and components, to ensure sustainability thanks to the resource-saving and environmentally friendly system, to provide high efficiency in production, to increase flexibility in production and to ensure a production without stopping and to reduce production costs, advances such as the development of new service and business models are emerging. On the other hand, with the takeover

of the production by robots, developments such as decreased need for human power and decreased number of employees in companies are emerging. With the development of AI in Industry 4.0, it is discussed that robots that can code and design themselves take over production in the future ([Oncul & Ates, 2019](#)).

As a preventive of this disadvantage of robot technology, the concept of robot tax has been proposed by some researchers. It is thought that taxing robots, eliminating the destructive effects of automation, which expands the economic gap between capital owners and workers, and at least can provide tax neutrality among productive inputs. Thus, companies will decide to use robots when they will be more efficient, not to pay less tax and will reduce economic inequality by slowing the growth of unemployment due to the increasing economic advantages of employing workers ([Turan, 2020](#)).

Reporting of financial and non-financial data resulting from the activities of enterprises by recording and analyzing will create an even faster and more complex process in the future. Company managers and financial affairs responsible for this process need to have new equipment to adapt to changing business methods with Industry 4.0. It will be inevitable for the traditional accounting approach to enter a process of change with the reduction of the labour force in accounting transactions with the communication of the objects. The recorded information creates large databases in businesses. Accessing useful meaningful information and providing useful and timely information to information users require expertise. Therefore, there is a need for system designer accounting engineers who can use technology ([Yardımcıoğlu, Karahan, & Yörük, 2019](#)).

Accountants have always been involved in the financial management of companies. Important activities for companies such as data processing and preparation of financial reports are included in the expertise of accountants. Short and long term financial strategies are prepared by accountants. Therefore, all accountants, whether under company management or not, have to expand their areas of expertise and keep up with new developments. Otherwise, with the developing technology, financial reports will be made by

supercomputers and they will dismiss accountants ([Damayanti, 2018](#)).

Thanks to information technologies, accounting and auditing can be done much more comprehensively and quickly. Accounting and auditing, which evolve day by day, will be replaced by computer systems in the future, just as in the professions where robots replace humans. Although this situation still contains many uncertainties, it is inevitable that things will not be as before. On the other hand, Industry 4.0, which will be subject to major changes, will create many new opportunities and only those who are eager to develop themselves will be able to take advantage of these opportunities. In today's reality, information technologies are indispensable for competition and development and are a must for companies to survive ([Kruskopf et al., 2019](#)).

Industry 4.0 is a system in which all management and control systems are transferred to digital. In this period when human intervention will be minimized, the risks will belong to the digital world. The concept of security is vital for these systems. Accountants should have the skills to perform risk analysis to solve security problems. In this respect, accountants will emerge as equipped professionals. They can set long-term goals for companies and advise on technological investments and return on investment by making analyses. They can also assist companies in increasing productivity in production, setting price policies and contracts ([Chenne, 2019](#)).

The answer to the question of what properties accountants should have in the period of Industry 4.0 is revealed by the accountants' duties in factories and the definition of new skills required by these duties. The use of information technologies in the factories of the future forms the basis of these features. In this context, the deficiencies of accountants should be defined correctly, and these deficiencies should be constantly eliminated through training such as accounting education and vocational education in order to train qualified individuals required by the competitive market ([Ohridski & Todorova, 2019](#)).

Accounting practices will change radically with Industry 4.0. Due to advance in technologies which comes with Industry 4.0, big data

that could not be analyzed before will be able to analyze in real time. Performing data analysis in real time will provide advantages such as making faster decisions, preventing loss of time and making the factory much more efficient ([Hart, 2017](#)). Industry 4.0 will also enable much more accurate and reliable analysis. As smart machines can operate autonomously without being attached to humans in smart factories, the role of accountants in everyday operations will decrease and accountants will deal with larger tasks such as resource efficiency, determination of long-term strategies and management of monetary policies. But all this will be in direct proportion to the compliance of accountants with information technologies and their ability to compete ([Ghani & Muhammad, 2019](#)).

The most important and biggest element in the information network of businesses is accounting information. This information is used in companies' planning, financial analysis, audits and reporting used by many other departments. Moreover, thanks to accounting information, companies determine their long and short term strategies and implement their plans under the guidance of this information ([Slyozko & Zahorodnya, 2017](#)).

Industry 4.0 contributes to the development of accounting practices. The fact that information technologies will become more important will increase the importance of accounting practices and companies will develop their accounting systems ([Badem & Kilinc, 2019](#)).

Industry 4.0 brings new professions with new technologies and applications. Accountants must read the future well and develop themselves correctly in order to turn this into an advantage. One of the important technological developments for future accountants is block chain technologies ([Ohridski & Todorova, 2019](#)).

Block chain is an encrypted database that can be changed on documents simultaneously by users connected to the system. Only documents defined as block chain are included in this definition and each block chain is a ledger. So every ledger is not a block chain. In order to make changes on a block chain, all defined users must approve the changes. In the system called Digital Ledger

Technology, this situation is continuously improved ([Antoney & Augusthy, 2019](#)).

In accounting transactions, a block chain is made in triple entries called “triple entry bookkeeping.” In an accounting application of the example, two transactions are made as debt and credit. As a third transaction, the cryptographic signature is entered as the confirmation of these two transactions. The basic logic here is that three users can see these transactions in real time ([Brandon, 2016](#)).

In this respect, a block chain provides big advantages to the companies such as more comprehensive and large data processing, maximizing security, preventing loss of time and low cost. In addition, control of block chain transactions and new professions related to block chain may arise ([Kruskopf et al., 2019](#)).

Although it is thought that there will be no need for accountants with advanced technologies such as AI, Internet of Things (IoT), block chain and cloud computing, these technologies will not completely replace accountants. It is important to note that these technologies will enable larger and more complex data processing. This means that the way accountants work and their roles change. Complex numerical operations and calculations can be made by technological systems offered by Industry 4.0, but interpreting this will remain the job of accountants ([Small, Yasseen, & Salie, 2019](#)). Especially, Industry 4.0 has produced many facilitators in accounting professions, such as support for auditor evaluation procedures, financial statements analyses of accountants and clients, as well as external parties. However, the barriers faced by career associations were competition for qualified workers, and adoption depended on the quality of human resources, changes in macro environment and technology capabilities. Each accountant and auditor should therefore be prepared to keep abreast of technology in order to maintain a competitive climate, and the development of their digital skills was certainly necessary for accounting professional growth ([Khanh, 2018](#)).

The development of the accounting profession has been in line with the development of the sectors in particular. As for Industry 4.0, developing technology and digitalization will broaden the scope of

traditional financial management of accountants and include technological and human resources factors ([Damayanti, 2018](#)).

Since all accounting transactions like accounts receivables and account payables will be instantly controlled by smart systems, the employment of accountants dealing with these works will be compromised. Accountants will have to deal mostly with paperwork. All economic activities can be carried out and followed up by the RPA (Robotic Process Automation) program ([Kaya, Türkyılmaz, & Birol, 2019](#)).

Production and management in smart factories require special software tools to continue without interruption. Industry 4.0 plans the activities related to measurement, analysis and reporting of the information arising from the usage costs especially in integration with the enterprise resource planning software (ERP). ERP systems are software that provides coordination between all units of businesses. In this way, a synchronized planning is made within the enterprise. With ERP, information can be seen in an integrated way by all units at the factory ([Oztemel & Gursev, 2020](#)). These outputs help the management in the decision-making process by establishing operational profitability and performance criteria. The use of AI and the Internet in the accounting information system will remove the input errors originating from the user. ([Rasgen & Gönen, 2019](#)).

Within the scope of inventory activities in the enterprises of the future, many activities such as stock orders and controls, warehouse counts and controls, completeness and accuracy of physical assets, depreciation times and amounts related to physical assets and follow-up of purchases and sales can be carried out without human factors. Payment methods will also change. Since cash concept will disappear as a result of cashless and cashier-free cash registers, audit activities of cash and similar monetary assets will not be required and bank reconciliations will be easily overcome with information transparency and many audit cycles will disappear. Thus, electronic applications will be emphasized to prevent informality. Invoices will automatically fall into the system of the customer and accounting, and coordinated accounting records will be made ([Kablan, 2018](#)).

It suggests that in a scenario such as autonomous ordering by smart machines, automatic delivery of the first substance and material to the factory with the drone or driverless vehicles, realization of the sales in a virtual environment with the customers to try and select the products with augmented reality, realization of the production plans by the cyber physical systems by instant evaluation according to the demand and transportation by drone or driverless vehicles, it will be complicated to distinguish investments on the value chain or indirect expenses in the value chain on the basis of activities, which control, monitor and improve the value chain ([Okangökten, 2018](#)).

Direct Labor Expenses, which is an important part of production costs in traditional cost accounting, is thought to be replaced by indirect expense with the concept of qualified workmanship considering that robots will be replaced by direct workers working in the factory. With Industry 4.0 era, as a result of the replacement of the direct workmanship with robot, the indirect expenses such as energy, repair and maintenance for robots, software experts who can qualify like the foreman who will direct these robots, etc. expenses will need to be evaluated as indirect workmanship ([Arslan & Demirkan, 2019](#)).

In order to adapt the new way of thinking that will prevail with Industry 4.0, enterprises have to develop new strategies including financial accounting. Traditional cost calculations and expenditures will not be sufficient for high technology adaptation and competition. For this reason, companies should allocate large budgets for the development and change of their RPA systems ([Kaya, Türkyılmaz, & Birol, 2019](#)).

Today, the impact of Industry 4.0 on companies can be seen in the production planning and strategies of operating companies. Financial indicators such as profitability of companies and cash flows are accounting data in which this effect is felt more clearly ([Ungerman & Dedková, 2018](#)).

Companies operating with a competitive spirit are naturally open to development and economically strong. Since their management mechanisms are shaped accordingly, they have a high ability to respond to new developments ([Wardini, 2018](#)).

Finally, the more the needs of human increase, the more new enterprises are established day by day to meet these needs. Enterprises' owners desire to continue this process with the minimum waste and cost while producing. It is known that technology and digitalization are continuously utilizing from these developments in the industry. The aim of enterprises is to reach high profit numbers by producing waste-free and low-cost production. In order to achieve that, they must follow the constantly changing conditions and innovations, and also utilize from technological developments in order not to lose their competitiveness. Industry is in continuous development. When these developments are summarized, first of all, the use of steam power in the operation of the machines in the first industrial revolution led to a significant increase in production and a revival of the world economy. In the second industrial revolution, mass production started with the use of electricity in factories. In the third industrial revolution, information technology has emerged and companies that can utilize from this technology correctly and respond quickly to the diversity of human needs have come to the forefront in competition. Finally, enterprises are in the fourth industry which is called as Industry 4.0, a popular topic being studied in literature today.

AI concept has come to the fore with Industry 4.0. Human intelligence is still an important decision-maker as well as labour force when the operating mechanisms of the established enterprises are considered. With the concept of AI, this decision-making mechanism and labour power will be replaced by AI. Digitalization should be used within the enterprise as a whole, if not, it may be interrupted in the system of the production, management, marketing and accounting-finance functions of the enterprises. In particular, the complex financial transactions taking place in enterprises are recorded by the accounting department with the help of information technologies. Along with AI, the accounting department's tasks will change and the data will be processed in real time by intelligent systems. It brings on the agenda to accounting employees evaluating data results that can only be interpreted by human intelligence. Instead of the wasting time due to time consuming transactions, professional accountants will be able to evaluate their

time by analyzing data, thinking on more productive and strategic decisions. It is believed that Industry 4.0 will provide a positive effect on the entire system of the enterprise in such gains that will provide to the accounting field. The study will focus on the innovations to be brought into the field of accounting beside Industry 4.0, and the scope of tasks that are considered to change the professional accountants will be examined.

Conclusion

The Industry 4.0 can be known as a smart factory. According to the recent researches for this system have both advantages and disadvantages on the enterprises. One of the advantages of Industry 4.0 provides faultless of production process and then delivers the products to customer quickly and perfectly. While this provides many different utilities for enterprises where cost of labor, production and time will decrease, on the other hand, it brings various undesired results, for example, unemployment. One of the professions that are thought to be negatively affected is accountants. The role of accountants in everyday operations will decrease, and accountants will deal with larger tasks such as resource efficiency, determination of long-term strategies and management of monetary policies. But all this will be in direct proportion to the compliance of accountants with information technologies and their ability to compete ([Ghani & Muhammad, 2019](#)). Basically, this means the need of educated people will increase. However, this people must obtain multiple abilities. Thanks to smart systems, all financial events will be carried out quickly and that's why accountants don't have to enter financial issues to the systems. It is thought that the need of accounting profession only reviews financial results. For example, it would base if researchers would investigate, when Industry 4.0 systems apply on whole of the enterprises what kind of abilities obtained for professions can be searched. Finally, there is no doubt that this system should proceed step by step and must be established a good base. Employees should be selected and trained in accordance with the characteristics of the system. The company must have a good

financial structure and the whole of the organization must be reorganized.

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Chapter 9

Biomimicry and Agile Leadership in Industry 4.0

Ayşe Meriç Yazıcı

Abstract

Biomimicry is an interdisciplinary approach inspired by the living beings in nature while searching for solutions to solve mankind's problems. This new approach emerging in the late 1990s has been quite innovative while dealing with basic problem solving processes in a business environment. Biomimicry is a creative solution for such processes as design, transformation, organization and sustainability in business enterprises. The objective of this work is to offer model samples that build a bridge between the nature and business organizations. The principles in nature offer many strategies for a sustainable business performance and thus help us maintain optimization and effectiveness in business management through cooperation.

Keywords: Agile Leadership; biomimicry; nature; business organization; sustainable performance; creativity

Introduction

Change is a fact of human lives and happens in all every part of the world where people live. Specifically, the nonstop changes in customers requests and needs cause it inescapable for supervisors and managers of organizations to continually adjust with changes to keep their organizations in a high competitive environment ([Akkaya & Tabak, 2020](#)). Change is an indispensable fact of our lives. Change happens in organizations, in people behaviors and ideas and in every area of the world we live in ([Akkaya, 2019](#)). Mankind has always been in search for solutions either for their needs or to solve their problems. However, as these needs have changed, their search for solutions also has changed. Furthermore, they force human to be open for change and innovation ([Akkaya, 2020](#)). This new type of generating solutions has brought new quests for innovation and utility of different resources. However, it took sometime for us to realize that the answers we are looking for are, indeed, really close and they are in nature. The sense of innovation brought to us by nature has enabled a systematic progress and boosted our intellectual capacity. Today, we are at a certain point in technology where we base our inventions on nature, ranging from space technology to agricultural production. Biological systems are a product of 3.8 billion year old biological process. With its databank of 3.8 billion year-old, it provides us the greatest research development data that have ever been collected.

In modern science, studying biomimetic applications has been transformed into a scientific discipline and thus innovation based on biomimicry has been a subject of systematic studying. The basic methodology of biomimicry is based on a comprehension of the basic principles on biological process or biological adaptation as well as on solving specific technical difficulties while adapting these principles into bio-inspirational product applications ([Luke, 2014](#)).

Biomimicry finds its inspiration in nature ([Benyus, 1997](#); [Passino, 2005](#)). The characteristic features of many living beings reflect themselves on innovative designs. There are several models inspired by the technology in living beings such as increasing the aerodynamic efficiency for Shinkansen 500 Bullet Train by modeling

kingfisher's beak design; Speedo's swimsuit design inspired by shark skin, Velcro technology known for its splendidly durable grip (Dündar, 2019), self-cleansing lotus leaves (Altun, 2011; Özdoğan, Demir, & Seventekin, 2006; Primlani, 2013), adhesives inspired by gecko feet (Autumn, Dittmore, Santos, Spenko, & Cutkosky, 2006; Volstad & Boks, 2012), studying paper making wasps for producing paper out of wood fiber, and unmanned aerial vehicles inspired by landing methods of bumblebees (Holbrook et al., 2010).

Historically speaking, the age of renaissance offer us the first products of designs inspired by nature. Leonardo Da Vinci (1505–1508) was a prominent figure in interpreting himself as a philosopher bridging the nature and man and thus he conducted several experiments and collected all his notes in a notebook. Da Vinci designed a tank mimicking the durability of the turtle shells by observing Roman soldiers using hard turtle shells as a shield to protect themselves from the arrow strikes during a battle. His most well-known designs include the flight machines inspired by bats, flying fish and dragon flies (Genç, 2013).

Biomimicry is a scientific field including all kinds of “man-made materials, tools, mechanism and systems inspired by natural systems” (Altun, 2011). Replacing the questions “how we might come up with better solutions to our problems” with the question of “how we can do better” has been an approach biomimetic thinking has brought to us by modeling nature for scientific and technological problems (Primlani, 2013).

With the advent of technology, the new social world creates a demand for new organizational structures in order to communicate and to produce globally efficient results. How these immense societies will be able to organize themselves is not a question that is solely relatable to human beings. Some social insects such as termites, bees and ants have already been engaged in huge social organizations for 200 million years (Fewell, 2015). These social groups of insects have been following a series of general organizational principles so that they can overcome the problems of organizational life.

As time goes by, observations have been replaced by a rather systematic way of thinking. As times goes by, such observations get

transformed into a more systematical way of thinking. In 1920s, Ludwig Von Bertalanffy explained the relevance between the business management and biology and he put forward a theory called General Systems, which was later developed by others such as Weiner and Boulding ([Kaban, 1994](#)). Although its basic principles are based on biology, physics and engineering, it also had a profound impact on social sciences. General System Theory is an outcome of two paradigms following one another once the metaphysical difference between the part and the whole replaces the system and environment within the concepts of sustainability ([Ertong, 2011](#)).

This particular improvement in biology has led to adapting a contingency approach to the field of management. According to this approach, there is no such thing as the best organizational structure. The best organizational structure is determined by such factors as the contingency and the conditions. It was studied how certain Dynamics as modal technology or a combination of various concepts could influence the organization. During 1950s, such terms of thermodynamics as entropy and negative entropy have gained an importance, as well. The system approach puts forward that the corporations assess their environment against the effects of entropy in order to challenge these effects and thus to survive ([Celep, Özbebek Tunç, & Düren, 2017](#)). This chain of ideas leads to a strong sense of competition in business enterprises in order to secure their future.

In the various problems faced by businesses today, there is no single management model to run the business successfully. Business environments are now in a complex and chaotic system theory. Biomimicry can provide convenience in this context in the solution of these complex adaptive systems. In presenting the principles of nature and their results on businesses in a concise and accessible manner, biomimicry offers us suitable examples for a sustainable business.

Biomimicry

Some of the greatest inventions in design and technology have been inspired by those looking up to the natural world. This source of inspiration is called “biomimicry.” Biomimicry or biomimesis is a study of systems, procedures and elements in nature. Its purpose is to improve the quality of business, social and economic life by providing high efficiency solutions for future use.

The term derives from the Greek word “bios”(life) and “mimesis”(imitation), and it refers to a discipline of design suggesting that a man has to model nature in order to solve the problems he has to face ([Çırpı & Sev, 2015](#); [Minsolmaz Yeler & Yeler, 2017](#); [Volstad & Boks, 2012](#)).

The term “biomimicry” was coined by Otto Schmitt in 1957. It was first used as a general term referring to cybernetics and bionics. Therefore, the term “bionics” was also used for the purpose of referring to the same field as biomimicry. A thorough comprehension of functions and principles in organic structures will eventually lead to productions and designs of materials and commercial interests in such fields as engineering, material science and chemistry ([Zorica Kasandra, Tadic, & Ante Omazic, 2015](#)). However, it was in 1970 when the term “biomimetics” was introduced to scientific literature upon an integration of technology into biology ([Kallioğlu, Karakaya, & Durmuş, 2013](#)).

In order to solve a problem or to meet a specific need, one must study the organisms and ecosystems in nature and thus they must transform their observations into man-made designs. Biomimicry is a combination of biology, nature and architecture into a single composition ([Maglic, 2012](#)). Although it has a wider area of application in architecture and industrial engineering, it might actually include all kinds of nature based improvements that might lead to any product and service.

Biomimicry examples may vary from the engineers getting inspired by a kingfisher’s beak design to boost the aerodynamic efficiency of Japanese Shinkansen 500 Bullet train, and a swimsuit manufacturer company called Speedo mimicking shark skins for the purpose of lowering the frictional force of water on the suit, toward durable and easily attachable Velcro tapes ([Dündar, 2019](#)) and the self-cleansing textile materials inspired by the lotus flower ([Özdoğan,](#)

[Demir, & Seventekin, 2006](#)). Biomimicry is a discipline offering answers to many daily design problems so that sustainable solutions might be discovered by looking at nature. The difficulties in designs might vary from product design, machine engineering to medicine, business and even to architecture. Biomimicry is a heterogenic approach encompassing biologists on the design desks studying the technologies existing in a 3.8 billion database of nature ([Orrú, 2018](#)).

Nowadays, biomimicry also helped shaping design and business management in organizations by combining itself with innovation as it did in many other sectors. By making observations in nature on animals and plants living in nature, various designs ideas are collected.

According to what Volstad and Boks suggest, biomimicry can be used in two different ways. The first model is from biology to design and the second is from design to biology ([Volstad & Boks, 2012](#)). In some cases, the future of an organism discovered by the biologists might trigger an innovation in an industrial and technological design while in others, examining the organisms and the ecosystem in nature and learning from the existing database, there might provide an answer to a design problem one might come across with when developing a product.

The Application and Samples of Biomimicry in Business Enterprises

Innovation is a currently a trending topic in business enterprises. It is indispensable for the business enterprises pursuing sustainable leadership to look for innovation and to establish an innovation-based company culture.

Biomimicry offers a strategic advantage to develop sustainable product design, research and development. Institutions studying nature as a source of innovation has led to drastic changes and transformations for several industries in various ways. Making use of the unique approach offered by biomimicry to design and research will indeed result in profit boosting, risk lowering, cost-effective,

lucrative and sustainable technologies ([The Biomimicry Institute, 2015](#)).

Biomimicry is a comprehensive methodology to innovate a range of industries from agriculture, architecture, manufacturing and transportation to medicine software, material science and robotics. Nowadays, there are a number of examples for natural inspiration in business organizations. The most common organizational biomimicry models have been the cooperation and collaboration between the living organisms.

For instance, fire ants work together to build a raft when they need to cross water. This is a good example in business organizations to prove that team-work might indeed help the members of an organization overcome a challenge ([Mlot, Tovey, & Hu, 2011](#)).

Dragonfly is known to be the only insect to fly in any direction by coordinating all four wings. This symbolizes the importance of integrated influence and ripple effect. It is a term to describe how minor changes might create major events in economics, sociology and psychology. The Dragonfly Effect shows us how synchronized ideas can be used to cause rapid transformations over social media ([Aaker & Smith, 2011](#)).

The paradigm for developing technology creates an assumption on the creation of sustainable competition and optimization in business and thus it leads to more growth-focused approaches. However, this is not how nature works. Instead of maximizing a single compound or feature, nature optimizes the whole. In business, successful social enterprises make use of a shareholder theory in order to optimize the price rate with which every shareholder will be satisfied. This optimized rate can only be possible through multi-functional designs in order to adapt future market dynamics ([Patel & Mehta, 2011](#)).

Biomimicry is a comprehensive methodology for innovation in various industries such as agriculture, architecture, production, transportation, medicine, software, material and robotics. Today, more and more business organizations are being inspired by nature. Cooperation and collaboration in nature are the most common forms of biomimicry examples modeled in business environments.

The Organization Model in Bumblebees

Bumblebees are the smartest living beings in our ecosystem. While fulfilling their tasks, they follow a common intelligence in terms of ecommunication and interaction ([Akay & Karaboga, 2017](#)). Within this perspective, bumblebees are the living beings operating within the scope of efficient work, clear results and skilled participants.

While searching for food, they perform social organization behaviors within the scope of certain algorithms. [Karaboga \(2005\)](#) refers to the process of systematic search for food in the honey making process. During this search for food, bumblebees perform a splendid behavioral pattern while making an analysis of the source for the best food. This behavioral pattern is studied in three different groups. There are three types of bees in bee population. These are workers, scouts and onlookers ([Akdağlı, Biçer, & Ermiş, 2011](#); [Banharnsakun, Achalakul, & Sirinaovakul, 2011](#); [Brajevic & Tuba, 2013](#); [Nayyar, Le, & Nguyen, 2019](#)). Workers are responsible for collecting information on food resources for the bees living in the hive. Assessing this information, the onlooker bees start to use these food resources. If these resources are consumed recklessly, these resources are considered abandoned and the onlooker bees are replaced by the scouts ([Akbari, Hedayatzaheh, Ziarati, & Hassanizadeh, 2012](#); [Nayyar, Le, & Nguyen, 2019](#)). Bumblebees are smart swarms wisely and collectively working while searching for food. They have optimized their abilities to transfer information, to assess their parameter, to keep and share information and thus to make collective decisions ([Bansal, Sharma, & Jadon, 2013](#)).

In their quest for leadership, organizations can simply model the decision-making processes in bumblebees. [List and Vermeule \(2010\)](#) state that the decision-making process in bumblebees has three basic prerequisites such as open-ended meeting agenda items, high quantity and high opportunity cost. In order to reduce the risks, bumblebees make use of distributed decision making. They base their decisions on local clues and information. In doing so, they make sure that their hives are much stronger. Unlike central systems causing problems, they are capable of making quicker decisions in a better quality thanks to their distributed decision-making mechanisms. While making big decisions, they adopt a technique

that is similar to Delphi. The diversity in their genetic structure is also effective while they evaluate their options. The more diverse their DNA becomes, the more sensitive they become when faced with different conditions and the more options they will have in their hives ([O'Malley, 2012](#)).

Amoeba Organization

It is common to think that the activities of those engaged in workplace organization for enterprises occur in a wide organizational framework. In that respect, the design of workplace environment is highly important for the efficiency and effectiveness ([Eryilmaz, 2015](#)). When the literature on biomimicry for business is studied, there are arguments against a single leadership within an organization. Amoeba organization can be suggested as an exemplary model for such a leadership. The concept of amoeba organization was first introduced by Wilbert L. (Bell) Gore. The main features of the organization are consisted of self-organization and centralized hierarchy. The organizational structure of amoeba includes teams of small or organizational units that self-organizing themselves without any hierarchy but with team leaders. The real force behind this theory is that the amoeba metaphor symbolizes a talent to adapt and change. Perhaps, the most efficient sample for the use of amoeba model in business might be its being used as a way of solution to bring 80 local divisions of the US Green Building Council to overcome possible management issues. The amoeba model was designed for the council as an inspiration for the symbiotic relationship between mycorrhiza fungi and such environment resources as sun, food and water ([Schatten & Žugaj, 2011](#)). This model has been being applied in several successful companies, including Kyocera. The company founded by Kazuo Inamori in 1959 owes a great deal of its success to the amoeba management system. Inamori made use of this management system in order to make sure that the company employees who have no background in business and financing contribute to the company growth. With this model, Kyocera managed to obtain a successful business concept “at its smallest and most elemental level.” Studying the success story

of Kyocera, [Adler and Hiromoto \(2012\)](#) define the amoeba organization as such: “In other words, amoeba management is intended to offer a spontaneous, homeostatic response to a business world characterized by rapid, dynamic change”. Small units of organization concentrate on a common objective for the company but follow an independent operation.

The common objective might be corporate values or a profitable prospect. Thanks to this decentralized management model, the company gains market agility, dynamism as well as the commitment of its employees ([Adler & Hiromoto, 2012](#)).

Artificial Neural Network Organization

Artificial neural networks are the new generation computer software designed to operate in a way similar to human brain. This software is made up of operational units called neurons. Each operational unit can send and receive signals to the other units. Some scientists think that these units are highly similar to what we call “communication.”

This theoretical idea of organization dates back to 1945 when Warren McCulloch conducted neuropsychological studies on human brain. According to the scientists, the human brain has not a hierarchy but a heterarchy model. They model this organization as a neural network for parallel information technology ([Reihlen & Rohde, 2002](#)). The notion of heterarchy organization depends on an organization made up of organizational units mutually connecting each other. It is organized mutually and independently. Most of the time, it operates internally and externally with a common objective. According to the fractal organizational principle, there might be organizational units, teams, individuals, divisions, departments and even for all organizations and organization groups ([Žugaj, Šehanović, & Cingula, 2004](#)). If this concept is applied to an organization, the relationship cannot be defined; on the contrary; it leads to a self-organizing active structure depending on a certain situation.

Artificial neural network organizations are also used in business as AI-based knowledge acquisition tools. They are used to allow

easier decision making for the management. In doing so, they enable better solutions in information systems requirements analysis ([Walczak, 2016](#)).

Swarm Intelligence Organization

For years, such tiny creatures as ants, bees, termites and wasps have been an inspiration to poets as well as natural scientists. What astonishes them is the secret behind certain mechanisms in these tiny organizations. It is quite fascinating to study how these organizations are managed, how smoothly the chain of command works, who or what makes the plans-or, in other words, what keeps everything in balance. In a social insect colony, a worker does not necessarily perform all the tasks it is in charge of. Instead, it works hard to acquire expertise depending on its age and morphology. This cooperation between the members of the colony took place in harmony with the others. Modeling the self-organizational structures in the insect world might actually help designing self-organizing artificially problem solving tools and swarm-smart systems. The purpose of the swarm intelligence is to create multi-agent smart systems, inspired by social insects, school of fish and flock of birds and others ([Deepa & Senthilkumar, 2016](#)).

There are many flocks, herds and swarms in our ecosystem. Although they might not be all called smart, their intelligence varies from species to species. A smart swarm must have certain principles such as basic space and time calculations, responding to the surrounding quality factors, adapting its behavior in every movement and even perhaps changing its mode of behavior ([Karaboga, Gökemli, Öztürk, & Karaboğa, 2014](#)).

An insect colony is a decentralized problem solving system consisted of many members with a basic interaction. Among the routine problems solved by the members of a self-organizing colony are construction works, search for food, sharing the labour with others and responding to the external challenges. The most important feature of these insects can be their ability to function and to keep the colony strong by adapting constantly and finding flexible solutions despite some members failing to fulfil their duties.

Swarm intelligence is typically consisted of simple swarm population that interacting and communicating with another and their surroundings. Swarms follow basic rules and they do not possess a central structure of control. Even so, interaction between the members of the swarm eventually leads to smart global behavior ([Zhang, Agarwal, Bhatnagar, Balochian, & Yan, 2013](#)). Moreover, the efficiency of logistics might be enhanced by mimicking a task assignment model observed in these social insects. The term “bucket brigades” has been used to describe this assignment model, and it was first coined by John Bartoldi and Donald Eisenstein. In this operation model, each worker is in charge of a particular task and the worker can only start working once the previous task is completed by another co-worker. Some swarms are also adopt unique operational models when clustering the dead colony members and classifying the larvae. This method of classification and clustering is being adopted in banks, when the data for new partnerships between their clients are being analyzed. In general, the advantages of the swarm intelligence can be summarized in three categorized:

1. Flexibility: The group is flexible enough to adapt itself to the rapidly changing conditions in its environment.
2. Durability: The group is capable of sustaining its performance and tasks despite the failure of one or more of its members.
3. Self-organizing: The group is capable of managing its organization on its own.

Nevertheless, the swarm theory and self-organization might be challenging to apply in newly emerging companies, especially in the swarm intelligence concept ([Bonabeau & Meyer, 2001](#)).

Fish Swarm Organization

Artificial fish swarm algorithm suggested by Li Xiao Lei in 2002 is based on a motivated stochastic population seen in the intelligent collective behavioral patterns of fish swarms such as hunting, transporting and following ([Azizi, 2014](#)). Unlike lions and apes, fish

represents a species, in which there is no specific leader and every member has its own self-organizing behavior. They have no information about their group surroundings, and they act randomly depending on the information they obtain from the neighbouring member. This interaction between the members makes the fish swarm even more complicated. They usually live in the areas where they find plenty of food, and they move onto a richer area either by following the other members or searching food on their own. The more crowded the fish population in an area is, the more nutritious it is. The next behavior of every fish depends on the local environment and the existing conditions. Fish swarm is under the influence of the environment, the individual behavior of its members and the interaction between them ([Nayyar, Le, & Nguyen, 2019](#)).

Fish swarm algorithms contribute to a better business management by combining individual efforts of the members so that it can create a greater value for the company.

Super Organic Organization Models Inspired by Termites

Termites have been living in our symbiotic planet for more than 300 million years, and they have a reputation as “super organisms.” The term superorganism was coined by Herbert Spencer in the nineteenth century. Super organisms account for a compound organism, each of which is consisted of other individual organisms. The members of super organisms are known to have examples of super organisms as ant colonies, hornet hives, coral reefs, bacteria and fungi colonies; termite mounds are one of the greatest examples for symbiotic super organisms in which no partner can survive without the other ([French & Ahmed, 2011](#)). These tiny living beings have become the source of inspiration to the design of two buildings: Eastgate Center Building in Zimbabwe and Council House 2 in Melbourne, Australia. Termites are complicated creatures who are capable of constructing mounds of a few meters high, and they adopt a sophisticated architecture model while designing their buildings ([Turner & Soar, 2008](#)). The termite mounds have been mimicked to optimize different structural features ([Andara, Griffon, & Jaffe, 2015](#)). Mimicking the architectural models adopted by the

termites are mimicked to boost the energy efficiency in buildings ([Tonn, 2015](#)). As seen in Council House 2 and Eastgate Center, the ventilation systems inspired by the termites “allow buildings breathe freely” and in comparison to traditionally ventilated buildings, the buildings with the termite ventilation systems consume 10% less energy ([French & Ahmed, 2011](#)).

Termites are also modeled in business organizations. Their communication methods are especially analyzed. They have simple rules of communication. These simple rules are of particular importance for the sustainability of the organization as any mistake will be obvious during direct management. Termite mounds are well-organized. The leaders decide what to do as well as how to do it, and this is how modern business organization works. Therefore, leaders work hard to create a termite-like organization, in which people can work together and create a superorganism out of individual ones ([Tofield, 2002](#)). Jim Donehey, the CIO of Capital One, tried to adapt this termite-like organization structure and developed simple rules to replace the existing chain of commands and thus to shape the organization’s behavior. He prepared four basic guidelines to ensure that everyone in the organization works for the same goals as such:

1. IT activities should be planned in line with the general objective of the company.
2. A good financial decision-making is essential.
3. Flexible, out of the box thinking should be developed.
4. Altruism at work place should be encouraged.

Donehey established these rules to encourage the employees to make their own decisions ([Bonabeau & Meyer, 2001](#)). These company cultures, in which the individuals are responsible, are especially suitable for extraordinary team work, quality and customer services. Besides, it is also important that business enterprises are synergic and holistic ([Tofield, 2002](#)). In other words, termites represent a source that we can use to optimize our results ([Srinivasan, 2018](#)).

Termites, one of the best architectures and workers in nature, provide a great business model in many ways as an outcome of more than 300 million years of experience. The organizational models they follow are well-organized enough to be adapted by the twenty-first century business organization so that efficiency can be optimized and business performance can be sustainable.

Conclusion

Ever since the rise of mankind, nature has always been a subject of inspiration. This article offers a brief summary on certain concepts inspired by nature and bringing about significant improvements in the theory and practice of modern organization. Another objective of this work is to provide an analysis on relation between biomimicry and organization management. Biomimicry is the art of mimicking nature and an important scientific field of application in social sciences. A great deal of biomimetic applications in modern organizations is about metaphors. The use of metaphors can be a tangible biomimetic application in the organization theory. The development of these applications might provide a better analysis and improved optimization in modern organizations.

An enterprise formed by nature encourages a reliable cooperation between the members of the business ecosystem. Optimized teams quickly get together to focus on certain targets such as work environment, department, organization or expertise thanks to a strong sense of collaboration. Creating devotion and cooperation will eventually make a difference for a sustainable business environment. This difference will eventually provide viability for the business organization.

The findings obtained from amoeba organization, artificial neural networks, swarm intelligence, bumblebee algorithm and fish swarm are applicable for the organizations, which can provide an optimization prediction model and offer different methods.

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Key Terms

Biomimetics: It is an imitation of model, system and nature elements to solve the complex problems of people.

Homeostasis: It is the ability to maintain a relatively stable internal situation that continues despite changes in the outside world.

Stochastic: Adjective, which means variable, random.

Superorganism: It is a group of synergistically interacting organisms of the same species.

Holism: It is the philosophical view that the whole is more than the sum of its parts.

Chapter 10

The Role of Agile Leadership in Organisational Agility

Melis Attar and Aleem Abdul-Kareem

Abstract

The present business environment of accelerated complexity, ambiguity, uncertainty and volatility seems not to be settling down soon. Today's business world is consistently bombarded with turmoil and unprecedented change. This makes it very difficult for organisations to accurately predict possible future opportunities and threats. To overcome this scenario, organisations need to fully embrace and implement agility in their operations. However, for organisations to develop a considerable level of agility that corresponds to the current hypercompetitive and volatile environment, premium needs to be placed on agile leadership. Agile leaders are capable of setting the guiding principles, develop strategies and build mechanisms that will lead to smooth transition to organisational agility (OA). These leaders give directions on how to make organisations agile and ensure that there is a leadership culture that models and promotes a holistic agility in the organisation. The primary purpose of this chapter is to establish the position of agile leadership in determining OA. The study departs from previous studies by using empirical reviews to depict the

significance of agile leadership and its key roles on OA. Specifically, the concept, evolution and characteristics of OA are explored, as well as its benefits and multidimensional nature are discussed. Furthermore, the chapter sheds light on agile leadership, levels of leadership agility and competencies of successful agile leaders. The chapter concludes that the leadership style suitable for OA is agile leadership. This calls for business investments in agile capacity building and development of frameworks suitable for agile leadership.

Keywords: Agility; organisational agility; agile leadership; volatility; uncertainty; complexity

Introduction

The incessant need for organisations to swiftly respond to uncertainties, radical changes, complexities in both internal and external environments in this dynamic and knowledge-based business world requires business entities to embrace agility and consistently build their agile capability ([Joiner, 2009b](#)). Today, due to the turbulent, complex and dynamic nature of business environment, companies operating in emerging markets face certain problems ([Akkaya & Tabak, 2017](#)). The advancements in technology, volatility of the business setting, tumultuous socio-political and economic environments are among the factors prompting and sparking the recent discourse and attention on agility ([Chatwani, 2019](#)). There are unexpected changes and threats in the current order of business world; for organisations to cope, survive and take advantage of these drastic changes, emphasis should be placed on flexibility and adaptability to the new business environment ([Sharifi & Zhang, 2001](#)). Thus, organisations that fully embrace the idea of creating and developing agile capacity are in a better position to successfully traverse in this turbulent business environment.

Organisational agility (OA) is basically an organisation-wide capability to proactively and relentlessly anticipate, respond, react

and capture unique market opportunities in the quest to thrive in this current unpredictable, volatile and global competitive business environment. Organisations need to effectively use dynamic and flexible agile abilities. It has become inevitable for enterprises to change the organisational structure and become more flexible. Agile organisations have to keep up with technological innovations so as to overcome competition (Koçyiğit & Akkaya, 2020). The relevance of agility to contemporary organisations in terms of its contribution to business performance, competitive advantage, effectiveness, customer satisfaction, among others cannot be overemphasised, and there are appreciable amount of global surveys alluding to this assertion. A survey conducted by McKinsey revealed that nine executives out of ten ranked OA to be increasing in significance with passage of time and critical to organisations (Sull, 2010). [Allied Consultants Europe \(2010\)](#) posits that most businesses understand that agility is needed for successful running of their operations in the future. In 2009, the Economist Intelligence Unit carried out a study on 249 executives globally and found that 88% of them indicated that agility is either somewhat important or extremely important to business. Similarly, an agile organisation survey conducted on 12 countries in three continents by Capgemini Consulting in 2017 on 119 respondents from different industrial sectors showed that participants have the notion that broadening their implementation scope of agility to day-to-day business activities could result in significant benefits to their organisations ([Kroll et al., 2017](#)). However, despite the knowledge and understanding of the need for agility in an organisation, a large proportion of European companies surveyed expressed dissatisfaction to the present level of their business agility and intend to implement measures to improve it ([Allied Consultants Europe, 2010](#); [Brandt, Kjellström, Andersson, & Hallencreutz, 2015](#)).

The high level of volatility and complexity of the current global business environment places high demands on organisations to institute agile leadership approach in order to stay ahead of competition and effectively respond to rapid environmental changes. The type of leadership required by business executives to confront the ambiguous nature of today's business environment is agile

leadership. Agile leadership relates to the ability to adopt an adaptively flexible strategy based on external and internal dynamics and employs broader perspectives to recognise and examine different circumstances and swiftly respond to them. Agile leaders possess the skills of adaptability and versatility to lead in a variety of situations that are challenging and unpredictable. In the view of [Joiner \(2009b\)](#), human resource managers need a clear understanding of agile leadership in order to ensure that there is a leadership culture that promotes an all-inclusive agility in the organisation. Agile leaders are visionary, develop strategies and guide to make the whole organisation agile.

The primary objective of this chapter is to establish the position of agile leadership in OA. Thus, to determine the critical roles leadership agility plays in agile organisations. The study has been necessitated due to the gap identified in the surveyed literature, where there is a handful of research work focusing on establishing how agile leadership interplay with OA. Even though, these two concepts (OA and agile leadership) arguably are new to the literature, most of the existing studies employed a univariate approach to examine each of them separately. The current study is therefore relevant to the extent that it adopted a bivariate approach to present detailed empirical reviews on these two concepts and demonstrated how agile leadership is crucial for OA. The next immediate section discusses agility and OA in broader perspective. Subsequent sections present detailed reviews on agile leadership and its place in OA. In the final section, conclusion of the study is presented.

The Concept of Agility

Agility as a concept is not unique or limited to the field of information technology (IT), engineering but spreads its tentacles to the field of management and business. Agile alliance officially introduced agility into the field of IT in its manifesto on agile software development ([Beck et al., 2001](#); [Denning, 2018](#)). Initially, in the 1990s, agility has been connected to software development, methods of process improvement, just-in-time supply chain and lean production

([Holbeche, 2015](#)). For over 20 years, agility has been applied in IT field, mainly in the project world ([Kroll et al., 2017](#)). In management literature, agility has become a buzzword and appears to be highly glorified by organisations in recent years. Synonymously, agility denotes an organisation that is lean, innovative, customer oriented, flexible, adaptable and quick to respond to changes. Despite the high interest shown on agility in recent times, the concept still lacks a generally accepted definition ([Chatwani, 2019](#)). Agility is a complex concept that has the tendency of taking several forms ([Holbeche, 2015](#)). Nevertheless, herein some scholars' views on agility will be highlighted.

Agility describes exploitation of productive and lucrative opportunities in volatile market environment using virtual corporation and market knowledge ([Mason-Jones, Naylor, & Towill, 2000](#)). Agility focuses on swift responsiveness to customer demands, in-depth understanding and mastering market turmoil as well as possessing particular capabilities such as 'lean thinking' ([Hoek, Harrison, & Christopher, 2001](#)). In this thinking, the aim is to provide more benefits to various business stakeholders by removing unnecessary waste and cutting down costs. Agility encompasses the capability to effectively react speedily to an environment that has the propensity to vary radically. It refers to the ability to respond quickly to unpredictability and profitability ([Erande & Verma, 2008](#)). The concept emphasises an organisation wide capability to adopt information systems, organisational structures and right mindsets in particular ([Dahmardeh & Banihashemi, 2010](#)). After critical evaluation of various views expressed on agility, [Yusuf et al. \(2014\)](#) infer that in manufacturing context, agility describes a successful adoption of competitive forces to deliver a customer-driven product and services in an unpredictable business environment. These competitive bases comprise of flexibility, quality, innovation proactivity, profitability and speed. These are integrated with best facilities and practices in a highly knowledge-based environment to provide best value to an uncertain market condition. Agility is the way to success in market conditions, where change and uncertainty are effective. Agility is a method used to provide flexible, inexpensive, precise, fast and effective responses to uncertainty

(Çamoğlu, Akbakır, Yücalar, & Bayraklı, 2010). It appears as though the descriptions of agility given by these authors strike on distinct aspects of agility, but also there are certain commonalities. Embedded themes in most of the definitions are quick, fast and speedy response to changes, adaptability, synchronisation of operations, flexibility and rapid reaction to uncertain market setting.

The kind of business environment present today is not absolved from volatility, uncertainty, complexity and ambiguity. This can be shortened using the first letters of these words into an acronym known as VUCA. This originated from US military in early 1990s to signify the new security environment due to the end of the cold war and the collapse of the Soviet Union and has since been a trendy managerial term in business literature as well. VUCA describes a business environment characterised with four different kinds of challenges that require new orientations, distinct responses and approaches to management (Horney, Pasmore, & O'Shea, 2010). Volatility refers to the nature, speed and degree of changes in an environment. It illustrates a business situation that is persistently unstable day-by-day, dynamic in nature, events occurring in an entirely unexpected ways and their durations may not be known due to fluctuations in extreme directions. Leaders have knowledge of volatility and not necessarily difficult for them to understand it. Uncertainty relates to the inability of a business to anticipate major future changes that may occur due to inadequate availability of information. Consequently, uncertainty blurs business decision-making processes, causes delays in projects execution and affects strategic planning abilities. Although cause and effect relationship is known in uncertain situations, the realisation of a change is not guaranteed. Complexity denotes a situation where the business world is inundated with problems that are multifaceted and their ramifications are tougher to comprehend. The different parts of a problem are intertwined with both direct and indirect relationships making it practically impossible to choose a 'one-size-fits-all' solution. Uncertainty creates a condition, where cause and effect relationships of events are totally unclear to business leaders and existing data are difficult to interpret. In this circumstance, leaders move from an unknown situation to known, deal with events that

have no precedence and shrouded with too much risks ([Bennett & Lemoine, 2014](#); [Horney et al., 2010](#)). In a VUCA world, the business atmosphere is dynamic in nature, and it is difficult to predict the scope and speed of change. The change is so rapid that it makes it challenging to understand your direction as an organisation. This new business order is as a result of factors such as globalisation, political and economic instabilities, digitilisation, increasing in customers' demands and other environmental factors. In VUCA situations, leaders are handicapped in determining the appropriate strategies and how to successfully implement strategies in order to attain competitive edge and ensure long-term business success. Currently, it is this VUCA environment that is necessitating for organisations to embrace agility in their operations and rethink of the kind of leadership styles/theories appropriate for such circumstances. Thus, building agile capabilities and instituting frameworks that accommodate agile leaders can be helpful to organisations while overcoming VUCA challenges.

The Emergence of OA

The history of OA can be traced back to counter-measures taken by US Congress in the early 1990s to combat industry downturn and to revive American manufacturing industries competitiveness ([Yeganegi & Azar, 2012](#)). In 1991, the US Congress upon reflecting on the deleterious consequences the 1980s economic downturn had on the economy and realising that the prevailing mass production manufacturing era is gradually coming to an end and would be dominated by agile manufacturing approaches, instructed Department of Defence to set up a body to investigate and analyse the country's manufacturing sector with an objective of making it regain its global marketplace leadership in manufacturing. The US Department of Defence then charged Iacocca Institute of Lehigh University the responsibility of producing a succinct vision that would encapsulate long-term manufacturing objectives ([Nagel, 1992](#)). The Institute served as a facilitator to formulate strategies leading to a successful agile manufacturing system. Iacocca Institute set the foundation for an implementation plan developed by a core group of

industry leaders. This group consists of 15 industry-executives from 13 companies. The group was supported and counselled by more than 150 members from 11 professional bodies and 77 business entities ([Prakash & Attri, 2017](#)). The vigorous efforts of the group culminated into a two-volume report captioned '21st Century Manufacturing Enterprise Strategy'. The well-detailed report stipulates the fierce global competition US manufacturing firms will encounter as it transits in the next 15 years through to the twenty-first century and the required resources needed to compete and sail through this global competitive environment. For the purpose of meeting the demands created by the new production era and highly competitive business environment, the authors of the report strongly recommended an invention of a new system of production anchored on OA ([Charbonnier-Voirin, 2011](#)). Besides, this Enterprise Strategy Report resulted in the creation of Agile Manufacturing Enterprise Forum in December, 1991 and officially introduced agility into research literature ([Dahmardeh & Banihashemi, 2010](#); [Yeganegi & Azar, 2012](#)). Among other functions, the Forum is established to work with industries, academics and government, in order to sensitise people on the word (agility) and to deepen their understanding on agile competition. It fundamentally seeks to build mechanisms that would lead to smooth transition to agile enterprise and to restore the global leading position of American manufacturing sector through the implementation of agile manufacturing strategies ([Nagel, 1992](#)). The concerted efforts of Lehigh University's Iacocca Institute and the work of the Enterprise Forum led to the popularisation of agility and agile organisations ([McCurry & McIvor, 2002](#)). Hereby, agile mindset and practices are progressively going into the management mainstream.

As [Lee and Xia \(2010\)](#) postulate that a lot of organisations embrace agile development processes but fall short of properly understanding factors they can regulate to affect it and how the concept is defined and measured. Hence, organisations need to clearly define reasons for becoming agile. Organisations are learning how to fully exploit the talents of their workforce, engage consumers at each stage of product development and create consumers-tailored innovations ([Denning, 2018](#)). OA is a highly touted attribute and

normally considered worthy of implementation ([Teece, Peteraf, & Leih, 2016](#)). Agility in an organisation could be attained by adopting the right practices and processes to react appropriately to business environment changes ([Sharifi & Zhang, 2001](#)). Agile organisations have the ability of detecting both external and internal environmental dynamics by searching for important cues and deriving sense out of them. This sensing ability continually engenders proper knowledge flows in the organisation that leads to timely responsive actions ([Chatwani, 2019](#)). OA is defined as an invaluable capability of a firm in a turbulent market condition to identify and capture opportunities more swiftly than competitors do ([Sull, 2010](#)). OA is a proactive management approach aiming at eliminating wastage, constant improvement, judicious utilisation of resources, flat organisational structure and effective supply chain management. It is the capability of an organisation to accomplish its goals by updating the knowledge of its workforce, developing demand-driven products and lightening its movement in a speedily varying environment ([Nafei, 2016](#)). OA is expressed as the response given and developed by the organisation to be efficient in an environment characterised by turbulence, uncertainty and complexity ([Charbonnier-Voirin, 2011](#)). Organisational agility is the ability of a company to quickly and effectively respond to unanticipated changes in both internal and external environments and to exploit opportunities available in the market by meeting customers' needs ([Akkaya & Tabak, 2018](#)). It requires not just efficient knowledge management and learning ability to be an agile organisation but also effective decision-making process and timely implementation of solutions in attempt to react to changes ([Trinh, Molla, & Peszynski, 2012](#)). [Chung, Lee, and Kim \(2014\)](#) argue that an agile organisation has the ability to respond to changes in the external environment and to benefit from these opportunities in an effective and efficient manner by recognising market opportunities. This type of an organisation meets customer demands, provides novel products and goes into strategic alliances. On one side of the coin, the basic reason necessitating OA is the search for core capabilities, the other side of it focuses on identifying and seizing opportunities in the business environment ([McCarthy, Lawrence, Wixted, & Gordon, 2010](#)). The ability of an organisation to

respond to and adapt to imminent threats or opportunities determines the agility of the organisation in question. Work is carried out successfully and simultaneously in organisations with high level of agility (Sağır & Gönülölmez, 2019). According to Chakravarty, Grewal, and Sambamurthy (2013) OA is achieved when the organisation's function and where everyone fits in the hierarchy of the organisation are known, as well as understanding and delivering the outcomes customers need. To achieve OA, companies should put in place a number of internal and external linkages such as integrated teams, flexible supply chain, virtual working mode, strategic alliances with key stakeholders including competitors, among others. These linkages enable companies to rightly detect changes in the business environment and rapidly react to them. The kind of organisational structure used in an agile company is characterised with more flexibility, decentralisation and less hierarchical in shape that permits delegation of decision-making power, cross-departmental communication, functions, collaboration and cooperation to occur. This organisational structure as opposed to the conventional static type provides a fertile ground for successful execution of agility in an organisation. In this kind of structure, good interpersonal skills, effective communication, flexible leadership styles, self-learning and development are required (Holbeche, 2015).

When the studies on OA are examined in the literature, it is seen that the concept is viewed and discussed by scholars from different perspectives. This review buttresses the deduction made by some researchers on OA having no generally accepted and precise definition as it stands now (Almahamid, Awwad, & McAdams, 2010; Anca-loana, 2019; Ebrahimpour, Salarifar, & Asiaei, 2012; Su, 2011). However, despite the lack of consensus on a common meaning of OA, it appears from the review of existing literature that most authors describe the concept as an organisation-wide capability to proactively and relentlessly anticipate, respond, react and capture unique market opportunities in the quest to thrive in this current unpredictable, volatile and global competitive business environment. Also, for an organisation to be fully agile, every single individual and each and every department in addition to the

executives/management need to understand OA and contribute towards meeting dynamic expectations of the VUCA world.

Characteristics and Multidimensional Nature of OA

For an organisation to be regarded as an agile enterprise, there are a number of distinctive features it requires in order to detect and deal quickly with changes, uncertainties and unpredictability in a volatile business environment.

In a conceptual model offered by [Zhang and Sharifi \(2000\)](#), speed, flexibility, competency and responsiveness are identified as the requisite capabilities of agile organisations. In the same vein, a four-component framework presented by [Lin, Chiu, and Tseng \(2006\)](#) captures flexibility, responsiveness, competency and quickness as the necessary elements of agile enterprises. Responsiveness describes the capability of an organisation to anticipate, determine and respond quickly to changes as well converting them to profitable ventures. There is the need for organisations to rightly and timely respond to changes in consumers' needs, tastes and preferences resulting from changes in environmental forces and technologies. Organisations will derive competitive advantage if they suitably adapt to these dynamics ([Akkaya & Tabak, 2020](#)). Competency indicates the organisation's ability to effectively and efficiently accomplish its goals. Competency is a comprehensive list of abilities that ensure productive use of resources towards successful realisation of business objectives. It encompasses internal and external integration, cooperation, competent workforce, right technology, quality products, among others ([Sharifi & Zhang, 1999](#)). Flexibility focuses on the usage of analogous facilities to ensure smooth flow of different processes and attain diverse objectives ([Lin et al., 2006](#)). With flexible manpower, organisational structure, product model and other alternative resources, businesses will be able to expand their market shares, meet consumers' needs and increase profitability ([Akkaya & Tabak, 2020](#)). Lastly, speed/quickness measures the ability of a firm to perform its activities within the shortest duration ([Jain, Benyoucef, & Deshmukh, 2008](#); [Zhang & Sharifi, 2000](#)). It encapsulates the ability

of an organisation to make and implement decisions to swiftly respond to environmental changes and promptly deliver products and services to markets. The issue of speed and flexibility is also espoused in the work of [Prater, Biehl, and Smith \(2001\)](#) in relation to agility. They defer slightly in their opinion on these agility characteristics. Their point of departure is on linking speed to time spends on shipping or receiving a product, while flexibility is connected to the extent to which an organisation can adjust the time it takes to ship or receive products.

Having taken cognisance of the works of previous researchers on agility, [Charbonnier-Voirin \(2011\)](#) proposed three main characteristics of agile organisations and labelled them as agile capabilities. The firm's ability to marshal swift response to a change that is grounded on judicious utilisation of current resources and reactive flexibility is regarded as the first agile capability of an organisation. The aptitude to read and understand the business market is considered the second agile capability. This capability helps the organisation to forecast, detect and discover potential and up-and-coming events in the market by using its monitoring capabilities ([Sanchez & Nagi, 2010](#)). The third agile capability emphasises on efficient integration and management of organisational learning/knowledge. Thus, it boils down to the capability of an organisation to properly manage its human capital and relate it to the strategic perspective of the firm.

[Worley and Lawler \(2010, p. 194\)](#) in their agility framework postulate that 'a robust strategy, an adaptable organisation design, shared leadership and identity, and value-certain capabilities', characterise an agile organisation. The capability of an organisation to produce outcomes in different environmental situations determines the robustness level of its strategy. The second characteristic (adaptable design) is somewhat similar to what the above researchers refer to as flexibility. This feature of an agile organisation describes its ability to possess flexible structures and processes to quickly respond to both internal and external forces. The third characteristic, which is, shared leadership and identity focuses on decentralisation of decision-making power, activities and information throughout the organisation in order to develop the

managerial or leadership skills of several workers. The last characteristic: value-certain capabilities defines the requirements to meet if an organisation wants to persistently maintain its agility.

The content of OA is worth more in-depth studies in order to propose its dimensions. The latent and multidimensional nature of OA coupled with lack of precision concerning its characteristics have constrained suggestions on its dimensions and measure ([Charbonnier-Voirin, 2011](#)). Emphatic efforts are made to comb the existing literature on dimensions of OA and varied views of prior researchers are presented below.

In the view of [Yusuf et al. \(2014\)](#), OA can be summarised into four main dimensions: customer sensitivity, process integration, network integration and leveraging the effect of information and people. The first dimension emphasises on ensuring high customer satisfaction by providing quality goods, superior value, fast and timely delivery of products to them. Process integration focuses on risk taking attitude of the enterprise, innovation, proactivity and fast decision-making capability. The third dimension accentuates the firm's supply chain management processes, competitiveness and creating of alliances to respond to rapid changes. The authors' last dimension describes the accessibility of information, team spirit, support and fair reward system. [Yusuf et al. \(2014\)](#) propositions on OA dimensions can be juxtaposed with that of [Zain, Rose, Abdullah, and Masrom \(2005\)](#) and inference on the two being almost similar could also be made. [Zain et al. \(2005\)](#) after critically examining the overall list of 87 questions [Goldman, Nagel, and Preiss \(1995\)](#) suggested could influence agility of an organisation, developed four major dimensions of OA. They labelled these dimensions as enriching customers, mastering change and uncertainty, cooperating to enhance competitiveness and lastly, leveraging people and information. [Zain et al. \(2005\)](#) after considering these four principal dimensions proposed 78 questions to measure OA in Malaysian context.

With the aim of developing a valid measurement scale for OA, [Charbonnier-Voirin \(2011\)](#) carried out extensive qualitative and quantitative research and came up with a logical set of eleven dimensions for OA. These are proactivity, reactivity, communicating the strategic vision, recognising and assessing performance,

development of skills and sharing of knowledge, creativity and continuous improvement, delegating responsibilities, knowledge of customers, internal cooperation, external cooperation and finally, foreseeing change relating to customers. Charbonnier-Voirin did not hesitate to point out her uncertainty on whether the scale developed really captured all the facets representing OA and the need for this scale to be practically tested on samples from other domains. [Hamad and Yozgat \(2017\)](#) having critically evaluated these eleven dimensions condensed them into four dimensions before adopting into their study. The naming of their dimensions seems different but basically touch on elements of those eleven dimensions. The first dimension of [Hamad and Yozgat \(2017\)](#) focuses on practices the organisation uses on environmental scanning, flexible structure, research and development, among others in the pursuit of developing capabilities to deal with change and uncertainty. They called this dimension 'Practices Directed towards Mastering Change'. Their second dimension labelled as 'Practices Valuing Human Resources' hammers on instituting programs to train, empower and develop the workforce, participative decision making, practices regarding knowledge sharing, improving creativity of workers and providing the necessary resources to them. The emphasis of the third dimension, which they called 'Cooperative Practices', centres on devising strategies to establish alliances or partnerships with other parties in the supply chain including suppliers and competitors. Largely, this dimension focuses on fostering both internal and external cooperation to gain synergy. Their fourth and last dimension is about practices the organisations adopt to provide high and better value to customers, meeting customers' expectations and addressing their problems through innovative products and solutions. They referred to this dimension as 'Practices of Value Creation for Customers' ([Hamad & Yozgat, 2017](#), p. 409).

In an OA survey conducted in 2010 by Allied Consultants European (ACE) on 533 top European executives in both private and public sectors, six dimensions are identified as key enablers for OA. These are leadership and management, innovation, strategy, culture, structure and finally learning and change ([Allied Consultants Europe, 2010](#)). ACE acknowledged that OA is a broad subject and for the

purpose of its survey, the focus is on internal human aspects of OA. The dimensions that are highly ranked as more important by the participants are 'Leadership and Management' and 'Strategy', while, 'Innovation' is the least evaluated dimension. This six-dimensional approach of measuring OA is also affirmed and captured in a research carried out by [Brandt et al. \(2015\)](#) on agile organisations.

Another different dimensional approach of OA worthy of presenting in this section is that of [Araza and Aslan \(2016\)](#), who after thoroughly reviewing research works on OA proposed six dimensions of the concept. They dubbed these dimensions as 'proactivity orientation, responsiveness orientation, strategic flexibility orientation, speed orientation, internal stakeholder orientation and external stakeholder orientation'. [Baker \(2017\)](#) suggests seven dimensions of an agile organisation after considering Kanter's three-dimensional approach to speed (innovation, processing and recovery speed). Baker added four other dimensions: continuous improvement, customer responsiveness, problem solving and changing direction.

In a methodology presented by [Sharifi and Zhang \(1999\)](#), a three-component model for practically attaining agility in manufacturing enterprises is offered. This model, comprising of three main parts, assists manufacturing businesses to develop strategic policies in the quest to successfully adopt and implement agile manufacturing in their operations. These parts are agility drivers, agility capabilities and agility providers. Agility drivers are changes that encourage a company to critically re-evaluate its strategies in order to gain a new market position and derive competitive edge from business environment. Agility capabilities are vital abilities necessary for a company to possess so as to become agile by properly responding to changes and transforming them into opportunities that benefit the business. These capabilities characterise the nature of an agile enterprise and include responsiveness, speed, flexibility and competency. The last part, which is agility providers describe the techniques and practices for obtaining the necessary capabilities in an agile manufacturing. Agility providers outline the means or guidelines in which capabilities required for agile manufacturing can be accomplished. These agility providers should be pursued through

a holistic integration of innovation, people, (information) technology and organisation ([Sharifi & Zhang, 2001](#); [Zhang & Sharifi, 2000](#)). In an attempt to adapt OA capabilities scale developed by [Sharifi and Zhang \(1999\)](#) into Turkish context, [Akkaya and Tabak \(2018\)](#) conducted an empirical study on managers and employees of SMEs and Technopark firms in Izmir and Manisa. The results of validity and reliability tests carried out reveal that a scale of four-factor structure is suitable to Turkish context in the suggested sample.

A survey of the literature presented above reveals that there are no clear-cut dimensions of OA. Other researchers in this regard have also earlier on expressed their concerns on this issue. For instance, [Yusuf, Sarhadi, and Gunasekaran \(1999\)](#) note that agility is a complex concept and its content has not yet penetrated the mutual consciousness of practitioners and academics. The vague nature and multidimensional structure of agility combined with perceptions of human beings make it complex to provide precise dimensions of OA. Even, the use of algebraic formulae could not arrive at the dimensions of agility ([Tsourveloudis & Valavanis, 2002](#)).

Benefits of Being an Agile Organisation

Nowadays, it does appear that OA has gained ground and recognition in most fields of study including management. The importance and impact of OA on many performance indicators and other organisational variables such as effectiveness, business success, competitiveness, efficiency, customer and employee satisfaction, among others have been documented and asserted by many researchers (see [Balaji et al., 2014](#); [Charbonnier-Voirin, 2011](#); [Hamad & Yozgat, 2017](#); [Wendler & Dresden, 2014](#); [Yang & Liu, 2012](#)). In the view of [Rigby, Sutherland, and Takeuchi \(2016\)](#), an agile organisation stands a better chance of attaining more success and benefits as compared to an organisation that still adopts and implements traditional management approaches in its operations.

Agility is vital for organisational competitiveness due to its propensity of improving a firm's capability to provide quality goods and services ([Crocitto & Youssef, 2003](#)). Agile organisations have the ability to distinguish themselves in a unique way from

competitors and provide superior customer value in this hyper-competitive business environment. Agility serves as a differentiator and helps organisations to be proactive in gaining competitive advantage ([Harraf, Wanasika, Tate, & Talbott, 2015](#)). An agile organisation has an edge to increase its market share, effectively and efficiently confront unexpected changes and in the process gain a holistic competitive advantage in the global market ([Ganguly, Nilchiani, & Farr, 2009](#)). Greatest competitive advantage could be attained by an agile enterprise due to its flexibility nature ([Goldman, 1993](#); [Nejatian & Zarei, 2013](#)). OA is an absolute requirement for organisations willing to be competitive in their field of operations ([Anca-loana, 2019](#)).

The ability of an agile organisation to understand the dynamics of customers' expectations makes it possible to provide services or design well-tailored products that customers love and value. One of the goals of an agile organisation is ensuring customers' satisfaction ([Balaji et al., 2014](#)). This kind of organisation enhances customers' satisfaction by introducing into the market products that are highly valued and preferred most by customers as well as constantly adapting to changing priorities of customers. An agile organisation does not only focus on customers satisfaction but also increases employee satisfaction by promoting team work, participative decision making, building mutual trust and respect ([Rigby et al., 2016](#)). It is important that the information should be distributed to employees by leaders with an accurate perception of knowledge for corrective decisions ([Akkaya & Üstgörül, 2020](#)). The stronger innovation focus of agile organisations and ability to respond swiftly to customer needs will eventually improve customer experience and satisfaction ([ACE, 2010](#)).

In contemporary business environment, agility is regarded as a key performance factor and a core contributor to business success. About 90% of 533 top European executives polled in a survey affirm to this point ([Allied Consultants Europe, 2010](#)). Thus, being agile helps an organisation not only to survive but also successfully to thrive, improve performance and overall business success in an unpredictable competitive environment ([Chatwani, 2019](#)). OA is a precondition for the survival and success of organisations ([Hamad &](#)

Yozgat, 2017). Agility is undoubtedly the current criteria for effective performance and agile organisations thrive in spite of ambiguity and uncertainty (Baker, 2017). The empirical findings of a study conducted by Yang and Liu (2012) reveal that an organisation's agility capability is an essential source of competitive strategy for firm performance, and by strengthening enterprise agility, the organisation could gain the benefits of first mover advantage, reduce losses and properly deal with unforeseen risks. Likewise, Ebrahimpour et al. (2012) demonstrate in their quantitative research results that there is a significant positive relationship between a firm's agility capabilities and performance. Their results depict how OA significantly influences company's performance indicators such as market share, profit, and return on investment, among others. OA is proven to be a great tool of accelerated significance on organisational performance. Alhadid (2016) reports a significant positive impact of OA on organisational performance in a research carried out on 12 large companies.

OA increases efficiency and team productivity by minimising if not eliminating the unnecessary waste embedded in redundant meetings, routine plans and other bureaucratic procedures as well as maintaining open communication and providing constant feedback to team members (Rigby et al., 2016).

Agile Leadership

The high degree of complexity of today's global business setting coupled with its volatile and ambiguous nature places high pressures and demands on not only private sector businesses, but also on public organisations to adopt agile leadership approach in their operations so as to smoothly and effectively react and deal with continuous changing environment. Leadership experts and researchers use varied range of names to denote agile leadership. Names such as leadership agility, leadership versatility, among others are sometimes used by leadership scholars to refer to agile leadership. However, in the view of De Meuse, Dai, and Hallenbeck (2010), the evolving consensus on this construct (agile leadership) is that it reflects leadership effectiveness, capability to properly

respond across diverse and dynamic organisational conditions and requirements. On the concept of agile leadership or leadership agility, Joiner and Joseph can be credited for their marvellous work done so far. These two authors appear to be highly noted scholars in this field of study, and have broadly written on the concept. As noted by [Lediju \(2016\)](#), the scope of studies on agile leadership is thin and Joiner and Joseph can be referred to as the creators of the concept. According to [Joiner and Josephs \(2007\)](#), agile leadership is ability of a leader to make judicious and effective decisions amidst complex, volatile and swiftly changing environment. It describes the capacity to take intelligent and prudent actions under situations characterised by high uncertainty, complexity and rapid change. Agile leadership is the ability to learn new styles of leadership and flexibly switch the already mastered leadership styles in an effort to quickly respond to dynamic and changing circumstances ([Meyer & Meijers, 2017](#)). Due to the inevitability of change and its impact on all levels of organisational hierarchy, leaders in different organisational sizes and industries are currently searching for ways to effectively apply agile thinking philosophy ([Prange & Heracleous, 2018](#)). Agile leadership is more about embracing an adaptively flexible strategy based on external and internal changes and creating a value-connected experience between all stakeholders by empowering organisational actors ([Chatwani, 2019](#)). In this type of leadership, different perspectives are employed to identify and analyse events and flexibly respond to them. It is not about strictly following bureaucratic or rigid structures or processes of decision-making. Agile leaders have the capability to adapt, switch and lead in a variety of situations that are ambiguous, new, difficult and unpredictable. This skill of adaptability and versatility of agile leaders is critical for twenty-first century business entities to survive in this hyper-competitive and volatile business setting. The findings of a study done by [Akkaya \(2020\)](#) showed that top managers exhibiting agile leadership can directly or indirectly boost dynamic capabilities of firms through the creation of a favourable organisational climate that motivates employees to practise open-mindedness, live exemplary life and being innovative. Furthermore, these leaders create appropriate conditions for their followers to reveal their knowledge and skills and

use them effectively as well as encourage them to be creative at developing new methods and techniques ([Akkaya, Koçyiğit, & Tabak, 2018](#)).

Agile leaders possess the mental faculty to analyse and understand the complexity of matters as well as using broader perspective to draw the line of distinction between these matters. Such leaders in decision-making process are able to utilise the views of others that contradict theirs and listen to the advice of people with expertise on issues where they are lagging behind. Hence, agile leadership entails self-assessment, awareness, understanding one's weaknesses and strengths and the impact of one's behaviour on other people ([McPherson, 2016](#)). Agile leaders have the abilities to lead in complex situations, determine internal and external needs of organisations, build good corporate culture and engage in relationship and change management as well as being talent developers and managers ([Holbeche, 2015](#)). Agile leaders are aware of the source of momentum building in an organisation, know how to support others and effectively lead to serve the needs of customers and other stakeholders ([Meyer, 2016](#)). Agile leaders are communicative, have intellectual curiosity, receptive to change, willing to tap experiences of others and able to determine when obsolete methods can no longer withstand the test of competitive forces ([Coleman, 2017](#)). Agile leadership describes the ability of a leader to be quick, adaptable and flexible in responding to unforeseen event in an unfamiliar circumstance. It comprises of dealing with and managing both internal and external environments as well as one's personal life ([Fernández, 2006](#)). Agile leaders repudiate the chain of command principle, inspire others, show empathy to others, practise shared management, regularly communicate and strengthen common vision and persistently seek the development of employees, organisations and the society as a whole ([Medinilla, 2012](#)). Agile leadership involves analysing events from a bigger picture, stepping back from the focus and putting new insights into what is next to be done ([Joiner, 2009a](#)). Agile leaders possess the ability to recognise and react timely to changes in a business milieu with actions that are swift and flexible ([Horney et al., 2010](#)).

Levels and Competencies of Agile Leadership

In the literature, although there are studies proposing that some leadership styles or capabilities are needed as key drivers for a resilient agile organisation, most of those neither propose a model nor provide a clear definition of agile leadership. In a conceptual framework developed by [Coleman \(2017\)](#), consistency and agility are proposed as pillars for strategic leaders to effectively implement the core values of their business and adapt to market changes. In the view of [Holbeche \(2015\)](#), the foundations and resilience of OA hinge on two main leadership theories; values-based leadership and shared leadership as key drivers. These studies are fruitful to shed light on the importance of agility in the leadership literature. However, development of Agile Leadership as a separate leadership model can be diagnosed in the work of [Joiner and Josephs \(2007\)](#). [Joiner and Josephs \(2007\)](#) after gathering data from over 600 managers came up with five different levels of agile leadership. These levels are hierarchical in nature and as leaders ascend to the next level, the skills acquired in the current level of agility are carried to the succeeding level. Agile leaders are required to determine which level(s) is needed for a specific circumstance ([Joiner, 2019](#)). These levels of agile leadership are briefly explained below.

- **Expert:** In this first level of agile leadership, leaders are tactically oriented and have the ability to analytically solve problems. Due to this, the expert level is most suitable for environments in which attainment of success is possible by introducing additional improvements to current strategies. In this level, leaders are valued and their subordinates follow them because of their position in the organisational hierarchy and expertise. These leaders excessively engaged in their subordinates tasks and deal with reports directly on one-one basis. Hence, they end up separately developing individuals instead of a team.
- **Achiever:** Leaders in this second level of agile leadership are strategic in their thinking and outcome oriented. These leaders together with their subordinates direct their efforts towards developing plans and strategies that will help in achieving the intended results. They accept suggestions and feedback from

subordinates, as well as motivate them through challenging and scintillating tasks to contribute to attaining the organisational goals. Thus, these leaders hold the view that power does not only emanate from expertise or position one occupies but also through motivating other people.

- **Catalyst:** In this level, leaders' orientation is on facilitation and vision. These leaders are capable of building participative culture that empowers and inspires others to actualise the vision of an organisation. They engage stakeholders in diverse range of issues in order to improve the effectiveness of decisions taken. They facilitate congenial environment where open discussions and exchange of different views can take place. This is the first level of agile leadership where sustained success can be realised in volatile business situation.
- **Co-Creator:** Leaders in this fourth level of agile leadership believe in common good, interdependence and collaboration. These leaders are capable of setting up their own organisations, where corporate social responsibility is embedded in it as a vital practice. These leaders are committed to providing service to the benefit of mankind at large. They are capable of creating collaborative team and strong stakeholder relationships rooted on high dedication to shared purpose. Co-Creators possess emotional stability, ability to engage in open dialogue and produce creative solutions to impromptu problems.
- **Synergist:** This is the final level of agile leadership where leaders at this level are holistically oriented. These leaders have the ability to shift between leadership styles and detect which style is more appropriate for a particular circumstance. They focus not only on leading others to accomplishing the objectives of the team but also transfer leadership experiences to them. Leaders at this level can alter or increase the energy rate of subordinates at workplace to lead to outcomes that are beneficial to all. Their present-centred awareness enables them to succeed amid challenging and chaotic conditions.

Through the study conducted by [Joiner and Josephs \(2007\)](#) on hundreds of managers, four key competencies successful agile

leaders display in unstable business environment are identified. These four competencies of agile leadership are briefly presented below.

- **Context-Setting Agility:** This refers to the ability of an agile leader to use broader perspective to scan both internal and external aspects of an organisation, forecast significant changes that may occur in both short and long run and determine the necessary initiatives to take in achieving the desired outcome.
- **Stakeholder Agility:** After laying the context for initiatives to be taken, the next competency is the capability of a leader to identify core stakeholders that would be affected by the initiatives and seek for their views and support on the initiatives. Leaders seek the suggestions and contribution of these stakeholders not just to gain their acceptance but as well to improve the effectiveness of the initiatives. Seeking for stakeholders' ideas or input could influence and lead to quality strategies.
- **Creative Agility:** This entails the capability of a leader to identify novel opportunities and transform difficult problems into preferred outcomes. The problem encountered needs to be examined from a bigger picture and clearly defined taken into consideration its underlying assumptions, new alternatives and viewpoints of others.
- **Self-leadership Agility:** This is a competency an agile leader required to possess in order to effectively lead others towards organisational vision. Leaders with this capability are able to accelerate their personal and professional development, proactively seek ideas and decide on the kind of leaders they aim to become. These leaders use their initiatives as avenues to achieve their leadership aspirations, make the necessary adjustments and learn from their experiences.

The Place of Agile Leadership in OA

In OA discourse, agile leadership is generally recognised to be playing a significant and critical role. This explains why the focus in management literature is currently gearing towards fostering or developing a leadership approach oriented towards agility and a

business mindset suitable for transformation and constant change. It is therefore imperative for agile leaders to be considered as an essential component of a business assets and serious attention should be given to this asset ([Chatwani, 2019](#)).

The foremost step towards building an agile organisation is a leadership mentality or perspective for agility. Agile leaders step forward to lead and empower others but not to command and exercise control over them in pursuit of meeting customers' needs and achieve business objectives. The right leadership agility mindset will build the best momentum and cause the finest technology, business systems and structures to have greater effect on the organisation. The success of an organisation is contingent on the extent to which individuals and leaders give attention to agility ([Meyer, 2016](#)).

The resilience of an organisation to consistent dynamic environmental change and attainment of long-term success is under the ambit of agile leaders. Today's business setting needs leaders who epitomise a constant degree of agility to create organisations and teams capable of overcoming unexpected turbulence. Being an agile leader signifies that one embraces and represents agile concepts and principles and ready to propagate these across the entire organisation ([Medinilla, 2012](#)).

Leadership agility is directly related to organisational agility. Agile leaders develop strategies and give directions on how to make organisations agile. To ensure that there is a leadership culture that depicts and promotes a holistic agility in the organisation, human resource managers need a deeper understanding of the meaning of agile leadership and how it can be evaluated and augmented. Agile leaders are visionary, tactical and capable of effectively leading in a rapidly changing business environment where success demands a wide range of multiple perspectives and priorities ([Joiner, 2009b](#)).

Agile leaders are capable of creating a conducive working environment, where employees are satisfied to give out their best. This kind of leaders has a crucial role in building a culture that promotes creativity, collective identity and innovation in this ever-changing global environment. Agile leaders act as change agents to help organisations detect and react quickly to uncertainties and

drastic changes. Forecasting and tolerating changes is one of the qualities of agile leaders which enable them to develop strategic agility vision to aptly adapt and turn those changes into opportunities for organisations (Crocitto & Youssef, 2003; Holbeche, 2015). For the purpose of getting a clear understanding of how agility affects business performance, one needs to first recognise and appreciate leadership agility (de Oliveira, Valentina, & Possamai, 2012).

Conclusion

It is not an easy task to become and sustain an agile organisation. It is perhaps a perpetual journey that is saturated with upheavals and ambiguity that demands a concerted efforts and a well-designed plan to make it a reality. There is an increased in the need for organisations to boost their capabilities in proactively reacting to changes in volatile business environment. The decision-making process of agile organisations is flexible and easily adaptable to unforeseen external forces.

The present study sheds lights on the position of agile leadership in OA. The study departs from the previous studies by using empirical reviews to depict the significance of agile leadership and its key roles on OA. This chapter argues that the leadership style suitable for OA is agile leadership. Hence, the need for organisations to invest in building capacities and developing frameworks that can appropriately accommodate agile leaders. It is also necessary to recruit and train people with agile leadership qualities or potential of becoming agile leaders.

Agile leaders are capable of thinking outside of the box to perfectly align an organisation with its internal and external environments. These leaders are able to unearth the latent strengths of others, utilise their views in transitioning into the change the organisation desires and capitalise on opportunities in the change while trying to minimise its negative repercussions on the organisation. Organisations that incorporate agility into their operations through agile leaders are more successful at responding swiftly to change and providing superior business value to their stakeholders. With leadership agility, organisations will be in a better

position to quickly detect developments in business environment and attain agility with little resources.

This study is limited to the extent of it not expanding its tentacles to cover the impact or significance if any, of other leadership styles on OA. It only vehemently contends that the best leadership style for agile organisations is agile leadership. This gap opens an avenue for future researchers to introduce and evaluate the importance of other leadership styles on OA. There is also an opportunity to conduct empirical quantitative research and more qualitative studies on this area. Finally, when one accepts that organisational agility and agile leadership as a theory lacking clarity, future studies may dwell upon the core theories themselves and suggest a more precise portraiture.

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Chapter 11

Strategic Leadership in Perspective of Industry 4.0

Semra Tetik

Abstract

Industry 4.0 and the new technological transformation process it represents pose major challenges for the world today. Now, a new period has been entered, which will affect every point of daily life, from production to trade, and from health to entertainment. In terms of its scope and complexity, this period is unlike any that humanity has experienced before. The concept of Industry 4.0 is an external reflection of the innovations and applications made and to be made not only in today but also in the production methods of the future. Because this change and development occurs as a result of a certain accumulation and continues to occur continuously, it is evident that strategy is of great importance in today's constantly changing and developing world. It is of great importance for leaders to develop strategies for evaluating the opportunity areas emerging by anticipating change in the Industry 4.0 process and to carry organizations and society into the future. Strategic leaders can do this effectively. At this point, strategic leadership is an important issue. Based on this idea, in this

study, the strategic leadership issue from Industry 4.0 perspective will be examined in a theoretical context.

Keywords: Industry 4.0; organization; strategy; strategic leadership; strategic management; Strategic Leadership Characteristics

Introduction

The world is in economic, industrial, and political changes. Although there are many factors that cause these changes that deeply affect every area of life, it can be said that the changes and developments occurring especially in the industrial field are among the essential elements of the world reaching a very different point. The innovations that emerged with the Industrial Revolution for the first time in the industrial sense have come to the fourth stage today and reveal the concept of Industry 4.0. Industry 4.0 and the new technological transformation process it represents pose major challenges for the world today. With Industry 4.0, not only technology or industrial production, the world has entered a completely strategic change process. Strategic leadership is one of the most influential factors that determine the survival efforts of organizations in an environment of intense uncertainty and competition caused by rapid and continuous change. Thus, it can be more successful in diagnosing change in time and implementing the right strategies. On the other hand, leaders with creative management skills that produce and implement creative and innovative ideas have been effective in the emergence of the Industry 4.0 phenomenon. It is of great importance for leaders to evaluate the opportunity areas that arise during the Industry 4.0 process, to direct their investments and to produce strategies by predicting the future, and to carry their organizations and society into the future. With Industry 4.0, strategic leadership emerges as an important issue in an environment, where dizzying digital transformation is experienced. In this context, in this study, the strategic leadership approach from Industry 4.0

perspective is discussed within the theoretical framework. Accordingly, information was given about the concept of Industry 4.0, its features and importance, and then detailed information about strategic leadership was given. In the last part, the relationship between these two concepts is examined.

Industry 4.0

Industry 4.0 started as a technology project supported by the German Government and enables the transition from the traditional production method to the new production model with computer and internet support. The goal of the project, providing an increase in resource efficiency is the integration of its customers in the production process ([Bulut & Akçacı, 2017](#), p. 53). Industry 4.0 refers to the improvement of existing processes by using the technological possibilities developed in all supply chain stages from the procurement of raw materials to the final product delivery ([Pamuk & Soysal, 2018](#)).

Industry 4.0, machine power: It can be defined as being able to manage production processes spontaneously by replacing human power. The fact that the machines can be coordinated thanks to new developments in computers and internet technologies has created the new industrial revolution. Thanks to this new system, which is known as the “Internet of Things” concept, advanced technology has been advanced with the fact that factories are self-managing ([EBSO, 2015](#), p. 7). Industry 4.0 is not just about the communication of machines, it can be said to be more comprehensive. Progression can be made simultaneously in all fields that will affect all kinds of scientific fields from genetics to information technology. The difference of this revolution from the previous ones is that the developments in technology are intertwined by each other, they act in a coordinated manner, and all fields are influenced together ([Schwab, 2016](#), p. 17).

In general, a comprehensive and explanatory definition of Industry 4.0 is made on a relevant platform as follows:

Industry 4.0 defines the Fourth Industrial Revolution, a new level in the organization and management of the entire value chain in the life cycle of products and production systems. This cycle focuses on increasingly individualized customer requests and includes services that include the entire chain, starting with the idea phase, from product development and production order to distribution and recycling of a product to the end user. ([Pamuk & Soysal, 2018](#), p. 4)

The concept of Industry 4.0 is an external reflection of the innovations and applications made and to be made not only in today but also in the production methods of the future. This change and development occurs as a result of a certain accumulation and continues to occur continuously. Industry 4.0 is a system that removes the bricks from the factories, regardless of time and place, is produced with completely smart systems, ready for change and expectations at the same time, and at the same time, its information is kept by cloud systems and big data methods and prepared for the future ([Yüksel & Genç, 2018](#)).

Three major industrial revolutions have occurred throughout the history of civilization. The First Industrial Revolution occurred with the use of steam systems. In this period, it changed from workshop-style production to factory-style production. In this way, a great increase has been achieved in the production amount. When we look at the Second Industrial Revolution, with Henry Ford's use of production line technology, this technique was applied in other sectors to increase productivity in production. The use of electricity is considered the beginning of the Third Industrial Revolution. In this period when electricity was used in serial production, developments in mechanical and electronic fields occurred and devices and information technologies that could be programmed thanks to digital technology emerged ([Brettel, Friederichsen, Keller, & Rosenberg, 2014](#), p. 37; [Köse & Bilginer, 2019](#), p. 124).

All these developments affect the sectors and reveal new business models. Established institutions are changing and production, consumption, and logistics systems are being reshaped.

There is a paradigm shift in the social dimension, in the ways of working and communicating, and in the way individuals express themselves, inform and have fun. Many systems, such as governments and institutions, as well as education, health, and transportation, are being reshaped. These changes and transformations are of historical importance. The table presented by Industry 4.0 in terms of entrepreneurship includes a change and development compatible with the general picture. It is moving toward a period when the traditional ways of doing business are rapidly transforming. Besides existing business models and jobs, different business models, different jobs, and professions emerge. This situation brings new opportunities and challenges for entrepreneurs ([Soylu, 2018](#), pp. 54–55).

In order to understand Industry 4.0, it is necessary to examine the development of industrial history. The “Industrial Revolution” is the turning of small workshops into industrial workshops, which gradually grow from the late eighteenth century to the mid-nineteenth century. It is the technological change that mobilizes the revolution and enables the discovery of machines that can produce faster and more efficiently than small craftsmen working in small workshops ([Frederick, 2016](#), p. 9). With the technological development and change in process, the successive industrial revolution has passed from past to present.

In the first industrial revolution, called Industry 1.0, the use of machines powered by water and steam has generated mechanical production ([Aydın & Demiral, 2019](#), p. 1978). The use of steam power has become increasingly widespread, with the replacement of wood and biofuels with coal. The change in the production structure has changed the social structure as well as the economic structure so much such that the average life expectancy and quality of life of the society have increased ([Efeoğlu & Bozkurt, 2018](#), p. 291). The transition from tool production to machine production, from workshops to factories took place in this period. For example, thanks to a textile machine manufactured with steam power, the production capacity of workers has increased 40 times. The increase in production contributed significantly to economic growth. For this

reason, this period was described as a revolution for the history of humanity ([Aydın & Demiral, 2019](#), p. 1978).

The Second Industrial Revolution, called Industry 2.0, is a process that started in 1870 and continued until 1989, the collapse of the eastern bloc. In this period, the effectiveness of Petrol in industry and transportation was discovered and used. The effectiveness of globalization has also increased in this period, and progress in transportation has developed rapidly ([Görçün, 2016](#), p. 51). Another innovation in the Industry 2.0 era is the use of developing electrical technology in production lines. This technology, which is superior to steam power, has allowed machines to develop further and increase production. Thus, the world got acquainted with the concept of mass production. The best known example of mass production in this period is Ford Motor Company ([EBSO, 2015](#), p. 5). In addition to the use of electrical energy in the second industrial revolution, the development of communication tools such as telegraph, telephone, radio, and typewriter caused the communication to take place faster and more effectively. Thus, the long distances between people have decreased ([Pamuk & Soysal, 2018](#), p. 42).

Industry 3.0 has benefited from the electronic and information technologies in its production since the 1970s in the third industrial revolution and thus automation has been achieved in production. Digital technology has developed and information technologies have emerged. Access to information has increased and information has now started to count among production factors ([Arklan & Taşdemir, 2008](#)). The widespread use of the internet in this period has increased communication channels extraordinarily. Perhaps access to information that has been freed more than ever before has become much easier. At the same time, the spread of electricity has radically changed the daily lives of people, the use of natural gas has shaken the distribution of sources of energy production structures and the production processes have gained a different dimension ([Sarikulak, 2018](#), p. 10).

Today, we are experiencing the transition to the fourth industrial revolution, which came to the agenda at the Hannover Fair in 2011, which was expressed as Industry 4.0, which was prepared by the Working Group formed by the German Government in 2012 and

announced with the final report presented in 2013. According to the final report, the eight stages required for the fourth industrial revolution to succeed must be completed. These stages are: (1) identification and standardization of the reference hardware architecture, (2) managing complex systems, (3) providing a comprehensive and high-speed communication infrastructure to the industry, (4) safety and security, (5) work organization and design, (6) training and continuity of professional development, (7) adaptation of existing legislation, and (8) efficient use of resources (Kagermann, Wahlster, & Helbig, 2013, pp. 49–50; Kılıç & Alkan, 2018, p. 32). The developments in the Industry 4.0 process have a significant impact in the manufacturing industry (Stock & Seliger, 2016, p. 537).

If we look at the basic elements of Industry 4.0, we can list as smart factory, smart production, future of production, new systems in the production of new products and services, self-organization, smart product, new systems in distribution, cyber-physical systems, smart cities, and digital sustainability. This change may lead to redefinition of jobs, to keep existing professions up to date, and to cause some to disappear. Many occupations that do not exist with this change are expected to emerge in the near future. Industry 4.0 technologies are growing at an extraordinary rate. As well as the positive aspects of sub-level jobs, such as the transfer of jobs that threaten human health to machines/robots, a relatively narrowing in employment is among expectations (Saatçioğlu, Kök, & Özispa, 2018). Demand is expected to contract in the blue collar workforce. At this point, getting the necessary training on new technologies is a necessity in terms of employment. However, the need for the workforce to be in the office environment for long hours will be eliminated. (Preuveneers, Joonsen, & Illie-Zudor, 2017). Especially, wide-ranging internet networks and cloud computing applications allow working independently from the office (Aydın & Demiral, 2019, p. 1979).

The changes caused by Industry 4.0 can be listed as follows: (EBSO, 2015, p. 24)

- More complex and smart products can be produced with the Internet of Things and cyber-physical systems.

- From the mass production to the special production according to the customer needs.
- Production facilities and products are connected to data and information exchange in real time.
- The amount of data and information required for product design, development and production reaches huge volumes.
- While raw material and resource consumption are minimized, efficiency increases and the green energy period is started.
- Thanks to self-organizing production methods, the need for resources required for production (energy, human, machine, etc.) is reduced and robots are managing their production processes.
- As the spread of robots reduces the need for the workforce, new employment policies are being developed.
- While the duration of the product-life cycle is shortened, the margin of error in production is minimized.
- Occupational health and safety is provided better with robots.
- As robots speed up production processes, working hours become much more flexible.
- With 3D printers, all kinds of raw materials, intermediate products, and final products can be produced anywhere in the world.
- The share of firms that adapt to Industry 4.0 is growing in the global market, and production is beginning to intertwine with the virtual world.

Briefly, as a result of these developments, an information-based social structure has emerged. In parallel with the developments in the communication sector, computers experienced revolutionary developments in the information field and computers were integrated into this process. This situation deeply affected the economic structures of developed countries. With the information revolution, it entered a new structuring process, and with the acceleration of globalization, this situation spreads all over the world. The last point of the information society is expressed as Industry 4.0 ([Bulut & Akçacı, 2017](#), p. 52). With Industry 4.0, strategic leadership emerges as an important issue in an environment where dizzying digital transformation is experienced.

Strategic Leadership

The concept of strategy is a Greek term identical to the word “strategos”, which refers to the art that generals practice in military actions as a military term ([Luecke, 2008](#)). The concept of strategy is thought to consist of the words “stratos” (army) and “ago” (manage) in this language ([Söyler, 2007](#), p. 104). The concept of strategy is used in almost every sector today. The name has become a frequently mentioned concept. The concept of strategy has emerged as a concept that expresses strategic action in the military field. Later, in the private sector organizations in the field of management, it has found itself as the decisions and goals that enable companies to make more profit, to make a difference to their competitors, and to determine the policies they will implement according to these goals ([Güven, 2014](#), p. 65). The strategy is to identify a broad perspective, stance or position an organization has developed on how it will compete against its competitors ([Porter, 1980](#)).

In the constantly changing environmental conditions management processes, it develops new methods and new perspectives; it also changes the organization of organizations. These new developments reveal new leadership types and skills. Today’s postmodern reality requires leaders to understand their strategic conditions and to be confident, talented and flexible leaders to adapt their institutions to these conditions. Coping with these rapid and complex changes also requires strategic leadership ([Ergen, 2011](#), p. 2). It is only possible for organizations to keep up with changing environmental conditions in their sectors and to have a superior advantage over their competitors. Today companies operating in emerging markets face with certain problems especially in turbulence, complex, and changeable environment ([Akkaya & Tabak, 2017](#), p. 119).

There are different opinions about the emergence of the strategic leadership theory. Strategic leadership has emerged as an important process that develops in parallel with strategic management and affects success in practice. Some authors, such as Vera and Crossan, argue that the strategic leadership theory was developed from the original Echelon Theory developed by [Hambrick and Mason \(1984\)](#). On the other hand, authors like Neumann and Neumann

emphasize that strategic leadership is derived from transformative leadership theory. John Adair argues that he was the first to introduce the concept of strategic leadership in the 1980s. When the researches are examined, it is understood that the concept of strategic leadership emerged in the 1980s. It is not known exactly who used it first ([Ergen, 2011](#), p. 30). When the literature on strategic leadership is examined, it is seen that different authors define strategic leadership in different ways.

Strategic leadership is defined as the ability to see the future, create vision, provide flexibility, think strategically, and work with others to initiate changes that will create an achievable future for the organization ([Uğurluoğlu & Çelik, 2009](#), p. 124). According to [Hitt, Ireland, and Hosisson \(1999\)](#), strategic leadership is the ability to see the future, be flexible and empower those who work to create the necessary strategic change. Strategic leadership: to predict the future, to create the necessary strategic management understanding for this, to empower other managers and employees in this direction, to direct them to innovative and creative goals in line with the corporate vision, and to provide strategic change in a complex global competition environment when necessary ([Altinkurt, 2007](#), p. 11).

Strategic leaders drag people with their visions. There are three basic dimensions of behavior for this: seeing the road, walking on the road, and being the road. The primary behavior of strategic leaders is to see the way. For this, to look at the picture of tomorrow today, it is necessary to clearly state the policies and targets for the future. Just seeing the road is not enough. It is also very important to walk on the road. The third behavior dimension of strategic leaders is to be the way. The strategic leader should be the way to remove the obstacles on the road and enable the followers to follow him easily ([Sevinç, 2009](#), p. 61).

According to [Sullivan and Harper \(1997\)](#), it is to manage and oversee a logical and well-thought-out action that creates the organization in its most basic sense, such as strategic leadership, purpose, culture, strategy, core identities, and critical processes. [Ireland and Hitt \(2005\)](#), pp. 63–67) define strategic leadership as the ability to anticipate the future, to adapt to changing conditions, to think strategically, and to work in harmony with others, to initiate

changes that create a future that can be applied to the business. Strategic leadership is to take an organization, institution or structure from a certain level and move it to a much higher level and to take the individuals behind it in that direction ([Kara, 2018](#), p. 423). Strategic leadership includes not only working successfully today, but also making deep and lasting change and transformation established in the core of the organization ([Uğurluoğlu & Çelik, 2009](#), p. 127).

Today, the world we live in is constantly changing in terms of businesses as in every field. It seems possible that leading companies, especially in the field of technology, may lose their market leadership in the future and perhaps end their assets. There are many businesses that have been in the first place in terms of market share in their sector in the past, but are not able to adapt to changing environmental conditions now ([Akkaya & Tabak, 2018](#), p. 186). Leaders create appropriate conditions for their followers to reveal their knowledge and skills and use them effectively and encourage their creativity to develop new methods and techniques ([Akkaya, Koçyiğit, & Tabak, 2018](#), p. 187). While change and transformation are so serious, leaders must educate themselves as people who learn to learn, can access information in the fastest and most accurate way and manage their knowledge well. In this context, leaders who closely monitor change and transformation and are positive in adaptation will be able to analyze, direct, and be a successful strategic leader in this information age ([Ünal, 2012](#), pp. 136 and 137). According to [Dinçer \(1998\)](#), the strategic leader is the person responsible for the management process, which involves creating, implementing, and evaluating an organization's strategy. [Adair \(2005\)](#) defines the strategic leader as "the person who successfully fulfills what is expected in the leadership positions of the organization in times of change." [Pisapia \(2009\)](#), on the other hand, has defined the strategic leader as the wise person who has high decision-making skills in areas related to goals, behaviors, and tactics in environments where change is experienced rapidly and chaos is high. It is important that the information should be distributed to employees by leaders with an accurate perception of knowledge for corrective decisions ([Akkaya & Üstgörül, 2020](#), p. 53).

Strategic leadership, having multiple leadership roles, is the type of leadership that displays the appropriate ones at the right time and conditions (Thompson & Strickland, 2001, pp. 430–431). The characteristics of an effective strategic leader should be summarized as follows (Ülgen & Mizre, 2004, p. 43); be able to see the future, create a vision, be flexible, cope with uncertainties, empower employees, influence others' feelings, thoughts and behaviors positively and positively, manage human resources effectively, establish good relations with stakeholders, constantly question and develop their own paradigms and talents, and brave according to environmental conditions to make decisions. According to Davies (2004), organizational skills that strategic leaders should have are: having the strategic wisdom and strong relationship network, directing its environment, implementing the strategy, organizing the environment of the organization, intervening when necessary, developing the capabilities of the organization, questioning and struggling constantly, having a high level of communication and social skills, and prioritizing learning.

As can be understood from the above information, the concept of strategic leadership first appeared in the 1980s and gradually began to come to the fore in our rapidly changing world, because a strategic leader was needed in which organizations could sustain their existence and manage future uncertainties. This reveals the importance of strategic leadership. Based on definitions, we can define the strategic leader as the person who can provide a competitive advantage, guide the future, be successful by making change in time, analyze the internal and external environment using his knowledge and experience, and prepare his own organization for the future well (Derin, 2019, p. 51).

Successful strategic leadership is based on the effective implementation of some key functions. In organizations, these functions are defined as S5K. These key functions can be listed as STRATEGIC: Editing-Position-Control-Staff-Culture (Eylem, 2002; Ergen, 2011, pp. 32–33).

Strategic fiction is the determination of the organization's cause (Mission and Vision) and its objectives. In general, the time perspective of this function is long-term. Here, it is determined how

the organization should take shape and its orbit during this period of time. A philosophy is created for the target that is desired to be achieved and a strategic fiction is put forward.

The strategic position, the harmony between structure and strategy determines the strategic position of the organization. At this stage where the strategic position is determined, leaders determine the targeted customer base of the organization and which goods and services will be produced in line with the demands of these customers. However, it is determined how this process will be effective and efficient for the organization. Specific strategies envisage specific organizational structures and their management processes.

Strategic control is the creation and maintenance of a balanced and stimulating strategic control system. The organization uses this function to reach the level of performance it wants to reach in the future, to reach the efficiency and efficiency of the strategies it has set, and to test its reliability.

Strategic staff is the development of human resources and integration with the strategy. The performance of all employees in both the management and operational staff of the organization is the main factor that determines the organization's strategy design and implementation success.

Strategic culture is the creation and dissemination of an effective organizational culture based on deontology. An organizational culture is the shared values, ideals, symbols, and insights shared by the whole organization. Organizational culture largely determines how business is carried out and how its strategies are implemented. The strategic leader is obliged to create an effective organizational culture.

It is also possible to explain the strategic leadership functions with three ring models. Three roles are basically expected from a strategic leader in the model. These three general functions ([Adair, 2005](#)):

- achieving the common goal;
- building and maintaining a team; and
- motivating and improving the individual

Adair (2005) took strategic leadership functions into seven dimensions based on these three roles. These: To determine a direction for the entire organization, to follow a successful strategy and policy, to implement strategies, to organize and reorganize, to reveal the spirit of the organization, to connect the organization to other organizations and society, to choose today's leaders, and to raise the leaders of tomorrow.

When these functions of strategic leadership are evaluated in terms of three ring models, it is seen that there is a close relationship between them. Strategic leaders operate in a team spirit by achieving a goal, by thinking strategically, maintaining the balance between all and parts and motivating individuals (Ergen, 2011, pp. 31–35). In light of the explanations made so far about strategic leadership, it is possible to say that the strategic leader basically determined the purpose and vision of the organization has the ability to think strategically, determines the basic capabilities of the organization, develops manpower, and can create strategies to adapt the organization to rapidly changing environmental conditions. Therefore, strategic leadership plays an important role in the formation of Industry 4.0, and Industry 4.0 plays an important role in the formation of a successful and promising strategic leadership.

The Relationship Between Industry 4.0 and Strategic Leadership

Heraclitus said, "Everything is in flow and nothing is at rest." With the word, it is accepted that the process of understanding, perceiving, and adapting to change has begun. This complex phenomenon, called change, deeply affects the culture and management styles of all social systems, together with rapid information growth, rapid development, and rapid communication. The phenomenon of "leadership" plays an important role in creating and achieving the obligation of this change process. The fact that the traditional leadership understanding is not sufficient to realize the change has revealed the necessity of structuring a change-based leadership

approach today, when the impact of change is felt intensely. It is claimed that this leadership approach is “Strategic Leadership.”

Global competition and accompanying social, cultural, technological, economic, human, and digital world developments have made it compulsory for countries to develop cooperative relations. Industry 4.0, which Germany first brought to the agenda, aims to apply information technology to traditional industrial processes, thereby completely changing existing industrial processes. In Technology-Intensive new generation factories, businesses can develop the technology they need with internal R&D studies as well as transfer them from outside the organization ([EKOIQ, 2014](#), pp. 1–2). Industry 4.0: Thanks to the innovations in the technological field, the developing cost corresponds to the idea of production and marketing based on efficiency, speed, and innovation. On the one hand, it defines the production systems that adapt to the changing needs of the consumer instantly, and on the other hand, it defines the automation systems that are in continuous communication and coordination with each other, and also encourages close cooperation between various disciplines in product development ([Yıldız, 2018](#), p. 548).

Fast process of technology and digital world are taking place in all organizational authoritative in all area and in all kind of sectors, because the business world is transformed by the postmodern revolution – fourth industrial revolution ([Akkaya, 2019](#)). The digital revolution has transformed the lives of people in the way that people communicate with technology rather than with each other. This change powers managers to comprehend these improvements in environment and try to adapt them ([Akkaya & Tabak, 2020](#), p. 1).

Industry 4.0 has completely transformed the world into strategic changes. These changes deeply affect societies/countries. Societies consist of basic units called organizations. Organizations are extremely sensitive to these changes due to their nature. Changes and developments in technological, economic, political and socio-cultural fields, and the diversity and increases in the demands and needs of the society formed accordingly affect the organizations deeply. Organizations that cannot manage and foresee these beforehand and cannot properly regulate the interactions between

the external environment harmony and the sub-systems that make up the internal environment, and therefore the societies they create, can face serious problems. Since frequent problems cause significant harm to organizations and society, strategic leadership is increasing in importance ([Krakaya, 2004](#), p. 226).

In this new period, production systems and processes communicate with each other via the internet. These structures, which enable the communication of objects with each other, are called "Internet of Things." The structures that include the communication and coordination between the cyber world and the physical world are named as Cyber-Physical Systems. Industry 4.0 includes the developments in the field of resource saving in production, where the coordination of the production process is realized simultaneously, the production information required for the enterprise is easily obtained from outside the enterprise, losses and waste are eliminated by sensors ([Köse & Bilginer, 2019](#), p. 125). With Industry 4.0, strategic leadership emerges as an important issue in an environment where dizzying digital transformation is experienced.

Although today's age is the age of information and technology, which has a great influence on the lives of individuals, it is an era that is changing and developing day by day and this change and speed cannot be stopped. At the same time, these changes and developments have affected the characteristics of individuals who follow individuals. While there was an autocratic and sovereign leadership approach in the past, a more scientific leadership approach that started with the first industrial revolution (Vapor Power Discovery), which could be considered the first of the industry 4.0 stages, was adopted. Today, the definition of leadership has changed a lot and has become more democratic and participatory by becoming transparent with modern management approaches. With the birth of Industry 4.0, the increasing speed, complexity and intricacy cannot embrace autocratic and individualistic leadership and make strategic leadership almost imperative ([Tüzmen, 2017](#)).

The world is completely in strategic changes. These changes deeply affect every area of life. Strategic leadership is one of the most influential factors that determine the survival efforts of

organizations in an environment of intense uncertainty and competition caused by rapid and continuous change. Thus, in Industry 4.0 perspective, it can be more successful to diagnose change in time and apply correct strategies. At the same time, leaders with innovative management skills who had innovative ideas and brought these ideas to life were influential in the emergence of the concept of Industry 4.0. In this context, strategic leadership plays an important role in the formation of Industry 4.0, and Industry 4.0 plays an important role in the formation of a successful and promising strategic leadership.

Today, knowledge is one of the leading organizational inputs, but it has become very easy and inexpensive. All organizational strategies, missions specifying the purpose of establishment, visions specifying future goals are concentrated and continue to focus on knowledge-based products and services. Based on this concentration, new organizational values are being created for the most effective development, use, sharing and dissemination of information, which can only be achieved through the understanding of Industry 4.0 revolution and leaders who understand this revolution ([Memduhoğlu & Yılmaz, 2017](#)).

Industry 4.0 is the main dimensions of the transfer of information, adopting solutions appropriate to the period, encouraging creativity, directing to common organizational or social goals, strategic leaders who are change managers, and directing the viewers to common ideals during the four industrial revolution periods. Industry 4.0 and beyond relate directly to the existence of strategic leaders as visionaries. The development of societies and the achievement of humanity to a higher level of prosperity, realization of common dreams will be accompanied by encouraging strategic leaders. Valuable, unique skills and abilities, inimitable, properly organized, and managed human resources will be the main source of competitive advantage in Industry 4.0 period. Creating and maintaining such a human resource is directly related to strategic leadership ([Yüksel & Genç, 2018](#)).

As a result, the world we live in is an extremely dynamic environment where hundreds of changes occur every day. Globalization process, international competition, technological,

scientific, economic, political, and social developments in many areas cause the era to be characterized as an era of change. The speed and scope of these changes have reached incredible dimensions compared to previous periods, because the changes that can be realized in a very long time in the past are reached in a very short time today. Organizations need to adapt to environmental change and even affect environmental change in order to survive and improve themselves in an environment of change. The important thing now is to be a pioneer: To go beyond being managed by change and to create active approaches to manage change. This requires strategic leaders who create and share a shared vision, develop creativity with intellectual incentives, inspire respect and trust in employees with charismatic influence, attach importance to teamwork, take personal risks, mobilize employees with inspirational motivation, and realize radical changes. In other word, leaders should empower their teams, inform them from technology and environmental changes, adapt them to those changes, provide self-training to employees, and allow them to work on a specific project in order to get maximum profit and value in this competitive era ([Akkaya, 2020](#), p. 391).

The leadership needed in today's businesses is quite different from that needed in the previous 20–30 years. Businesses and their environment have changed. Therefore, leaders need to find the correct answers to the “what” and “how” questions in choosing a strategy. As expressed in an ancient Chinese proverb, the proverb “Teaching a fisherman is better than giving it fish” reinforces this statement. Strategic leaders generally work to develop strategies appropriate to their own organizations, in which they work and to train members attached to the organization ([Sütçü, 2008](#), p. 62). The strategic leader is a visionary leader who accepts people as the most valuable element of the organization, who is proactive to changes that value innovation and creativity, and who has a vision. It constantly takes into account the moral values of the employees. Believing that the values will lead to results, the leader constantly presents the values that will increase the commitment, satisfaction, and performance of the employees. With all these features, the transformative leader will be able to challenge the big waves and

chaos of change in the information and communication age, turn these situations into opportunities, and thus design new organizations required by the age.

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Chapter 12

Industry 4.0 and Agile Firms

Erdal Şen and Necmiye Tülin İrge

Abstract

Today humanity is facing a time period, in which the speed and rate of change in different fields are faced. The rapid and effective change in the technological, economic, military, social and cultural fields all over the world in recent years has had important results in many topics such as informatics, interactive communication, production, data production and sharing and forms of consumption and perception. Changes and developments at very high speeds and rates affect the personal life, social life and work life in a deep manner, especially after the 1980s, paving the way for many concepts such as globalization, Industry 4.0, digitalization, new economy, new world order and digital transformation age to be the leading variables in every field of social sciences.

Production, consumption and communication forms at the global level are becoming more and more in depth and predicting the future is becoming more and more difficult and valuable. Understanding, analyzing and predicting the future for national and international companies and organizations directly and indirectly affect all economic, military and political variables. At this point, competition stands out as one of the most critical concepts for survival and growth for profit-oriented companies.

Based on this, the topics and contents selected in this study were created to cover the field of management and strategy. Indeed, in this study, the concept of organizational agility is explored theoretically from a historical perspective on the concept of Industry 4.0 and especially emphasized the dimensions and components of agility used in today's world.

Within the scope of this study, the aim is to define and present the general framework of the Industry 4.0 and agile firms, which aim to combine the conceptual and theoretical infrastructure with the implementation of the agile firms within the field. In this respect, the study defines the situation coming from the past together with the world of today in the relationship of the technological and economical dimensions of globalization with the digitalization and latest theoretical view for the agile firm's structure and management. On the other hand; in this study, the possible effects of the recent COVID-19 pandemic crisis on the management of organizations are interpreted in the context of Industry 4.0 and agile companies. This aim will provide a foresight for the future periods, with the conclusions added with the theory of senism, which present the dominant value of the study.

Keywords: Industry 4.0; agile firms; globalization; digitalization; governance; senism

Introduction

Many different dimensions and indicators of progress and development come to the fore in the known history of humanity. When evaluated from this point of view, production and consumption patterns throughout history have been examined in different fields of social sciences and historical analyses have been made according to this methodology. One of the main reasons for this methodology is based on having knowledge about the past and understanding the present according to objective and scientific bases. This may provide

the way to make possible scenarios and forecasts related to future changes and transformations ([Şen, 2020](#)). Today's world for social sciences is defined and analyzed mostly with the concept of globalization. Global issues directly and indirectly related with the changes and transformations of the information, communication, production and consumption technologies, and techniques are the basis of different management fields. These fields are generally argued as if they are only the variables of business administration. On the other hand, all managerial processes starting from the individual (self-management) to the organizations and systems are the basic dimensions of the management field ([Şen, 2000](#)). The basic importance of the change and transformation is relatively high speed and rate and the interactions of all the related variables within the systems. Digitalization is the basis for the renewal of the global system, and globalization has been started to be effected directly from digitalization effect since the mid-1990s ([Şen & Batı, 2020](#)). The current phase is the beginning of the new production revolution. The change and transformation that take place in industrial and production activities are unprecedented in terms of scale and effect. It prepares the environment for the emergence of very important innovations in the business structure of the technological developments and in the labour market ([Kergroach, 2017](#)). Today's technologies such as the cyber-physical systems (CPS), Internet of Things (IoT) and cloud computing are the basic transformation sources of automation technologies in the manufacturing industry. In this transformation, virtual processes and the physical world are integrating with embedded systems, semantic machine-to-machine communication, IoT and CPS technologies ([Germany Trade & Invest \(GTAI\), 2014](#)). The new generation industrial systems, such as smart factories, enable to manage the complex structure of production in the cyber-physical environment effectively and efficiently. The radical changes and transformations in production and consumption ways and methods have affected not only production processes or industry, but also the social demographic structures, cultures and economic conditions of the countries and even caused the redrawing of the countries' maps. At this point of view, the historical overview for the

industrial changes and transformations defines and shows both the conceptual and practical variables affecting today's world.

Industry 4.0: Historical Overview

The last phase of the process that started with the first industrial revolution, which occurred approximately 260 years ago, the fourth industrial revolution also known as the Industry 4.0 was founded on the studies related to unification of man and machine. In other words, it aims to reduce costs and increase productivity by using a new production tool that will realize human-machine cooperation with industrial robots, which can make human brain power and machine work force together ([Özkan, Al, & Yavuz, 2018](#)).

Historically, major changes and developments in production systems have paved the way for the emergence of four industrial revolutions, which have been considered a breaking point in world history. With these developments, the structure of societies has also changed. The transformation experienced with industrial revolutions led to technological, economic, social and cultural changes and changes in the class structure of societies ([Yazıcı & Düzakaya, 2016](#)).

The new paradigm of the new economy must be defined and measured in the parameters of Industry 4.0. Industry 4.0 can be defined as the fourth Industrial Revolution that is merging the real and virtual works on the basis of cyber-physical production systems. Some of the main new implementations like Digitalization, Data Analytics, Artificial Intelligence, Autonomous Systems, IoT, Machine Learning and Advanced Algorithms directly affect the competition among the firms within the global marketplace. For the firms of Industry 4.0, it is necessary to redesign and implement all the necessary structures and processes in order to manage this change to create a surplus-value and competitive advantage with its own dynamics and processes like manufacturing plant, global facility insight, customer site and global operations within smart factories.

The first industrial revolution took its place in factory production between 1760 and 1830 with the discovery of machines operating with water and steam energy and the use of them in mechanical production systems. At the time of the first industrial revolution, the

human factor was a quantitative variable in productivity equations. In other words, more employees or more hours are the definition of productivity. On the other hand, attracting people to the system, making sure that they continue their work not only with their bodies but also with their minds and thoughts will contribute to efficiency and, more importantly, contribute to the existence of the firm ([Katz, 1964](#)).

With the discovery of electricity, the establishment of division of work and serial production logic, the second industrial revolution took place between approximately from the 1850s to the second half of the 1970s. The Industry 2.0 revolution played an important role in all other industrial revolutions and paved the road to modern life due to steam power. It is also the revolution that started the machine age. These technical advances that took place in the second half of the nineteenth century, industries based on inventions such as electric motor, internal combustion engine, electric bulb, telephone, radio and telegraph changed the world economic system. With Industry 2.0, which is also called the Second Industrial Revolution period, very important innovations entered the individual, social and industrial life. The emergence of electric and internal combustion engines continued for many years and has taken its place as an important building block in today's world ([Günay, 2002](#)).

The third industrial revolution was a process involving the establishment of automation systems and the establishment of mass production logic between 1974 and 2011 ([Gabaçlı & Uzunöz, 2017](#)). Many new systems and techniques such as computer numerical control, industrial robots, manufacturing systems, computer-aided manufacturing and computer-aided processing planning are being used in this process thanks to innovations such as the use of electronics, the introduction of the internet, information and technology and the use of transistors ([Kabaklarlı, 2016](#)).

As the industrial revolutions continue, new approaches have been researched, used and new technologies have emerged. New technologies have also shaped industrial revolutions. Autonomous systems are pioneers of the third industrial revolution. The driving force of the fourth industrial revolution is considered to be CPS. The fourth industrial revolution was announced by the German government in 2011. The fourth industrial revolution emerged as a

result of the rapid transfer of cyber physical systems and knowledge in production from 2011 until today ([Bartodziej, 2015](#)).

Industry 4.0 is a collective set of technologies and concepts of value chain organizations. This structure contributes greatly to the vision of intelligent factories. Industry 4.0 cannot happen without an IT infrastructure. Smart production processes are implemented with the established IT infrastructure. Smart production processes will be strengthened by legal infrastructure and will continue to develop with new business models ([Selek, 2019](#)).

Industry 4.0 is also mentioned as a new concept or an understanding considered with developing technology and know-how. Companies that are not continuously developed with this understanding will gradually lose their competitiveness in the market. The technologies used in the right place will reduce production costs and increase the capacity and quality. In short, the limited resources of the world will be used more efficiently with Industry 4.0 in production and other fields of industry ([Dengiz, 2017](#)).

Components of Industry 4.0

Existing and proven technologies in every process used in Industry 4.0 are combined with new technologies and applications to address production problems. In this context, considering the major components of Industry 4.0, industry standardization 4.0 is a good start and is related to major technologies such as the IoT and CPS supporting industry 4.0 ([Xu, Xu, & Li, 2018](#)).

Other components of Industry 4.0 are: Autonomous Robots, Layered Production, Augmented Reality, Cloud Computing Systems, Big Data, Simulation and Horizontal/vertical Integration ([Kagermann, Helbig, Hellinger, & Wahlster, 2013](#)).

Fast process of technology and digital world are taking place in all organizational authoritative in all area and in all kind of sectors, because the business world is transformed by the postmodern revolution – fourth industrial revolution ([Akkaya, 2019](#)). Industry 4.0 enables all systems, processes and applications in data collection, processing and interpretation to be carried out in a digital environment much faster, flexible and low cost manner. In digital

businesses, it is seen that an existing business process has been changed or transformed, or a new business system and process has been created. The facilities provided in the production, collection, protection and interpretation of information increase the efficiency and speed of decision-making processes, thus making it easier for companies to access information in production processes and customer expectations. For example, Big Data and Cloud systems contribute to the organization's performance by being used in the relative processes of the firms. With these contributions, the efficiency and productivity of the enterprise is increased. Thanks to their flexibility, speed and adaptability of the digital businesses that emerge through digitalization, traditional processes and the chain of command orders are replaced by the current management and trend management processes used in the market.

Big Data show the whole database used to increase productivity by analyzing the big data in the data storage (Cloud) systems that provide the flexibility of access to information and working. Systems where integration is prominent in all processes, both in-house (Vertical) and supplier-to-customer (Horizontal), where not only products but also factory processes are analyzed with simulation techniques create value as a combination of all these applications. With the analysis and use of big data, it is aimed mainly to increase efficiency, decrease error rates and increase flexibility ([Pamuk & Soysal, 2018](#)).

The major technologies, structures and systems used in digital businesses include Industry 4.0, numerous technologies and associated paradigms. Some of these emerging paradigms include the following list: Radio Frequency Identification (RFID), Enterprise Resource Planning (ERP), IoT, Industrial Internet of Things (IIoT), CPS, Cloud Based Manufacturing (CBM), Smart factory, Smart product, etc. Thus, through new structures, systems, business practices and applications of companies have changed and a new type of business has emerged.

These features are not only significantly related to internet technologies and advanced algorithms, but also indicate that Industry 4.0 is a value-added information processing and industrial value-added process ([Yıldız, 2018](#)).

Fourth industrial revolution starts to be introduced into production processes, related tools and agricultural processes through progress toward becoming a fully digital enterprise with the fourth industrial revolution, the emergence of unmanned production networks and as one of the main driving forces of the CPS, large sensor networks (or Internet of various objects today – including IoT, solutions) which is a key feature of this revolution.

Radio Frequency Identification (RFID) is a relatively new automated identification system that allows you to collect information about people and objects and transfer them to information systems unmanned. Automatic identification technologies include barcodes, optical character recognition, magnetic strips, smart cards and biometric recognition systems ([Feng, 2011](#)).

Enterprise Resource Planning (ERP) deals with a wide range of planning, operation and accounting functions in an integrated manner such as accounting, finance, logistics, production planning, inventory management, purchasing, production, marketing, quality management, maintenance/repair, human resources and customer relationship management ([Levine, 1999](#)).

The Internet of Things (IoT) and the Industrial Internet of Things (IIoT) are a new technology paradigm designed as a machine and a device network that can interact with each other ([I. Lee & Lee, 2015](#)). IoT which is one of the main concepts and applications within the scope of Industry 4.0 refers to the information and communication infrastructure through which objects can connect to the Internet and send or receive instant information. With its first use, the Internet helps people communicate with each other, despite the distance between the continents. In addition, the internet, the main building block of globalization, has changed the structure and operating strategies of existing markets. It is clear that this communication, which is currently being carried to the dimension of objects, will affect the structure of the market and the production and marketing strategies of the enterprises. Today, many big technology and software companies create products related to IoT ([Pamuk & Soysal, 2018](#)).

Cyber-Physical Systems – CPS, the concept of Cyber Physical Systems is one of the most important and essential components of

Industry 4.0. The reason for this is the unification of the physical and virtual world with Industry 4.0 (Soylu, 2018). In a general framework, with a programming to be made at the beginning of the system, the entire process of the system can be performed automatically without any intervention or additional labor. In this process, automation comes into play primarily. Many machines, including learning robots, are involved in the production process. Today, it is known that robots and tools learning especially in the automobile industry are already involved in some production processes (Pamuk & Soysal, 2018). In this context, the combination of the physical and virtual world is made possible by the CPS. CPS is the integration of computation and physical processes. Embedded computers and networks monitor and supervise the physical processes, often through feedback loops where physical processes affect computations (Lee, 2007). The development of CPS can be characterized by three stages. In the first generation CPS, storage and analytics are provided as a central service. Second generation CPS are equipped with sensors and actuators with a limited number of functions. Third generation CPS can store and analyze data, it is equipped with multiple sensors and actuators and is compatible with networks (Hermann, Pentek, & Otto, 2015).

Autonomous Robots describe intelligent machines that use robots that are able to work and learn like people on assembly lines. Augmented reality includes applications that enable you to send information to the desired location in a meaningful manner and real time with Virtualization. Whereas, Additive Manufacturing covers systems that can quickly develop and produce prototype and personalized products with 3D printer technology. Manufacturers in various industries have been using robots for a long time in their operations. Robot technology in the world is now becoming more autonomous, flexible and cooperative as well as reducing the cost of ownership by improving its competencies. In the coming period, robots are predicted to increase their interactions with each other, work more safely with humans side by side and improve their learning skills (TÜSİAD Raporu, 2016).

Cloud Based Manufacturing – CBM is another emerging concept that will contribute significantly to the success of Industry 4.0. The

CBM definition is as follows: It is a production model connected to a network with the ability of leveraging on-demand access to a shared set of shared, diversified and distributed production resources to create temporary, reconfigurable cyber-physical production lines that increase efficiency, reduce product life costs and allow optimal resource allocation to respond to variable demand generated by the customer (Yıldız, 2018).

Industry 4.0 in Different Sectors

The Turkish Industrialists' and Businessmen's Association (TÜSİAD) and Boston Consulting Group (BCG) in the report developed in our country in 2016 with the joint work and the support of industry stakeholders conducted a study on Industry 4.0 awareness with a total of 25 Turkish manufacturing companies operating in six different sectors in our country. These sectors are: automotive and automotive supply industry, machinery, domestic appliance, food and beverage, textile and chemistry. According to the results of the research, the awareness rates of the selected industry leaders about Industry 4.0 are quite high; however, they are already implementing some practices in their own companies. Indicators of Industry 4.0 applications and examples in different sectors are summarized in Table 1 (TÜSİAD Raporu, 2016).

Since the world of the future is a world, where Industry 4.0 will dominate in production and digital detailed information in consumption, the condition to be in this world will be to fulfill the requirements of keeping up with the digital age (Eğilmez, 2018). Organizational agility is one of the prominent concepts in the ability of firms to gain competitive advantage and sustainable growth in the global conditions that emerge with Industry 4.0. Firms need to manage the rapidly changing indicators that are vitally important, such as performance, quality, sustainability and customer satisfaction, as flexible, fast and adaptable. In this context, the concept of agility comes to the fore. Organizational agility for companies has an important place in management processes such as determining, implementing and controlling an integrated organizational strategy that prepares the ground for innovation initiatives, as well as for

employees to seize opportunities individually. It also makes the firm strong against the business challenges of making profits from global markets.

Table 1. Industry 4.0 Applications in Pilot Sectors.

Industry 4.0 Indications	Sector	Examples
1. Integrated, automated and excellent production flow	Domestic Appliance	Integrated quality management The waste is reduced and the processing process is improved by monitoring the products during the production process and based on the error data obtained during the pre-production tests
	Machine Systems	Integrated design data In the production process, vertical data integration is used from design to the end of the production line to optimize the operations
	Domestic Appliance	Horizontal data integration By providing suppliers to see some ERP data, the integrated production process is approached
2. Virtual Product Design	Automotive	Virtual factory and product design A joint solution that integrates factory and product design is offered to optimize production with factory simulations based on real production needs
3. Flexible Production	Domestic Appliance	Flexible production robots A production line that communicates with RFID-based smart products and adjusts the tools and production tasks according to the product range was launched
4. Smart and Automated Logistics	Automotive	Laser guided automatic guided vehicle (AGV) Laser-guided logistics systems are used, where the server checks the inventory and production schedule and deliveries of the computer and directs the AGVs
5. Learning Processes	Chemicals	Process flow that optimizes itself

By using the IT algorithm, the problems in the basic material mixture are recognized and the quality of the final product process is optimized

Source: TÜSiAD Raporu (2016).

Agility and Organizational Agility

The concept of agility is defined within the theoretical framework, from the perspective of dynamic capabilities. In this context, agility as a dynamic capability is defined as “the organization’s ability to integrate, construct and restructure internal and external competencies to address rapidly changing environments” (Teece, Pisano, & Shuen, 1997). Organizational agility is defined as being able for a company to react swiftly to inevitable and unforeseen changes in its internal and external business environment. According to another definition, agility is beyond flexibility, which is the ability to move from one job to another quickly, meeting unpredictable and changing customer expectations with varying amounts of products, in short lead times, high quality and low cost (Vokurka & Fliedner, 1998). After the concept of agile production emerged, it was an organizational, strategic philosophy that attracted great attention among managers, practitioners and academics, and with its simple and flexible perception, was an organizational, strategic philosophy that attracted much attention among the managers in industries (Dahmardeh & Banihashemi, 2010). When evaluated in terms of Industry 4.0, the rapid and radical change in recent years necessitates the re-evaluation of basic management concepts and theories in both theoretical and practical fields. Innovations and developments in all application areas, especially information, communication and production technologies, increase the importance of agility day by day in change management activities. When evaluated in this respect, there are different types of agility concept at the application stage.

Types of Agility

Agility includes operational, partnership and customer agility, as defined as a dynamic capability. Operational agility is defined as “the ability of organizations to take advantage of innovation and competitive action opportunities of business processes, to achieve speed, accuracy and cost economy.” Partnership agility is expressed as “the ability to use the assets, knowledge and competencies of suppliers, distributors, contract manufacturers and logistics providers through alliances, partnerships and joint ventures.” As another type of agility, customer agility is considered as “cooperating in researching and using opportunities for customers’ innovation and competitive action moves” ([Sambamurthy, Bharadwaj, & Grover, 2003](#)). Operational agility covers the internal environment of the organization within the framework of the innovation of processes and actions in operational steps. Partnership agility is concerned with all internal and external stakeholders, both inside and outside the organization. Customer agility, on the other hand, determines the framework of the organization’s agility in communication and relationship management with customers.

Organizational Agility and Agile Firms

Organizational agility means increasing the speed and flexibility of company-level inclusive processes, decisions and concerns. It deals with high-level process improvements to facilitate activities for efficiency and validity. All aspects of the business such as human capital management, operations, production/manufacturing, relationships (supplier or customer) and IT are reviewed and improved. Infrastructure and business culture play an important role to support changing business environments. A business must be flexible to anticipate or respond to changes, but there must also be standards and protocols to provide a model for rapid response ([Lu & Ramamurthy, 2011](#); [Zain, Rose, Abdullah, & Masrom, 2005](#)).

Organizational agility emerges as being quick in the face of changing market conditions, taking a position that can challenge these conditions and adopting a new way of doing business. However, it is also noted that a communication network should be established between management and employees in order to

respond rapidly to constantly changing market and customer demands ([Zain et al., 2005](#)).

[Akkaya and Tabak \(2018\)](#) also define organizational agility as the ability of an enterprise to adapt to the rapid changes of internal and external environmental factors, and the ability to respond quickly and effectively to customer demands and needs through technological developments.

It can be considered as organizational agility to identify changes in the internal and external environment of the organization and to realize the difficulties and opportunities that will create them. Their ability to find and use appropriate ways to respond to these changes in time explains agility. In this context, agility is a general outward-oriented capability that allows the firm to quickly adapt its operations to cope with unexpected changes and fluctuations in customer needs and market conditions ([Braunscheidel & Suresh, 2009](#)).

Agility in an organization can be handled in different ways. Organizational agility, which can provide different benefits to the firm, can be realized through two different application forms, entrepreneurial approach and adaptive approach ([Chakravarty, Grewal, & Sambamurthy, 2013](#)). The first one is a proactive (entrepreneurial) approach, which is seen as an opportunity of change that can positively affect the results of a company so that they act accordingly. The second adaptive approach is a reactive approach that occurs after an event occurs to request a change. Higher market volatility affects both agility types, reduces benefits and/or denies them altogether.

Agile organizations will need to provide products and services that can meet changing all stakeholders' demand patterns ([Aghina et al., 2018](#)). From another point of view, organizational agility is to build a unified strategy, paradigm, system or management practice on multi-faceted abilities, not just narrow scope as a skill ([Brannen & Doz, 2012](#); [Cao & Dowlatshahi, 2005](#); [Dahmardeh & Banihashemi, 2010](#); [İleri & Soylu, 2010](#); [Koçyiğit & Akkaya, 2020](#); [Li, Chung, Goldsby, & Holsapple, 2008](#); [Mehdibeigi, Dehghani, & Yaghoubi, 2016](#); [Sharifi & Zhang, 1999](#); [Yusuf, Sarhadi, & Gunasekaran, 1999](#)). [Zhang and Sharifi \(2000\)](#) developed a methodology for production businesses to

successfully implement agility and supported the methodology with the questionnaire they created.

According to the model, “Agility Drivers” means changes and pressures in the business environment that require businesses to look for different and new ways to maintain their competitive advantage. “Agility Capabilities,” on the other hand, are the basic abilities required for the enterprises to respond positively to the changes and turn them into an advantage. The tools needed to achieve these abilities are “Agility Providers.” A manufacturing company experiences various “Agility Drivers,” that is, various changes and pressures in its business environment, and this prompts businesses to investigate their “Agility Capabilities.” More specifically, the enterprise is exploring a variety of tools and ways to improve its capabilities or to have the skills it needs.

As seen in [Fig. 1](#), the main components of agility skills are responsiveness, competence, flexibility and speed ([Zhang & Sharifi, 2000](#)). The ability to respond means that how an enterprise detecting changes, responding quickly to changes and exiting these changes with an edge. This capability means recognizing, perceiving changes, being prepared in advance of the changes and responding instantly by including the changes in the system and exiting the changes with a manner of superiority. Competence capability is when an enterprise achieves its goals while providing efficiency and effectiveness. The elements that make up a large part of the competence ability are the following: strategic vision, appropriate technology or sufficient technological ability, product/service quality, cost effectiveness, high rate of new products to market, changing management, knowledgeable, expert and empowered employees, efficiency and effectiveness of transactions (simplicity), collaboration and integration inside and outside the business. Flexibility ability means that a business conducts different jobs with the same utilities and achieves different goals. Elements included in this ability are the quantity flexibility, product model flexibility, organizational flexibility and employee flexibility. Speed capability is when an enterprise performs tasks and operations as soon as possible. The elements included in this capability can be listed as being fast in introducing new products

to the market, fast and timely product/service delivery and being fast in operations which can be result as, short operational delivery time.

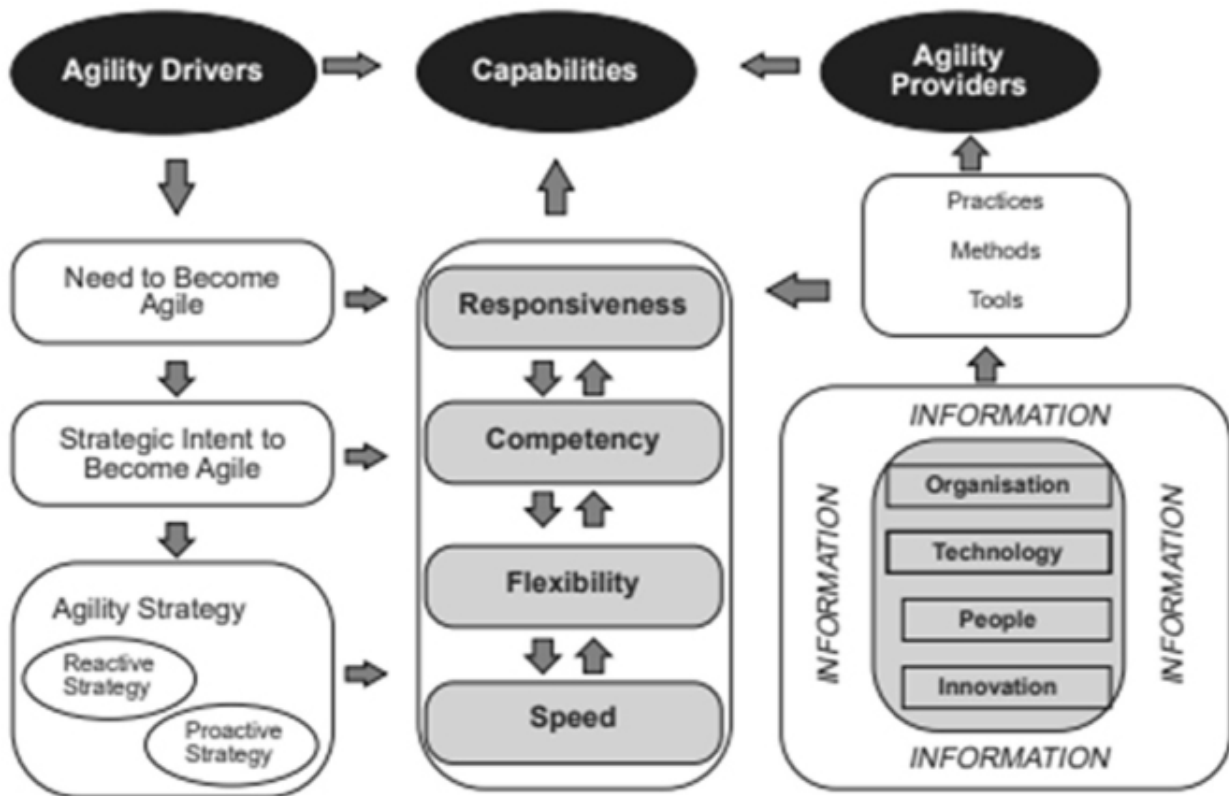


Fig. 1. The Conceptual Model for Implementing Agility. Source: [Zhang and Sharifi \(2000\)](#).

Businesses are affected at different levels by changes due to the differences in their environments. Therefore, enterprises have to have various talent combinations in order to respond to these changes in line with the strategies they set. This means that the level of agility that each business needs will be different ([Jin-Hai, Anderson, & Harrison, 2003](#), p. 176; [Zhang & Sharifi, 2000](#), p. 497). For example, the production of consumer goods and hardware and software that have reached the end of its development requires a high level of agility ([James-Moore, 1995](#)). According to [Sharifi and Zhang \(1999\)](#), agility drivers, the strategies and capabilities adopted by manufacturing companies have been classified into four different dimensions. These four major categories of capabilities are: responsiveness, competence, flexibility and speed. Responsiveness

can be defined as: feeling, perceiving and predicting changes, responding instantly by including changes in the system and recovery after change. Competence includes the strategic vision, appropriate technology (software and hardware), adequate technological skills, product and service quality, cost efficiency, new product offering rate, change management, knowledgeable, adequate and empowered individuals, activity effectiveness and efficiency (simplicity), internal and external cooperation and compliance. Flexibility is based on production volume flexibility, product model and shape flexibility, organizational flexibility, organizational issues flexibility and individual flexibility. Speed includes the speed of delivering new products to the market, the speed and punctuality of shipment of products and services and the fast operating time.

[Sharifi and Zhang \(1999\)](#) highlighted the four agility skills that are classified in [Table 2](#), and emphasized that these four features should be necessary for an enterprise to be agile. According to Sharifi and Zang, responsiveness is defined as the ability to detect changes and to respond quickly to these changes; and competence is the comprehensive ability to achieve efficiency and productivity in achieving organizational goals and objectives. While flexibility is stated as the ability to process different products and services with the same starting facilities for many different purposes; and speed is defined as the ability to perform tasks and functions at the maximum speed that can be achieved ([Zhang & Sharifi, 2000](#)).

Model of Organizational Agility Dimensions

In the model of [Meredith and Francis \(2000\)](#), organizational agility dimensions are defined as the Agile Wheel Reference Model (AWRM), which expresses the Organizational Agility components by showing them as a wheel, when a gear is missing, broken or fragile, the wheel weakens. In other words, if any of these 16 components are underdeveloped in agility, the organization's agility capacity is weakened. The purpose of the wheel is to help organizations control their agility capabilities, identify agility bottlenecks and develop a focused development plan ([Meredith & Francis, 2000](#)).

Components of Organizational Agility Model and AWRM

As seen in [Fig. 2](#), [Meredith and Francis \(2000\)](#), the “Organizational Agility Reference Model” focuses on the four basic dimensions of agility. These are Agile Strategy, Agile Operations, Agile Relations and Agile People.

Table 2. Agility Capabilities.

Responsiveness	Sensing, perceiving and anticipating changes Immediate reaction to changes Recovering from changes
Competency	Strategic vision Appropriate technology, or sufficient technological capability Products/service quality Cost-effectiveness High rate of new products introduction Change management Knowledgeable, competent, and empowered people Operations efficiency and effectiveness (leanness) Co-operation (internal and external) Integration
Flexibility	Product volume flexibility. Product model/configuration flexibility. Organisation and organisational issues flexibility People flexibility
Speed	Quickness in new products time-to-market Quickness and timeliness in products and services delivery Quickness in operations [short operational lead-times)

Source: [Sharifi and Zhang \(1999\)](#)

Quarter I: Agile Strategy. Agile Strategy, which is the first quarter of the wheel, focuses on four basic strategic applications of agility.

1. Wide Deep Scan: It concerns the organization’s procedures for collecting information from the outside environment. Multiple external factors such as competitors strategies, technological developments, customers’ needs, changes in the economic and market situation need to be monitored in terms of change management.

2. Strategic Commitment: It means that senior teams in agile firms must be adhered to the agile ideal and develop responsive, reconfigurable capabilities. In this way, it is possible for companies to change their own processes in order to quickly formulate and enforce their strategies in a quest to fulfill their unique needs of the customers instantly.

3. Full Deployment: It is about the level of agile policy and practices being adopted by all departments, groups, project teams and cells of organizations.

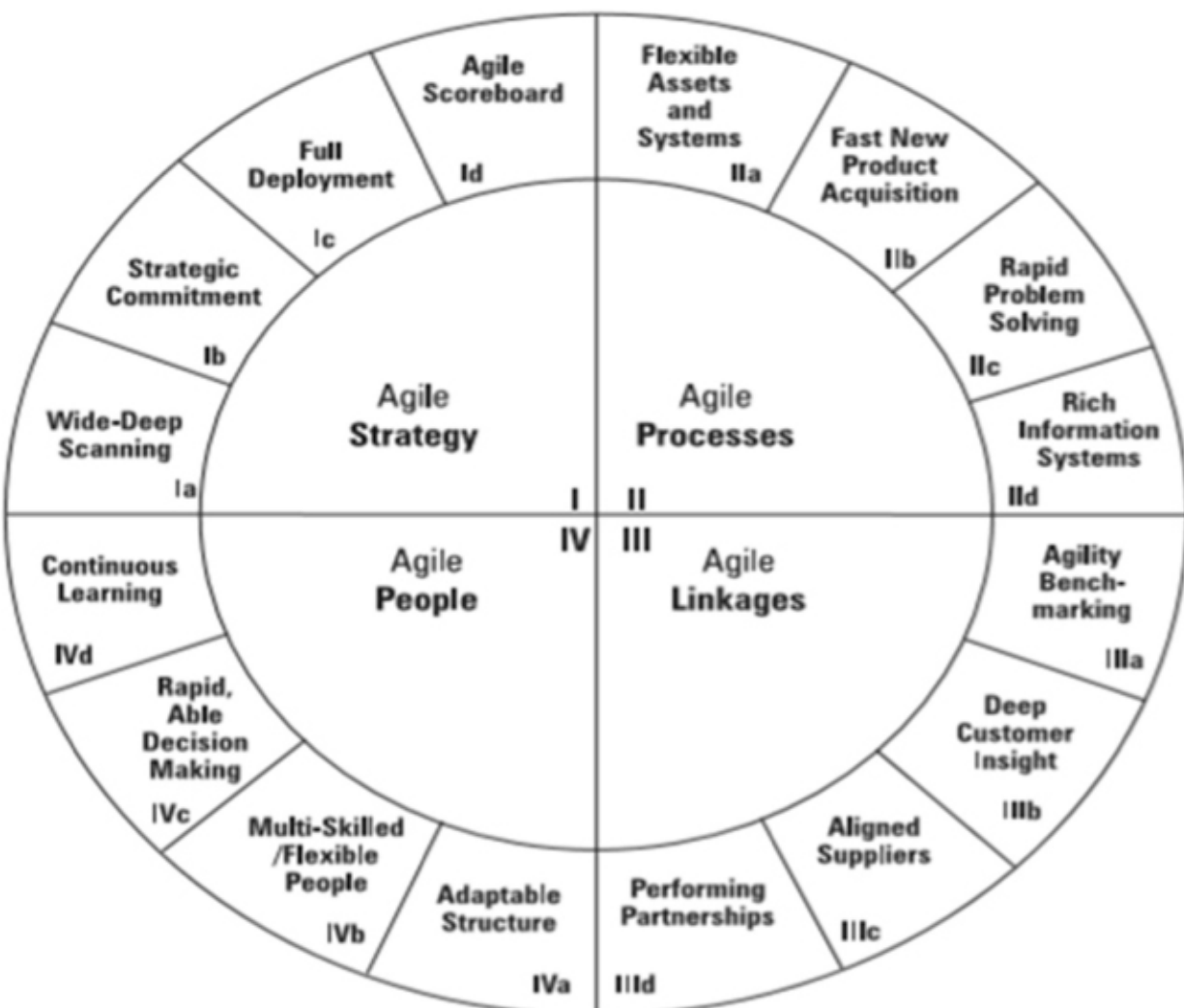


Fig. 2. Agile Wheel Reference Model (AWRM). Source: [Meredith and Francis \(2000\)](#).

4. Agile Scoreboard: It refers to the extent to which the organization's performance management system supports agile policies and practices. Agility needs to be measured and monitored in all process and performance outputs. This plays an important role in defining improvement goals.

Quarter II: Agile Processes. Agile Processes, the second quarter of the wheel, focus on four applications that support agility.

1. Flexible Assets and Systems: It covers all facilities, systems, software, buildings, services, production order, technologies, IT, equipment and control systems in the organization. It aims to increase the effectiveness of all assets and systems and processes.

2. Fast New Product Acquisition: Product "acquisition" refers to the relatively smaller scale companies' ability to obtain ideas, licenses and formats from others instead of new product development or improvement activities. When evaluated from this point of view, it is an important aspect of agility especially for SMEs.

3. Fast Problem Solving: If a firm is slow in identifying and solving problems, its creative energy is depleted by correction and minor improvement, not toward evaluating opportunities. Rapid response to problems must become a way of life. Problem symptoms should be identified quickly, and sufficient resources should be allocated to find an effective solution that can be implemented quickly.

4. Rich Information Systems: It helps ensure that decision making is shared and effective. In the decision-making process, both the information systems used in making the decision and participation have vital importance.

Quarter III: Agile Connections. Agile Connections, the third quarter of the wheel, focuses on the four policies/practices of external links:

1. Agility Benchmark: In determining agile targets, planning is done by benchmarking method, and the location of the companies in terms of flexibility, responsiveness, market position, technological innovation, technology application and human resources capacity is provided. Agile firm aims to be the first to act and does not wish to be a follower. The benchmark provides a source of ideas and examples.

2. Deep Customer Understanding: Companies that pursue an agile strategy strive to have built-in adaptability to meet changing

customer needs quickly and to work with customers to develop products or tackle problems. It is very important to respond quickly to changes regarding customer requests and needs, but it is not sufficient alone. The essence of agility is the rapid assessment of opportunities, an understanding of the signals in the market and the creation of sales opportunities.

3. Aligned Suppliers: For companies, suppliers are an integral element in the implementation of an agile strategy. Eliminating barriers between the firm and the supplier enable the company to share its goals, build long-term partnerships and share high levels of integration and accrued benefits.

4. To form a Partnership: Collaboration offers firms new or expanded capabilities. Firms can be linked together to create “virtual businesses.” Agility and flexibility can increase when companies partner to share their core competencies. Thus, they can develop projects or products that none of them can achieve alone. Such “virtual relationships” can be an effective way for companies to develop new technologies and products, supply critical resources, research new markets and complete core competencies.

Quarter IV: Agile People. Agile People, the fourth quarter of the wheel, focus on four policies/practices related to the company's human resource management:

1. Adaptable Organization Structure: Adaptability of the organizational structure required to grasp opportunities in organizational agility practices is very important. Due to the organizational flexibility achieved in this way, it will have an administrative application ground that will provide competitive advantage.

2. Multi-Talented, Flexible People: The existence of multiple talent and flexible people creates opportunities for companies to progress by developing systems with agile management (MBO, TQM, JIT, etc.). Although agile firms are less dependent on systems, it is an important requirement that the people's skills, knowledge, judgment, experience and intelligence be used to their full capacity.

3. Fast and Powerful Decision Making: An integrated organization where rich information systems allow information flow across the company helps decision making based on information, fast

and effective. When evaluated from this point of view, decision makers must be skilled in accordance with organizational agility in order to manage the information used in the entire system actively, effectively and efficiently.

4. Continuous Learning: It refers to the rate of employees in the firm being open to new ideas and adopting new practices. Learning speed, which is one of the most important competitive advantage factors, is required for sharing and increasing knowledge and experience ([Meredith & Francis, 2000](#)).

Industry 4.0 and Agile Firms

In today's competitive world, it is becoming increasingly difficult for companies to achieve production success, survival and sustainable growth. This fact stems from the emergence of a new business period with "change" as one of its key features. This critical situation has led to a major overhaul in the business priorities, strategic vision and feasibility of traditional and even relatively contemporary models and methods developed to date. Emphasis is now a proactive way to approach market and customer needs, with emerging collaboration methods such as adaptability to change in the business environment and virtual organization. The resulting paradigm is agile production, which is a strategic approach in the concept, which is a step in the production of new tools for better performance and success of the business and in practice, considering the new conditions of the business environment. Responding to changes, taking advantage of innovative and competitive opportunities in business processes, the strategic use of management and production methods and tools are the basic concepts of agile production ([Sharifi & Zhang, 1999](#)).

The digital age refers to a constant change and transformation that affects our entire life. However, this phenomenon should be handled not only for businesses but also for individuals and communities. It should also be discussed how to adapt to this formation and what the individual, social and cultural effects of all these transformations are. Whether in terms of transformation of economic habits or transformation of social behavior, individuals will be at the center of all these transformations ([Irge, 2020](#)).

Agile production is defined as the ability of the supply chain to adapt to change. While a business can be lean alone, agility is only possible if the entire supply chain is agile. Accurate, fast information flow is provided through early recognition with market sensitivity, simultaneous engineering with Strategic Supply Relations, Corporate cooperation with Process Integration not with operational, strong IT Infrastructure, thanks to the main Agile Manufacturing processes and applications.

As shown in [Fig. 3, Zhang and Sharifi \(2000\)](#) defined a multi-layered structure as a tool that shows the analysis the need of a company to be agile. At the top layer, the marketplace, competition, customer requirements, technology and social factors are classified as the five business aspects of business environment. Supplier problems are the variable for the complexity of external conditions, and the internal complexity of the company has been shown as the last aspect of the tool. Within the second layer, the 32 different sub-areas are broken down. For example, technology aspect has been broken down into two sub-areas such as technology change and introduction of new technologies. This tool is one of the most efficient tools that show the internal and external variables for the firms to define the needs for the agility.

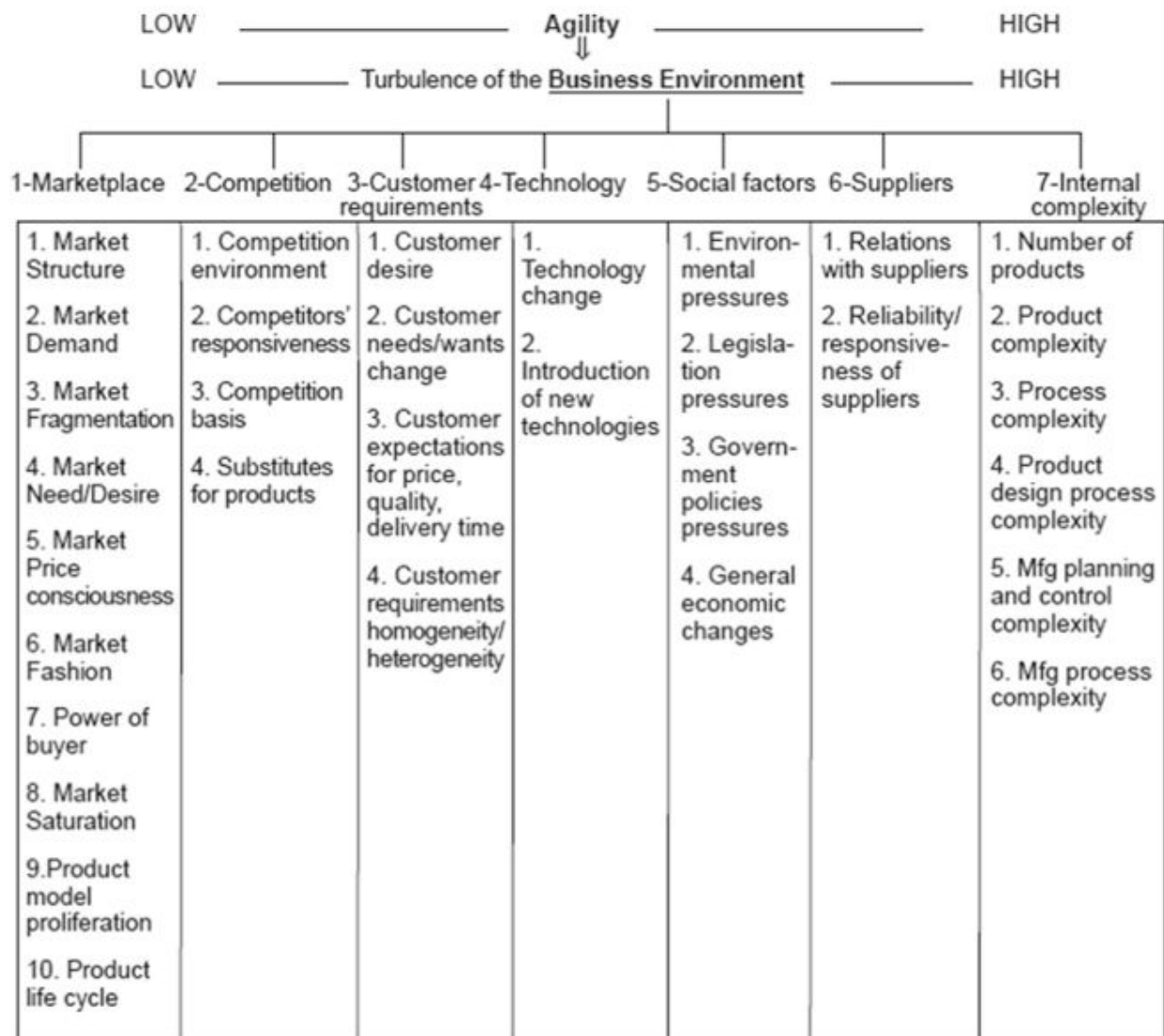


Fig. 3. Analyzing the Need of a Company to be Agile. Source: [Zhang and Sharifi \(2000\)](#).

Two characteristic features of agile companies are that they are fast and flexible. Quickness refers to the speed of completion; flexibility refers to the ability to adapt to changes in the environment. Agility means working quickly and being able to adapt immediately to changes in the environment. In other words, agility means being able to be both fast and flexible at the same time ([Sekman & Utku, 2009](#)).

In this context, for an organization to be agile, the adaptation of all human resources to the agile organizational structure and strategies, starting from the management level in internal environment is critical. At this point, the main difference between inert and agile companies

emerges. However, processes should also be designed to support the agility of human resources and functioning. Making agile processes is to bring processes that are fluent, simple and suitable for human nature. The criteria to be used for this should be definability, fluency, measurability, simplicity and conformity to human nature, flexibility, rapidity and repeatability ([İleri & Soylu, 2010](#)).

With the changing structure of the marketplace, competition, consumer profile and personalization of products and services, ability to respond immediately to changes in the external environment of the organization, at the same time, thanks to the organizational agility described as the ability to effectively perceive and react to opportunities in the market, organizations can foresee opportunities in the target market and take competitive steps by preparing the necessary knowledge and skills for these opportunities and can follow ways to improve their products and services ([Chung, Young Lee, & Kim, 2014](#), p. 609). While firms are doing so, they should also act with consideration for the organization's employees and stakeholders, not just their own goals and objectives ([Cegarra-Navarro, Soto-Acosta, & Wensley, 2016](#), p. 1544). In this respect, governance and stakeholder theory stand out in terms of both for agile firms and new management practices, when considered in this regard.

Conclusion

In today's world where the effects of globalization on all institutions and organizations are heavily discussed, the importance of sub-dimensions such as economical, technological, political, military, social, cultural and legal dimensions are emphasized in studies in different disciplines in the field of social sciences. The rapid and effective change in information, communication and production technologies indicates a new ecosystem or even a paradigm shift. The world lived in before the latest century was radically different in many aspects from the today's world ([Şen, 2019](#)).

Today, the fourth industrial revolution triggered by digital technologies is discussed. It is observed that nine technologies such as smart robots, big data, IoT, 3D printing and cloud technology have a very important role in triggering this revolution. The concept of

Industry 4.0, which emerged with this revolution, is now defined as the integration of parts which is beyond the automation of the pieces of value chains. The most important feature of integration is that all value chain steps are in real-time and are in continuous communication with each other, and by doing so, coming to the fore as the vision of reaching an intelligent and adaptive industry process. This vision defines a faster, more flexible, higher quality and more efficient industrial journey ([TÜSİAD Raporu, 2016](#)).

Organizations should adapt to a changeable environment and develop their strategies to deal with rapidly changing conditions to sustain growth in this critical Industry 4.0. This, certainly, can be achieved with the help of using their dynamic capabilities which help them stay competitive in the long term and sustainability ([Akkaya & Üstgörül, 2020](#)). Along with Industry 4.0, firms should not only be fast due to changing internal and external factors, but also include concepts such as flexibility, innovation and accountability. In addition, organizational agility has also started to be seen as an important factor in terms of the productivity of the organization as it will increase the organization's high quality product and service ratio ([Mehdibeigi et al., 2016](#), p. 96). Therefore, companies should not only aim to be faster, but they should also focus on building agility and adaptability. Companies aiming to gain competitive advantage by adapting to dynamic environmental factors such as technology and rapidly changing customer demands should bring their agility levels to the top ([YaşarUğurlu, Çolakoğlu, & Öztosun, 2019](#)). In other words, companies need to adapt and update their manufacturing and management system to preserve their potential for efficiency and innovation for today's economical purposes. It may be succeeded by dynamic capabilities in the companies ([Akkaya, 2020](#)).

In order to manage and survive the dynamic and changing business environment, the managers and leaders of the businesses have sought new solutions and have placed as much emphasis on adaptation and flexibility in this dynamic environment. Successful leading technology companies may sustain their high performance only if they keep being agile in that environment ([Akkaya & Tabak, 2020](#)). As a result of all these developments, the concept of agility, which is expressed as responding to the changes quickly and in

place, has been established. In this context, “organizational agility” has gained even more importance, which will keep the business in a competitive environment and meet customer demands and expectations quickly and on-site. The fact that both small and large enterprises adapt to the environment that changes so fast, adapt to the changes in this environment, avoid uncertainty and, most importantly, be prepared for unexpected developments, is one of the important problems faced by managers today. For this reason, companies must have dynamic abilities in order to ensure organizational agility and turn them into a holistic value (Akkaya, Kayalidere, & Tabak, 2019; Akkaya & Tabak, 2017).

Agility is seen as the most important success factor for businesses in today’s competitive and rapidly changing environments (Sharifi & Zang, 1999; Sherehiy, Karwowski, & Layer, 2007). Agility has become one of the most important business issues lately, especially in terms of digital business life and digitalized business manners. Although the software is more frequently mentioned in the data processing literature, agility can be considered as a character feature. Agility can be seen as being agile, quick and with the ability to adapt quickly to changes. In addition, being able to learn from failures and rise up again is an element of agility. Competition comes with both winning and losing. Being agile to engage in a good and strong competition for a firm means being ready for success and failure. The more organizational culture has the characteristics that will allow agility to emerge, the easier it will be for this character to rise. Review studies on what kind of agile business cultures should be at the organizational culture level are discussed and in the light of these studies, the characteristics of agile business cultures are determined (Cicerali, 2019).

As a result, with Industry 4.0, the profound effects and consequences of all the changing and developing structures, systems and applications around the world appear more and more. The data used in the field of corporate management level can be the subject of stakeholder theory. On the other hand, implementations like “Social Score/Credit” applications can be the examples of the social and political usage of data, databases, artificial intelligence, database management and all related new technology and programs used at

political governance and in various fields, all over the world (Şen, Gerni, & Ateşoğlu, 2018). In addition to the discussion that a major transformation in this scale is revolution or evolution, the high rate of change makes it difficult to analyze the period experienced. In this context, all stakeholders that are influential all over the world must make the necessary attempts and efforts to manage this transformation for the benefit of nature and humanity. The studies to be carried out with the focus of the “individual” at the center of the system, structure and institutions will enable societies to live in much higher prosperity and happiness in the periods after Industry 4.0. In this context, both the manufacturer and the consumer are of great importance (Şen, 2017). Both in strategic and operational terms, the necessity of structures and organizations where needs and practices are determined and discussed in the global arena comes to the forefront as an undeniable reality. In this context, the effects and consequences of the change and transformation that will occur in individuals, institutions and systems after the COVID-19 pandemic crisis in the field of management sciences should be evaluated (Şen, 2020). According to the possible new world of “Post-COVID-19,” the definition of the “New Normal” that may be needed to be renewed may base on three basic dimensions. The importance of “Organizational Agility,” “Intelligence and Intellectual Capital of the Organizations” and “Governance and Ethical and Moral issues for Organizations” may have the potential to be most dominant variables according to the all management theories and implementations. As the perspective of Senism, all the systems, organizations and individuals must be managed according to these three dimensions while all the radical changes and transformations are realizing in the field of new production, communication and information technologies. The importance of the defining individual within all these fields, within various contexts may be the most dominant scientific area of the future.

Recommendations for Future Studies

By the year 2020, the world was changing relatively very fast and all the factors that are causing for this rapid change and transformation

was questioned in detail within all scientific studies and other research reports and paper. With the starting of the pandemic from the new type of coronavirus COVID-19 in 2020 for almost all over the world, the change and transformation have been started to be in a much different structure and way. The concepts related with the globalization and all the technological, economical, political, social, cultural, military and ecological dimensions have started to be analyzed much with digitalization and concepts related with it. Agility, intelligence, ethics, competition, labour, strategy and all related terms and literature may be needed to be revised in the long term. This “New Normal” may continue with defining the individual’s role and existence within the production and consumption systems. Therefore, future studies may need to focus on the “individual” as the “new” starting variable for the social sciences fields.

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Chapter 13

Leadership and Industry 4.0 As A Tool to Enhance Organization Performance: Direct and Indirect Role of Job Satisfaction, Competitive Advantage and Business Sustainability

Kittisak Jermsittiparsert

Abstract

The primary objective of the current study is to examine the role of leadership styles and Industry 4.0 in organization performance. In addition, the direct and indirect effect of job satisfaction, competitive advantage and business sustainability is also examined. For this purpose, a cross-sectional research design was selected. Population of the study were based on the manufacturing Small and Medium Enterprises (SMEs) of Thailand. Employees of these SMEs were selected to collect the data. Therefore, respondents of the study were employees of Thai manufacturing SMEs. Various close ended questions related to the concerned study were asked on a five-point Likert scale. Furthermore, this study analyzed the data with the help of

Partial Least Square-Structural Equation Modeling (PLS-SEM). Results of the study revealed that both leadership styles and Industry 4.0 have significant role in organization performance. Leadership styles and Industry 4.0 influence positively on job satisfaction and competitive advantage, respectively. Further, job satisfaction and competitive advantage influence positively on business sustainability. Finally, business sustainability significantly improves organization performance. Therefore, the current study has vital importance for practitioners to enhance organization performance through leadership styles and Industry 4.0.

Keywords: Leadership style; Industry 4.0; organization performance; job satisfaction; competitive advantage; business sustainability

Introduction

Performance is significant element of every organization ([Altaf, Hameed, Nadeem, & Arfan, 2019](#); [Jermsittiparsert, Sutduean, & Sutduean, 2019](#)), which is the basis for survival in a competitive market. A certain level of performance is compulsory for every organization to meet the survival requirements in the present competitive environment. As the competition among organizations is growing day by day, therefore, organizations required higher level of performance to compete with their competitors. Therefore, organization performance is a vital part of organizations ([Arda, Bayraktar, & Tatoglu, 2019](#)) for success and survival.

More specifically, the performance of SMEs has more importance for different countries as SMEs play a vital role in various nations ([Jermsittiparsert & Rungsisawat, 2019](#)). SMEs are the significant contributor to the economic development ([Ayandibu & Houghton, 2017](#); [Jermsittiparsert, Siriattakul, & Wattanapongphasuk, 2019](#)). Particularly, in Thailand, SMEs also has important role in economic development. Especially, the manufacturing SMEs has vital role in

various nations. In the context of Thailand, manufacturing SMEs are described as firms with less than 200 employees and these SMEs should have fixed capital of 200 million baht, equivalent to 5.6 million USD. [Fig. 1](#) shows the contribution of Thai SMEs. In the year of 2019, Thai SMEs contributed 42% to the gross domestic product (GDP) of Thailand. These SMEs provided 78% of employment opportunities. Therefore, Thai SMEs has major role in economic development.

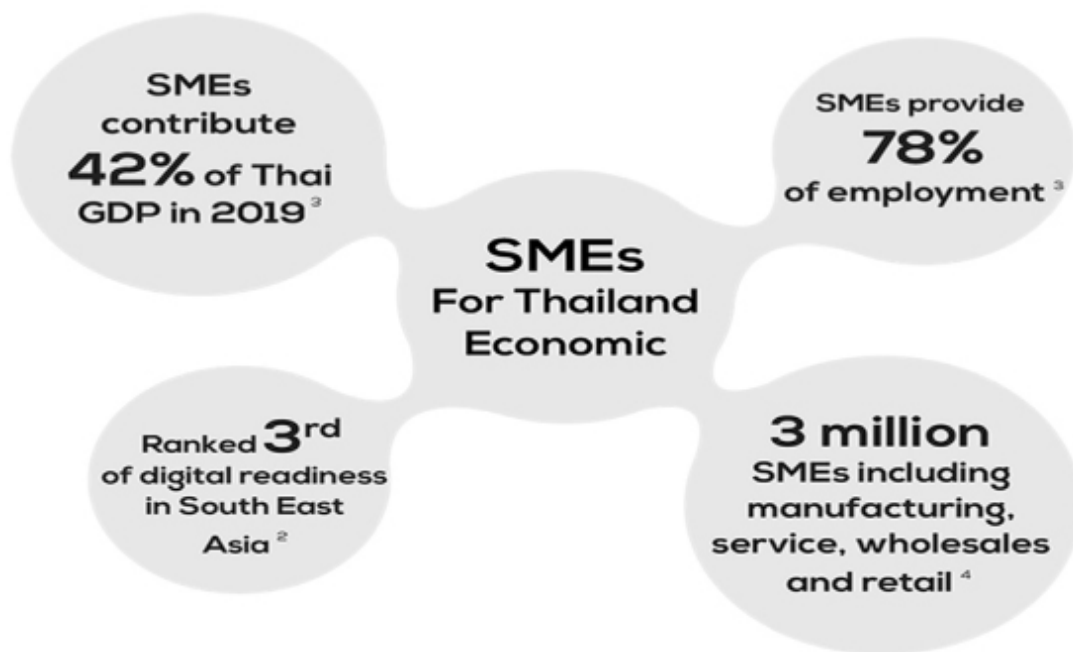


Fig. 1. Contribution of Thai SMEs. Source: [CEBIT ASEAN Thailand \(2019\)](#).

However, these SMEs are lacking with better leadership. It has significant influence on the performance of these SMEs, because leadership has important role in performance ([Miao, Eva, Newman, & Cooper, 2019](#); [Saleem, Bhutta, Nauman, & Zahra, 2019](#)). Better leadership can improve job satisfaction level and increases employee performance, which ultimately increases the business sustainability among the organizations. Various studies are examined the role of leadership in firm performance ([Chen, Sharma, Zhan, & Liu, 2019](#); [Hayat, Latif, Humayon, Ahmed, & Azeem, 2019](#)), however, these studies did not examine the business sustainability. Therefore, this study filled literature gap and examined the role of leadership on

organization performance through job satisfaction and business sustainability.

In addition, Thai SMEs are lacking with technological skills, for instance, Thai SMEs lacking with 17.9% to manage customer data, 15.7% lacking in digital skills, and talent and 14% in digital mindset, as shown in Fig. 2. However, these issues can be resolved with the help of leadership and Industry 4.0. As the Industry 4.0 technologies has key advantage to store big customer data. It is clear from the literature that big data have important role to store data (Gupta & Singla, 2019; Nidzwetzki & Güting, 2018). Therefore, the issue of customer data can be resolved with the help of big data technology of Industry 4.0. The other issues related to the digital skills and digital mind set can also be handled through Industry 4.0 and leadership.



Fig. 2. Issues in Thai SMEs. Source: [CEBIT ASEAN Thailand \(2019\)](#).

Scholars and practitioners are continuously working to explore various ways to enhance organization performance. Various studies investigated various methods to enhance organization performance. However, this study introduced the integration between leadership and Industry 4.0 as a tool to enhance organization performance and to resolve aforementioned issues. Various authors have explored the phenomenon of leadership and Industry 4.0 (Chen et al., 2019; Kamble, Gunasekaran, & Dhone, 2019; Kim & Brymer, 2011; Lalic, Majstorovic, Marjanovic, Delić, & Tasic, 2017; Shin, Sung, Choi, & Kim, 2015), however, these studies did not emphasized on job

satisfaction, competitive advantage, business sustainability along with integration between leadership and Industry 4.0. Thus, the current study contributed to the literature by developing a framework for Thai SMEs organization performance. Fig. 3 is the theoretical framework of the study showing the relationship between leadership style, Industry 4.0, job satisfaction, competitive advantage, business sustainability and organization performance.

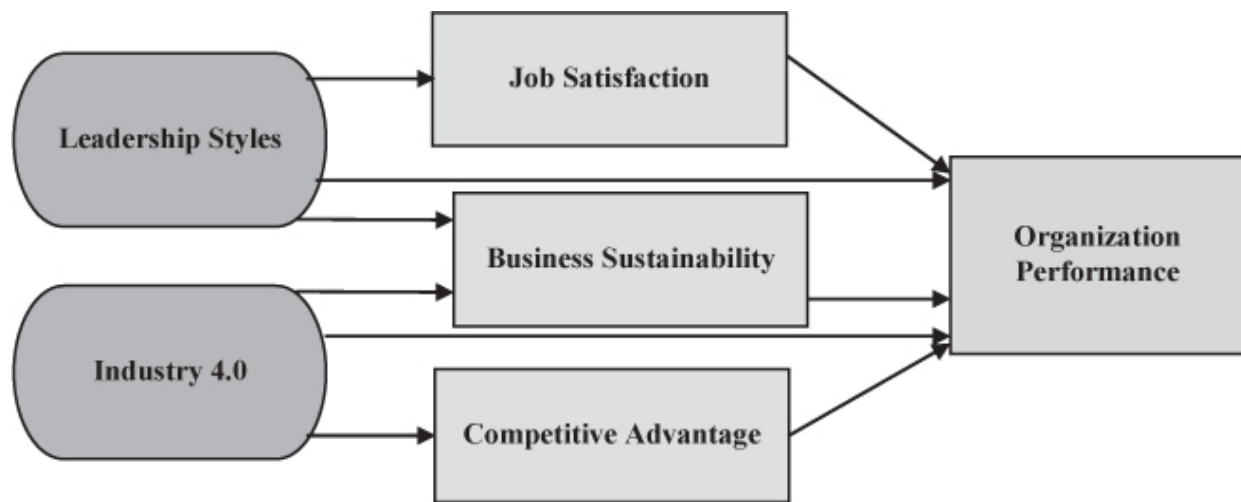


Fig. 3. Theoretical Framework of the Study Showing the Relationship Between Leadership Style, Industry 4.0, Job Satisfaction, Competitive Advantage, Business Sustainability and Organization Performance.

Literature Review

Leadership Styles

Leadership has significant contribution in various organizations (Ul-Hameed, Mohammad, Shahar, Aljumah, & Azizan, 2019). Leadership has many styles; however, transformational and transactional leadership are the major leadership styles. A leader who possesses autocratic leadership style will require followers to follow the orders without explanation. Meanwhile, for democratic leaders, they tend to be more persuasive in their leadership. There are some other leadership's styles which can also be found in organizations such as directing, participating, delegating, transformational and transactional.

However, competence leaders not only possess one single type of leadership style but will possess mixed elements of each style in their leads. Leaders which only use one single type of leadership style will be distinguished as inflexible and will face obstacles in some circumstances, where the chosen style can't be applied. Each of the leadership styles (directing, participating, delegating, transformational and transactional) will be illustrated more as follows which is applicable in various organizations. [Fig. 4](#) shows various leadership styles.

1) Directing Style: A leader who applied directing style is a leader centred where he/she will not request any input such as opinion from the followers. This type of leader will only give specifically instructions regarding the way of completing the task given. The communication style for directing leadership style is one of the ways, where all of the decisions will be made by the leader and delivered to followers for completion. The directing leader will supervise followers closely to avoid misconduct tasks. [Manning \(2004\)](#) stated that this leadership style is best to be used at the situation, where there is limited time to accomplish the given tasks. The directing leadership style may also be suitable to lead an individual or a team which is inexperienced as they are not yet being given training for performing particular task. Some people may misunderstand of such leadership style which it means for demeaning language or abusive authority including intimidation and threats to followers.



Fig. 4. Leadership Styles.

2) Participating Style: Second leadership which can be found in the organizations is participating style. This style centres both of the team and leader. The leaders of participating style will make final decision after obtaining recommendations, information and inputs from the followers (Manning, 2004). This style is applicable for a leader, who has plenty of time to have discussion with experienced followers for decision making. Followers tend to be given opportunities to participate in decision making for a plan. The approach which lies behind this type of leadership style is the team building approach. Participating style will also increase the followers' confident while performing the plan as they are involved.

3) Delegating Style: A leader who possess delegating style will grant followers authority in decision making and problem solving (Mwaisaka, K'Aol, & Ouma, 2019; Warsono & Riduwan, 2019). The followers can make the decision where problem arise without have to acknowledge the responsible leader. When a team consists of followers who are highly experienced, skilled, matured and with the given authority, it will create a good learning experience for the followers. Compared to other leadership style where a leader normally will take the responsibilities regarding the decision being made with the taken actions, delegating leader will let the followers' team leader to hold responsible for their group's actions.

4) Transformational Leadership: A leader who possess transformational style will tend to focus on the change and inspiration compared to transactional leadership style which punishment and reward become centre of attention. According to literature, followers will be challenged to rise above their self-interest and needs. Transformational leadership will emphasize on organizational improvement development of an individual. Transformational leaders create appropriate conditions for their followers to reveal their knowledge and skills and use them effectively and encourage their creativity to develop new methods and techniques ([Akkaya & Tabak, 2020](#)). Characteristics of transformational style are including mentally stimulating subordinates and empowering ([Manning, 2004](#)).

5) Transactional Leadership: According to [Manning \(2004\)](#), there are some of the techniques which used by the leaders who implemented transactional style such as offering rewards or punishment to motivate followers. A leader who solely relies on transactional style will receive followers' temporary commitment with discouragement in making innovation and risk taking. Although there are circumstances in organizations which make transactional style more preferable, but in long run, it will deprive followers' opportunities to grow. However, it has central role in various organizations ([Akkaya, 2020](#); [Delegach, Kark, Katz-Navon, & Van Dijk, 2017](#); [Masa'deh, Obeidat, & Tarhini, 2016](#); [Puspitasari, Minarsih, & Saputra, 2019](#)).

Organizational Performance

The dependent variable of organizational performance has become permanently interest by many researchers and scholars in any of the management's area. In stiff competition, business environment, the capital, inputs and customers' market competition are crucial factors for a firm to be considered for better success and survive. There are many studies done by researcher which realized that the performance for an organization is encompass multi areas such as customer related performance domain, operational and financial. Meanwhile, organization performance in business environment also covers some particular domains such as shareholder return, product market performance and financial performance which mentioned by

[Richard, Devinney, Yip, and Johnson \(2009\)](#)). Performance had been widely used as dependent variables in many researches. Majority of total articles from the *Administrative Science Quarterly*, the *Academy of Management Journal* and *Strategic Management Journal* which took period of three years' time to be evaluated are using the performance as dependent variable in respective research. It is an unpredictably open question which lacks of studies related in measure and defines organizational performance. Meanwhile, the definition and structure of performance which usually being used in management research are based on assumption regarding its form and appropriateness and seldom been clearly justified. In this research, there is different approaches for the organizational performance will be used which is based on operational performance.

Leadership Style and Organization Performance

Many scholars and researchers have the strong interest in analyzing and evaluating the association between the leadership style and the organizational performance ([Obiwuru, Okwu, Akpa, & Nwankwere, 2011](#)). Most of their findings reveal that there is existence significant connection between leadership style and performance of organizational. Depending the variables used by researchers, the result obtained included the leadership style has the negative and positive direction correlation with the organizational performance. Leadership style has the impact and significant relationship to the performance either in profit or non-profit organization. This is support by a research done by [Sun \(2002\)](#), where the researcher has analyzed the possible correlation direction between the leadership style and performance in enterprise and school. Result illustrates that there is existence positive correlation relationship between the two variables. One of the leadership styles, which had been found to be has the impact to the firm performance, is the transformational leadership where it shows a positive correlation ([Huang, 2006](#)).

For the business environment, it is found that administrative supervisors' leadership efficiency has the significant impact to the overall organization. Besides, the employees' work performance had been strongly influencing by the leadership style of a leader when

lead them toward achieving goals. Subordinates or followers have the responsibilities to provide feedbacks which related to the types of the leadership that a leader should have in order to better lead them. In the two way communication, both followers and leaders will be beneficial where leader may know the demands from followers and followers may know their role in affecting leader's style in leading.

Consequently, a leader should prepare in awarding followers to rise their job performance and indirectly to overall organizational performance. There are many researches' results showing that there is positively significant connection between the leadership style and the organizational performance. [Elenkov \(2002\)](#) described that the use of leadership style is to motivate and guide followers carefully and properly in order to achieve expected or beyond expected outcomes. When followers were being given enough respects, cares and concerns by the leaders' leadership style, they will tend to be work with better performance in the place of work ([Campbell, 1977](#)).

1) Relationship between Transactional Leadership and Organization Performance: Transactional leaders tend to increase the level of readiness within followers in an organization in order to achieve the projected outcomes with the rewards which equal to the efforts given. Therefore, transactional leaders have the influences to determine the organizational performance. A leader with skilled transactional leadership style will be able to execute appropriately empowerment, which is a powerful tool in leading within an organization. [Wall, Cordery, and Clegg \(2002\)](#) defined empowerment is a set of applications which involve responsibility entrustment to the lower hierarchical level in an organization, in which the subordinates will have more authority in decision making regarding their given work tasks. Meanwhile, other studies characterize empowerment as an interference activity, which purposely increases mental health, physical, proactively and motivation. A leader who possess transactional leadership style should behave in the sense of reduce followers' conflict to management actions, implement decisions and goal substitution with the purpose of sustain organization's performance quality and quantity.

One of the important organizational leaderships is the transactional leadership, where it is essential in order to be effectively in the

efficiency improvement and cost reduction competition. Nevertheless, depending only on the skills of transactional leadership is not sufficient for better efficiency and execution, and there is no guarantee that the survival of organization in such stiff, rising and rapid changing environment. Contingent reward that stressed in transactional leadership has the effective influences in creating lower order change which can bring significant improvement for the organizational achievement and excellent performance.

2) Relationship between Transformational Leadership and Organization Performance: There are researches which had been done to evaluate the relationship between transformational leadership style and performance either from objective measures (volume of sales, percentage of operational of use, etc.) or subjective measures (perception regarding leader by followers). [Pritzker \(2002\)](#) stated that common empirical studies are being completed by using the subjective measures to evaluate the correlation between transformational leadership and performance. It is important to evaluate the link between transformational leadership and performance as it is involved in presenting attractive convincing and future organizational vision which will influence the motivation level of the followers. This kind of leadership style will make changes for the followers and organization in order to achieve better performance which is beyond the expectations. There are enough evidences in past studies to show that there is existence connection between transactional leadership and organization performance. There is also experiential evidence that the four transformational leadership elements which are perfect influence, inspiring motivation and rational stimulation have statistically significant influence on numerous indices of performance, firm performance and business results.

From all of the evidences which are including empirical evidences by some of the researchers reveal that there is important relationship between transformational leadership and organization performance. Therefore, the transformational leadership style will be used as one of the variables to analyze the impact to the performance. There are also a number of studies available which show that leadership has positive relationship on organizational performance ([DeConinck, Moss, & Deconinck, 2018](#); [Hartnell, Kinicki, Lambert, Fugate, & Doyle](#)

[Corner, 2016](#)). Therefore, by following the above discussion, below hypotheses is proposed:

H1: Leadership styles has positive effect on organizational performance.

Leadership Styles, Job Satisfaction and Business Sustainability

Job satisfaction is explained as the degree to which an employee of organization feels self-motivated, more satisfied with job. It happens at the time when an employee feels job stability in concerned organization, significant career growth as well as a happy work life balance. Business sustainability is based on the sustainable growth of the business. If the business is working well in the given environment and achieving its targets, it will be sustainable business. In the competitive environment, business sustainability is one of the crucial issues ([Dyllick & Muff, 2016](#)). There are various measures of business sustainability. Sometimes it is measured based on growth in sales of the company, profits, as well as returns on investment (ROI) and they are the main standards for assessing business sustainability. Sales growth means to the total change in percentage in the sales revenue from the beginning to the end in a stated period. Generally, it is planned by using the estimate products revenue, where the entire product unit shaped is reproduced with the price of sales ([Ishikawa, Fujimoto, & Tomoyose, 2010](#)).

It is clear from the literature that leadership styles have relationship with job satisfaction. Job satisfaction is most important in an organization ([Hussain, Rizwan, Nawaz, & ul Hameed, 2013](#)), which can be achieved with the help of leadership. Better leadership styles have the ability to increase satisfaction level among the employees of an organization. More the satisfaction is, more will be the performance of employees ([Purnama, 2017](#); [Yim, Zou, & Chan, 2018](#)) and an increase in employees' performance will definitely increase the business sustainability ([Lee & Ha-Brookshire, 2017](#)). Various studies are also available, which shows that leadership style has positive effect on job satisfaction ([Chandra, 2016](#); [Olaniyan & Hystad, 2016](#)). Moreover, various studies are also available which shows that

job satisfaction has positive role in business sustainability. An increase in job satisfaction has the ability to boost business performance (Al Idrus, Ahmar, & Abdussakir, 2018; Kanyurhi & Bugandwa Mungu Akonkwa, 2016). Therefore, it is clear that, leadership styles have positive effect on job satisfaction, and job satisfaction has positive effect on business sustainability. Thus, these discussion lead to the following hypotheses:

H2: Leadership styles has positive effect on job satisfaction.

H3: Job satisfaction has positive effect on business sustainability.

Industry 4.0

Fourth industrial revolution (Industry 4.0) is a term for the present trend of automation as well as computerization in various newest technologies. Industry 4.0 is much prevalent and widely debated topic in numerous social groups, scholars, industry as well as in the academic field of diverse scholars (Sae-Lim & Jermsittiparsert, 2019). First revealing in 2011 by the German federal government, with the time it has developed a top superiority for several research centres, companies as well as universities. Now it has great importance for businesses. The logic behind rapid growth is that it has been evaluated that the benefits of Industry 4.0 will donate maximum to the newest technology. Now, the Industry 4.0 technologies are widely adapted in most of the developed countries such as United States, Germany and Australia. With the receiving of this idea as a developing research topic for the very first time, an industrial revolution is being forecasted, and practitioners and academics have covered the true meaning as well as definition of Industry 4.0 (Bauernhansl, Ten Hompel, & Vogel-Heuser, 2014). Particularly, for the first time it is studied with leadership styles. Although there are numerous definitions upon which “experts” agree, but there are numerous concepts as well as criteria are required to discuss like the emphasis of this study with the relationship of Industry 4.0 with leadership styles, job satisfaction, business sustainability and

organization performance ([Haseeb, Hussain, Slusarczyk, & Jermisittiparsert, 2019](#)). Fig. 5 shows the emergence of Industry 4.0.

1) Industry 4.0 and Organization Performance: In the recent years, Industry 4.0 has importance in the era of mechanization ([Fitzgerald, Wang, & Karanassios, 2019](#)). Many organizations are now concentrating on Industry 4.0 to encourage total performance. It has important benefits for the economy of each nation. It has modern technologies which aid for organization performance to work efficiently with lowest time. Rise in the accuracy as well as decrease in time has significant role to contribute to the performance. Now, Industry 4.0 technologies are the highest priorities of companies to enhance performance.

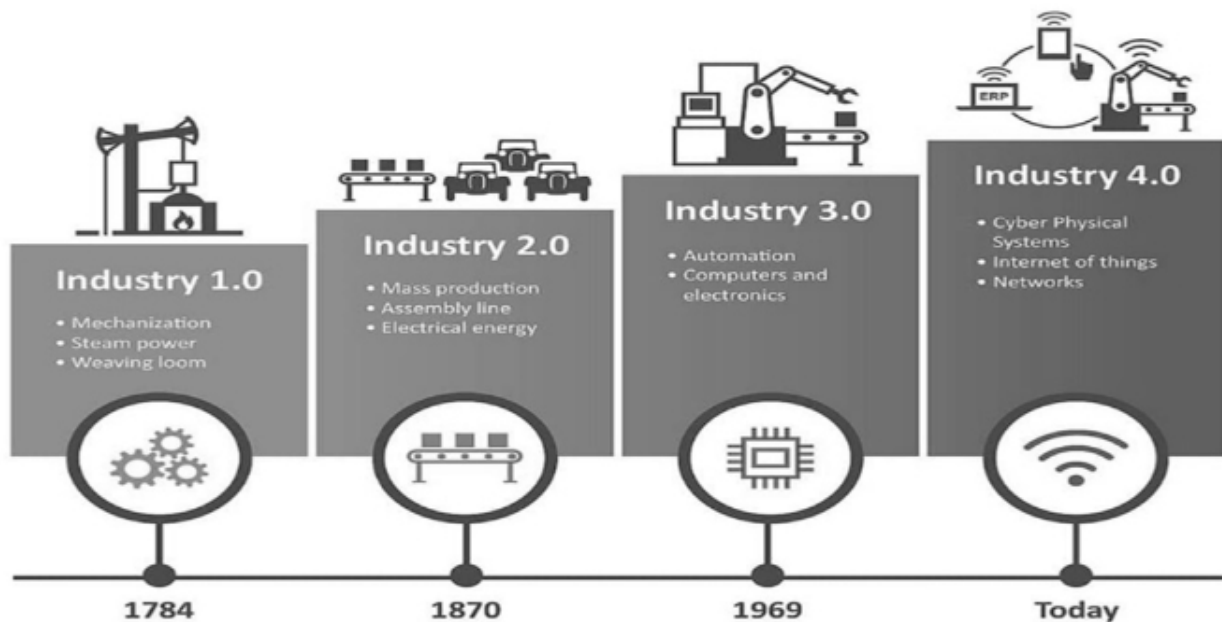


Fig. 5. Emergence of Industry 4.0. Source: [Chaudhary \(2019\)](#).

Industry 4.0 has five major components, which has major role in economic development of various countries. Five main components comprise technologies connected to the internet of things (IoT), big data, Cyber-physical system, smart factory and interoperability. These elements have noteworthy consequence on business growth which influence positively on growth of companies. Previous studies are also highlighted that Industry 4.0 has positive role in organization

(Bhagawati, Manavalan, Jayakrishna, & Venkumar, 2019; Petrisor & Cozmiuc, 2020; Rane & Narvel, 2019). Moreover, latest technologies always show positive role in the business performance (Malhotra, 2005; Tsui & Malhotra, 2005). Thus, following hypothesis is proposed:

H4: Industry 4.0 has positive effect on organization performance.

Industry 4.0, Competitive Advantage and Business Sustainability

Fast process of technology and digital world are taking place in all organizational authoritative in all area and in all kind of sectors, because the business world is transformed by the postmodern revolution-fourth industrial revolution (Akkaya, 2019). In a business environment, a competitive advantage is the characteristic that permits an organization to outperform its competitors. Generally, competitive advantage is based on both tangible and intangible resources. Tangible and intangible resources are the major properties which help to gain competitive advantages. Good skills of the employees of an organization could be the competitive advantage. Good environmental condition also could be the basis of competitive advantage (Ploenhad, Laoprawatchai, Thongrawd, & Jermisittiparsert, 2019). Apart from all these elements, latest technology is one of the major competitive advantages. And latest technology could be handled with the help of Industry 4.0 factors. Industry 4.0 has significant contribution to attain competitive advantage and survive in a competitive environment. Previous studies also mentioned an important relationship between Industry 4.0 and competitive advantage (Adamik & Nowicki, 2018, 2019; Majeed & Rupasinghe, 2017; Zhou, Liu, & Zhou, 2015). As the technologies are the fundamentals of competitive advantage (Mao, Liu, Zhang, & Deng, 2016; Nainaar & Masson, 2018) and Industry 4.0 is only based on to the introduction of new technologies, Industry 4.0 has positive role to enhance comparative advantage. In addition, an increase in competitive advantage increases the business sustainability (Haseeb, Hussain, Kot, Androniceanu, & Jermisittiparsert, 2019). Various studies proved that competitive advantage has positive role in business performance (Pereira-Moliner et al., 2015; Saeidi, Sofian,

Saeidi, Saeidi, & Saaeidi, 2015) and business performance increases the business sustainability among the organizations. Competitive advantage provides the strengths to the organization to sustain in a competitive condition. Thus, competitive advantage has an important role in business sustainability (Cantele & Zardini, 2018). Accordingly, from previous studies, below hypotheses are proposed:

H5: Industry 4.0 has positive effect on competitive advantage.

H6: Competitive advantage has positive effect on business sustainability.

Business Sustainability and Organization's Performance

Business sustainability and organizational performance are closely related to reach others. Both sustainability and organizational performance lead to the higher efficiency, and both these elements are important to survive in a competitive environment. Performance can be measured in both financial and non-financial ways. Various studies have considered performance in financial measures with the accounting-based performance measures such as related to market shares, product and profits (Richard et al., 2009). It has significant relationship with business sustainability (Morioka & de Carvalho, 2016). Generally, an increase in business sustainability increases the organization performance both in financial and non-financial terms. Previous studies also investigated a significant link between the business sustainability and performance (Hanelt, Busse, & Kolbe, 2017; Schaltegger & Wagner, 2017). Therefore, this study also proposes that business sustainability has positive effect on organization performance.

H7: Business sustainability has positive effect on organization performance.

Additionally, from above discussion, literature proved that leadership styles, Industry 4.0, job satisfaction, competitive advantage, business sustainability and organization performance

have significant relationship with each other. In this situation, by following the instructions of [Baron and Kenny \(1986\)](#), situation is ideal to take job satisfaction, competitive advantage and business sustainability as mediating variables. All three paths mentioned by [Baron and Kenny \(1986\)](#) are significant to propose indirect hypotheses. Thus, to examine the mediation effect, following indirect hypotheses are proposed:

H8: Job satisfaction mediates the relationship between leadership styles and business sustainability.

H9: Competitive advantage mediates the relationship between Industry 4.0 and business sustainability.

H10: Business sustainability mediates the relationship between job satisfaction and organization performance.

H11: Business sustainability mediates the relationship between Industry 4.0 and organization performance.

Research Methodology

This section of the study deals with a method to achieve end results based on empirical evidences. Based on the above discussion with the help of previous studies, the relationship between leadership styles, Industry 4.0, job satisfaction, competitive advantage, business sustainability and organization performance was examined, and 11 hypotheses were proposed. To test the hypotheses, primary data were collected from the employees of manufacturing SMEs in Thailand. The reason to select manufacturing SMEs is that manufacturing process requires more technological and innovative methods to develop new products. As compared to the service sector, manufacturing sector requires more technology related to the fourth industrial revolution.

Population of the study was based on the manufacturing SMEs of Thailand. Employees of these SMEs were selected to collect the data. Therefore, respondents of the study were employees of Thai

manufacturing SMEs. Thus, unit of analysis was individual. For this purpose, cross-sectional research design was selected, and primary data were collected with the help of questionnaire. This study carried out a research survey to collect primary data. A survey questionnaire was developed by using the previous studies. Various close ended questions related to the concerned study were asked on a five-point Likert scale. Questionnaire was based on two major sections: the first section was representing the profile of respondents and second section was based on the scale items related to key variables: leadership styles, Industry 4.0, job satisfaction, competitive advantage, business sustainability and organization performance.

Sample size of the study was determined by using [Comrey and Lee \(1992\)](#) instructions. [Comrey and Lee \(1992\)](#) provided the sample in a series of inferential statistics.

Sample having less than 50 participants will be observed to be a weaker sample; sample of 100 size will be weak; 200 will be adequate; sample of 300 will be considered as good; 500 very good whereas 1000 will be excellent.

Therefore, five hundred (500) sample size was selected. Five hundred (500) questionnaires were distributed among the Thai SMEs. Among these distributed questionnaires, three hundred and ten (310) valid responses were returned and used for data analysis. Furthermore, this study analyzed the data with the help of PLS-SEM.

Data Analysis and Findings

Before data analysis, data screening is most important to get original results. Therefore, before further analysis, this study carried out data screening to examine the missing value, outlier and normality of the data. It is evident from [Table 1](#), and the data have no missing value and free from any case of outlier. Thus, it is supposed that data are accurate to proceed for further analysis.

Suitable tool for data analysis is always a key to get original results. There are various techniques and software's are available to

analyze the data. However, to analyze the primary data, PLS is widely used tool. Generally, PLS-SEM is most popular technique to analyze the primary data. PLS-SEM is based on two major steps, including: measurement model and structural model ([Reinartz, Haenlein, & Henseler, 2009](#)). The measurement model is based on confirmatory factor analysis. The structural model is based on the hypotheses testing. [Fig. 6](#) shows the measurement model.

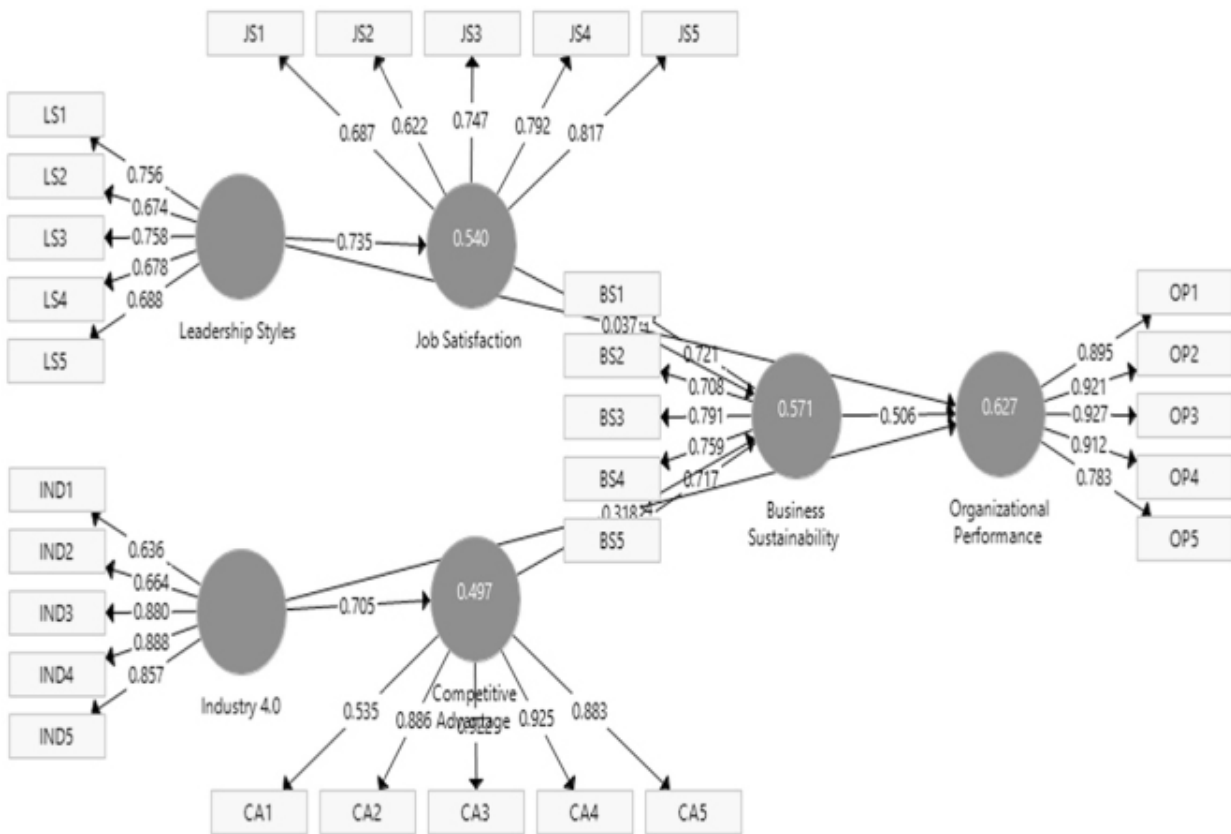


Fig. 6. Structural Model.

Confirmatory factor analysis is highlighted in [Fig. 6](#). All the factor loadings are presented in [Table 2](#). By following the recommendations of previous studies, 0.5 threshold level was considered for factors loadings. It is found that all the items have factor loadings above 0.5. Moreover, [Table 3](#) indicates the composite reliability (CR) and average variance extracted (AVE). All the constructs achieved the minimum level of CR (0.7) and AVE (0.5). For discriminant validity,

cross-loadings are highlighted in [Table 4](#) which confirmed that data are valid.

Table 1. Data Screening.

	Missing	Mean	Median	Min	Max	SD	Kurtosis	Skewness
LS1	0	3.961	4	1	5	0.968	0.555	-0.89
LS2	0	3.744	4	1	5	1.078	-0.246	-0.631
LS3	0	3.615	4	1	5	1.159	-0.581	-0.538
LS4	0	4.084	4	1	5	0.982	0.925	-1.118
LS5	0	4.181	4	1	5	0.836	1.696	-1.121
IND1	0	3.903	4	1	5	1.066	-0.175	-0.788
IND2	0	3.835	4	1	5	1.191	-0.402	-0.774
IND3	0	3.683	4	1	5	1.133	-0.527	-0.546
IND4	0	3.731	4	1	5	1.164	-0.423	-0.651
IND5	0	3.803	4	1	5	1.087	-0.336	-0.619
JS1	0	3.764	4	1	5	1.079	0.074	-0.714
JS2	0	3.961	4	1	5	0.891	1.435	-1.026
JS3	0	3.964	4	1	5	1.019	-0.356	-0.685
JS4	0	4.126	4	1	5	0.966	0.384	-0.949
JS5	0	3.987	4	1	5	0.962	0.054	-0.763
CA1	0	4.006	4	1	5	0.985	0.555	-0.951
CA2	0	3.625	4	1	5	1.071	-0.342	-0.509
CA3	0	3.77	4	1	5	1.05	-0.098	-0.675
CA4	0	3.786	4	1	5	1.082	-0.423	-0.629
CA5	0	3.725	4	1	5	1.129	-0.416	-0.624
BS1	0	3.896	4	1	5	1.025	0.743	-0.969
BS2	0	3.851	4	1	5	0.981	0.137	-0.753
BS3	0	3.867	4	1	5	0.92	0.163	-0.66
BS4	0	3.871	4	1	5	0.864	0.359	-0.563
BS5	0	3.683	4	1	5	1.245	-0.694	-0.574
OP1	0	3.673	4	1	5	1.241	-0.915	-0.462
OP2	0	3.828	4	1	5	1.152	-0.326	-0.721
OP3	0	3.825	4	1	5	1.169	-0.748	-0.608
OP4	0	3.825	4	1	5	1.216	-0.728	-0.639
OP5	0	3.311	3	1	5	1.231	-0.93	-0.243

Table 2. Factor Loadings.

	Business Sustainability	Competitive Advantage	Industry 4.0	Job Satisfaction	Leadership Styles	Organizational Performance
BS1	0.721					
BS2	0.708					
BS3	0.791					
BS4	0.759					
BS5	0.717					
CA1		0.535				
CA2		0.886				
CA3		0.922				
CA4		0.925				
CA5		0.883				
IND1			0.636			
IND2			0.664			
IND3			0.88			
IND4			0.888			
IND5			0.857			
JS1				0.687		
JS2				0.622		
JS3				0.747		
JS4				0.792		
JS5				0.817		
LS1					0.756	
LS2					0.674	
LS3					0.758	
LS4					0.678	
LS5					0.688	
OP1						0.895
OP2						0.921
OP3						0.927
OP4						0.912
OP5						0.783

Table 3. Construct Reliability and Validity.

	Alpha	rho_A	CR	(AVE)
Business Sustainability	0.803	0.817	0.858	0.548
Competitive Advantage	0.888	0.895	0.923	0.711
Industry 4.0	0.849	0.874	0.892	0.629
Job Satisfaction	0.786	0.794	0.855	0.543
Leadership Styles	0.758	0.768	0.837	0.507
Organizational Performance	0.933	0.933	0.949	0.791

Table 4. Cross-loadings.

	Business Sustainability	Competitive Advantage	Industry 4.0	Job Satisfaction	Leadership Styles	Organizational Performance
BS1	0.721	0.599	0.567	0.534	0.716	0.514
BS2	0.708	0.329	0.382	0.37	0.436	0.373
BS3	0.791	0.459	0.416	0.471	0.498	0.416
BS4	0.759	0.426	0.438	0.481	0.477	0.405
BS5	0.717	0.644	0.621	0.564	0.616	0.851
CA1	0.565	0.735	0.498	0.58	0.626	0.426
CA2	0.607	0.886	0.603	0.516	0.638	0.599
CA3	0.618	0.922	0.61	0.564	0.639	0.604
CA4	0.595	0.925	0.612	0.57	0.631	0.575
CA5	0.558	0.883	0.619	0.592	0.659	0.569
IND1	0.491	0.366	0.636	0.546	0.546	0.451
IND2	0.513	0.46	0.664	0.504	0.567	0.457
IND3	0.589	0.652	0.88	0.626	0.677	0.623
IND4	0.563	0.624	0.888	0.584	0.653	0.621
IND5	0.578	0.634	0.857	0.679	0.638	0.576
JS1	0.549	0.629	0.802	0.887	0.65	0.563
JS2	0.448	0.467	0.45	0.622	0.465	0.351
JS3	0.44	0.348	0.442	0.747	0.442	0.396
JS4	0.437	0.438	0.434	0.792	0.48	0.363
JS5	0.574	0.533	0.518	0.817	0.606	0.471
LS1	0.614	0.603	0.516	0.513	0.756	0.448
LS2	0.471	0.459	0.47	0.413	0.674	0.457

LS3	0.593	0.67	0.795	0.591	0.758	0.661
LS4	0.517	0.455	0.453	0.472	0.678	0.378
LS5	0.531	0.487	0.458	0.598	0.688	0.4
OP1	0.692	0.626	0.644	0.546	0.606	0.895
OP2	0.681	0.604	0.617	0.528	0.612	0.921
OP3	0.652	0.621	0.653	0.551	0.633	0.927
OP4	0.611	0.536	0.577	0.479	0.568	0.912
OP5	0.699	0.562	0.586	0.54	0.564	0.783

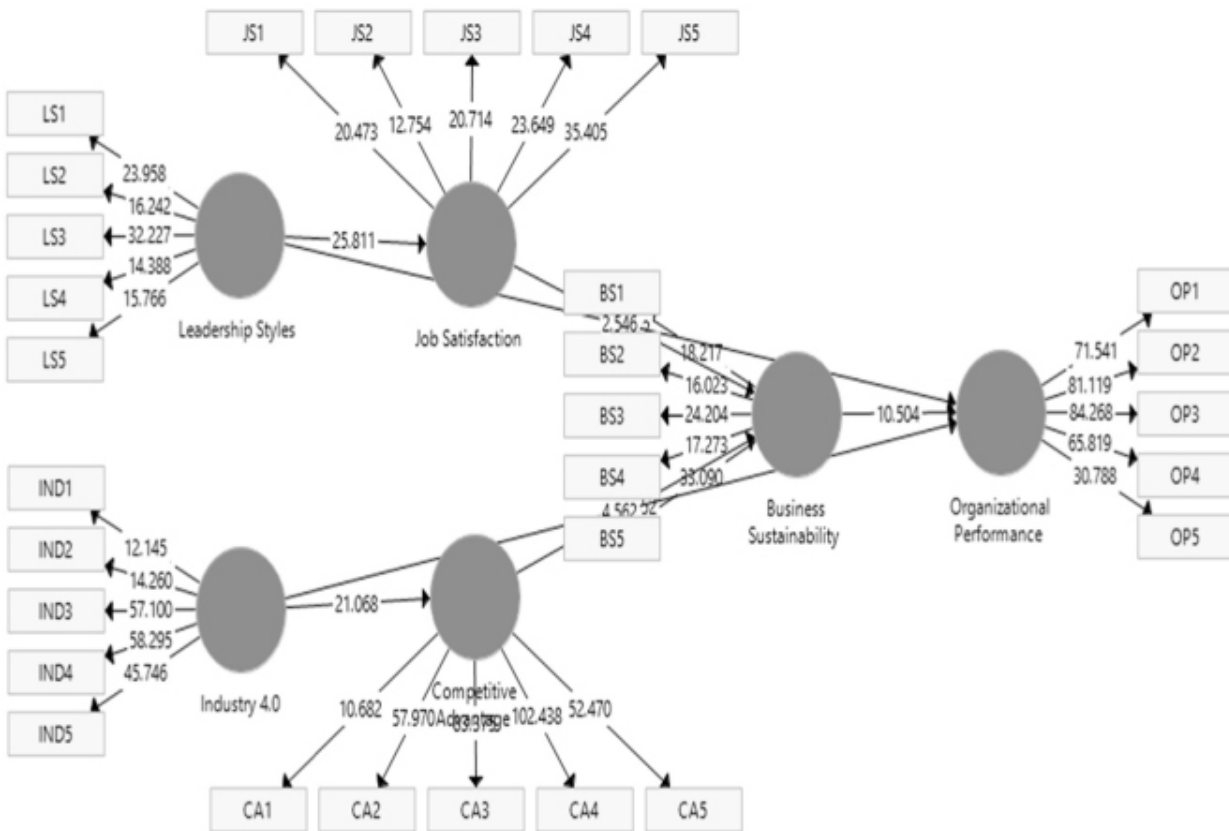


Fig. 7. Measurement Model.

After the assessment of reliability and validity, a structural model was carried out and presented in Fig. 7. In this step of PLS-SEM, the data were analyzed to test the hypotheses. Table 5 shows the results of direct effect. T-value 1.96 was considered to examine the significance level between various relationships. It is found that the

leadership style has positive effect on job satisfaction with t-value 25.811. Industry 4.0 has positive role to enhance competitive advantage with t-value 21.068. Furthermore, both leadership style and industry 4.0 have positive effect on organizational performance with t-value 2.546 and 4.562, respectively. Additionally, both job satisfaction and competitive advantage have significant effect on business sustainability with t-value 83.12 and 10.352, respectively. Therefore, all the direct hypotheses are supported (*H1, H2, H3, H4, H5, H6, and H7*).

Table 5. Direct Effect.

	(O)	(M)	SD	t Statistics	P Values
Business Sustainability -> Organizational Performance	0.506	0.509	0.048	10.504	0
Competitive Advantage -> Business Sustainability	0.454	0.456	0.044	10.352	0
Industry 4.0 -> Competitive Advantage	0.705	0.707	0.033	21.068	0
Industry 4.0 -> Organizational Performance	0.318	0.318	0.07	4.562	0
Job Satisfaction -> Business Sustainability	0.371	0.371	0.045	8.312	0
Leadership Styles -> Job Satisfaction	0.735	0.736	0.028	25.811	0
Leadership Styles -> Organizational Performance	0.037	0.037	0.015	2.546	0.01

Table 6. In-direct Effect.

	(O)	(M)	SD	t Statistics	P Values
Industry 4.0 -> Competitive Advantage -> Business Sustainability	0.32	0.322	0.034	9.44	0
Leadership Styles -> Job Satisfaction -> Business Sustainability	0.273	0.273	0.037	7.307	0
Competitive Advantage -> Business Sustainability -> Organizational Performance	0.23	0.232	0.032	7.117	0
Industry 4.0 -> Competitive Advantage -> Business Sustainability -> Organizational Performance	0.162	0.164	0.023	7.061	0
Job Satisfaction -> Business Sustainability -> Organizational Performance	0.188	0.189	0.026	7.127	0
Leadership Styles -> Job Satisfaction -> Business Sustainability -> Organizational Performance	0.138	0.139	0.021	6.57	0

After the assessment of direct effect, indirect effect was also examined. [Table 6](#) shows the indirect effect of different variables. It is evident that competitive advantage is a mediating variable between industry 4.0 and business sustainability with t-value 9.44. Moreover, job satisfaction is also a mediating variable between leadership styles and business sustainability with t-value 7.307. In addition, business sustainability is also a mediating variable between the competitive advantage and the organization performance with t-value 7.117. Finally, business sustainability is a mediating variable between job satisfaction and organization performance with t-value 7.127. Thus, all the mediation hypotheses are accepted (*H8, H9, H10, and H11*).

To examine the expected variances, r-square (R^2) was examined. It is shown in Fig. 4. The r-square (R^2) value is 0.627 which is substantial (Chin, 1998). Thus, all the variables, namely, leadership styles, Industry 4.0, job satisfaction, competitive advantage and business sustainability are expected to bring 62.7% change in the organization performance.

Discussion and Conclusion

The major objective of this study was to examine the role of leadership styles and Industry 4.0 in the organization performance. Moreover, along with major objective, the effect of job satisfaction, competitive advantage and business sustainability was also examined. A cross-sectional research design was selected, and the survey was carried out to examine the relationship between leadership styles, Industry 4.0, job satisfaction, competitive advantage, business sustainability and organization performance.

Findings of the study revealed that both leadership styles and Industry 4.0 have significant role in organization performance. Better leadership can increase the performance. These results are consistent with previous studies such as Chang, Chen, and Chiou (2015), Cho, Yi, and Choi (2018) and Masa'deh et al. (2016). Moreover, Industry 4.0 also increases the organization performance. Positive role of Industry 4.0 in organizations is also evident from previous studies (Bhagawati et al., 2019; Petrisor & Cozmiuc, 2020; Rane & Narvel, 2019). Furthermore, latest technologies always show positive role in business performance (Malhotra, 2005; Tsui & Malhotra, 2005). Thus, leadership styles and Industry 4.0 have significant role in the organization performance, and these results are in line with prior studies.

Moreover, leadership styles and Industry 4.0 influence positively on job satisfaction and competitive advantage, respectively. Increase in leadership styles and Industry 4.0 increases the job satisfaction and competitive advantage, respectively. More the satisfaction is, more will be the performance of employees (Purnama, 2017; Yim et al., 2018). Consistent with current study, Boamah, Laschinger, Wong, and Clarke (2018) also found the similar results. Prior studies also

highlighted an important connection between Industry 4.0 and competitive advantage ([Adamik & Nowicki, 2018, 2019](#); [Majeed & Rupasinghe, 2017](#); [Zhou et al., 2015](#)). The technologies are the basics of competitive advantage ([Mao et al., 2016](#); [Nainaar & Masson, 2018](#)), and Industry 4.0 is only based on to the introduction of new technologies. Hence, Industry 4.0 has important role to improve comparative advantage in organizations.

Further, job satisfaction and competitive advantage influence positively on business sustainability. Increase in employees' performance will definitely increase the business sustainability ([Lee & Ha-Brookshire, 2017](#)). As the job satisfaction leads to the higher employees' performance which effect positively on business sustainability. Various studies proved that competitive advantage has positive role in the business performance ([Pereira-Moliner et al., 2015](#); [Saeidi et al., 2015](#)) and business performance increases the business sustainability. Lastly, it is found that business sustainability significantly improves the organization performance. In line with current study, earlier research studies also investigated a significant link between business sustainability and performance ([Hanelt et al., 2017](#); [Schaltegger & Wagner, 2017](#)). Thus, leadership styles and Industry 4.0 have significant role in organization performance by enhancing job satisfaction and competitive advantage. Job satisfaction and competitive advantage increase the business sustainability which finally promotes organization performance.

Implications of the Study

The current study has valuable theoretical and practical implications. Theoretically, this study examined the valuable relationship between leadership styles, Industry 4.0, job satisfaction, competitive advantage, business sustainability and organization performance. This is a pioneer study which investigated the role of Industry 4.0 to enhance competitive advantage in Thai manufacturing SMEs. This is also the first study which examined the combine effect of leadership and Industry 4.0 on organization performance. Thus, it has significant contribution to the literature. Along with the theoretical implications, this study has vital importance for practitioners to enhance

organization performance through leadership styles and Industry 4.0. As this study proved that manufacturing SMEs performance can be improved with the help of integration between leadership and Industry 4.0, by implementing better leadership styles within the organization and implementation of Industry 4.0 technologies, practitioners can improve organization performance.

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Chapter 14

Factors of Successful Women Leadership in Asian: Moderating Role of Political Differences and Organization Culture

Kittisak Jermsittiparsert

Abstract

Objective of the study is to determine the factors effecting successful women leadership and to draw a comparison between Thailand and Malaysia. Primary data were collected from both countries. Quantitative research approach was used in this study by using a cross-sectional research design. Population of the study was based on the textile companies of Thailand and Malaysia. Employees of textile companies were selected as the respondents. Partial Least Square (PLS) was used for data analysis. Results of the study found that learning autonomy, emotional intelligence, political differences and organization culture (OC) are the major factor which effect on successful women leadership in both countries. Moreover, in both countries, learning autonomy, emotional intelligence and OC have positive role in successful women leadership, however, political differences effect negatively on successful women leadership. Additionally, there is a difference between

both countries, in case of moderating role of political differences and OC. In Thailand, OC has positive role to enhance women leadership through emotional intelligence. In case of Malaysia, political differences weaken the positive relationship between the emotional intelligence and successful women leadership.

Keywords: Women leadership; learning autonomy; emotional intelligence; political differences; organization culture; Asian countries

Introduction

Leadership is to teach, be a guide and help others in any possible way ([Au-Yong-Oliveira, Gonçalves, Martins, & Branco, 2018](#)). A leader takes good care of those that look to him for strength and direction, puts aside his personal life for the sake of the team, makes big decisions and guides the team, therefore, together they can make a difference. Leadership is always a group effort; it's stepping out of the spotlight sometimes and letting things go and doing it well. Leadership is inspiring others to become leaders. Leadership has significant role in organization's success ([Raziq, Borini, Malik, Ahmad, & Shabaz, 2018](#); [Ul-Hameed, Mohammad, Shahar, Aljumah, & Azizan, 2019](#)). Competent leaders always lead their company toward success ([Jingting Shao, 2018](#)).

Nowadays, leadership is one of the major concerns in most of the organizations ([Akkaya, 2019](#); [Hartinah et al., 2020](#); [Jermsittiparsert & Srihirun, 2019](#)), as leadership is considered as the major factor in the success and failure of the organization ([Sriyakul, Singa, Sutduean, & Jermsittiparsert, 2019](#)). Initially, men were prevailing as leaders in most of the organizations, however, in the current decade, the numbers of female leaders are increasing. Most of the women leaders are performing a great job in various organizations. Various studies are available in the literature on women leadership ([Glass, Cook, & Ingersoll, 2016](#); [Hoobler, Masterson, Nkomo, & Michel, 2018](#); [Randsley de Moura, Leicht, Leite, Crisp, & Gocłowska, 2018](#)),

however, previous studies ignore the discussion of factors that effect successful women leadership. Particularly, in the textile industry, the literature is silent to investigate women leadership.

Therefore, this study filled the aforementioned literature gap and examined the factor effecting successful women leadership in the textile sector of Thailand and Malaysia. Textile sector is the most important sector of both these countries ([Ali & Haseeb, 2019](#); [Crinis, 2016](#); [Leerojanaprapa & Bhundarak, 2017](#)). In these countries, textile industry has significant contribution in economic development. Therefore, effective leadership is required in this sector for further growth. For this purpose, this study highlighted various factors that influence on successful women leadership. These factors include; learning autonomy, emotional intelligence, political differences and organization culture (OC). [Fig. 1](#) is the theoretical framework of the study showing the relationship between learning autonomy, emotional intelligence, political differences, OC and successful women leadership.

Learning autonomy is a psychological social learning theory which refers to autonomous learners, who are explicitly responsible for their own learning. Previous studies on the psychological social learning theory reveal that the learning autonomy needs more opportunities concerning to learning experience and explain intentional behavior as a response to psychological need ([Schunk, 2005](#)). Therefore, organization should take part in creating the learning paradigm among leaders in order to increase learner autonomy among individual in the organization. Moreover, emotional intelligence denotes to the capability to understand others and to act intelligently in human relation. A women leader must have self-awareness, empathy and management interpersonal associations which rebuild and rehabilitate the fundamental of emotional intelligence. It has significant role in leadership activities which has important influence on women leader performance ([Elsagir, 2009](#)).

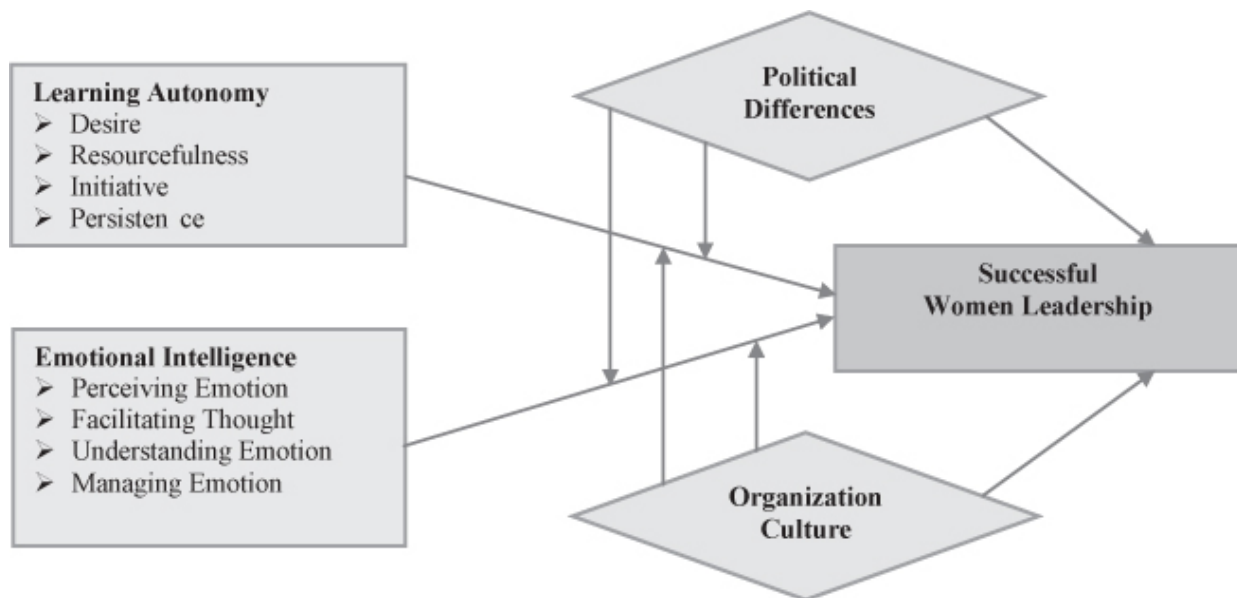


Fig. 1. Theoretical Framework of the Study Showing the Relationship Between Learning Autonomy, Emotional Intelligence, Political Differences, OC and Successful Women Leadership.

Moreover, man is a political animal by cooperating with more persons develop a network of connections called political. Differences based on politics has crucial role in organizations. Generally, increase in political differences create disturbance in the organization and effect women leadership. As previous studies point out, a relationship exists between leadership and political differences ([Alapo, 2018](#); [Højgaard, 2002](#)). Increase in this difference effect adversely on women leadership. Additionally, culture of the organization is also closely related to the politics. Generally, polticals is also one of the part of OC. However, this study investigated the effect of both of these factors separately for more accuracy. The characteristics of OC effect women leadership. Earlier studies also proved a relationship between OC and leadership ([Golden III & Shriner, 2019](#); [Jingting Shao, 2018](#); [Liu, Wang, & Chen, 2019](#)). Therefore, the current study has two objectives:

1. to determine the factors effecting successful women leadership and
2. to draw a comparison between Thailand and Malaysia.

Literature Review

Leadership is based on the skill encompassing the ability of an individual or organization to “lead.” Generally, there are various types of leadership at organizational level. Most prominent types of leadership are transactional or transformational leadership ([Cho, Shin, Billing, & Bhagat, 2019](#); [Ul-Hameed et al., 2019](#); [Woods, 2019](#)). There are a few criteria that have been used in order to describe the transactional or transformational leadership styles. The first criterion is the approaches that can be found in both of the leadership styles. According to literature, the transactional leadership style can be defined as the maker for exchanges with employees and as an agent for the changing, which will cause the productivity improvement for an organization. It is a mutual exchange style of leadership, that is, transactional leadership relies on a set of clearly defined exchanges between leaders and their followers ([Akkaya, 2020](#)). Meanwhile, transformational leadership is a leader who authorizes followers in achieving organization’s uttered vision, leads followers for professional and personal growth and increases followers’ job satisfaction level. Besides the responsibilities mentioned above, a leader also needs to transfer the leadership to followers within the group in an organization. The main reason is to modify the followers’ approach to be more tasks focused due to the organization’s needs or the negotiation process has become tougher.

According to [McCann, Langford, and Rawlings \(2006\)](#), leadership approaches are similar to the contingency or situational approaches. The contingency or situational model heavily pointed out that the need for an effective leader in order to adapt and change their respective leadership styles which depended to the particular time business needed with the consideration regarding the particular situation or circumstance’s demands. This kind of approach mentioned that there is no sole dominant way for a leader to lead an organization. Therefore, the best way for a leader to lead is to depend on the current situation within the organization. The responses or the developmental of the followers who are being led have the massive impact to the type of leadership’s style that needed to be equipped by a leader. These will help in determining whether a leader should be more toward relationship focused or task oriented while leading in an

organization. Besides considering the needs from the followers, leaders should also stand still to their own believe and needs which concerning the leadership style that suitable for them to lead a particular organization. An environment which provide rewards systems that persuade people to try new things, innovate and sense of safety while take risk. Different approach is a trait approach in leadership that concerning about the way of leader being born with the core traits and charisma. This theory recommends that a leader will distinguish him/her from the followers where they have some unique personal characteristic in order to become a leader. Some publications mentioned that the trait approach leadership that the personality of leadership style cannot be predicted and was abandoned. The leadership essence is to followership where all people in a group will tend to follow a person who believed will help them to achieve their desired needs and goals ([Ahmad & Ejaz, 2019](#)). Therefore, leadership can be considered as a set of practices which influence others in achieving the desired goals.

In organizations, leaders have various characterises. Women leaders are also leading the organizations effectively. Sometimes women leaders are also known as managers as well, however, sometimes women leaders are different and women managers or different. There are few authors who perceive leaders and managers of an organization are basically the same in the characteristic and some categorize leadership as one of the functions in management. On the other hand, there is a group of authors who suggest that it is essential to make a distinction between managers and leaders' term for better understanding and analysis. There are few sources which argued that a manager can create trough the academic studies in higher learning institution compared to leadership which is action oriented an interpersonal and most excellent to develop it is through the practices. It is essential for people not to make confusion whenever differentiate between leaders and managers of different organization. Besides, it is important to understand well involvement of leadership in the decision-making process. According to [Ismail \(2011\)](#), there are two major roles for a leader whenever being given task to accomplish: 1) to complete the given set of mission and 2) to preserve group members' harmonious who work together as a team.

The second role is referring to the importance of long-term relationship building and need for effective in a team to be built up between the followers and leader in the formed team and among the members in the same group while achieving goals. Obviously, for a leader who was considered as successful is the one that leads his/her group to complete the specified task. Furthermore, when a given task had been accomplished, instead of letting him or herself in relax circumstance, a successful leader will continue to motivate followers to face the next challenge that will be given by organization. Comparing to an unsuccessful leader, he/she will leave behind the group once the job is done ([Pietraszewski, 2019](#)). Some organizations may view two kinds of scenario between the successful and unsuccessful leaders. The first scenario is a group who is led by a successful leader which may not be successful in accomplishing the given task, but the strong bond among leader and followers to continue and challenging a new task will be more preferable by the organization. Second scenario is an unsuccessful leader leading a group and succeeded in completing the task but leaves the group in a messy condition will be unfavourable. There are various factors which effect on women leadership. The current study considered emotional intelligence, learning autonomy, OC and political differences.

Learning Autonomy

Learner autonomy refers to an intentional behavior of individual to learn. Most authors define learner autonomy based on their researches. [Holec \(1979\)](#) defined learner autonomy as:

... the ability to take charge of one's own learning, noting that this ability is not inborn but must be acquired either by "natural" means or (as most often happens) by formal learning.

[Dickinson \(1995\)](#) defined "learner autonomy as a capacity to make and carry out choices or an ability to perform rational decision-making processes over learning activities." These definitions capture the challenge of learner autonomy: a holistic view of a learner that

requires individual to engage with the cognitive, Meta cognitive, affective and social dimensions of learning and to worry about how they interact with one another. The practice of learner autonomy however, needs a approach, an optimistic attitude, ability for indication and a willingness to be practical in self-management and in communication with others. It has significant influence in leadership activities ([Dou, Devos, & Valcke, 2017](#); [Waaland, 2016](#)).

There are two universal arguments in favour of trying to explain the learners autonomous of individual employees. Firstly, learner autonomy fosters the efficiency and effectiveness in doing task and responsibilities, for instance, if the individuals are showing involved with their task and responsibilities it is likely to become more efficient because it is more personal as well as focused than otherwise; especially, what is learned is more likely to serve their specific tasks and responsibilities and directly achieve the organizational goals. Women leader is also the employee of an organization. Secondly, learner autonomy will increase motivation of individual leaders. For instance, if individual employees are committed to their task and responsibilities, the problem is by definition solved; although they may not always feel entirely positive about all aspects of their task and responsibility.

Most scholars' rapidly expanding literature has debated the term learner autonomy. Several authors have accepted that learner autonomy based on political, social and contemporary situations. However, the development of a theoretical model delivers a theoretical basis for understanding learner autonomy. [Park and Confessore \(2002\)](#) the father of the field proposed four conative factors for learner autonomy. The factors are desire, initiative, resourcefulness and persistence. These factors are critical for understanding individual employees who engage in independent and self-directed learning endeavours.

1) Desire to Learn: According to [Meyer \(2001\)](#), "desire to learn is the formation of intentions." Meyer separated precursors from intentions. There are three-sub-factor was inferred which are the factors construct of basic freedoms, power management and change skills. It means that, the individual's motivation is described as a desire to learn. Simultaneously, it should involve participation of

individual in a learning process. Thus, it will lead to life-long learning and increase individual performance as a women leader.

2) Learner Resourcefulness: According to [Carr \(1999\)](#), to be a learner resourcefulness individual must increase their knowledge and should have intentions to be source of references. [Carr \(1999\)](#) inferred a four-factor construct which are prioritizing learning alternatives, resolving conflict, future orientation and problem solving. Therefore, to be learner resourcefulness, the individuals need necessary skills, knowledge and abilities related to the leadership.

3) Personal Initiative in Learning: According to [Ponton \(2000\)](#), learner initiative is emphasized on intentions of individuals to start the learning activities. [Ponton \(2000\)](#), “inferred a five-factor construct, which are goal-directedness, action-orientation, overcoming obstacles, active approach and self-starting.” Therefore, it is clear that individuals have intentions and willingness should to start learning activities and women can perfume better in leadership.

4) Persistence in Learning: According to [Derrick \(2001\)](#), persistence involves the intentions of individuals to continue learning activities. It is a three-factor construct, which are volition, goal maintenance and self-regulation. Therefore, in simple words and in the context of individual employees in the organization, desire describes the individual’s motivation to participate in a learning experience to perform tasks and responsibilities to increase them performance, while resourcefulness means the learner’s intention to be resourceful. Initiative describes the person’s willingness to initiate tasks and responsibilities without any instructions from their immediate supervisors and persistence describes the person’s intention to continue doing the task and responsibilities.

Further, [Confessore and Park \(2004\)](#) explored the theoretical link between the learner autonomy and motivation. An increase in motivation increases the job satisfaction ([Hussain, Rizwan, Nawaz, & ul Hameed, 2013](#)), which influence positively on women leadership. In order to follow up to the study, researcher uses motivation theory to describe the learner autonomy. Motivation is the psychological processes that cause arousal, direction and persistence of voluntary actions that are goal directed. The motivation is a voluntary behavior. Therefore, the employees’ behavior will reflect consideration of

perceived consequences of their actions if they don't have a choice in any situations. Moreover, motivation concentrates on some process that affects behaviors such as persistence and effort. Nevertheless, motivation is always likely linked to the individual. This is because individual is unique in nature. Thus, motivation plays an important role in determining successful of individuals and organizations leadership of women. The successful factors of individual's women leadership are influenced by intentional behaviors of individuals such as desire, resourcefulness, initiative and persistence. A few previous studies explored the link between the learner autonomy and leadership, investigated a significant relationship ([Harrison, 2013](#); [Mensch & Rahschulte, 2008](#); [Norris, 2011](#)). Thus, below hypothesis is proposed:

H_1 : Learner autonomy has a relationship with successful women leadership.

Emotional Intelligence

The terms of emotional quotient used for the first time by [Mayer and Salovey \(2007\)](#), which denoted to intelligence in what people recognize feelings, motivate as well as manage emotions in life. Actually, emotional intelligence has basics concept of social intelligence, which was introduced by Thorndike in 1920, that denotes to the capability to understand others and to act intelligently in human relation. This intelligence comprises self-awareness, empathy and management interpersonal associations, which rebuild and rehabilitate the fundamental of emotional intelligence, that essentially as dimensions of social intelligence. Assessments are behavioral intelligence that based on emotional quotient has a significant impact on women leadership.

Found that ability in managing and maintaining good emotion will escort people reach beliefs, loyal, faith and commit in enhancing individual success. Competency such emotional quotient eases an individual adjustable with environment and changes. People with a high level of emotional quotient have ability in coping failures and motivated to move onward. Therefore, it also has relationship with

leading capabilities. On the other hand, people with a low level of emotional quotient are easy to get afraid, get angry and against other people (Indvik & Johnson, 2009). Still, competency as emotional intelligence is important to renew relationship with others within the organization and could increase satisfaction and raise the profit. People who involve in organization have to enhance their emotional quotient to increase initiative toward the change, depression and conflict within workplace. Emotional quotient is used to influence people in an exact place and in a certain good time. A wise people always ask themselves what to do in order to create success in the future. In addition, soft skills consist of several items those have to be attached in people who want to be successful in their life. Some of those items are self-development, excellent inter-personal skills, being strong in the face of sorrow, the ability to stay cool in certain crisis and being able to understand what other people's want and could handle any criticism. Besides, emotionally intelligent people identified as people who intrinsically accept many virtues as kindness, empathy and wisdom, without hoping any appreciation from it. Further, there are six personal strengths those can be used by people to be success in their life and effect on leadership which include wisdom and knowledge, courage, love and humanity, justice, temperance and spirituality.

In fact, emotional intelligent should be used by many people who want to be success in their entire life. Some researchers found that emotional intelligence has an important role in occupational settings, such as occupational and career assessment, job performance and satisfaction and coping with occupational stress. Further, in this case, people's emotion provides us invaluable information about ourselves, other people in work and some various dynamic transaction that shared inside organizational environment; and this information filtered through to us because people's feeling on the job reflect spontaneous emotional responses to the measurements and interpretations made in the workplace. Work-related emotional intelligence competencies are vital in successfully negotiating the demands, constraints and chances that necessary to success in organization. Therefore, emotional competence has been claimed twice than IQ and expertise for success. People those high in self-awareness can be able to

monitor themselves, observe and influence actions. Therefore, emotional intelligence has significant role in leadership ([Ibrahim, 2019](#); [Mullen, Limberg, Tuazon, & Romagnolo, 2019](#)). Thus, it also has significant effect on success of women leadership.

Emotional intelligence contents of five elements are as follows: (1) self-awareness, (2) self-regulation, (3) motivation, (4) empathy and (5) social skills. Emotional intelligence is especially so important in regulating the emotion of people, in the workplace, particularly for women leadership. The self-regulation of emotions in fact, include some various competencies such as restraining and controlling stimulus, decrease suffering, channeling negative affect effectively and draw out and maintaining pleasant and unpleasant. Besides, there is another facet of emotional intelligent, mentioned in literature, which is related to the theory of Big Five Personality that might be similarly as the role of emotional intelligent as a variable in leadership success. Still, those are trust, altruism, self-discipline, positive emotion and compliance. Therefore, another Big Five factors consist of mixed model elements of emotional intelligent as assertiveness, activity, positive emotions and sensation looking. This extraversion was very modestly forecast of success for people in management and marketing. A number of previous studies show that emotional intelligence and leadership have relationship with each other's ([Bhattacharjee & Rahman, 2016](#); [Dartey-Baah, 2019](#); [Haber-Curran, Miguel, Shankman, & Allen, 2018](#); [Lumpkin & Achen, 2018](#); [Wouters, Brüll, & Zafra, 2017](#)). Therefore, from the discussion on previous studies, it is proposed that:

H_2 : Emotional intelligence has relationship with successful women leadership.

Political Differences

The notion of politics is derived from the Greek word “politics” which means a city or state. It was initially utilized by Aristotle (384-322) in his textbook *Politics* to mean the matters of Greek city. In his direction, man is one of the political animals by cooperating with more persons develop a connection called political. According to [Lasswell](#)

(1951), politics is basically the struggle for positions of power as well as effect by which those who succeed in controlling such positions in society are capable to make choices that disturb the lives of several citizen within the country. On the other hand, in case of organization, decisions by leaders disturb the lives of employees which also effect adversely on leadership activities. Therefore, there are two categories of politics, namely: state politics and organizational politics. The current study is majorly discussed the organizational politics.

Within the organization, due to the politics, a difference is created among the employees and leaders of the organization. In this situation, the decisions of leaders do not remain rational and these decisions go against most of the employees (Omisore & Nweke, 2014). This situation not only hurts the employees, but it also hurts the leader's success and ultimately creates a hurdle for leaders to become a successful leader. Organizational politics make very uncertain replies because people look at the entire procedure both negatively as well as positively, dependent on the specific view they have in mind. But, those who get success in organizational politics incline to be observed positively perhaps, because they are successful opponents in other respects as well (Omisore & Nweke, 2014). However, political differences in organizations are not a good sign for both organizations and leaders. Previous studies highlighted that political differences have relationship with leaders (Alapo, 2018; Højgaard, 2002; Persons, 1993; Pullen & Vachhani, 2018). Thus, from the above discussion, following hypotheses are proposed:

H_3 : Political difference has relationship with successful women leadership.

H_4 : Political differences moderated the relationship between learning autonomy and successful women leadership.

H_5 : Political differences moderated the relationship between emotional intelligence and successful women leadership.

Organization Culture

[Allaire and Firsirotu \(1984\)](#) for identifying OC system argued that two interrelated sets of systems can have a great influence on an organization's culture. The first among them is the system, which is in-lined with typology of culture. This system consists of strategies, policies, structures and management practices of an organization and is aligned with the classic theory of management (CTM). The focus of this CTM has been on achieving the organizational goals with the focus on task orientation. The second system which influences OC is the organization's belief system consisting of ideologies and values. However, scholars suggest that the responsibility of the development of organizational culture is central to the top management ([Allaire & Firsirotu, 1984](#)). From setting organizational goals till communicating them effectively to all people concerned with an organization is the leader's responsibility ([Heck, Larsen, & Marcoulides, 1990](#); [Larsen, 2018](#)).

As per the research emphasis of [Allaire and Firsirotu \(1984\)](#), individuals are mentioned as important 'pillars' of organizational culture. Individuals are also including women leaders. In fact, individual interaction with each other in the organization is based on the beliefs, goals and attitudes as well as the organizational belief system. A strong organizational culture can be formulated only when organizational members get well with its definition of the firm. Importantly, the attitudes and values of employees are collectively considered as important determinants of organizational culture. Due to the significance of the role of organizational culture as an organizational performance determinant, researchers have paid a great deal of attention for examining its effects and potential limitations.

Since long time, the organizational culture has been reported as imperative construct having its great influence on many individual leaders' behaviors. Moreover, the organizational culture has played significant role in understanding individual variables, for instance, commitment, job-satisfaction, self-efficacy and collective efficacy ([Walumbwa, Lawler, Avolio, Wang, & Shi, 2005](#)). For example, while investigating the organizational culture's influence over employee behaviors in the workplace, reported that the clan and adhocracy types of culture enhance job satisfaction of employees. With the well-

established organizational culture, the organizations not only differentiate themselves from their rivals, but they also establish a sense of identity of an organization, resulting an increase in overall commitment to the organizational goals and objectives with strong women leadership. This notion states organizational culture as glue for binding all individuals, activities, and behaviors together through a set of standards that determine the acceptable sets of behaviors.

Beside the other assumptions of organizational culture, the subcultures also exist in organizations ([Jermier, Slocum Jr, Fry, & Gaines, 1991](#)). By talking about culture in an organization, study talks about the culture which is dominant in an organization and effect on women leadership. Large organizations have many departments and each of them might have different culture. Uniformity in interpretations on the basis of culture will not exist without any dominant culture, and there will be no judgmental uniformity about the appropriate and inappropriate behaviors. The culture's dynamic view has been studied by several researchers.

[Zheng, Yang, and McLean \(2010\)](#) stated that the dominant culture of an organization goes through the phases of inspiration, implementation, negotiation and transformation, while the organization is going through growth phases such as start-up phase, growth phase, maturity phase and revival phase. For a longer period of time, the organizational culture has been thought of unitary, whereas other researchers claimed that organizational culture is dynamic by challenging the earlier assumption of "unitary." Therefore, the gradual development of subcultures within organizations has received much attention in research having significant role to influence leadership qualities.

The past studies have explained the distinctions among strong and weak organizational cultures and their by-products at both individual and organizational level. Academic researchers and practitioners have widely argued that mainly the strong culture determines business performance by effecting leadership activities. Some researchers, focusing their studies on giant organizations like HP, IBM, P&G and McDonald, have attempted to address advantage of performance because of the practiced organizational culture in these organizations. These researchers attributed the superior performance

of these companies to the core value sets (HR practices & customer supplier relationships), which are established and maintained by their leaders. Customer satisfaction is most important ([Nadeem, Alvi, & Iqbal, 2018](#)), and leadership is important in this aspect. As a result of these management practices, the organizational performance is increased leading them to competitive advantage by strengthening the leadership activities. Companies imitate successful organizational cultures as a way for improving their employee's morale and work life quality and that eventually lead to improved overall organizational performance. To sustain superior performance of women leadership, OC should help to add economic value to an organization by supporting leadership activities. Failure in this characteristic will be failure in obtaining competitive advantage. Rare and valuable will help the organization for behaving differently than rivals. The second characteristic talks about culture-based competitive advantage, helping organizations to behave differently than its rivals. To sustain superior organizational performance, an organization's culture should be uncommon (rare) and valuable. Therefore, all the characteristics of OC effect women leadership. Previous studies also proved a relationship between OC and leadership ([Golden III & Shriner, 2019](#); [Liu et al., 2019](#); [Lok & Crawford, 1999](#); [Meng & Berger, 2019](#); [Zhen Shao, 2019](#)). Thus, from the above discussion, below hypotheses are proposed:

H₆: OC has relationship with successful women leadership.

H₇: OC moderates the relationship between learning autonomy and successful women leadership.

H₈: OC moderates the relationship between emotional intelligence and successful women leadership.

Research Methodology

The current study is based on two countries, namely; Thailand and Malaysia. Therefore, primary data were collected from both Thailand

and Malaysia. Quantitative research approach was used in this study by using a cross-sectional research design. Population of the study was based on the textile companies of Thailand and Malaysia. Respondents of the study were the employees of these textile companies.

In order to collect the data to achieve research objectives, a questionnaire was used as the primary instrument for data collection method from the respondents. The advantage of using questionnaire is it is relatively low in cost and reasonableness, where a questionnaire is fairly easy while at the same time makes available hustle in terms of its reporting. Questionnaire was divided into two sections, the first section was representing the general profile of respondents and second section was based to scale items related to the key variables of this study, namely, learning autonomy, emotional intelligence, political differences, OC and successful women leadership.

Sample size was selected by using the instructions of [Krejcie and Morgan \(1970\)](#). In both countries, the employees of textile industry are more than 100,000. In this case, the sample size should not be less than 382. However, for more accuracy, this study used 500 sample size. Thus, 500 questionnaires were distributed among the employees of textile companies in Thailand and Malaysia, separately. Area cluster sampling was used to distribute questionnaires. Email survey was used in this study to collect the data. First of all, the email addresses of employees were gathered from various websites. After that the questionnaire was emailed to respondents, and instructions were also explained in the email to fill the questionnaire. In addition, three reminders were sent after the gap of one week. From total questionnaires, 216 valid responses were returned from Thailand and 240 valid responses were returned from Malaysia. Finally, the data were entered in excel sheet and analysed by using PLS.

Data Analysis and Findings

While entering the data in excel sheet, it is the significant probability of mistake in relation to the missing value and outlier. These mistakes may lead to the different results. Therefore, before data analysis,

analysis for data screening was carried out. For both the countries; Thailand and Malaysia, data screening is shown in [Tables A1 and A2](#) (Appendix). It is found that in both cases, data are free from missing value and outlier. Moreover, normality of the data is also highlighted, however, PLS is most appropriate tool to analyse the non-normal data ([Hair, Hollingsworth, Randolph, & Chong, 2017](#); [Henseler, Ringle, & Sinkovics, 2009](#)).

Partial least squares structural equation modeling (PLS-SEM) was used for further analysis. Various steps of PLS-SEM were followed by the following previous studies such as [Hair Jr., Hult, Ringle, and Sarstedt \(2016\)](#) and [Henseler et al. \(2009\)](#). In this step, factor loadings of the items, composite reliability (CR) and average variance extracted (AVE) were considered to examine the reliability and validity for both countries. [Figs. 2 and 3](#) show the factor loadings, which are above 0.5 in both cases ([Hair, Black, Babin, Anderson, & Tatham, 2010](#)). Moreover, [Tables 1 and 2](#) show the alpha, CR and AVE which are also above 0.7, 0.7 and 0.5, respectively ([Hair et al, 2016](#)). Discriminant validity is examined by using AVE square root ([Fornell & Larcker, 1981](#)) and cross-loadings. AVE square root is highlighted in [Tables 3 and 4](#). Cross-loadings are highlighted in [Tables A3 and A4](#) (Appendix). In both cases of Thailand and Malaysia, the measurement model has satisfactory results, as highlighted in [Figs. 3 and 4](#), respectively.

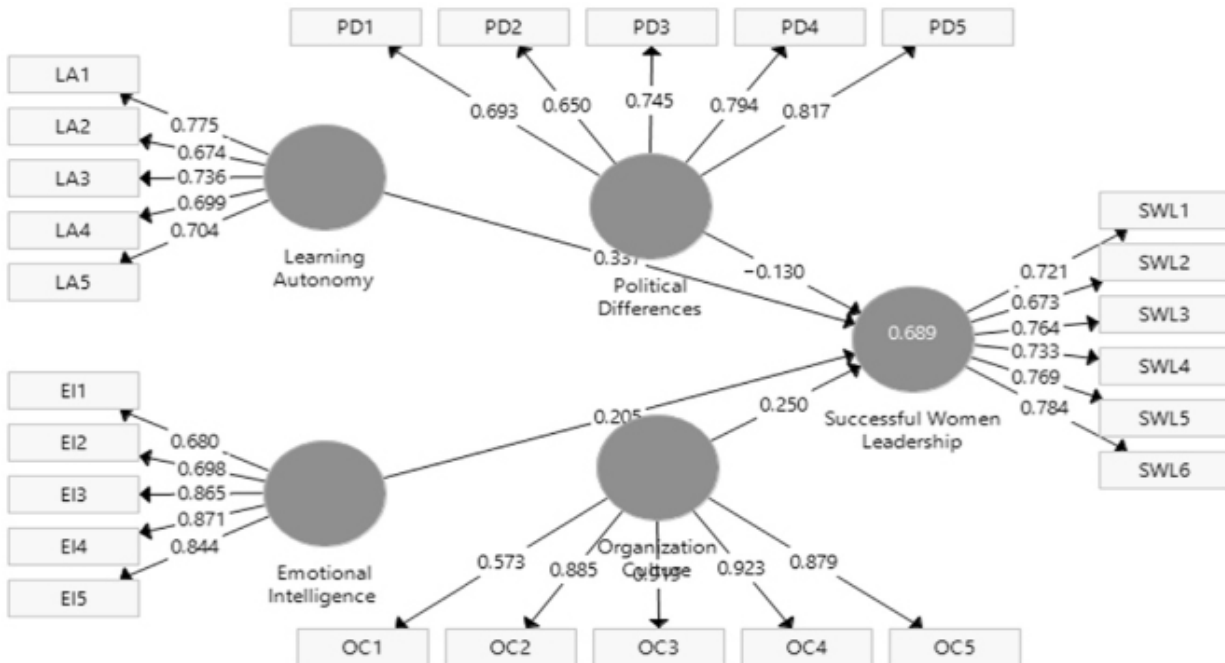


Fig. 2. Measurement Model (Thailand).

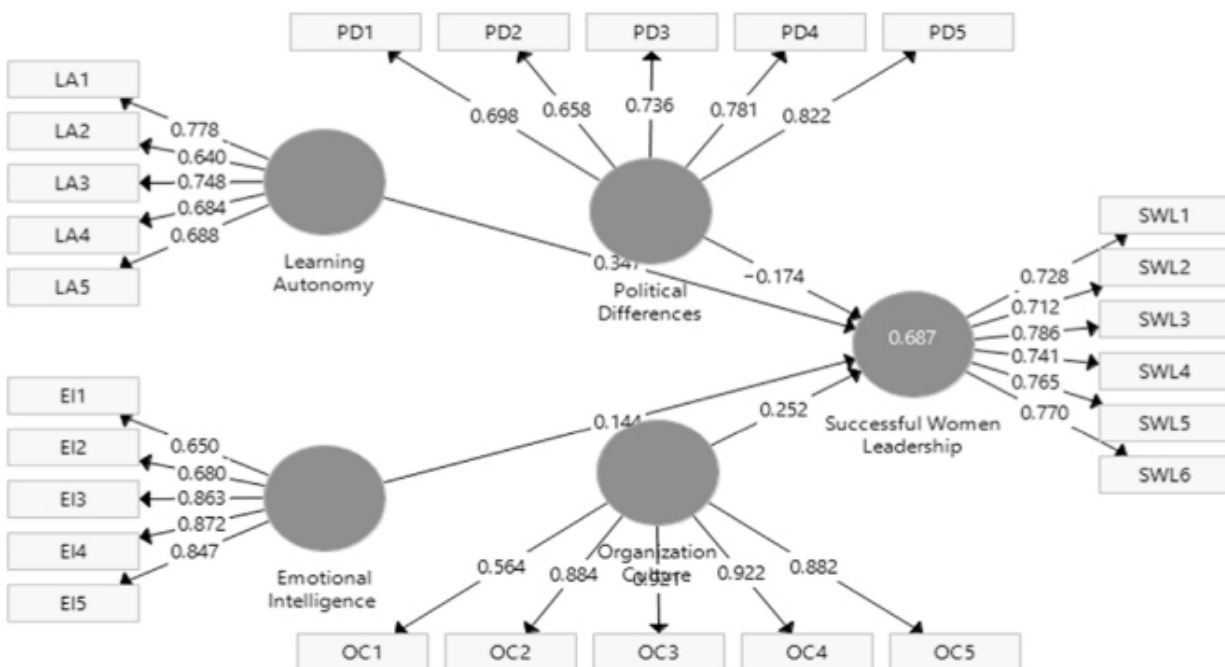


Fig. 3. Measurement Model (Malaysia).

After the measurement model, structural model was assessed both for Thailand and Malaysia. t-value 1.96 minimum significance level was considered. All the hypotheses having t-value less than 1.96

were rejected. It is found that learning autonomy, emotional intelligence and OC have positive effect on successful women leadership in both cases. However, political difference has negative effect on successful women leadership. Therefore, H_1 , H_2 , H_3 and H_6 are supported. Results for Thailand are shown in [Table 5](#), and results for Malaysia are shown in [Table 6](#).

Table 1. Construct Reliability and Validity (Thailand).

	Alpha	rho_A	CR	AVE
Emotional Intelligence	0.852	0.86	0.895	0.634
Learning Autonomy	0.766	0.772	0.842	0.516
Organization Culture	0.892	0.896	0.925	0.716
Political Differences	0.794	0.8	0.859	0.551
Successful Women Leadership	0.839	0.845	0.88	0.55

Table 2. Construct Reliability and Validity (Malaysia).

	Alpha	rho_A	CR	AVE
Emotional Intelligence	0.823	0.815	0.81	0.602
Learning Autonomy	0.732	0.863	0.874	0.633
Organization Culture	0.871	0.865	0.894	0.675
Political Differences	0.793	0.798	0.858	0.549
Successful Women Leadership	0.816	0.83	0.826	0.534

Table 3. AVE Square Root (Thailand).

	Emotional Intelligence	Learning Autonomy	Organization Culture	Political Differences	Successful Women Leadership
Emotional Intelligence	0.796				
Learning Autonomy	0.768	0.788			
Organization Culture	0.699	0.76	0.846		
Political Differences	0.737	0.738	0.689	0.742	
Successful Women Leadership	0.734	0.78	0.739	0.702	0.741

Table 4. AVE Square Root (Malaysia).

	Emotional Intelligence	Learning Autonomy	Organization Culture	Political Differences	Successful Women Leadership
Emotional Intelligence	0.789				
Learning Autonomy	0.671	0.809			
Organization Culture	0.708	0.741	0.806		
Political Differences	0.653	0.618	0.702	0.781	
Successful Women Leadership	0.621	0.781	0.784	0.765	0.651

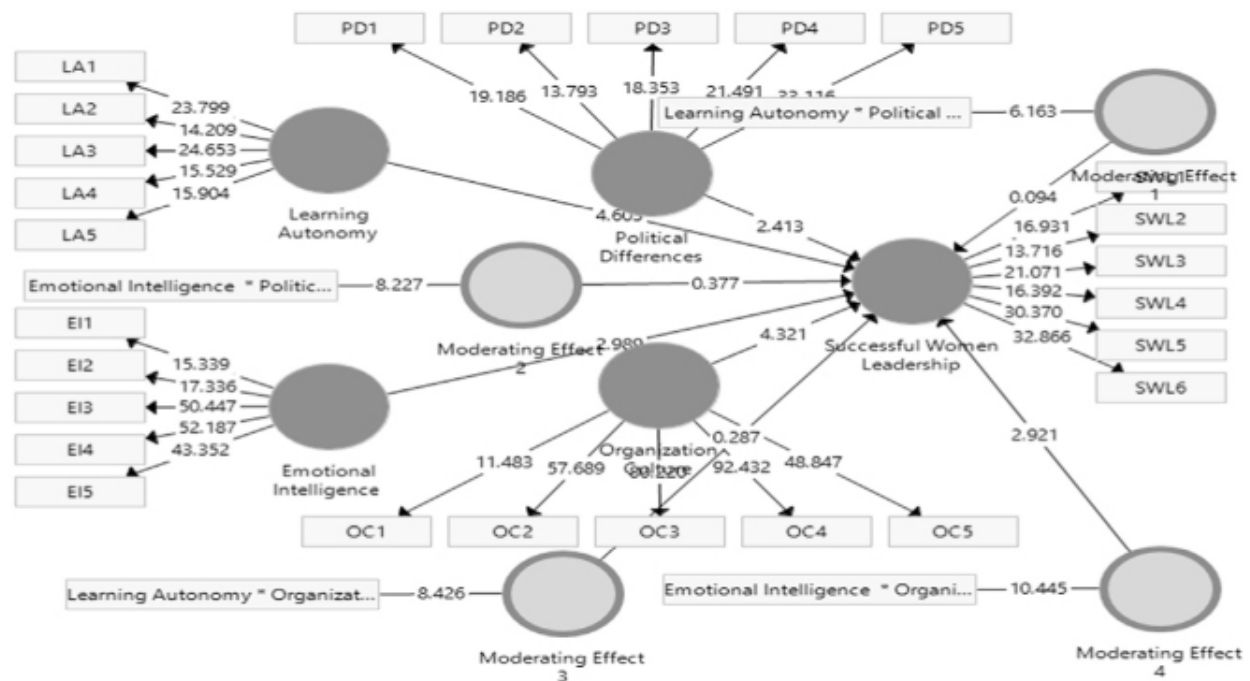


Fig. 4. Structural Model (Thailand).

While examining moderation effect, in case of Thailand, the moderation effect of OC is significant between emotional intelligence and successful women leadership. However, it is insignificant between learning autonomy and successful women leadership. Thus, H_8 is supported, but H_7 is not supported. Moreover, moderation effect of political differences is insignificant. In case of Malaysia, political differences moderate the relationship between emotional intelligence and successful women leadership. However, it is insignificant in other cases. Thus, these results supported H_5 and not supported H_4 . Finally, in case of Thailand, R-square is 0.689 and in case of Malaysia, it is 0.687. Thus, in both cases R-square is substantial (Chin, 1998).

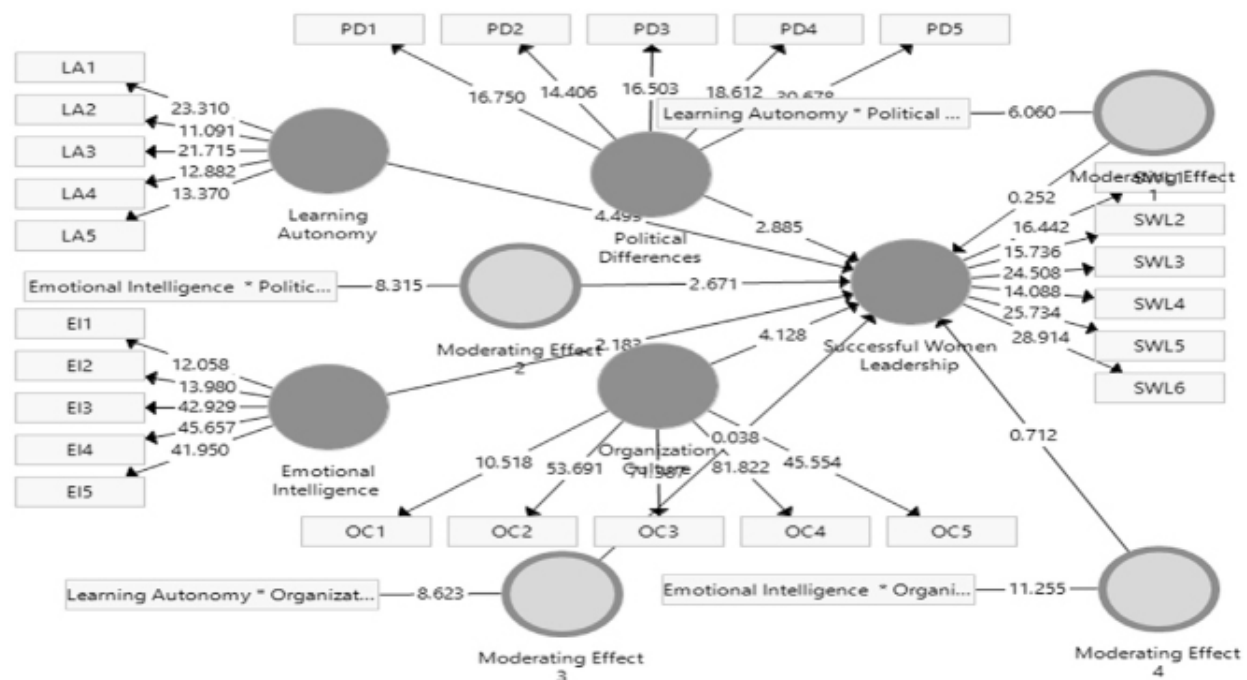


Fig. 5. Structural Model (Malaysia).

Table 5. Results (Thailand).

	(O)	(M)	SD	t Statistics	P Values
Emotional Intelligence -> Successful Women Leadership	0.192	0.186	0.064	2.989	0.003
Learning Autonomy -> Successful Women Leadership	0.337	0.337	0.073	4.603	0
Moderating Effect 1 -> Successful Women Leadership	-0.01	0.001	0.106	0.094	0.925
Moderating Effect 2 -> Successful Women Leadership	0.04	0.032	0.105	0.377	0.707
Moderating Effect 3 -> Successful Women Leadership	0.029	0.035	0.101	0.287	0.774
Moderating Effect 4 -> Successful Women Leadership	0.095	0.109	0.032	2.921	0.004
Organization Culture -> Successful Women Leadership	0.236	0.245	0.055	4.321	0
Political Differences -> Successful Women Leadership	-0.143	-0.142	0.059	2.413	0.016

Discussion and Conclusion

Objective of the study was to determine the factors effecting successful women leadership and to draw a comparison between Thailand and Malaysia. Quantitative research approach was used in this study by using a cross-sectional research design. Population of the study was based on the textile companies of Thailand and Malaysia. Employees of textile companies were selected as the respondents, and data were collected through survey questionnaire.

Table 6. Results (Malaysia).

	(O)	(M)	SD	t Statistics	P Values
Emotional Intelligence -> Successful Women Leadership	0.144	0.152	0.066	2.183	0.029
Learning Autonomy -> Successful Women Leadership	0.337	0.337	0.075	4.499	0
Moderating Effect 1 -> Successful Women Leadership	-0.03	-0.016	0.117	0.252	0.801
Moderating Effect 2 -> Successful Women Leadership	-0.079	-0.071	0.03	2.671	0.007
Moderating Effect 3 -> Successful Women Leadership	-0.004	-0.003	0.106	0.038	0.97
Moderating Effect 4 -> Successful Women Leadership	-0.074	-0.084	0.104	0.712	0.477
Organization Culture -> Successful Women Leadership	0.239	0.243	0.058	4.128	0
Political Differences -> Successful Women Leadership	-0.183	-0.174	0.063	2.885	0.004

Findings of the study found that learning autonomy, emotional intelligence, political differences and OC are the major factor which effect on successful women leadership in both countries. It is found that learning autonomy, emotional intelligence and OC has positive effect on successful women leadership in both cases. Improvement in learning autonomy, emotional intelligence and OC improves the women leadership. Therefore, textile companies of both countries should emphasize to improve learning autonomy, emotional intelligence and OC. However, political difference has negative effect on successful women leadership. Increase in political difference decreases the women leadership success. It is suggested that Thai and Malaysian companies should discourage political differences. Additionally, moderation effect for both countries is shown in [Figs. 6](#) and [7](#).

There is a difference between both countries in case of moderating role of political differences and OC. In Thailand, OC has positive role

to enhance women leadership through emotional intelligence. In case of Malaysia, political differences weaken the positive relationship between emotional intelligence and successful women leadership. Fig. 6 shows the moderation effect for Thailand. It is proved that moderation effect of OC strengthens the positive relationship between emotional intelligence and successful women leadership in Thai textile companies. Fig. 7 shows the moderation effect for Malaysia. It is evident that moderation effect of political differences weakens the positive relationship between emotional intelligence and successful women leadership. Thus, Malaysian textile companies should make various strategies to discourage political differences.

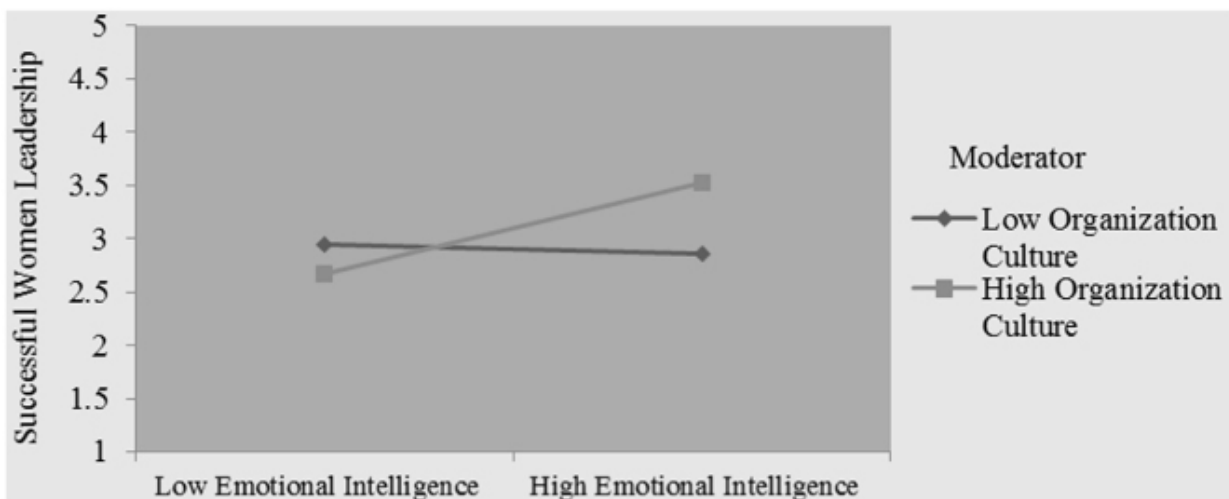


Fig. 6. Moderation Effect of OC Strengthen the Positive Relationship Between Emotional Intelligence and Successful Women Leadership (Thailand).

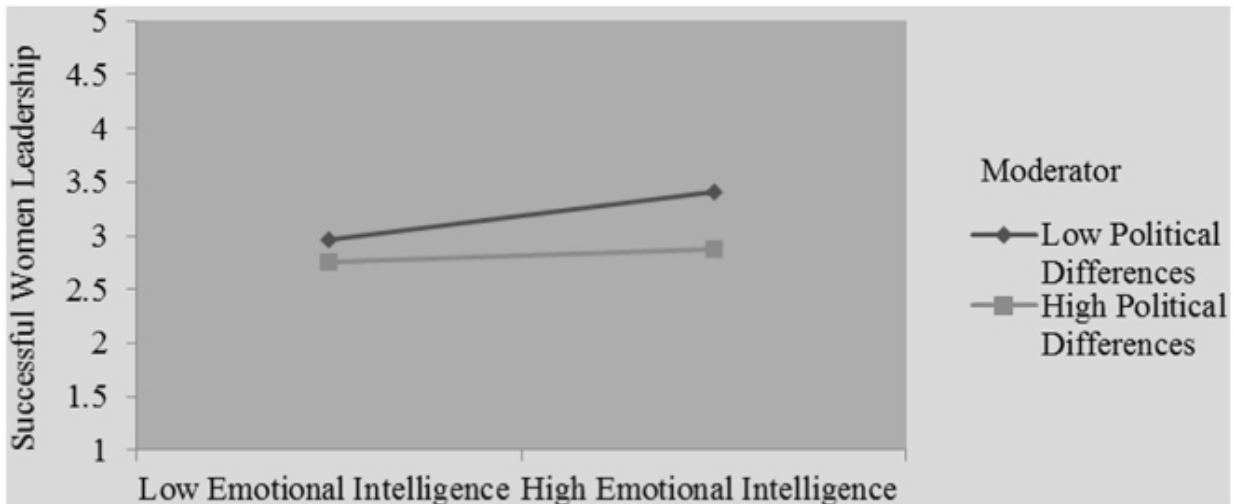


Fig. 7. Moderation Effect of Political Differences Weaken the Positive Relationship Between Emotional Intelligence and Successful Women Leadership (Malaysia).

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APPENDIX

Table A1. Data Screening (Thailand).

	No.	Missing	Mean	Median	Min	Max	SD	Kurtosis	Skewness
LA1	1	0	3.956	4	1	5	0.971	0.664	-0.918
LA2	2	0	3.705	4	1	5	1.087	-0.27	-0.605
LA3	3	0	3.622	4	1	5	1.164	-0.603	-0.548
LA4	4	0	4.065	4	1	5	1	0.887	-1.12
LA5	5	0	4.164	4	1	5	0.852	1.693	-1.136
EI1	6	0	3.865	4	1	5	1.089	-0.267	-0.767
EI2	7	0	3.796	4	1	5	1.204	-0.505	-0.721
EI3	8	0	3.684	4	1	5	1.159	-0.555	-0.572
EI4	9	0	3.749	4	1	5	1.172	-0.45	-0.66
EI5	10	0	3.796	4	1	5	1.103	-0.331	-0.636
PD1	11	0	3.756	4	1	5	1.087	0.084	-0.735
PD2	12	0	3.953	4	1	5	0.903	1.513	-1.067
PD3	13	0	3.956	4	1	5	1.008	-0.267	-0.683
PD4	14	0	4.12	4	1	5	0.967	0.502	-0.971
PD5	15	0	3.971	4	1	5	0.964	0.121	-0.773
OC1	16	0	3.96	4	1	5	1.006	0.416	-0.909
OC2	17	0	3.607	4	1	5	1.075	-0.284	-0.528
OC3	18	0	3.76	4	1	5	1.052	-0.013	-0.695
OC4	19	0	3.782	4	1	5	1.094	-0.4	-0.648
OC5	20	0	3.709	4	1	5	1.139	-0.387	-0.643
SWL1	21	0	3.895	4	1	5	1.013	0.76	-0.949
SWL2	22	0	3.84	4	1	5	1	0.08	-0.749
SWL3	23	0	3.865	4	1	5	0.93	0.148	-0.655
SWL4	24	0	3.873	4	1	5	0.887	0.27	-0.564
SWL5	25	0	3.636	4	1	5	1.262	-0.786	-0.512
SWL6	26	0	3.625	4	1	5	1.254	-0.99	-0.398

Table A2. Data Screening (Malaysia).

	No.	Missing	Mean	Median	Min	Max	SD	Kurtosis	Skewness
LA1	1	0	3.914	3	1	5	0.985	0.542	-0.881
LA2	2	0	3.644	4	1	5	1.105	-0.26	-0.612
LA3	3	0	3.532	4	1	5	1.176	-0.635	-0.494
LA4	4	0	3.995	4	1	5	1.046	0.552	-1.036
LA5	5	0	4.126	4	1	5	0.892	1.513	-1.134
EI1	6	0	3.802	4	1	5	1.109	-0.347	-0.717
EI2	7	0	3.716	4	1	5	1.218	-0.586	-0.663
EI3	8	0	3.599	4	1	5	1.173	-0.57	-0.544
EI4	9	0	3.64	4	1	5	1.192	-0.613	-0.544
EI5	10	0	3.698	4	1	5	1.133	-0.451	-0.565
PD1	11	0	3.676	3	1	5	1.12	-0.049	-0.688
PD2	12	0	3.914	4	1	5	0.933	1.144	-1
PD3	13	0	3.937	4	1	5	1.033	-0.201	-0.736
PD4	14	0	4.108	4	1	5	0.98	0.768	-1.056
PD5	15	0	3.928	4	1	5	0.975	0.221	-0.795
OC1	16	0	3.905	4	1	5	1.042	0.297	-0.892
OC2	17	0	3.55	3	1	5	1.097	-0.308	-0.519
OC3	18	0	3.707	4	1	5	1.082	-0.085	-0.684
OC4	19	0	3.721	4	1	5	1.116	-0.466	-0.604
OC5	20	0	3.635	4	1	5	1.154	-0.477	-0.581
SWL1	21	0	3.838	3	1	5	1.031	0.664	-0.935
SWL2	22	0	3.82	4	1	5	0.993	0.162	-0.771
SWL3	23	0	3.829	4	1	5	0.962	0.115	-0.689
SWL4	24	0	3.838	4	1	5	0.896	0.449	-0.621
SWL5	25	0	3.514	4	1	5	1.265	-0.831	-0.42
SWL6	26	0	3.491	3	1	5	1.262	-1.016	-0.284

Table A3. Cross-loadings (Thailand).

	Emotional Intelligence	Learning Autonomy	Organization Culture	Political Differences	Successful Women Leadership
EI1	0.68	0.542	0.365	0.544	0.519
EI2	0.698	0.57	0.461	0.501	0.535
EI3	0.865	0.661	0.655	0.623	0.634
EI4	0.871	0.632	0.62	0.571	0.6
EI5	0.844	0.641	0.643	0.681	0.624
LA1	0.52	0.775	0.603	0.541	0.609
LA2	0.484	0.674	0.447	0.438	0.482
LA3	0.701	0.736	0.666	0.592	0.637
LA4	0.459	0.699	0.476	0.465	0.516
LA5	0.476	0.704	0.506	0.6	0.538
OC1	0.517	0.647	0.693	0.58	0.586
OC2	0.601	0.633	0.885	0.535	0.638
OC3	0.597	0.63	0.919	0.577	0.645
OC4	0.601	0.628	0.923	0.589	0.626
OC5	0.612	0.655	0.879	0.614	0.603
PD1	0.792	0.643	0.639	0.693	0.593
PD2	0.473	0.504	0.497	0.65	0.47
PD3	0.441	0.44	0.364	0.745	0.454
PD4	0.436	0.496	0.457	0.794	0.458
PD5	0.522	0.606	0.548	0.817	0.586
SWL1	0.572	0.726	0.598	0.565	0.751
SWL2	0.418	0.462	0.354	0.392	0.673
SWL3	0.448	0.522	0.49	0.496	0.764
SWL4	0.459	0.5	0.449	0.492	0.733
SWL5	0.633	0.604	0.661	0.575	0.769
SWL6	0.664	0.595	0.646	0.558	0.784

Note: The [table A3](#) and [A4](#) of rotated components matrix are showing that almost all of the indicators are having factor loading more than 0.7, it means that all the indicators are eligible to be exposed to further hypothesis testing techniques, because all the factors are in suitable threshold level and all are in suitable and valid sequence and range, this data is good to go for further testing techniques, there is no cross loading in the data shown in the RCM. So, the data is reliable.

Table A4. Cross-loadings (Malaysia).

	Emotional Intelligence	Learning Autonomy	Organization Culture	Political Differences	Successful Women Leadership
EI1	0.65	0.512	0.326	0.523	0.479
EI2	0.68	0.552	0.448	0.513	0.522
EI3	0.863	0.659	0.67	0.638	0.617
EI4	0.872	0.645	0.633	0.59	0.594
EI5	0.847	0.656	0.661	0.687	0.616
LA1	0.543	0.778	0.619	0.553	0.623
LA2	0.464	0.64	0.413	0.406	0.447
LA3	0.779	0.748	0.684	0.605	0.636
LA4	0.436	0.684	0.469	0.437	0.503
LA5	0.466	0.688	0.5	0.586	0.531
OC1	0.516	0.525	0.599	0.572	0.594
OC2	0.603	0.642	0.884	0.542	0.643
OC3	0.609	0.654	0.921	0.594	0.648
OC4	0.608	0.649	0.922	0.612	0.625
OC5	0.629	0.658	0.882	0.621	0.603
PD1	0.598	0.651	0.651	0.698	0.591
PD2	0.492	0.533	0.513	0.658	0.503
PD3	0.46	0.422	0.368	0.736	0.459
PD4	0.448	0.481	0.465	0.781	0.462
PD5	0.533	0.597	0.552	0.822	0.595
SWL1	0.584	0.721	0.616	0.571	0.728
SWL2	0.448	0.509	0.414	0.435	0.712
SWL3	0.47	0.552	0.529	0.536	0.786
SWL4	0.468	0.516	0.47	0.525	0.741
SWL5	0.601	0.59	0.644	0.567	0.765
SWL6	0.635	0.587	0.623	0.561	0.77

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Chapter 15

Era of Industry 4.0 Technologies and Environmental Performance of Thailand's Garment Industry: Role of Lean Manufacturing and Green Supply Chain Management Practices

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Abstract

There has been remained a supporting linkage between the sustainable performance and the green supply chain through quality management practices. However, the main objective or aim of this research study is to understand the green supply chain and its role in lean manufacturing and analyze its impact on the performance of Thailand's garment industry, whereas there is a need to analyze the mediating role of lean manufacturing practices (LMP) and green supply management on environmental performance of garment industry. In this case, the data were assembled through a survey questionnaire, in which questions regarding LMP were asked. The selected

sample size for this research study was 411 employees that belonged to garment firms in Thailand, in which 56.4% were men and 43.6% were women. The researchers analyzed the data through different techniques and procedures such as confirmatory factors analysis, structural equation modeling (SEM), discriminant validity and other related techniques. The results of the research indicated that there is a significant impact of 4.0 technologies on environmental performance, whereas both of these, the green and the lean management practices also play a significant mediating role, because all the hypothesis are accepted. Moreover, it is recommended for future researchers to use other techniques to analyze data.

Keywords: Industry 4.0 technologies; green supply chain management practices; lean manufacturing practices; environmental performance; garment industry; Thailand

Introduction

Industry 4.0 technologies are driving the manufacturing systems to new heights of a digital, intelligent and automated platforms ([Akkaya, 2019](#); [Oztemel & Gursev, 2020](#); [Vaidya, Ambad, & Bhosle, 2018](#)). Inclusion of these technologies in the value chain of different manufacturing environments makes it possible for the delivery of real-time information between partners of a supply chain ([Jermisittiparsert & Boonratanakittiphumi, 2019](#); [Stock & Seliger, 2016](#)). In this research, the technologies of Industry 4.0 are promoted through the use of both green and lean sustainable management practices that derive the performance of environment. Lean manufacturing practices (LMP) are a set of techniques and technologies that are used for the purpose of improving the productivity and reducing the costs of manufacturing as well as removal of harmful impacts of such processes on the environment and an increased level of corporate social responsibility (CSR) processes ([Sanders, Elangeswaran, & Wulfsberg, 2016](#)). Green SCM (Supply Chain Management) has

become an effective tool for management that allows the prevalence of a sustainable environment and performance (Khalid, 2017). Literature shows positive effects of green practices of SCM on environmental performance (Ali et al., 2020; Cousins, Lawson, Petersen, & Fugate, 2019; Geng, Mansouri, & Aktas, 2017).



Fig. 1. Apparel Industry Retail Value in Thailand.

Thailand has a highly integrated textile industry and has numerous spinning, weaving and garment making enterprises. This country has emerged as a key player in the apparel and textile industry in the ASEAN region, which has seen continuous growth as evidently seen in Fig. 1. Thailand has over 2,000 enterprises that are working to produce garments, 82 companies registered as traders of garments and many retailers that are working in Thailand. This information is summarized in Table 1. These enterprises are a major source of generating gross domestic product (GDP) for Thailand, and they follow the green practices and use the industry 4.0 technologies.

Table 1. Number of Production Factories, Trading Companies and Retailers.

Practice	Number of Companies
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Production	Over 2000 garment and 2000 textile-oriented companies mostly located in the surroundings of Bangkok or the Eastern Thailand
Trade	82 garments export companies are registered
Retail	Many retailers such as Puma, Nike and others

Even though Thailand is at an advantage in the ASEAN region, its economy is still not in a favorable position in comparison to the Asian giants like India and China. This is the reason why the Thai people have comparatively lesser power to purchase. This leads to weak domestic purchase and demand. Thailand, just like any other developing countries has been implementing and formulating governmental policies for the revolutionary Industry 4.0 concepts. Thailand 4.0 has been launched since the year 2016 and this revolution has included various transformations of the Thai industries to create a value-based economy (Jones & Pimdee, 2017; Puncreobutr, 2017; Sae-Lim & Jermsittiparsert, 2019). However, these plans were created in one-size-fit-all based assumptions, reducing the overall effectiveness of these policies. The issues discussed here are not restricted to Thailand, but actually all the developing countries that are in the process of revolutionizing their industries face similar issues. Many researchers have conducted studies to see how green supply chain management can be used to enhance environmental performance (de Sousa Jabbour, Jabbour, Foropon, & Godinho Filho, 2018), lean manufacturing can be used (Mrugalska & Wyrwicka, 2016; Wagner, Herrmann, & Thiede, 2017) and how industry 4.0 technologies can be implemented. However, there is no study that has used green supply chain management and lean manufacturing mediates the effects of technologies that come under the umbrella of Industry 4.0 on the environmental performance. This research has been conducted in the context of garments industry of Thailand to fill this gap in research. The objectives of this research paper are outlined below:

- to analyze the effect of industry 4.0 technologies on the environmental performance of Thailand's garment industry;
- to find out the mediation role of lean manufacturing between the industry 4.0 technologies and the environmental performance of Thailand's garment industry; and

- to assess the mediation effects of green supply chain management practices between the industry 4.0 technologies and the environmental performance of Thailand's garment industry.

The scope of this study is to consider how the green SCM practices and LMP support the effects of Industry 4.0 technologies on the environmental performance in the garments industry of Thailand. The previous studies in this research area ([Bai, Satir, & Sarkis, 2019](#); [Helleno, de Moraes, & Simon, 2017](#); [Nadeem et al., 2017](#); [Sanders et al., 2016](#); [Vanalle, Ganga, Godinho Filho, & Lucato, 2017](#)) have had numerous theoretical, practical and policy-making benefits. They have contributed to the volume of literature content on green SCM, lean manufacturing and technologies of Industry 4.0. Moreover, these studies have served to provide guidance for the formation of policies in the garments industry for the implementation industry 4.0 technologies. Lastly, these studies have provided benefits on government level for enhancing the technological innovation levels in the country. This following section of the paper includes literature review, methodology, results and conclusion.

Literature Review and Theoretical Background

The environmental resources of the world are depleting slowly, and organizations can no longer ignore these raising concerns. Sustainability has become a high priority for organizations in recent times, as they realize their role in the depletion of environmental resources ([Dubey, Gunasekaran, & Ali, 2015](#)). The practices aiming toward environmental sustainability are becoming a common practice for developed and responsible organizations. Environmental sustainability is defined as the activities performed under the assumptions of a conscientious interaction with the environment, with the motive of avoiding exhaustion and dilapidation of resources of the nature, in turn allowing for long-standing environmental quality ([Lin & Chen, 2017](#)). Thus, organizations have started putting sustainability and environmental protection objectives at high priorities. In context of these practices, the notion of green intellectual capital has also sparked interest in academicians and practitioners alike ([Yong,](#)

Yusliza, Ramayah, & Fawehinmi, 2019). The vast knowledge pool and extant literature support the ideas of sustainability, and the business organizations are focusing on the different dimensions of sustainability (Rayner & Morgan, 2018). The environmental, economic and social dimensions of sustainability are being correlated with environmental performance of firms. Additionally, strategists realize the depravity of the situation and believe in the implementation of these strategies at the micro-levels in their organizations as well. In order to attain the goal of organizational sustainability, the activities of organizations internally need also be focused upon (Yahya, Arshad, & Kamaluddin, 2015). Researchers have stated before that the effective actions and working stipulations of the human capital and the production process has the potential of improving organizational efficiency and performance, whereas it also has a documented positive relation between green employee behaviors, environmental knowledge and technological adoption to attain these goals as proposed by Rayner and Morgan (2018). Management's dedication and orientation toward sustainability promote the adoption of revolutionary and sustainable production and management practices like green supply chain practices, LMP and these have a direct impact on the environmental performance and sustainability of the organization (Akhtar, Khan, Frynas, Tse, & Rao-Nicholson, 2018; Akkaya & Üstgörül, 2020). Therefore, the practices of the top management aimed toward green policies, and sustainable actions also motivate the employees in doing the same. The organizational structure, human capital and production or manufacturing processes are three tiers of a system that helps in the avoidance of environment harming practices and developing strategies that help reinforce this concept throughout the organization (Akhtar et al., 2018; Yong et al., 2019).

Industry 4.0 Technologies and Environmental Performance

Industry 4.0 is characterized as the procession of the digitized system of manufacturing that has been provided by the immaculate and effective incorporation of techniques, information technologies and production processes (Ahuett-Garza & Kurfess, 2018; Bukova,

[Brumerickova, Cerna, & Drozdziel, 2018](#)). The basic aim of industry 4.0 system is to focus on increasing the responsiveness and efficiency of the manufacturing system. Partners of the value chain exchange such real time exchange of the informative data and through this sharing, these systems are driven and are operated on the principles of vertical and horizontal integration ([Lin, Lee, Lau, & Yang, 2018](#)). Various ICT technologies like IoT, CoT, CC, RS, BDA, AM, AR and RS are branded as the significant technologies that are inculcated in the industry 4.0 system and promote process integration which ultimately leads toward sustainable organizational performance ([S. Kamble, Gunasekaran, & Dhone, 2019](#)). Cyber physical systems, cloud computing and internet of things are at the core of the industry 4.0 concept and are the main technological components that drive this process. CPS is characterized by the integration or conjunction of the physical systems, CNC, CPU or computer system, lathe machine, etc. that works in making all the factory equipment and arrangement in the factory adaptable ([Fatorachian & Kazemi, 2018](#)). Cloud Computing is a service provider that works through the involvement of the third party used for the storage of data or database over the internet. The cloud is recognized as a remote location where data are stored, not at the same location as where the manufacturing activities are carried out. CC offers organizations the facilities of economies in operations, speedy service and easy accessibility to sources of information. Internet of things, IoT, makes the interaction with machines without human intervention possible. It uses a network of device, where each device has its own latent network connection and an identification code verified and connected to a computer system. Manufacturing systems controlled by this technology are smart, as they have the capacity of efficiently and precisely connecting and controlling all devices ([Karaköse & Yetiş, 2017](#); [Lin et al., 2018](#)). I4 enabled manufacturing systems produce huge data therefore require specialized data assortment, storage, sharing and technologies to analyze. BDA contribute in achieving sustainable business objectives and performance and are also a measure of competitive advantage for the firms ([S. S. Kamble, Gunasekaran, & Gawankar, 2018](#)). AM, RS and AR are the utilization of technology for converting the manufacturing procedure into a simple, safe and efficient process for

the achievement of sustainable business performance objectives. I4 technologies are expected to play an important role in leading industrial and social organizations toward sustainable development (Gölzer & Fritzsche, 2017). These technologies facilitate a level of integration of various processes involved in the overall manufacturing and business development system that leads toward organizational performance in the sustainability dimensions of environment, society and governance. From the economic perspective, I4 technologies promote strong value creation, product customization, manufacturing procedural flexibility that ultimately leads toward improved customer satisfaction. The continuous changes in customers' demands and needs make it inevitable for leaders to constantly adapt to changes to keep their firms in a competitive environment (Akkaya, 2020). The features of automation and digitization of this system steer organizations in the direction of reducing time of lead, decreased costs of product making and supreme quality of services and products. From the environmental dimension perspective, the real-time data gathered from the incumbents aid the organization to distribute their resources used for manufacturing, in an efficient manner. I4 also supports the processes adopted for diminishing emissions of greenhouse gases, consumption levels of energy and efficient capacity usage of raw materials, reduction in the consumption levels of fuel and utilization of advanced monitoring and tracking systems. In the social dimension, these technologies offer employees novel techniques in learning new technologies, consequently improving proficiency levels and morale of the employees (Karaköse & Yetiş, 2017; Song, Fisher, Wang, & Cui, 2018). All of these factors contribute toward a better environmental performance which is the core essence of I4 technologies.

H1: There is a significant relationship between industry 4.0 technologies and environmental performance.

The Mediating effect of LMP

LMP are characterized as the manufacturing process, in which products are produced according to the customer's demands so that

waste production is reduced (Buer, Strandhagen, & Chan, 2018). In an I4 environment, the employees should necessarily be trained for analyzing the real-time information and taking corrective decisions from these data sources, thus the process of LMP is important for training employees existing in an industry 4.0 environment and helps in the acceptance of the technologies this process brings about. Successful integration of employees with the I4 technologies is a pertinent aspect of the LMP and this process also acts as a driver of organizational sustenance and efficiency (Kolberg & Zühlke, 2015; Latan, Jabbour, de Sousa Jabbour, Wamba, & Shahbaz, 2018; X. Zhu & Lin, 2017). LMP assists manufacturing firms in becoming competitively strong in the dynamic global market, these practices improve utilization of resources by 30%–70% by eliminating different waste materials and waste producing processes (Akkaya & Tabak, 2020; Kolberg & Zühlke, 2015; S. Kamble et al., 2019). It supports reduction in cost which promotes increased efficiency and productivity. Improved product quality and decreased inventory capacitance is associated with diminishing levels of pollutants, value stream mapping an infamous LMP works by mapping raw materials and contributes toward environmental protection. Thus, it can be propositioned that through LMP employees learn new technologies and processes they an inculcate in the minimization of waste produced, resources and fuel consumed and pollutants emitted in the environment (Latan et al., 2018), therefore the following hypothesis is proposed:

H2: There is a significant mediating effect of lean manufacturing processes on the association between industry 4.0 technologies and environmental performance of organizations.

The Mediating Effect of Green Supply Chain Management Practices

Green supply chain management activities or practices are characterized as the integration of the social and economic goals along with the sustainability orientation of the environment, by aligning the coordination among business processes and supply

chain partners with the objective of acquiring suitable economic performance of an organization and its supply chains. Green supply chain management (GSCM) practices include green purchasing, recovery of investment, intra-organizational environment management and supplier selection ([Suryanto, Haseeb, & Hartani, 2018](#)). There has been an ascertainment in the association among GSCM activities, economic and environmental performance of an organization. Environmental performance of a company considers the prevention of pollution and utilization of raw materials or products that will produce pollution, utilization of fewer resources from the natural environment, less consumption of fuel and emissions of greenhouse and biodegradable products in the environment, compliance with the environmental legislature of the country ([Irum et al., 2018](#); [Lin & Chen, 2017](#)). All of these concerns are being tackled by the introduction of industry 4.0 technologies, especially in the supply chain. The transference and storage of information is being made readily possible by technologies like CC and BDA. CoT and IoT (cloud of things and internet of things) are making the connections with machinery and equipment and control of these systems possible by minimalistic interference of the human component, mapping techniques focus on using lesser resources and are thus actively contributing in the procedural protection and performance of businesses regarding their environmental orientation ([Cousins et al., 2019](#); [Karaköse & Yetiş, 2017](#)). Application of these technologies is also lower costs and waste production which carries a significant economic benefit as well.

H3: There is a significant mediating effect of green supply chain management practices on the association between industry 4.0 technologies and environmental performance of organizations.

Research Model



Fig. 2. Research Model.

Methodology

Sample and Data Characteristics

For the purpose of this study, the researcher has chosen the garment industry of Thailand for a number of reasons. The textile industry in Thailand is wide-spread and a great level of proficiency in the design, production and sale metrics have been observed. The country is a globally established silk producer. It has also been doing exceedingly well in implementing the eco-friendly processes of dyeing, finishing and printing, all of which are according to the prescribed international standards. Presently, the garment sector of the country has employed over 1 million people whereas the textile sector has about 200,000 employees. The textile and garment sector are therefore viewed as the second most important drivers of economic and national growth. The garment sector has its sight set on expanding its global presence, while the government has recently become involved with the sector by providing vocational training to employees and domestic citizens and also has extended export credits to factory owners to allow for the commencement of expansion procedures. The government's initiatives regarding industry 4.0 orientation tie in well with the goals of the garment industry. A recent focus on eco-friendly production choices and programs, integration of technology in the manufacturing processes and a perspective of evolution in the country are what made the sector a premium choice for this study. In order to obtain data, the Chief operational officers and production managers of leading garment firms in Thailand were contacted,

whose names and other information haven't been mentioned in order to maintain respondent anonymity. A questionnaire-based survey method was developed, and the total number of respondents contacted for the purpose of this study was 411. The method of purposive sampling was utilized, because the study focuses on issues relevant to manufacturing and operations department. For this study, the researcher applied the item response theory following the criteria of ten responses against each item (question) of the measurement scale, that is $30 \times 10 = 300$ (Islam, Ahmad, Ahmed, & Ahmer, 2019).

Measures

The measurement scales for the designed survey utilized variable instruments from previously established and published literature that were found to have satisfactory psychometric properties. The responses for LMP, supply chain sustainability performance and sustainable organizational performance (environmental performance) were recorded on a five point Likert scale ranging from “strongly disagree to strongly agree” (Chen & Paulraj, 2004).

1) Lean Manufacturing Practices: The measurement scale utilized for measuring **LMP** was developed by Sanders et al. (2016) and Shah and Ward (2007). The sample items include “our company is always in close contact with our key suppliers.”

2) Environmental Performance: The measurement scale used for environmental performance used the scale developed by Q. Zhu, Geng, and Lai (2011). The sample items included “the decrease of consumption of hazardous/harmful/toxic materials.”

3) Industry 4.0 technology: The five point Likert scale used for the industry 4.0 technologies consisted of the following options;

not considering it, planning to consider it, representing initial stages of discussion with vendors, considering it currently for implementation, technology has been planned for and is on the way to be implemented, implemented successfully.

The measurement scale was devised in order to measure the level of implementation of 4.0 technologies in the focused organizations and also the maturity in the process of implementation. The scale was constructed by [Bibby and Dehe \(2018\)](#). The sample items included “Our firm is in the process of implementing or implemented robotic systems.”

4) Green Supply Chain Management Practices: The scale utilized for measuring green supply chain management practices was developed by [Q. Zhu, Sarkis, and Lai \(2008\)](#). The scale is divided into five subscales “internal environmental management, green purchasing, cooperation with customers, eco-design and investment recovery.” The sample items included “Environmental compliance and auditing programs,” “Cooperation with suppliers for environmental objectives,” “Cooperation with customers for cleaner production,” “Design of products for reuse, recycle, recovery of material, component parts” and “Investment recovery (sale) of excess inventories/materials.”

Research Results

Demographics

The sample consisted of 411 employees belonging to garment firms in Thailand. The sample comprised of 56.4% male and 43.6% female respondents. The disparity in gender is primarily due to the fact that more men are employed in the manufacturing sector. 75% of the sample was aged up to 35 and the experience of 72.5% of the sample was scattered between 2 and 8 years. The respondents were primarily managers; therefore the greater age and experience statistics have been observed in the sample.

Descriptive Statistics

The descriptive analysis of the data is presented in [Table 2](#). The mean values of all constructs seem to be approaching 4 which shows the agreement of respondents with the statements of the variables. The coefficient of skewness is a measure of data normality. All variables are within the threshold limit of $-1+1$, therefore the data can

be proclaimed to be distributed normally. The minimum and maximum values are used to check for outliers, extremely small or large values. As the minimum and maximum values of all variables are according to the minimum and maximum values of the Likert scale (1–5), therefore no outliers were detected in the data.

Table 2. Descriptive Statistics.

	N	Minimum	Maximum	Mean	Std. Deviation	Skewness	
	Statistic	Statistic	Statistic	Statistic	Statistic	Statistic	Std. Error
LeanManPr	411	1.00	5.00	3.1269	1.08459	−0.088	0.120
GreSCPr	411	1.00	5.00	3.3388	0.95494	−0.514	0.120
Ind4.0Tech	411	1.00	5.00	3.5307	1.13674	−0.576	0.120
EconoPerf	411	1.00	5.00	3.3631	1.14350	−0.511	0.120
Valid N (listwise)	411						

Convergent Validity and Factor Loading

Table 2 represents the individual component loadings of all scale items. The loadings of each scale item represent its contribution in the overall variance of the construct. A loading greater than 0.7 shows cases significant contribution. Loadings of all factors are significant as they are greater than 0.7. The factors haven't been observed to cross-load against each other as well. Convergent validity, a subtype of construct validity, is evaluated on the basis of CR and AVE values. CR, composite reliability, is a measure of the internal consistency possessed by all scale items of a construct. A CR value greater than 0.7 verifies the internal consistency, as demonstrated in Table 3 all of the CR values are greater than 0.7 and therefore internally consistent. AVE, average variance extracted is a measure of the variance an item has managed to explain. The AVE values are supposed to be greater than 0.5 to verify convergent

validity, and as [Table 2](#) demonstrates all AVE values of scale items are above the threshold value, ensuring the convergent validity of scale items.

Table 3. Factor Loading and Convergent Validity.

	Component				CR	AVE
	1	2	3	4		
LM1	0.749				0.918	0.788
LM2	0.767					
LM3	0.771					
LM4	0.741					
LM5	0.784					
LM6	0.932					
LM7	0.922					
LM8	0.912					
LM9	0.926					
LM10	0.915					
LM11	0.924					
LM12	0.927					
GM1			0.797		0.909	0.667
GM2			0.744			
GM3			0.773			
GM4			0.849			
GM5			0.820			
IT1		0.833			0.957	0.788
IT2		0.838				
IT3		0.834				
IT4		0.872				
IT5		0.896				
IT6		0.878				
EP1				0.829	0.912	0.723
EP2				0.846		
EP3				0.830		
EP4				0.826		

Discriminant Validity

Discriminant validity is a subtype of construct validity, and it is used to ensure whether the scale items included in a construct explain other items as well as they explain themselves. The correlation constants are evaluated to see if items explain themselves better than they explain other variables, that is if they converge with other items of the scale. The test measures the self-correlation and item-item correlation of all scale items. If the self-correlation statistics are greater than those of the item-item, then the discriminant validity of a scale is ensured. As [Table 4](#) represents, all items have high self-correlation values thus discriminant validity exists in the scale items.

Table 4. Discriminant Validity.

	GM	IT	LM	EP
GM	0.817			
IT	0.423	0.888		
LM	0.522	0.403	0.888	
EP	0.425	0.505	0.385	0.850

Model Fitness and KMO

Model fitness is verified by observing the GFI, IFI, CFI, RMSEA and CMIN values. Two rows are demonstrated in [Table 5](#), they showcase the limit threshold values and the observed values from the model. In accordance with the limits defined by the threshold values, all measures are in accordance, CMIN is under 3, GFI is greater than 0.8, IFI and CFI are greater than 0.9 and RMSEA is less than 0.08. KMO is a measure of the adequateness of the sample according to the model, a value between 0.8 and 1 represents adequateness of the selected sample. The KMO value for this model and sample is 0.955 showcasing that both our sample and model are fit and adequate.

Table 5. Confirmatory Factors Analysis and KMO.

CFA Indicators	CMIN/DF	GFI	IFI	CFI	RMSEA	KMO
Threshold Value	≤ 3	≥ 0.80	≥ 0.90	≥ 0.90	≤ 0.08	0.6–1.0
Observed Value	2.915	0.849	0.956	0.955	0.068	0.955

Table 6. Structural Equation Modeling.

Hypothesis	B-Value	SE	<i>p</i> -Value	Decision
IT→EP	0.320	0.050	0.000	Accepted
IT→LM→EP	0.169	0.049	0.000	Accepted
IT→GM→EP	0.186	0.055	0.000	Accepted

SEM

[Table 6](#) represents the results of SEM. A change of one unit in IT will produce an increase of 32% in EP. A change of one unit in It through LM and GM will produce a change of 16.9% and 18.6% in EP. The hypotheses are accepted as all the relationships are significant. [Fig. 4](#) represents the general path coefficients and fit of the general hypothesized structural model.

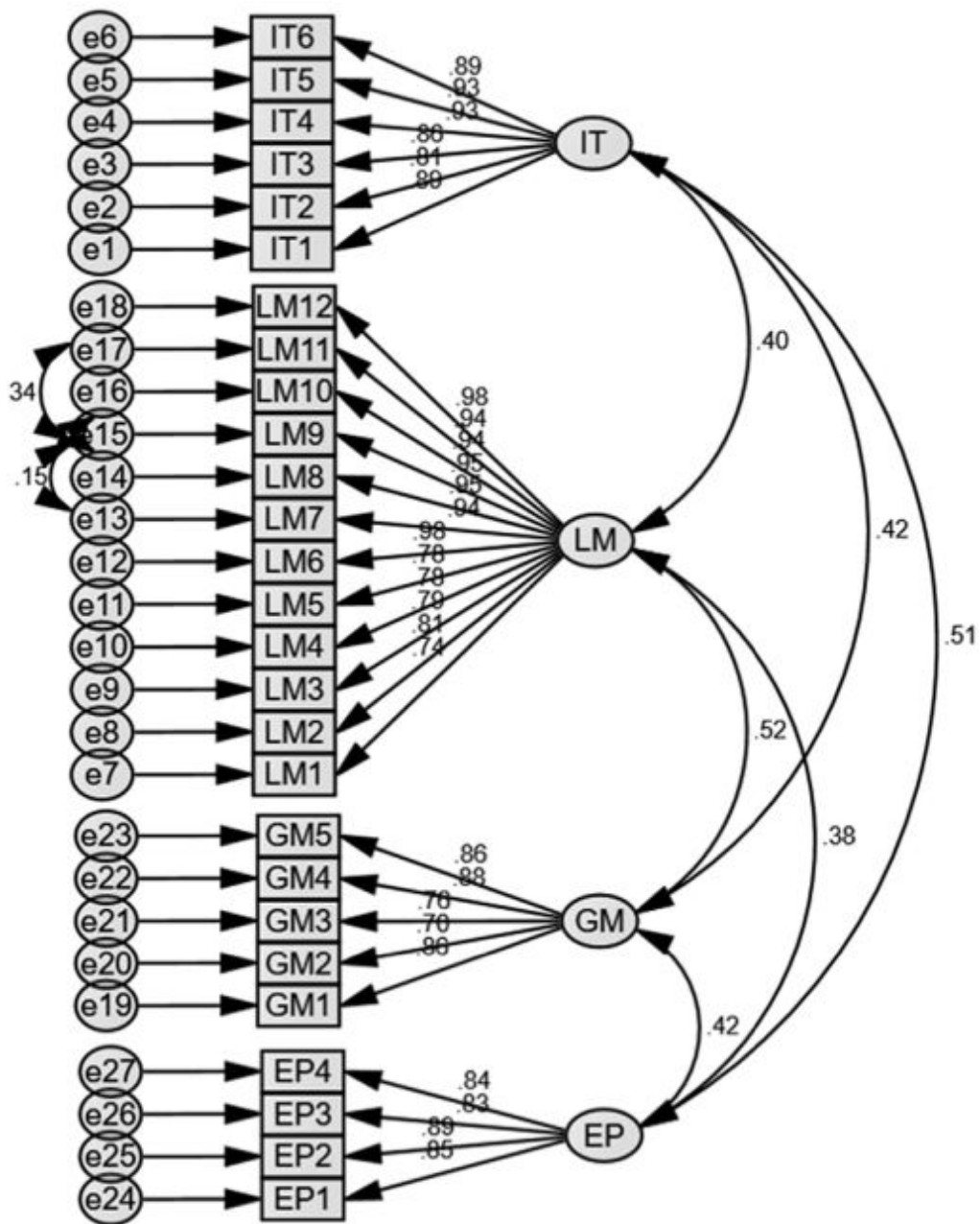


Fig. 3. CFA.

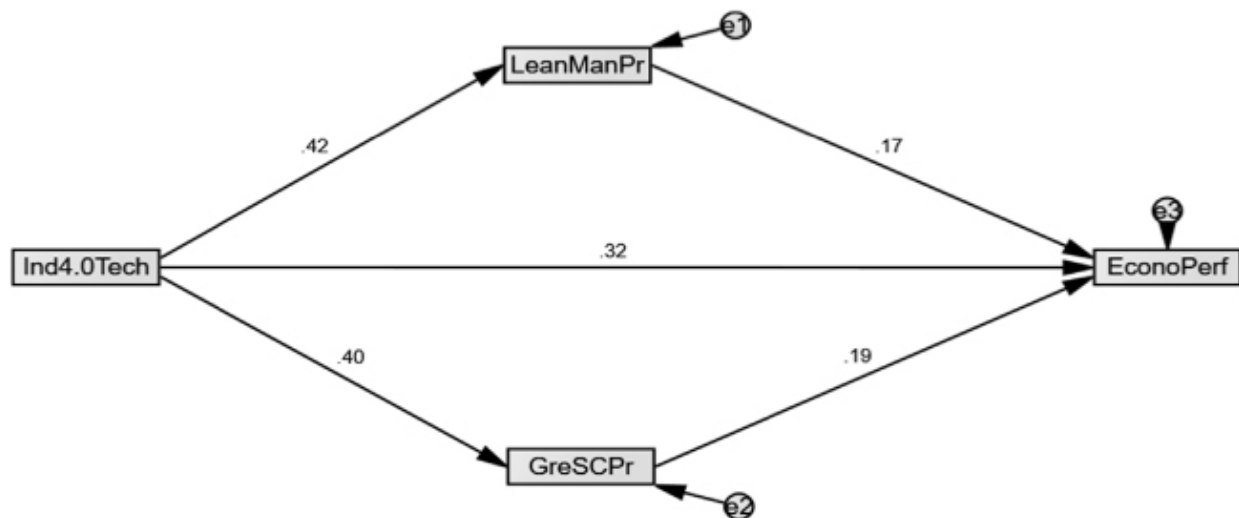


Fig. 4. SEM.

Discussion and Conclusion

Discussion

Industry 4.0 technology is a leading trend toward automation and also exchange of informative data in manufacturing processes and technologies, which comprise the IoT, cyber-physical systems, artificial intelligence, cloud computing and cognitive computing. This technology has provided many benefits to the garments industry, including enhanced efficiency and productivity, improved agility and flexibility and enhanced profitability. According to [Kelloway and Myers \(2019\)](#), it has also improved customer experience leading to sustainable environmental performance. The interconnections in industry 4.0 can lead to an improvement in environmental performance in garments companies, and due to this fact, the first hypothesis has been accepted.

LMP mainly include the elimination of wastage, mistake proofing, continuous improvement, level production and the detection of different defect with the help of automation ([Kumar, 2019](#)). These practices play a positive role in improving the concepts regarding industry 4.0 technology. It is proved through the research that LMP highly influence the trends of industry 4.0 technology and coax the garments industry to implement this technology in their products and

services. In this way, sustainable environmental performance is also ensured. Thus, the second hypothesis is also accepted.

Green supply chain management practices also have strong concerns regarding the influx of industry 4.0 technologies. These practices are derived from supply chain management practices, and their tenet is to eliminate and minimize wastages like emissions, solid waste, energy and hazardous chemicals. They mainly persuade the garments companies to reduce wastages and implement modern technologies in their operations, such as industry 4.0. Thus, environmental performance is enhanced. Hence, the third hypothesis is also accepted agreeably.

Conclusion

The determination behind the study is to examine the importance of industry 4.0 technologies in improving environmental performance and presentation of the garments industry in Thailand. The research also explores the mediating role played by green supply chain management practices and LMP in boosting the relationship between the environmental performance and the adoption of industry 4.0 technologies. The study comprises 411 participants, which include 170 females and 232 males. The principles of KMO and confirmatory factor analysis are used to evaluate results, and hypotheses are tested by using structural equation modeling.

The results show that the influx of industry 4.0 technologies has a positive influence on the environmental performance of the garments industry in Thailand. LMP also support the relationship between the industry 4.0 technology and the environmental performance. Lastly, green supply management practices also support industry 4.0 technology trends, which improve environmental performance.

Implications and Limitations

The research will provide useful implications for the garments industry in Thailand. It will help the industry to explore industry 4.0 technology and its impact on environmental performance. The study has implemented useful tests and approaches to identify the

significance of green and LMP. There are many opportunities for the garments industry of Thailand to use the prospects of this research in their operations and enhance their environmental performance. The outcomes will help the providers of green supply chain management practices in boosting their functions.

Future researchers and scholars can broaden and extend the approach of the study as it is limited to the garments industry of Thailand. The perspectives of this research are limited to only the positive facets of industry 4.0 technology. The present aspects can be studied in the garments industry of other countries as well. Other researchers can conduct longitudinal and exploratory studies instead of correlational research using the same variables and some new variables. Besides, they can also use modified sample sizing and data analysis methods to get improved results and broader perspectives related to the variables. The research can be conducted in countries and regions other than Thailand as well.

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Chapter 16

The Interaction Among Organizational Flexibility, Competitive Strategy and Competitive Advantage: A Path Analytic Study¹

Yeşim Koçyiğit and Akif Tabak

Abstract

Decisions about strategies to be implemented and organization design are important in having competitive advantage for companies. Although this statement widely takes place in management literature, very little research has been done on this subject in Turkey. The aim of this study is to examine the mediating role of competitive strategies in the impact of organizational flexibility on competitive advantage in order to extend about empirical literature of the structure-strategy-competitive advantage paradigm based on the resource-based view (RBV). In this context, the data were collected from the top 500 industrial enterprises in Turkey. The data were tested via structural equation model. The findings of our research show that the direct effect of organizational flexibility on competitive superiority was marginally significant and when three different

competitive strategies were included between these two variables in the model, the marginally significant effect of organizational flexibility on the competitive advantage disappeared. In other words, the full mediating role of applied competition strategies in the effect of organizational flexibility on competitive advantage has been identified. Our findings are similar to the findings of other studies that consider organizational structure as a valuable resource and examine the relationship between the structure and the strategy according to the RBV.

Keywords: Organizationl structure; organizational flexibility; competitive strategies; competitive advantage; resource-based view; structure-strategy relationship

Introduction

Nowadays, organizational structure is an unignorable variable to apply competitive strategies that are determined by considering situational factors and to gain a competitive advantage over enterprises' competitors. It is not enough just to formulate the strategies. To be able to implement, the formulated strategies are also very important. The resource-based view (RBV) emphasizes the internal qualities and considers the organizational structure as a source of competitive advantage rather than being an element in the implementation of the strategy of the organization ([Smirat, Abdullah, & Sharif, 2014](#), p. 2). Some studies emphasize the importance of organizational structure as a source in the implementation of strategies ([Pertusa-Ortega, Claver-Corte's, & Molina-Azorin, 2008](#); [Pertusa-Ortega, Molina-Azorin, & Claver-Corte's, 2010](#); [Peters, 2011](#); [Smirat et al., 2014](#); [Tanriverdi, Sekman, & Yilmaz, 2019](#)). Organizational structure is an internal factor that enterprises enable to be innovative and to implement the new ideas. As [Drucker's \(1974, p. 52\)](#) noted, who is one of the leading futurists in management, organizational structure is a means to reach the goals and objectives

of the institution. Because, organizational structure indicates to employees the areas of responsibility, what level they are in, who they will take orders from, who will work with them, and the authorities they have. Organizational structure also determines that an enterprise can move how quickly and flexibly. Increasing competition lately has forced enterprises to act quickly and flexibly. As markets have become more customer-focused, successful companies – from General Motors to IBM – have transformed themselves from the rigid bureaucracies of past eras to become much more responsive to their markets ([Simons, 2005](#), p. 9). Because modern enterprises operate in a rapidly changing environment that is hypercompetitive and turbulent, where customer preferences are volatile, and technology is transforming scenarios. Flexibility plays a key role in organizations to adjust the resources to environmental changes and sustain their existence in such a turbulent and unpredictable environment. In terms of competitive strategy, the firm can use its internal coordination mechanisms as a valuable resource to achieve a competitive advantage ([Akkaya & Tabak, 2017](#), p. 119; [Pertusa-Ortega et al., 2010](#), pp. 1282–1283). Organizational flexibility enables using tools such as organization, technology, human, and innovation, successfully developing new products and even outperforming and integrating external and well as internal business competencies for strategy practices to achieve a competitive advantage ([Akkaya & Tabak, 2020](#), p. 1; [Broekaert, Andries, & Debackere, 2016](#); [Lal & Bharadwaj, 2016](#); [Palanisamy & Sushill, 2003](#)). The intense competition in all sectors in recent years has necessitated adaptation to the environment by adopting different strategies and organizational structures rather than the traditional management activities of the enterprises. In this context, this chapter aims to determine whether organizational flexibility as a valuable resource influences the use of competitive strategies and competitive advantage in practice, and if so, to what extent.

These considerations and situations lead us to focus on organizational flexibility and competitive strategies for competitive advantage. This chapter has two main aims:

- To determine the organizational structures of enterprises in the industry sector of Turkey by going ahead from the level of organizational flexibility, the competition strategies they use, and the level of competitive advantage.
- Specific to Turkey's top 500 industrial enterprises is to reveal relationships and interactions between organizational flexibility, used competitive strategies and competitive advantage.

For the first purpose, descriptive statistics have been calculated by using SPSS Statistics software package and the results have been interpreted. For the second purpose, by handling the competitive advantage as a dependent variable, a model that tries to find out the mediating role of competitive strategies in the impact of organizational flexibility on this dependent variable has been developed. The model has been tested and evaluated within the framework of data obtained from senior management of Turkey's largest 500 industrial enterprises announced by the Istanbul Chamber of Industry.

The remainder of the chapter contains five sections: the literature section presenting a conceptual framework and reviewing on previous research to emphasize the model and hypotheses; subsequently, the purpose and hypothesis are described; the sample and measures are explained in the methodology section; the results are presented and interpreted; and conclusions are discussed and implications are presented in the final section.

Review of Literature

Organizational structure is the structure formed by authority, responsibility, and interrelation that form in the direction of the enterprise's goals. The structure of the organization includes formal communication channels through which information, orders, decision, and report flow. The structure that emerges at the end of the organizing process is called the formal organizational structure (Özgener, 2003, p. 143). The notion of being "organic" or "mechanical" put forwarded by Burns and Stalker to describe organizational structure is that the formality and centrality of an

organization are reflected in its structure. According to this idea, organizations can be ordered in two opposite-point planes. One of these points represents mechanical organizations that relations, the distributions of power and information take place in the vertical plane, highly centralized, and formalized. The other represents organic organizations that stand out decentralization, low formalization, horizontal relations, power, and information distribution. Organic organizations are flexible because they rely on rules less. It also allows less formality, greater clarity. This situation enables their employees to innovate (Wasti & Fiş, 2010, p. 14). These characteristics of organic organizations enable both high horizontal communication, and individual learning and organizational learning, which support to gain well-qualified staff with knowledge and skills. They also provide to use effectively all of the sources of the enterprise. By this means, enterprises can cope with all of the environmental changes. The notion of organic organization put forward by Burns and Stalker (1961) in this study will be considered as flexible organizations, which are a valuable resource rather than an element in implementing strategies. Organizational flexibility can be defined as firms being flexibility not only in a discipline or department but also in all disciplines or departments. It is defined as the ability to respond unpredicted change and the capacity to determine the level of firms may exercise in uncertain environments (Phillips & Wright, 2009, p. 1072). Teece, Pisano, & Shuen (1997) have handled the organizational flexibility as a combination of organizational and managerial capabilities. These capabilities allow organizations to adapt to environmental changes. It can be said that organic systems as flexible organizations enable enterprises to adapt to all changes that can change their strategies. Because these organizations are equipped with better skills by enabling employees to demonstrate their creativity and different abilities, they can use these skills to adapt to a dynamic, uncertain technology, and complex environment and gain a competitive advantage (Akkaya, 2019, p. 137; Koçyiğit & Akkaya, 2020, pp. 112–113).

There are some studies in the literature (Pertusa-Ortega et al., 2008; Pertusa-Ortega et al., 2010; Peters, 2011; Smirat et al., 2014; Tanrıverdi et al., 2019), which focus on mediating role of applied

competition strategies in the effect of organizational flexibility as a valuable resource on competitive advantage apart from studies that by using tools such as organization, technology, human and innovation, integrating external and well as internal business competencies for strategy practices, organizational flexibility enables to develop new products and services and gain competitive advantage (Broekaert et al., 2016; Lal & Bharadwaj, 2016; Palanisamy & Sushill, 2003). These arguments presented for the RBV approach previously lead also us to suggest that used competition strategies can play a mediating role in the effect of organizational flexibility on competitive advantage. As stated previously, this chapter tries to extend about empirical literature of the structure-strategy-competitive advantage paradigm based on the RBV. In this context, in order to examine both the effects of the organizational flexibility as a valuable resource on each competitive strategy and the effects of each competitive strategy on the competitive advantage, the model in the research method section and hypotheses over the model has been developed.

Research Method

The issue of this research, by dealing with the competitive advantage as a dependent variable, is to create a model that competitive strategies and organizational flexibility influence this dependent variable. The model will be tested and evaluated within the framework of data obtained from senior management of Turkey's largest 500 industrial enterprises announced by the Istanbul Chamber of Industry.

The research has two main objectives:

- By going ahead from the level of organizational flexibility, is to determine the level of the flexibilities of organizational structures of enterprises in Turkey industry sector, the competition strategies they use, and the level of competitive advantage.
- Specific to Turkey's top 500 industrial enterprises, is to reveal relationships and interactions between organizational flexibility as a valuable source, used competitive strategies, and competitive advantage.

The main focus of the study is to address the relationship between organizational flexibility and competitive strategies within the framework of resource-based perspectives and is to reveal the role of organizational flexibility as a fundamental resource and capability to support competitive strategies used for competitive advantage.

In the theoretical framework, it has been stated that competitive strategies should be determined according to opportunities and threats in today's complex and rapidly changing environment and that organizational flexibility should handle as a valuable resource to implement the strategies determined. So organizational performance will be positively affected, and relative superiority will be achieved. The research model has been established in line with these explanations, and hypotheses have been developed on the model. Fig. 1 has shown the research model.

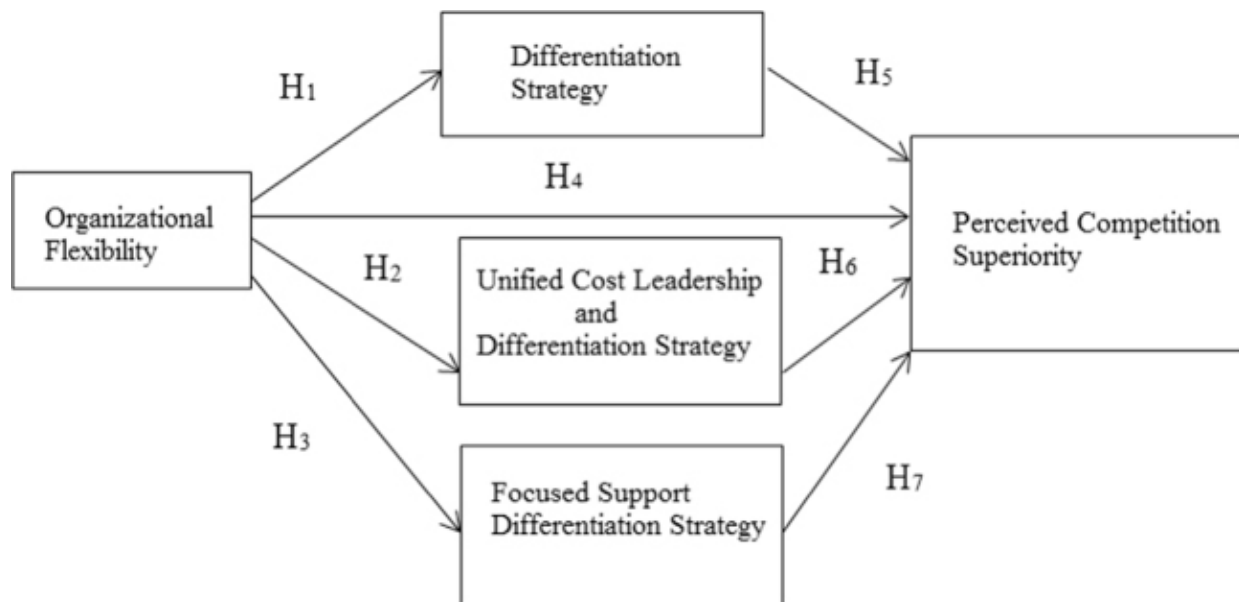


Fig. 1. Research Hypotheses.

H_1 : Organizational flexibility affects the application of the differentiation strategy.

H_2 : Organizational flexibility affects the application of unified cost leadership and differentiation strategies.

H_3 : Organizational flexibility affects the implementation of the focused support differentiation strategy.

H₄: Organizational flexibility affects the perceived competitive advantage.

H₅: The differentiation strategy affects the perceived competitive advantage.

H₆: Unified cost leadership and diversification strategy affect perceived competitive advantage.

H₇: The focused support differentiation strategy affects the perceived competitive advantage.

H₈: Competitive strategies have mediating roles in the effect of organizational flexibility on competitive advantage.

Path analysis was made by the structural equation model (SEM) to test the hypothesis of the research, reveal the mutual relations between the variables, and identify direct and indirect effects on them.

Sample and Data

The universe of the research is composed of Turkey's largest 500 industrial enterprises announced by the Istanbul Chamber of Industry. A simple random sampling method was chosen, giving all the firms that constitute the universe an opportunity to equally enter the sample. A total of 201 completed surveys were obtained from these 500 enterprises. This sample forms 40% of the mainframe.

Measurement of Variables

The questionnaire form, which is prepared to assess organizational flexibility levels and the competitive strategies they applied and to analyze the impact of these different competitive strategy components on the competitive advantage of Turkey's largest 500 industrial enterprises announced by the Istanbul Chamber of Industry, consists of four parts. The first part of the questionnaire is about reaching general information such as age, gender, educational status, organizational duties, the sector in which the business operates, the basic product character, and the number of employees. In the second

part, it is aimed to measure the organizational flexibility of the companies which participate in the survey with the rating scale consisting of 12 phrases. Also, it is aimed to measure by 22 phrases the competition strategies that they applied in the third part, and the last part aims to measure the perception of competitive advantage by 6 evaluation phrases. From the second part of the prepared questionnaire to the end of the questionnaire, the quinary rating was used in each question. Accordingly, the meaning of the rating used in the second and third part of the questionnaire is as follows:

1. Strongly disagree
2. Disagree
3. Neutral
4. Agree
5. Strongly agree

The rating used in the fourth section is as follows:

1. Lowest-in-sector
2. Below sector average
3. Sector average
4. above sector average
5. highest-in-sector

To avoid non-objective responding, 2nd, 7th, and 10th questions in the second part and 4th, 8th, 13th, and 20th questions in the third part were prepared negatively, opposite of the other questions prepared with the positive phrases. The prepared questionnaire has been applied via İkon Research Company to the administrators of Turkey's largest 500 industrial enterprises announced by the Istanbul Chamber of Industry. The questionnaire was not applied until ICI declared the list of the top 500 industrial enterprises. Fully filled out 201 surveys were obtained from operations on the list of the largest 500 industrial enterprises of the year 2014 published in July 2015. After the questionnaires were entered into the SPSS, the values of the answers to the opposite direction questions were changed from the Transform → Recode into Same Variables menu. This menu

performed a process on the same variable, aiming to replace the new encoding with the old one (Yazıcıoğlu & Erdoğan, 2014, p. 216). While determining the situation of competitive advantage, the direction of the answers was changed so that the question of “what is the level of your costs according to the competitors?” would not adversely affect the competitive advantage score. Exploratory factor analysis and confirmatory factor analysis were performed to test the reliability and validity of the scale, and Cronbach alpha coefficients were calculated for each variable and dimension.

Organizational Flexibility Variables

Statements for determining organizational flexibility of questionnaire, questionnaires of previous studies for determining organizational structure, and information in the literature were placed together by the researcher (Cosh, Fu, & Hughes, 2012, p. 302; Efil, 2009, p. 106; Ford, 1981, pp. 279–281). Organizational flexibility variable was evaluated as a single dimension. In the literature review, it was seen that the elasticity dimension of solid (mechanical) and flexible (organic) organizations did not move independently of each other and behaved like two poles which were opposite to each other (Wasti & Fiş, 2010, p. 14). Considering the arithmetic average of the statements that measure organizational flexibility, those with high values will be evaluated as flexible (organic) and those with low values will be evaluated as a solid (mechanical) organization. Exploratory factor analysis and confirmatory factor analysis of statements that are included in the questionnaire were conducted to measure the level of organizational flexibility. The questionnaire used in the survey is given in Annex 1. The analysis results are shown in the following tables and figures.

Table 1 presents the results of factor analysis for organizational flexibility dimension. The 0.759 value of **KMO** indicates that the sample is good enough for factor analysis. The significance of the p-value of the Bartlettsglobality test ($p = 0000$) indicates that the matrix formed by correlations is significant for factor analysis and that factor analysis can be performed. The factor loads of the questionnaires and the dependent items were examined by looking at the

Component Matrix table in the results of the factor analysis. When questions for determining organizational flexibility in factor analysis were analyzed with one-factor restriction, according to the results, questions 2, 7, and 10 were removed from the analysis as they had factor loads less than 30. For each question removal process, the rotating process was made with a varimax technique. The variables observed in this way are collected in one dimension. The factor loads and reliability values for this dimension are shown in Table 1. The total variance ratio explained by the organizational flexibility measurement tool is 31,666% and the α (Cronbach Alpha) coefficient is 0.71 as a result of reliability analysis.

Table 1. Factor Analysis for Organizational Flexibility Dimension.

Organizational Flexibility	KMO	Bartlett	Sig	df
	0.759	280.722	0.000	36
Factor Load				
Questions			Reliability Value (Cronbach Alpha)	Average
os5	0.679		0.709	3.76
os4	0.651			3.77
os9	0.648			4.21
os3	0.629			3.79
os11	0.581			3.50
os1	0.500			3.74
os6	0.469			3.68
os8	0.445			3.60
os12	0.379			3.23
Explained Variance		Total Variance Change		
31.666		31.666		

Fig. 2. shows the confirmatory factor model for organizational flexibility. The first-level factor model of the organizational flexibility

scale was tested by using the AMOS program. With five points Likert scale, the covariance matrix was constructed by using the maximum likelihood calculation method since the collected data from 194 subjects showed a normal distribution of the generated variable averages. For the organizational flexibility scale, the p -value of parameter predictive values of each of 9 items is statistically significant ($p < 0.05$). Table 2 presents the compliance indicators of organizational flexibility model. As seen in Table 2, the chi-square statistics obtained for the organizational flexibility model are significant ($p = 0.037$). However, as stated previously, since the chi-square value is very sensitive to the sample size, it is not sufficient to evaluate the harmony between the data and the model. For this reason, it is necessary to consider other harmonization criteria. When the ratio of the value of χ^2 to the degrees of freedom is examined, (χ^2/sd) this ratio is 1,547 and this value is below 2, indicating that there is a harmonization at a good level. According to the other goodness-of-fit indexes used in the valuation of the harmony between data and model, GFI value was 0.957, AGFI 0.926, CFI 0.929 and close to 1, RMSEA value was 0.053, smaller than 0.08 and very close to 0.05, indicating a good harmony between the model and the data set; NFI value was 0.830, also indicating a reasonable harmony. These results show that the harmony between this data set and the identified organizational flexibility measurement model is good.

Model	χ^2	Sd	χ^2 / sd	GFI	AGFI	CFI	RMSEA	RMR
Organizational Flexibility	40.224	26	1.547	0.957	0.926	0.929	0.053	0.039
	P							
	0.037							

Competitive Strategy Variables

Twenty two statements on the scale used in this study and to determine the competitive strategies applied by the enterprises were prepared by choosing the related statements on the scale of the previous studies ([Arıcı, 2008](#), p. 130; [Nayyar, 1993](#), p. 1660). Explanatory factor analysis and confirmatory factor analysis of the statement were used to determine the competitive strategies that the enterprises applied. The analysis results are shown in the following tables and figures.

Table 3. Factor Analysis for Competition Strategies.

Applied Competition Strategies	KMO	Bartlett		Sig	df
	0.867	988.222		0.000	136
Factor Load					
Questions	Differentiation. Str. (Mainly Quality, Image, Support Difference)	Unified Cost Leadership and Difference	Focused Support Dif-ferentiation	Reliability Value (Cronbach Alpha)	Avarege
cs18	0.788			0.861	4.22
cs16	0.774				4.01
cs10	0.721				4.35
cs15	0.719				4.31
cs12	0.703				4.23
cs19	0.671				4.24
cs17	0.651				4.20
cs11	0.431				4.26
cs3		0.651		0.667	3.55
cs13		0.611			3.81
cs4		0.600			3.86
cs20		0.558			4.04
cs9		0.511			3.85
cs5		0.476			4.03
cs21			0.776	0.561	3.55
cs22			0.748		3.74
cs14			0.571		3.97
Total				0.828	
Explained Variance				Total Variance Change	
	31.490	8.972	8.579	49.042	

[Table 3](#) presents the results of the factor analysis for competition strategies. When the factor analysis was conducted for questions to determine the competitive strategies used, it was seen that the value of **BMD** (KMO) was 0.867. This suggests that the sample is reasonably good enough for factor analysis. The significance of the p -value of the Bartlett's globality test ($p = 0000$) indicates that the matrix formed by correlations is significant for factor analysis and that it is possible to perform factor analysis. In the results of the factor analysis, the factor loads of the questionnaires and the overlapped items were examined by looking at the Pattern Matrix table. Firstly, the 8th question was removed from the analysis, because it has overlapped loads and the difference between the loads is less than 0.10. When the table in which the common variance values of the items are included is examined, it is seen that the contribution of the 7th question to the explained total variance is little (less than 0.4) and this item was removed from the analysis. As the reliability coefficient of the two questions (cost leadership) collected in the 4th dimension (cost leadership) was low, the effect of the 1st, 2nd, and 6th questions on the total variance explained was reduced. For this reason, they were omitted from the analysis one by one. For each question removal process, the rotating process was made with oblimin technique. As a result, the use of competition strategies was divided into three different factors. The factor loadings and reliability values for each dimension are shown in [Table 3](#). The total variance ratio explained by the assessment instrument used for measuring applied competitive strategies is 49.042%.

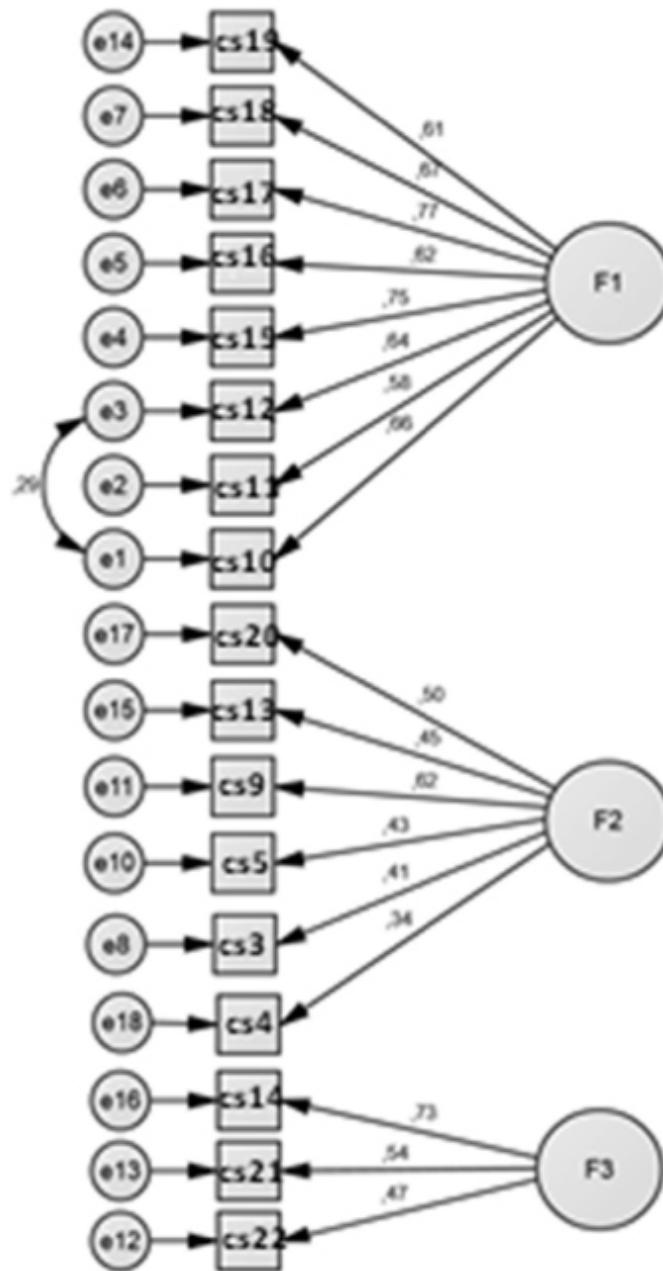


Fig. 3. First Level Confirmatory Factor Model for Competitive Strategies Used.

F1: Differentiation Strategy (Mainly Quality, Image, Support and Design Differentiation)

F2: Unified Cost Leadership and Differentiation Strategy

F3: Focused Support Differentiation Strategy.

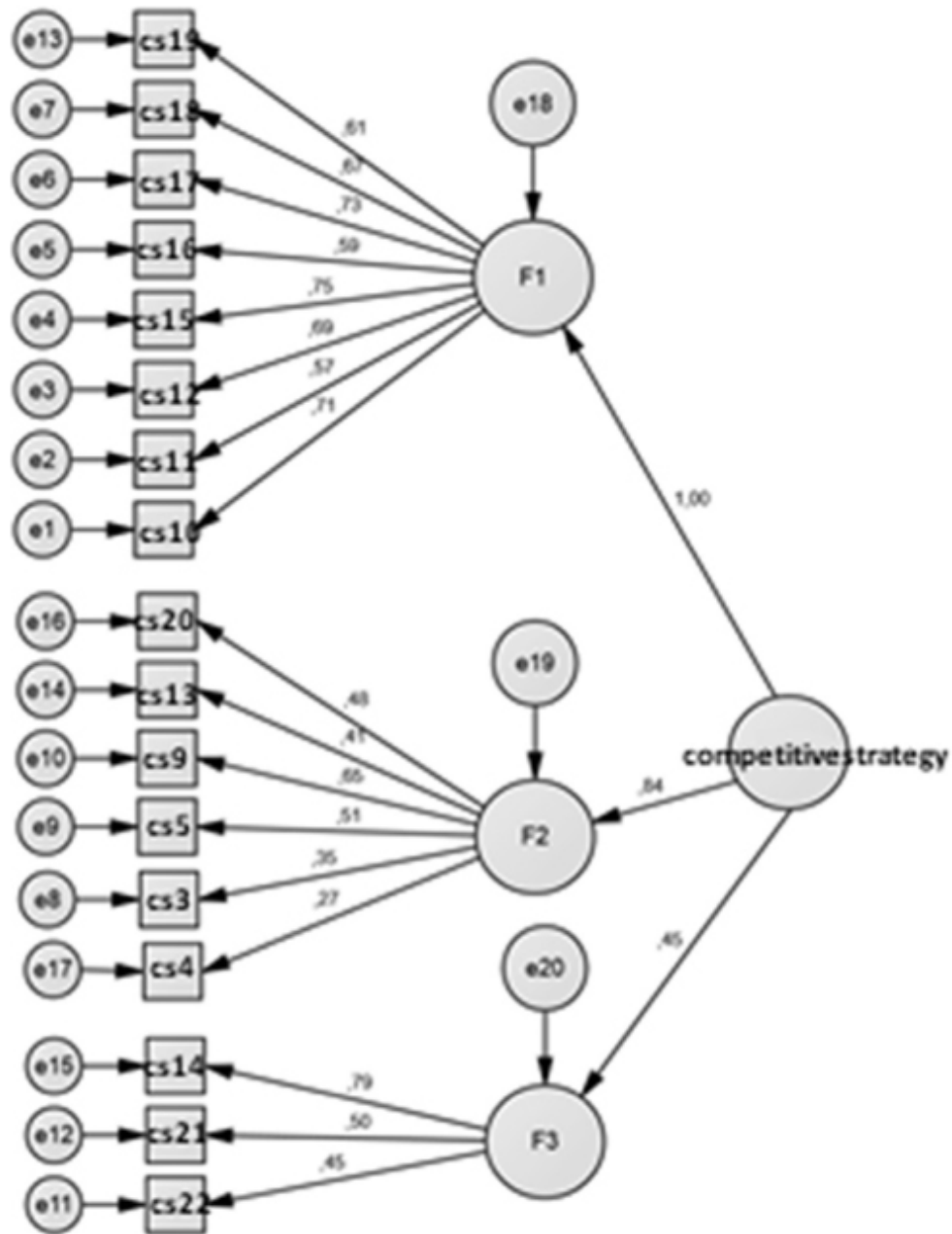


Fig. 4. Second Level Confirmatory Factor Model for Competitive Strategies Used.

F1: Differentiation Strategy (Mainly Quality, Image, Support and Design Differentiation)

F2: Unified Cost Leadership and Differentiation Strategy

F3: Focused Support Differentiation Strategy.

Fig. 3 shows the first level confirmatory factor model for competitive strategies used. It has been determined that the

observed variables for determining the competitive strategies used by the exploratory factor analysis are collected under multiple and unconnected factors; constituted first level unrelated multi-factor model was also tested by confirmatory factor analysis. For each of the three sub-dimensions, the p -value of the parameter estimation values for each of the 17 items is statistically significant ($p < 0.05$). Table 4 presents the coincident indicators of both the unrelated three-factor model and the second level factor model used in competitive strategies. As can be seen in Table 4, after covariance drawn for the 10th and the 12th questions that measure competition strategies used in the direction of the recommendation of improvement indices, the chi-square statistics obtained for the unrelated three-factor model are significant ($p = 0.000$). When the ratio of the χ^2 value to the degrees of freedom is examined for the Unrelated-Three Factor Model, (χ^2/sd) this ratio is 2.151 and the value shows that there is a harmony at a good level as it is small from 5 and very close to 2. According to the other goodness-of-fit indexes used in the valuation of the harmony between data and model, GFI value was 0.879, AGFI 0.844, CFI 0.835, and close to 1, RMSEA value was 0.077 and smaller than 0.08, indicating a reasonable harmony between the model and the data set.

Table 4. Coincident Indicators of Factor Model Used in Competitive Strategies.

Model	χ^2	sd	χ^2 / sd	GFI	AGFI	CFI	RMSEA	RMR
Unrelated Three-Factor Model	253.796	118	2.151	0.879	0.844	0.835	0.077	0.095
	$\overline{p}$							
	0.000							
Second Level Factor Model	χ^2	116	1.284	0.920	0.895	0.960	0.038	0.031
	$\overline{148.900}$							
	$\overline{p}$							
	0.021							

Fig. 4 shows the second level confirmatory factor model for competitive strategies used. The second-order multiple model is defined as a model that the observed variables are grouped under more than one and unrelated factors, that these factors also are gathered under a broader and more inclusive factor (Meydan & Şeşen, 2015, p. 23). When the goodness of fit of second-order multiple model has been examined in Table 4, it has been seen that the ratio of χ^2 to the degree of freedom (df) is 1.284, GFI is 0.920, AGFI is 0.895, CFI is 0.960, and RMSEA is 0.038.

These values indicate that the model and the data set have good harmony. The unrelated model will be included in the research model to see the effect of organizational flexibility on each competition strategy separately.

Competitive Advantage Variables

In measuring firm performance or competitive advantage, many authors have preferred subjective measures rather than financial measures (objective) such as profitability and return rates (Allen & Helms, 2006; Arıcı, 2008; Chen, Lai, & Wen, 2006; Lukas, Tan, & Hult, 2001; Pertusa-Ortega et al., 2008; Pertusa-Ortega et al., 2010; Powell & Dent-Micallef, 1997; Venkatraman & Ramanujam, 1986). Pertusa-Ortega et al. (2010) stated that objective measures could only reveal differences in firm performance because of the sector, not the actual differences between the firms and that various authors argued that subjective measurements were sufficient instead of objective measures in a multi-sectoral study (p. 1289). Yet, Venkatraman & Ramanujam (1986) pointed out that the use of secondary resources could be appropriate to functionalize financial performance in the studies focused on firm-level strategies versus strategic business units (p. 807). Subjective measurements are widely used in organizational research and preferred to financial tables. Because firms may have adopted accounting rules that vary in areas such as inventory valuation, depreciation, and officers' salaries, Allen and Helms (2006) determined firm performance based on performance perceptions of employees according to their competitors to reduce the sector's potential impact on performance and

generalize the findings. In this study, six of the eight items in the competitive advantage scale used by [Chen et al. \(2006\)](#) were used for the statements to determine competitive advantage perceived by the managers, and similar expressions like “AR-GE and innovation skills according to the competitors” and “corporate image status” were not included in the competitive advantage scale since they were included in the scale of competition strategies, and it was not desirable that the expressions would be of the same type.

The explanatory factor analysis and confirmatory factor analysis of statements to identify the competitive superiority perceptions of business managers participating in the survey were conducted. The analysis results are shown in the following tables and figures.

Table 5. Factor Analysis for Perceived Competitive Advantage.

Perceived Competitive Advantage	KMO	Bartlett	Sig	df
	0.811	282.520	0.000	10
Factor Load				
Questions	Reliability Value (Cronbach Alpha)		Avarage	
ca1	0.816		0.80	4.12
ca6	0.806			4.04
ca4	0.709			4.33
ca3	0.703			3.86
ca5	0.670			3.58
Explained Variance		Total Variance Change		
55.254		55.254		

[Table 5](#) presents the results of factor analysis for perceived competitive advantage. For the perceived competitive advantage variable, the KMO value was 0.811 and the p -value of the Bartlettsglobality test was significant ($p = 0.000$). The factor loads of the questionnaires and the overlapped items were examined by

looking at the Component Matrix table in the results of the factor analysis. In factor analysis, questions for determining the level of perceived competitive advantage were analyzed with one-factor restriction. Question 2 was removed from the analysis because it had a factor load of less than 30. This question was removed, and the rotation process was performed with the varimax technique. The variables observed in this way were collected in one dimension. The factor loads and reliability values for this dimension are shown in [Table 5](#). The total variance ratio explained by the assessment instrument used for measuring the perceived competitive advantage was 55,254%. In the reliability analysis, α was 0.80. This value indicates that the scale is highly reliable. The α value of the competitive advantage scale of [Chen et al. \(2006\)](#) was found 0.8006.

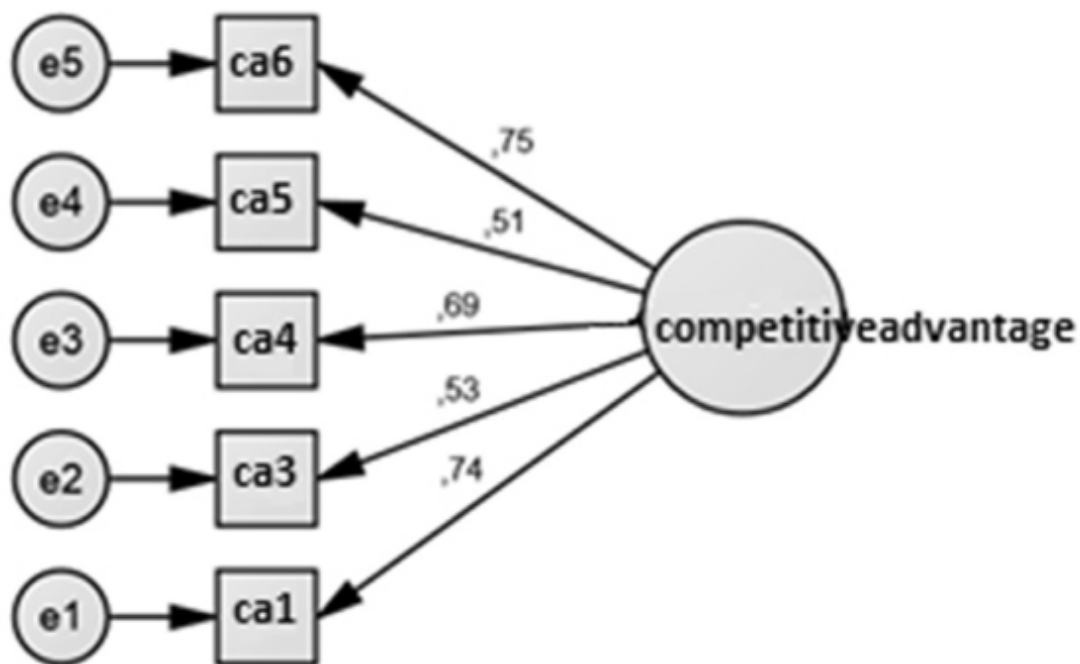


Fig. 5. Confirmatory Factor Model for Perceived Competitive Advantage.

Table 6. Coincident Indicators of Perceived Competitive Advantage Factor Model.

Model	χ^2	sd	χ^2 / sd	GFI	AGFI	CFI	NFI	RMSEA	RMR
Perceived	9.026	5	1.805	0.982	0.946	0.983	0.964	0.065	0.023
Competitive	p								
Advantage	0.108								

Fig. 5 and Table 6, respectively, show the confirmatory factor model for perceived competitive advantage and the coincident indicators of perceived competitive advantage factor model. As seen in Table 6, the level of significance of the chi-square statistic obtained for the perceived competitive advantage model was greater than 0.05 ($p = 0.108$). When the ratio of χ^2 value to the degree of freedom is examined, this ratio is 1.805 and since this value is below 2, it shows a good harmony. According to the other goodness-of-fit indexes used in the valuation of the harmony between data and model, GFI value was 0.982, AGFI 0.946, CFI 0.983. NFI value was 0.964 and very close to 1, RMSEA value was 0.065 and smaller than 0.08, indicating a good harmony between the model and the data set. These results show that the harmony between the data set and the competitive advantage measurement model is good.

Analysis and Findings

The Mahalanobis distance coefficient and the probabilities of these coefficients in the χ^2 distribution were calculated in SPSS to examine multiple normal distributions for the organizational flexibility, differentiation, unified cost leadership, and differentiation, focused support differentiation and perceived competitive advantage variables determined by exploratory factor analysis. Seven surveys the probability values of which were less than $\alpha = 0.001$ were removed from the data set since they have extreme values. 194 questionnaires were used in the analyses. Afterward, Kolmogorov-Smirnov and Shapiro-Wilk tests were performed for organizational flexibility, differentiation, unified cost leadership, and differentiation, focused support differentiation and competitive advantage variables, skewness and kurtosis values were calculated and the histogram and

Q-Q plot graphs were examined and it was concluded that the data showed normal distribution. By examining, averages within the descriptive statistics of variables, standard deviations, and range of points, the levels existing at the level of dimensions were determined. The average, minimum, and maximum values and standard deviations that pertain to variables used in the study are shown in the table below.

Table 7. The Descriptive Statistics Related to Variables.

	Organiza- tional Flexibility	Differen- tiation	United Cost Leadership and Differentiation	Focused Support Differentiation	Perceived Competitive Advantage
Average	3.72	4.23	3.87	3.77	4.01
Standard Deviation	0.45	0.47	0.50	0.69	0.59
Minimum	2.44	3.13	2.33	2.00	2.80
Maximum	5.00	5.00	5.00	5.00	5.00

Table 7 presents the descriptive statistics related to variables. The arithmetic average ranges in the five-point Likert are interpreted as: Very Low: 1–1.79, Low: 1.80–2.59, Medium: 2.60–3.39, High: 3.40–4.19, and Very high: 4.20–5.00 (Özdamar, 2003, p. 32). Accordingly, looking at the values in Table 7, we can say that the averages of the organizational flexibility (average = 3.72), unified cost leadership and differentiation strategy (average = 3.87), focused support differentiation strategy (average = 3.77), and perceived competitive advantage (average = 4.01) are high and also the average of differentiation strategy (average = 4.23) is very high.

To test H_1 , H_2 , H_3 , H_4 , H_5 , H_6 , H_7 , and H_8 hypotheses which were improved to reveal the direct effects on variance in variables, the power of arguments of independent variables on dependent variables and possible mediating roles in existing effects, a SEM was developed and the relations were tried to be analyzed in this model. For this purpose, a path analysis was made with the AMOS 21.0

program. The dimensions of the competitive strategies used while constructing the model are considered as independent variables from each other, and each sub-dimension was drawn as a latent variable in the model. Undoubtedly, the most fundamental reason for a demonstration in this way was to reveal separately the predictive situation of each competition strategy of organizational flexibility. For this reason, the first-level unrelated multifactor structure was included in the model for the competitive strategies used.

To determine the mediation relationship of competition strategies used between organizational flexibility and perceived competitive advantage, a path analysis was performed with AMOS 21.0 software. In the mediation test with SEM, first the direct model which the mediator variable is not included in is tested. At the next stage, it is checked whether the relationship between the dependent and independent variables has become insignificant by adding the mediator variable to this model (Gürbüz & Şahin, 2016, p. 351).

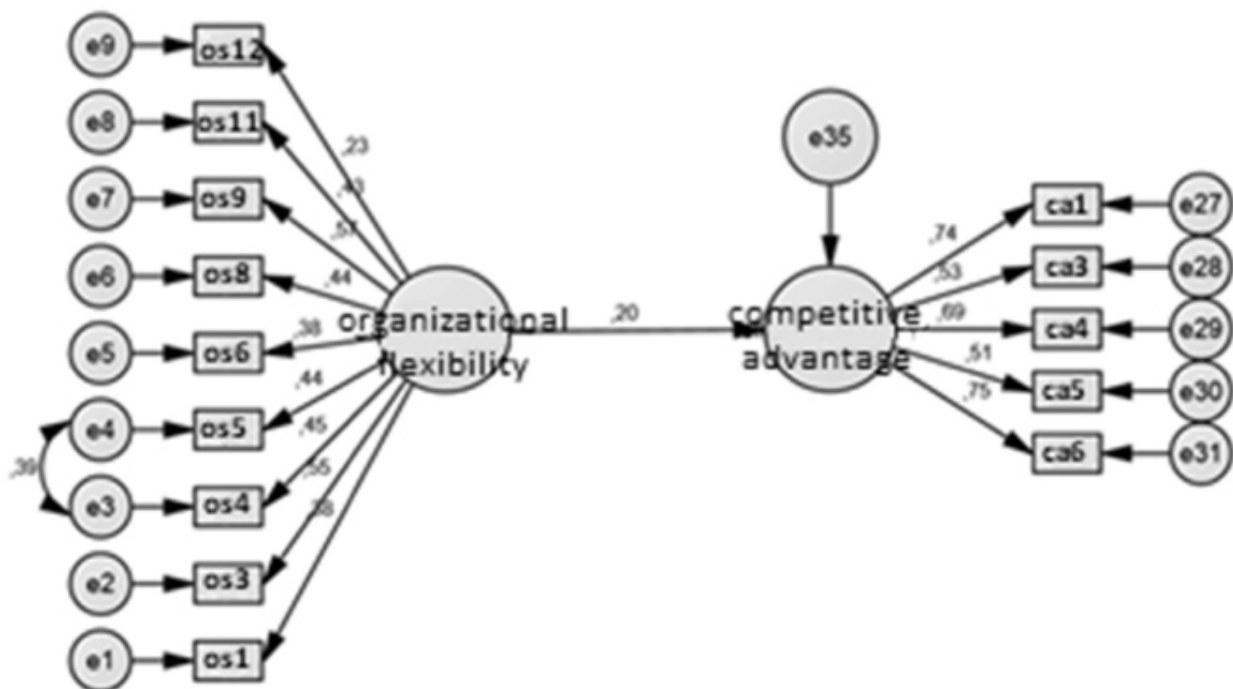


Fig. 6. Model of Impact on Perceived Competitive Advantages of Organizational Flexibility.

Fig. 6 shows the model of impact on perceived competitive advantages of organizational flexibility. It has been seen that the path coefficient value is marginally, in other words approximately, significant in the path analysis conducted to see whether organizational flexibility which is the independent variable influences the perceived competitive advantages which is the dependent variable (0.205; $p = 0.055$). In this model, the p -value was very close to 0.05. After examining 1535 articles, Pritschet, Powell, and Horne (2016) confirmed that researchers marginally denominated p values in the range from 0.05 to 0.18. However, they also stated that more than 90% of the marginally denominated p values were between 0.05 and 0.10. In this study, p values between 0.05 and 0.10 are considered to be marginally significant, depending on the majority.

Table 8. Impact Way Coincident Indicators of Organizational Flexibility on Perceived Competitive Advantages.

Model	χ^2	sd	χ^2 / sd	GFI	AGFI	CFI	RMSEA	RMR
Competitive Advantages	104.585	75	1.394	0.931	0.904	0.935	0.045	0.043
	p							
	0.014							

Table 8 presents the impact way coincident indicators of organizational flexibility on perceived competitive advantage. As seen in Table 8, the chi-square statistics obtained for the impact model on the perceived competitive advantage of organizational flexibility are significant ($p = 0.014$). The ratio of χ^2 value to the degree of freedom was 1.394, GFI 0.931, AGFI 0.904, CFI 0.935, and very close to 1, RMSEA value was 0.045, indicating a good harmony. According to these results, the harmony between the data set and this identified model is good.

The values between the variables after testing the model in which the mediator variables are included are shown in Fig. 7, and the

improvement indices were revised to improve the adaptation values of the model.

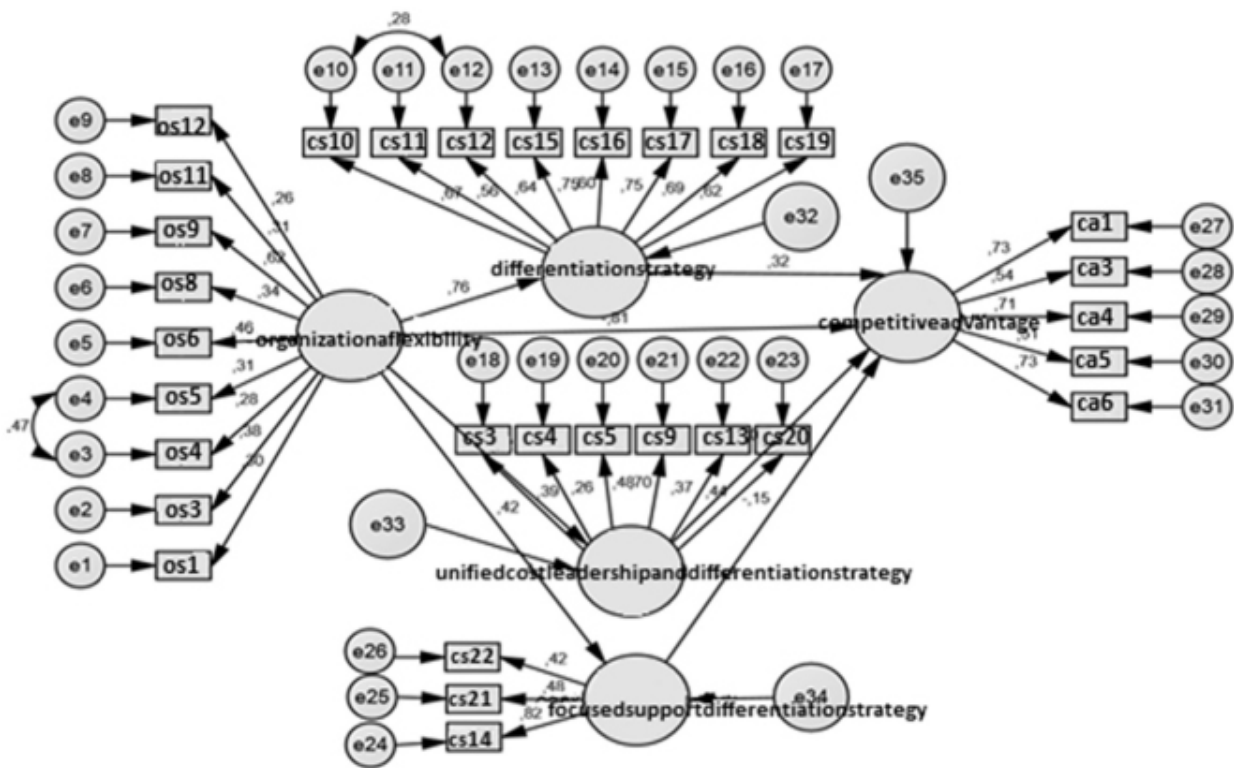


Fig. 7. Relations Between Variables in the Detailed Model.

Table 9. Regression Weights for the First Model.

Tested Way	Expectations	Standard Error	Critical Ratio	<i>p</i>
Focus support differentiation <--- flexibility of the organizational structure	0.420	0.393	2.999	0.000
Differentiation <---flexibility of organizational structure	0.758	0.374	3.560	0.003
Unified cost leadership and differentiation <---flexibility of organizational structure	0.923	0.420	3.072	0.002
Competitive advantage <---focused support differentiation	-0.145	0.095	-1.343	0.179
Competitive advantage <---differentiation	0.319	0.243	1.832	0.067
Competitive advantage <---Unified cost leadership and differentiation	1.020	1.478	1.210	0.226
Competitive advantage <---flexibility of organizational structure	-0.806	2.186	-0.904	0.366

Table 9 presents the regression weights for the first model. As seen in Table 9, the competitive advantage < --- focused support differentiation, competitive advantage < --- unified cost leadership and differentiation and competitive advantage < --- organizational flexibility ways are not statistically significant. It is seen that “competitive advantage < -- differentiation strategy” coefficient value is marginally, in other words, approximately significant (0.205; $p = 0.067$).

At this stage of analysis, the non-significant pathways must be removed from the model and a reanalysis must be performed. However, it is preferable to remove one by one instead of removing suddenly all of the insignificant paths in a model and by repeating the analysis after each one, the model harmony values and estimates should be reviewed (Meydan & Şeşen, 2015, p. 109). First of all, the competitive advantage which has the lowest significance level < ---

the organizational flexibility strategy was removed from the analysis. After this step, it was seen that the other low level of significance marginally reached to its significant level. As a result of these changes and analyzes, the model is shown in Fig. 8. Fig. 8 includes the model showing significant relations.

The values related to statistically significant ways are given in Table 10, and the adaptation values of the final state of the model are given in Table 11.

As a result of the path analysis in Table 10, it was seen that the values of path coefficient on the differentiation strategy of organizational flexibility (0.758; $p < 0.01$), the unified cost leadership and differentiation strategy (0.909; $p < 0.01$) and focused support differentiation strategies (0.424; $p < 0.01$) were significant. Besides, road coefficients on perceived competitive advantage of differentiation, unified cost leadership and differentiation, and focused support differentiation strategies were marginally significant ($p \leq 0.10$).



Fig. 8. Model Showing Significant Relations.

Table 10. Regression Weights Related to Significant Relations.

Hypothesis-Testedways	Expectation	<i>p</i>	Hypothesis Testing
H_1 – Differentiation strategy <--- org. flexibility	0.758	0.000	Supported
H_2 – Unified cost leadership and differentiation . <---org.flexibility	0.909	0.002	Supported
H_3 – Focused Support Differentiation <---org. flexibility	0.424	0.003	Supported
H_5 – Competitive advantage. <--- differentiation	0.232	0.090	Supported
H_6 – Competitive advantage. <---unified cost leadership and differentiation	0.308	0.059	Supported
H_7 – Competitive advantage <---focused support differentiation	–0.176	0.100	Supported

Table 11. The Consequences of Goodness of Fit of Significant Relationship Model.

Model	χ^2	sd	χ^2 / sd	GFI	AGFI	CFI	RMSEA	RMR
Significant	659.676	426	1.549	0.830	0.802	0.844	0.053	0.048
Relationship	<i>P</i>							
Models	0.000							

When the harmony indices of the significant relationship model are examined in [Table 11](#), it is seen that all the goodness of fit indexes are within accepted threshold values. According to the results of this analysis, in the effect of organizational flexibility on perceived competitive advantage, differentiation, focused support differentiation, and unified cost leadership and differentiation strategies create a full intermediary effect.

Results and Discussion

As a result of the analyses performed, it was seen that organizational flexibility affected competition strategies significantly. Organizational flexibility most influences unified cost leadership and differentiation strategy. Even one unit change in organizational flexibility causes also approximately one unit change in unified cost leadership and differentiation strategy ($\beta = 0.91$). From strategies implemented, the impact of focused support differentiation on perceived competitive advantage is negative ($\beta = -0.18$). Namely, the 1-unit increase in the focused support differentiation strategy leads to a decrease of 0.18 units in the competitive advantage. Medium-sized and large firms use focus-based strategies in combination with differentiation or cost leadership strategies. However, when the customers have different preferences and there are no rival firms in the niche market, the benefits which are obtained from focus strategies reach to the maximum level (From [Allen & Helms, 2006](#), p. 436; David, 2000). Reducing the effect of focused support differentiation strategy on competitive advantage might be that there are competitors in the niche market and the growth rate of the market is low. If so, it may be advisable for the companies to leave the focusing-based strategy.

Besides, it was observed that competition strategies used have a mediation effect on the effect of organizational flexibility on perceived competitive advantage. This result is similar to the results of studies that consider the situation approach from a different perspective and deal with the organizational structure within an internal resource and capability to increase the success of competitive strategies applied ([Hermanto & Yulianto, 2016](#); [Pertusa-Ortega et al., 2010](#)). Also, research by [Olson, Slater, and Hult \(2005\)](#) with over 200 top managers revealed that business strategies harmonized with organizational structure strongly impacted overall firm performance. [Wheelen and Hunger \(2005\)](#) showed that enterprises that change their strategies and structures according to the environment in their work perform better than those that do not. In that case, we can say that the organizational structure will increase the effectiveness and efficiency of competition strategies when considered as an element to strengthen human capital and organizational capital. If the strategy is based on the basic power of the organizational structure, it will be

accomplished and be able to achieve a competitive advantage. Managers should analyze their basic organizational strength before choosing their strategy. Analyzing the organizational structure in the first place will make it easier to produce proactive solutions for the obstacles that may arise during strategy implementations.

In their study of the effect of business strategies harmonized with organization structure impact on overall firm performance, [Olson et al. \(2005\)](#) approached their business strategies as behavioural strategies in the form of prospectors, analysts (their competitors and those following the market), low-cost advocates, and differentiated advocates. These are Miles and Snow's generic competitive strategies. In the following studies, the relationship between behavioural strategies of Turkish firms and organizational structure and the impact of these strategies on the financial and non-financial performance of the firms may be investigated. Besides, other than the organizational structure, other factors that may influence the implementation of the identified business strategies may also be a research topic.

Note

1. This study has been derived from the doctoral dissertation of Yeşim Koçyiğit, which has been named as “Firmaların Örgütsel Esnekliği, Kullandıkları Rekabet Stratejilerive Algılanan Rekabet Üstünlüğü Arasındaki Etkileşim: Türkiye’nin En Büyük 500 Sanayiİşletmesinde Bir Uygulama.”

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